



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

32-00-00-001

LANDING GEAR, GENERAL

Introduction

The landing gear supports the weight of the aircraft on the ground. It is also used to maneuver the aircraft on the ground.

General Description

The landing gear is electrically controlled, hydraulically operated and mechanically locked. The hydraulically operated nosewheel steering gives directional control during taxiing, take-off and landing. Each main wheel has hydraulically powered anti-skid brakes. There is also an emergency/park brake system.

The tricycle gear is a retractable dual wheel installation. The main gears retract aft into the nacelles and the nose gear retracts forward into the nose section. Gear doors completely enclose the landing gear when it is retracted and partially enclose the gear when it is extended. Flight compartment advisory lights show position of gear doors and downlocks. An aural warning sounds if the gear is not extended and the aircraft is in a landing configuration.

A Proximity Switch Electronics Unit (PSEU) monitors and controls the operation of the landing gear components.

An alternate landing gear extension method can be used to extend the gear if the primary extension method fails. There is also an alternate downlock verification system.

The nosewheel is steerable by a flight compartment hand control or by the rudder pedals. The main wheels are equipped with anti skid multiple disc brakes. The brakes can be controlled by the brake pedals or the EMERG BRAKE lever.

The Landing Gear System has the following subsystems:

- Main Landing Gear and Doors (32-10-00)
- Nose Landing Gear and Doors (32-20-00)
- Landing Gear Extension and Retraction (32-30-00)
- Wheels and Brakes (32-40-00)
- Steering System (32-50-00)
- Position and Warning (32-60-00)

Detailed Description

[Refer to Figure 1.](#)

The main gear retracts aft and the nose gear retracts forward and has steerable nosewheels. The landing gear is operated by the No. 2 hydraulic system and is controlled by the landing gear selector lever on the LANDING GEAR control panel. There is an alternate means of extension for the landing gear in the event that the primary system is not functioning. Advisory lights give extension and retraction fail/safe information.

Nose Gear Ground Lock Control Handle

[Refer to Figure 2.](#)

You can find the ground lock control on the left side of the external nose fuselage. The nose gear ground lock control handle has a



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push-button. When you push the push-button, the nose gear control handle releases from the fuselage, and the control handle moves out. You must pull the control handle out fully from the sprung position for it to turn.

When the control handle is in and flush with fuselage, the nose gear unlocked.

When the control handle is fully out and rotated clockwise, the nose gear is locked.

Nose Gear Lock Position Indication

[Refer to Figure 3.](#)

On Aircraft with ModSum 4Q428586 or SB84-32-45 or SB84-32-155 incorporated a ground lock position indication system is connected with the ground lock control system. When you engage the ground lock system, an advisory indication shows on the pilot's side panel if the aircraft buses have 28 V DC applied.

The ground lock position indication system has the components that follow:

- A cable switch assembly
- A lever assembly
- A striker plate
- An advisory assembly

The 28 V left main circuit breaker panel supplies power to the nose gear ground lock position indication system. A 7.5 amp circuit breaker (with the label LDG GEAR WOW2 + CONT) prevents damage to the system.

When you engage the ground lock system, the limit switch, (this is a part of the cable assembly) pushes the roller against the striker plate.

When the roller pushes on the limit switch, a circuit from 3261-P30 pin C through the limit switch to 3261-P30 pin A (ground) is completed. When the circuit is completed, the NOSE GEAR LOCK light on the pilot's side panel comes on.

Landing Gear Control Panel

[Refer to Figure 4.](#)

The landing gear position is controlled by the LANDING GEAR selector lever. The selector lever has a LOCK RELEASE button which must be pushed down before moving the LANDING GEAR selector lever. Moving the lever to the UP or DN position completes the electrical circuit to the hydraulic selector valve to retract or extend the gear. Gear proximity switches send signals to the Proximity Switch Electronic Unit (PSEU) that prevent gear retraction when the aircraft is on the ground.

Moving the LANDING GEAR selector lever to the UP position starts landing gear retraction sequence. Moving the LANDING GEAR selector lever to the DN position starts the landing gear extension sequence. This also causes the No Smoking signs to come on automatically. An amber light in the selector lever comes on when the LANDING GEAR selector handle does not agree with position of landing gear.

[Refer to Figure 4.](#)

LANDING GEAR ADVISORY LIGHTS

Gear and door position is shown by nine advisory lights on the LANDING GEAR control panel, and an amber light in the selector lever. Three green lights on the LANDING GEAR ALTERNATE



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EXTENSION panel, on the flight compartment floor, give backup visual verification of gear downlocking.

The L. DOOR, N. DOOR, or R. DOOR amber lights come on when the related hydraulic gear door is open. The LEFT, NOSE, RIGHT green lights come on when the related gear door is down and locked. The LEFT, NOSE, RIGHT red lights come on when the related gear is not locked in selected position. When the gear is up and locked all lights on the LANDING GEAR panel go off.

[Refer to Figure 5.](#)

LANDING GEAR ALTERNATE RELEASE DOOR

If the LANDING GEAR INHIBIT switch is set to the INHIBIT position, the gear will not retract or extend with the LANDING GEAR selector. If the LANDING GEAR ALTERNATE RELEASE door, located in the ceiling of the flight compartment, is opened, a bypass valve isolates the No. 2 hydraulic system. Pulling the LANDING GEAR ALTERNATE RELEASE handle opens the MLG doors and releases the MLG uplocks. Tension springs in the MLG door mechanism pull the doors open and the MLG free-falls into the down position.

[Refer to Figure 6.](#)

LANDING GEAR ALTERNATE EXTENSION DOOR

Opening the LANDING GEAR ALTERNATE EXTENSION door, located in the flight compartment floor, operates the NLG alternate selector valve and reveals the handpump socket. If the MLG does not reach the down and locked position, the handpump is used to fully extend the MLG. This pumps hydraulic fluid from the emergency hydraulic reservoir to the auxiliary main landing gear actuators. The pump handle is located behind copilot's seat.

Pulling the LANDING GEAR ALTERNATE EXTENSION handle unlocks the NLG forward doors. Tension springs in the NLG door mechanism pull the doors open. When the doors are fully open, the NLG uplock is released by further pulling the handle. The NLG free-falls to the fully extended position. Springs installed on the NLG drag strut move the lock links into the down and locked position.

LANDING GEAR DOWNLOCK VERIFICATION LIGHTS

Three green lights on the LANDING GEAR ALTERNATE EXTENSION panel, on the flight compartment floor, give backup visual verification of gear downlocking. The downlock verification toggle switch must be set aft for the lights to operate. The NG, LH or RH green light will come on when the related gear is locked down.

[Refer to Figure 7.](#)

ANTI SKID SWITCH

If the ANTI SKID switch is in the OFF position, the anti skid system is deactivated. The INBD ANTISKID and OUTBD ANTISKID caution lights come on. If the ANTI SKID switch is in the ON position, the anti skid system is activated. The INBD ANTISKID and OUTBD ANTISKID caution lights will go off if there is no system failure. If the ANTI SKID switch is held in the TEST position while in the air, with gear down and locked, the INBD ANTISKID and OUTBD ANTISKID caution lights come on for 3 seconds then go off. If the ANTI SKID switch is held in the TEST position while on the ground, the INBD ANTISKID and OUTBD ANTISKID caution lights come on for 6 seconds then go off. The INBD ANTISKID and OUTBD ANTISKID caution lights will come on if there is a system failure.

[Refer to Figures 8 and 9.](#)

EMERGENCY/PARKING BRAKE LEVER



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If the EMERGENCY/PARKING BRAKE LEVER is in the PARK detent, the parking brake is set if parking brake hydraulic pressure is more than 1500 psi (10342.5 KPa). The PARKING BRAKE caution light also comes on. The emergency brake system uses No. 2 hydraulic or accumulator pressure.

[Refer to Figures 10 and 11.](#)

NOSEWHEEL STEERING

Nosewheel steering is by the steering tiller or the rudder pedals when the steering switch is set to STEERING. The steering tiller is a self centering unit that lets the pilot steer the nosewheels for forward taxi at low speed. The steering tiller is installed in the flight compartment on the pilot's left side console. The steering tiller gives full authority steering command signals of the nosewheel position to the Steering Control Unit (SCU). It can steer the nosewheels ± 70 degrees to the left and right of the center position.

The rudder pedals are self centering and allow the crew to steer the nosewheels for high speed taxi, take off, and landing purposes. The rudder pedals can steer the nosewheels to a maximum of ± 8 degrees left and right of the center position. The potentiometer for the rudder pedals is installed on the support structure of the control quadrant of the rudder system. A linkage assembly connects the potentiometer to the control quadrant of the rudder system. The potentiometer gives limited authority steering command signals of the nosewheel position to the SCU.

Landing Gear

[Refer to Figures 12 and 13.](#)

Each main gear has a pair of forward and aft doors hinged to the nacelle side structure. When the gear is up, all doors enclose the

main wheels. With the main gear down, the forward door on each main gear stays open.

[Refer to Figure 14.](#)

The nose gear has a pair of forward and aft doors, that completely enclose the nose gear when the gear is up. With the gear down, the forward nose doors are closed, while the aft doors stay open.

The Proximity Switch Electronic Unit (PSEU) controls the landing gear and hydraulically operated gear doors and related advisory lights, and monitors proximity and Weight-On-Wheels (WOW) sensors. WOW signals prevent gear retraction while on the ground. Failure of a WOW sensor turns on a WT ON WHEELS caution light. Redundancy is built in to ensure landing gear operation if there is a failure.

An audible warning tone sounds, when the gear is not down and locked and the aircraft is in a landing configuration.

[Refer to Figures 15, 16 and 17.](#)

Ground lock pins are supplied for the main gear and an integral ground lock mechanism is controlled from outside the aircraft for locking the nose gear. The main gear lock-pins may be kept in the forward compartment of the forward passenger door. With the gear extended, the pins are inserted into the main gear stabilizer brace assemblies. There are also landing gear door lock pins for the nose and main hydraulic doors. These prevent the hydraulic gear doors from closing.

Gear Operation

Landing gear operation is controlled and monitored from the LANDING GEAR control panel, adjacent to the Engine Display (ED) in the flight compartment. The landing gear is selected UP or DN



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(down) by moving the landing gear selector lever. A LOCK RELEASE selector lever must be held down to let the gear selector lever move in either direction.

An alternate downlock verification system confirms downlock engagement if the primary downlock indication is in doubt. Three green downlock verification lights are located under the LANDING GEAR ALTERNATE EXTENSION panel in the flight compartment floor.

Gear Warning Tone

A landing gear configuration warning tone sounds over the flight compartment speakers and to the pilot's and co-pilot's headsets if:

- If the HORN toggle switch, on the landing gear panel, is set to the TEST position.

OR

- If all the landing gears are not down and locked.

AND

Condition 1 (Flaps and Engine torque):

- The flap lever is set to more than 8.5 degree
- AND
- ENG1 torque OR ENG2 torque is less than 50%
- AND
- PLA 1 AND PLA 2 are not in the rating power detent.

OR

Condition 2 (Airspeed, Radio Altitude and PLA setting):

- Indicated airspeed less than 156 Knots (288 km/h), AND RAD ALT 1 Invalid
- OR
- Indicated airspeed less than 156 Knots (288 km/h), AND RAD ALT 1 Valid, AND Above Ground Level (AGL) altitude is less than 1053 feet
- AND
- PLA1 AND PLA2 are less than FLIGHT IDLE plus 12 degrees.

OR

Condition 3 (Airspeed during single engine operation and PLA setting):

- Indicated airspeed less than 156 Knots (288 km/h), AND RAD ALT 1 Invalid
- OR
- Indicated airspeed less than 156 Knots (288 km/h), AND RAD ALT 1 Valid, AND Above Ground Level (AGL) altitude is less than 1053 feet
- AND
- PLA1 OR PLA2 is less than FLIGHT IDLE plus 12 degrees
- AND
- HORN MUTE switch is not latched to MUTE
- AND
- PLA 1 AND PLA 2 are less than the rating power detent.



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If the landing gear configuration warning is muted, the mute will reset in the conditions that follow:

- The down lock indication indicates all gear down and locked
OR
- The alternate down lock indication indicates all gear down and locked
OR
- PLA 1 AND PLA 2 are greater than OR less than FLIGHT IDLE plus 12 degrees
OR
- All the landing gears are not down and locked
AND
The Flap lever is set to more than 8.5 degree
AND
ENG1 torque OR ENG2 torque is less than 50%
AND
PLA 1 AND PLA 2 are not in the rating power detent.

Retraction Sequence

[Refer to Figure 18.](#)

When the landing gear selector lever is selected to the UP position, hydraulic pressure from No. 2 system is applied to the retract side of the system through the solenoid selector valve. This opens the nose

gear forward doors and retracts the nose gear, it also opens the main gear aft doors and retracts the main gear. The aft nose gear doors are mechanically linked and close with the retracting nose gear. After nose gear retraction, the forward nose gear doors close hydraulically. The forward main gear doors are mechanically linked and close with the retracting main gear. After main gear retraction, the aft main gear doors close hydraulically.

The advisory light sequence during landing gear retraction starts with the LEFT, NOSE and RIGHT red unsafe lights and the amber selector handle light coming on. Then the amber door advisory lights coming on to show the hydraulically operated gear doors are open. Then the green LEFT, NOSE and RIGHT lights go off to show the gear is not locked down. When the landing gear is retracted and locked in the up position, the amber selector handle light and red advisory lights go out. Finally, the amber gear door advisory lights go out to show all the hydraulic gear doors have closed. No advisory lights should be on if the gear is up correctly.

The solenoid selector valve is then de-energized, hydraulically isolating the landing gear system. The main and nose gear are held in the up position mechanically with uplocks.

Extension Sequence

[Refer to Figure 18.](#)

When the landing gear selector lever is moved to the DN position, hydraulic pressure is applied to the extend side of the system through the solenoid selector valve. The main and nose hydraulic doors open, and the main and nose gear extend. The hydraulic forward nose and aft main gear doors close after the gear is down and locked.



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The advisory light sequence during extension starts with the LEFT, NOSE, and RIGHT red unsafe lights and the amber gear selector handle light coming on. Then the amber door advisory lights coming on to show the hydraulically operated gear doors are open. When the landing gear is fully extended and locked in the down position, the red unsafe lights, and the selector handle light goes out. Then the green LEFT, NOSE, and RIGHT advisory lights come on. Finally, the gear door advisory lights go out when the hydraulically operated doors are closed.

The solenoid selector valve stays powered to allow for continued hydraulic pressure acting on the gear when down and locked, however primary downlock is by the overcenter locks.

If a landing gear hydraulic sequencing valve fails, or the PSEU is unable to control it, the LDG GEAR INOP caution light comes on.

The landing gear selector must not be used to extend the landing gear when the LDG GEAR INOP caution light is on.

Alternate Gear Extension

The alternate extension system gives a means of extending the landing gear when:

- The LDG GEAR INOP caution light is on
- The Landing gear indication fails
- There is a Loss of No. 2 hydraulic system pressure

The landing gear extension INHIBIT switch is installed in the flight compartment ceiling, adjacent to the main LANDING GEAR ALTERNATE RELEASE door. It removes electrical power from the landing gear selector.

When the main LANDING GEAR ALTERNATE RELEASE door on the flight compartment ceiling is opened it mechanically opens a bypass valve in the normal hydraulic extension system, and gives access to the MAIN L/G RELEASE handle. Pulling the handle releases the main landing gear doors and uplocks. The main gear will free fall but may not fully extend.

[Refer to Figure 19.](#)

The LANDING GEAR ALTERNATE EXTENSION door, on the flight compartment floor, must then be fully opened giving access to the alternate extension handpump and the NOSE L/G RELEASE handle. Opening the door mechanically operates the NLG alternate selector valve. If the MLG does not reach the down and locked position, the extension pump handle, located behind the copilot, is inserted into the pump handle socket and operated to complete main gear extension and subsequent downlock. Both the LANDING GEAR ALTERNATE EXTENSION door and the MAIN LANDING GEAR ALTERNATE RELEASE door must be left fully open after alternate landing gear extension.

When the NOSE L/G RELEASE handle is pulled, the nose gear uplock and doors are released and the nose gear free falls to a down and locked position, assisted by the airflow

Nosewheel Steering

[Refer to Figures 20 and 21.](#)

Directional control on the ground is by the nosewheel steering (NWS) system, powered by the No. 2 hydraulic system. Steering control is by either the Steering Hand Control or the rudder pedals. The Steering Hand Control turns the nosewheel up to 70 degrees either side of center for low speed taxi. Steering with the rudder pedals turns the nosewheel up to 8 degrees either side of center for



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high-speed taxi, take-off and landing roll. After take-off, the nosewheel automatically centers before retraction.

The hand control, located on the Pilot's Side panel, is self-centering and operates when the STEERING switch is set to the STEERING position. The nosewheel must be within 70 degrees of center for the steering to work. An index mark on the hand control shows the relative position of the nosewheel against a fixed STEERING RANGE decal. With the STEERING switch set to STEERING, power is directed to the nosewheel Steering Control Unit (SCU), if the nose gear is down and locked with weight-on-wheels.

The nosewheel will revert to a passive shimmy dampened castoring mode if the:

- Nosewheel angle is greater than 70 degrees
- SCU fails
- STEERING switch is set to OFF

In the passive mode, the nosewheel will castor up to 120 degrees either side of center. The NOSE STEERING caution light comes on if the nose wheel is forced more than 70 degrees or if the SCU fails. The NOSE STEERING caution light does not come on if electrical power is removed from the SCU. Differential braking and/or power may be used for directional control in the passive mode.

When taxiing in reverse, the STEERING switch must be set to STEERING. The tiller or rudder pedals must not be used during reverse taxiing.

Nosewheel steering using the hand control is limited to forward taxiing only, with the STEERING switch set to the STEERING position.

Do not set the STEERING switch to STEERING if the tow bar is connected to the nose gear.

Normal Braking System

[Refer to Figure 22.](#)

Each main wheel is equipped with a multiple disc brake unit powered by the No. 1 hydraulic system. An Anti Skid Control Unit (ASCU) modulates the application of brake pressure to each brake unit. Brake pressure is applied by pushing the pilot's or copilot's brake pedals.

The anti skid control unit has the functions that follow:

- Monitors wheel speed
- Modulates the brake pressure applied to each brake unit to prevent wheel lock-up
- Gives maximum braking at all levels of runway friction.

The ANTI SKID switch on the copilot's glareshield panel operates the anti skid system when set to the ON position and the wheel speed is more than 10 knots. A start up self test of the anti skid control circuits is made when the switch is moved to the ON, or momentary TEST. The start up self test is prevented with wheel speed more than 17 knots. A self-test, with the aircraft on the ground, causes the INBD ANTISKID and the OUTBD ANTISKID caution lights to come on for six seconds and then go out. A self-test in the air, with the landing gear extended and locked, will turn on the related caution lights for three seconds. If the ASCU senses a fault in the system, it turns on the related ANTISKID caution light continuously.

The PSEU supplies weight-on-wheels and gear up and locked signals to the ASCU, to make sure that the brakes are off until the



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aircraft has touched down and the wheels are spinning. In conditions of low runway friction, the main gear wheels may not spin up before the aircraft's weight is fully on the wheels. In this case, the ASCU gives a 5 second delay before brake pressure is applied. This delay is cancelled when wheel speed is more than 35 kt (65 km/h).

Emergency Brake System

[Refer to Figure 23.](#)

The emergency/parking brake system lets the brakes be applied if the normal brake system fails, or for setting the parking brake. An EMERG BRAKE lever on the Engine Control quadrant operates the system. The emergency/parking brake system is powered by No. 2 hydraulic system, or by parking brake accumulator pressure. Hydraulic pressure to the emergency/parking brake system is shown on the PK BRK indicator on the Permanent Systems Area (PSA) of the Multi Function Display (MFD).

The EMERG BRAKE lever operates against a spring to produce a resistance proportional to the brake pressure applied as the lever is pulled back. There is no differential braking and no anti skid protection when using the EMERG BRAKE lever. If the emergency/parking brake system is used with No. 2 hydraulic system inoperative, the accumulator supplies brake pressure. Pumping the brakes with the EMERG BRAKE lever will rapidly decrease the remaining accumulator brake pressure; therefore brakes should be applied with a steady continuous pressure. A fully charged accumulator is sufficient for approximately four full applications.

The parking brake is engaged by pulling the EMERG BRAKE lever all the way back to the detent PARK position. This turns on the PARKING BRAKE caution light on the caution and warning panel.

The button on the side of the handle must be pushed to release the lever from the PARK detent.

With the parking brake set, application of engine power will cause the take-off warning horn to sound.

A hand pump located in the right main wheel well can be used to increase the park/brake accumulator pressure. Minimum pressure required for setting the park brake is 1,500 psi (10342 kPa).

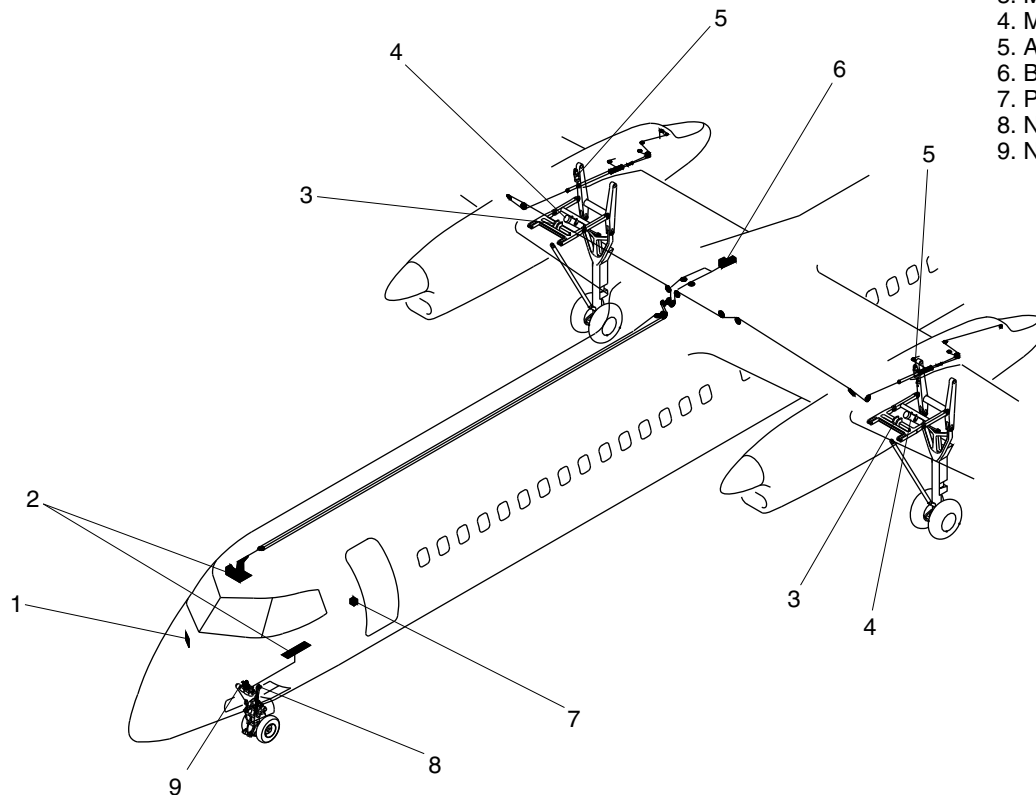


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LEGEND

1. Landing Gear Selector Lever and Control Panel.
2. Alternate Release Handles.
3. MLG Lock Actuator.
4. MLG Retraction Actuator.
5. Alternate Extension Actuator.
6. Bypass Valve.
7. Proximity Sensor Electronic Unit.
8. NLG Lock Actuator.
9. NLG Retraction Actuator.



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Landing Gear Locator
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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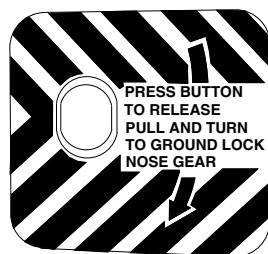
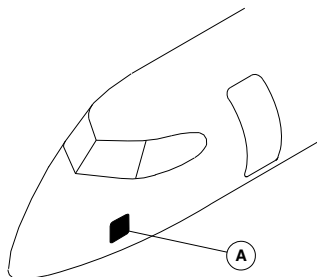
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DISENGAGED
(FULLY RETRACTED)



ENGAGED
(FULLY EXTENDED)

A

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Nose Gear Ground Lock Control Handle Locator
Figure 2

PSM 1-84-2A
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LEGEND

1. Cable Switch Assembly.
2. Lever Assembly.
3. Striker Plate.
4. Advisory Light.

Nose Gear Lock Position Indication System
Figure 3

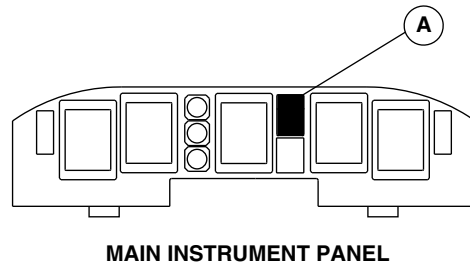
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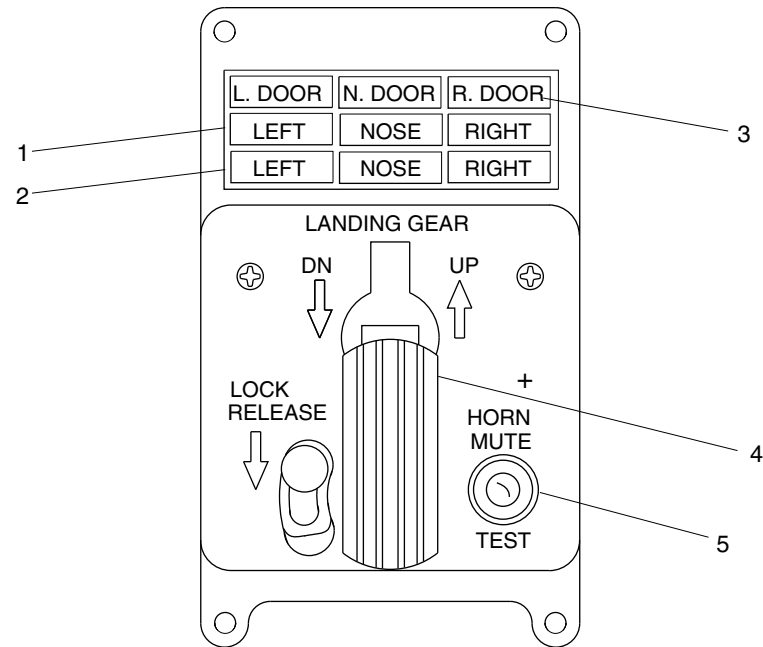
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LEGEND

1. "Green" Gear Downlocked.
2. "Red" Gear Not Uplocked Or Downlocked.
3. "Amber" Door Open.
4. Handle "Amber" (Concurrent W/"Red" Unsafe Lights) Gear In-Transit.
5. Warning Horn Mute/Test Switch.



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Landing Gear Control Panel Detail
Figure 4

PSM 1-84-2A
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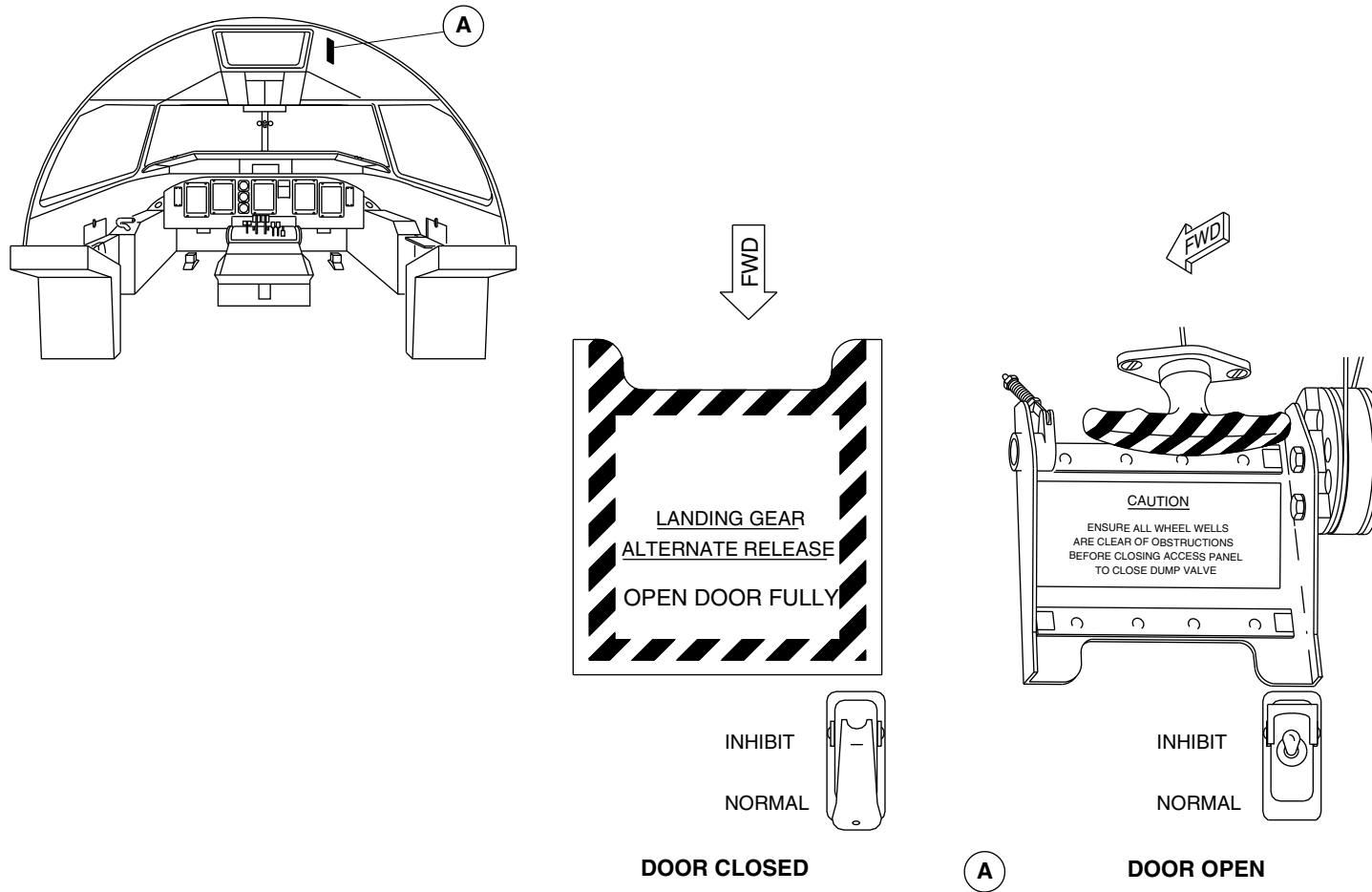
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Landing Gear Alternate Release Controls Locator
Figure 5

PSM 1-84-2A
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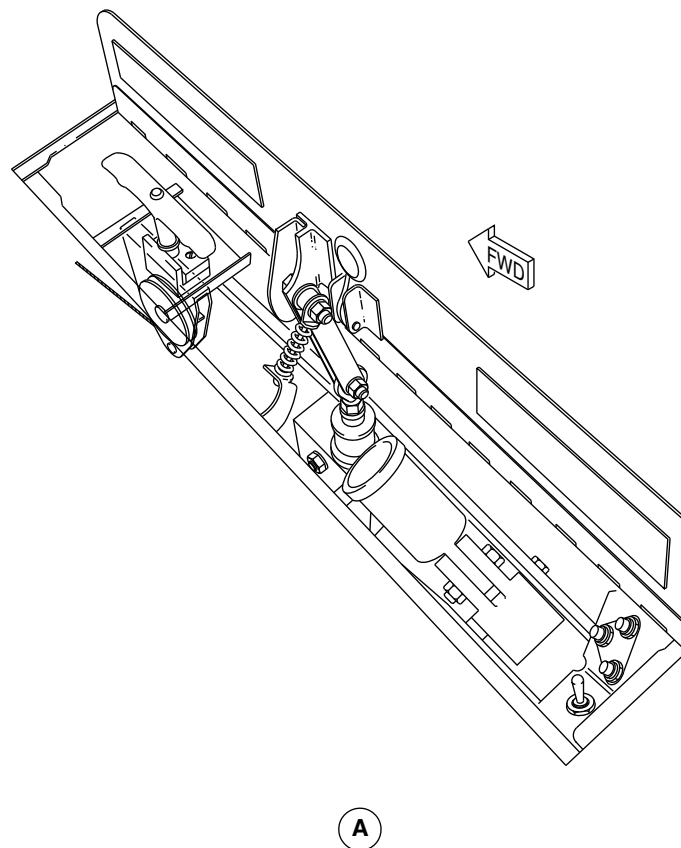
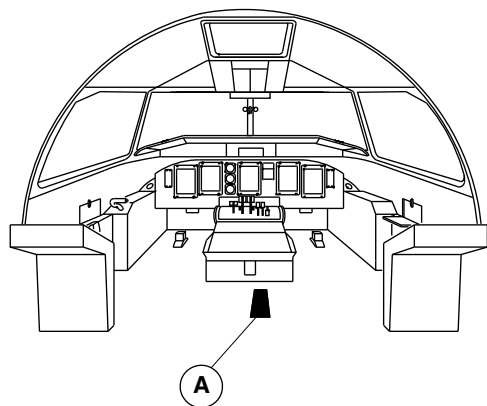
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Landing Gear Alternate Extension Door Locator
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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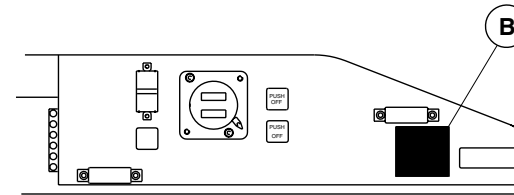
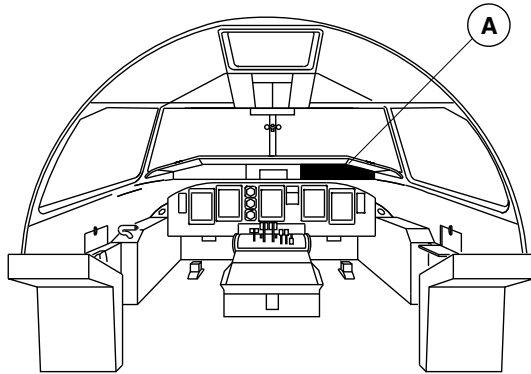
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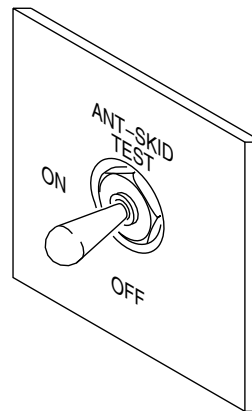


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A RHS GLARESHIELD



B

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Anti Skid Switch Locator
Figure 7

PSM 1-84-2A
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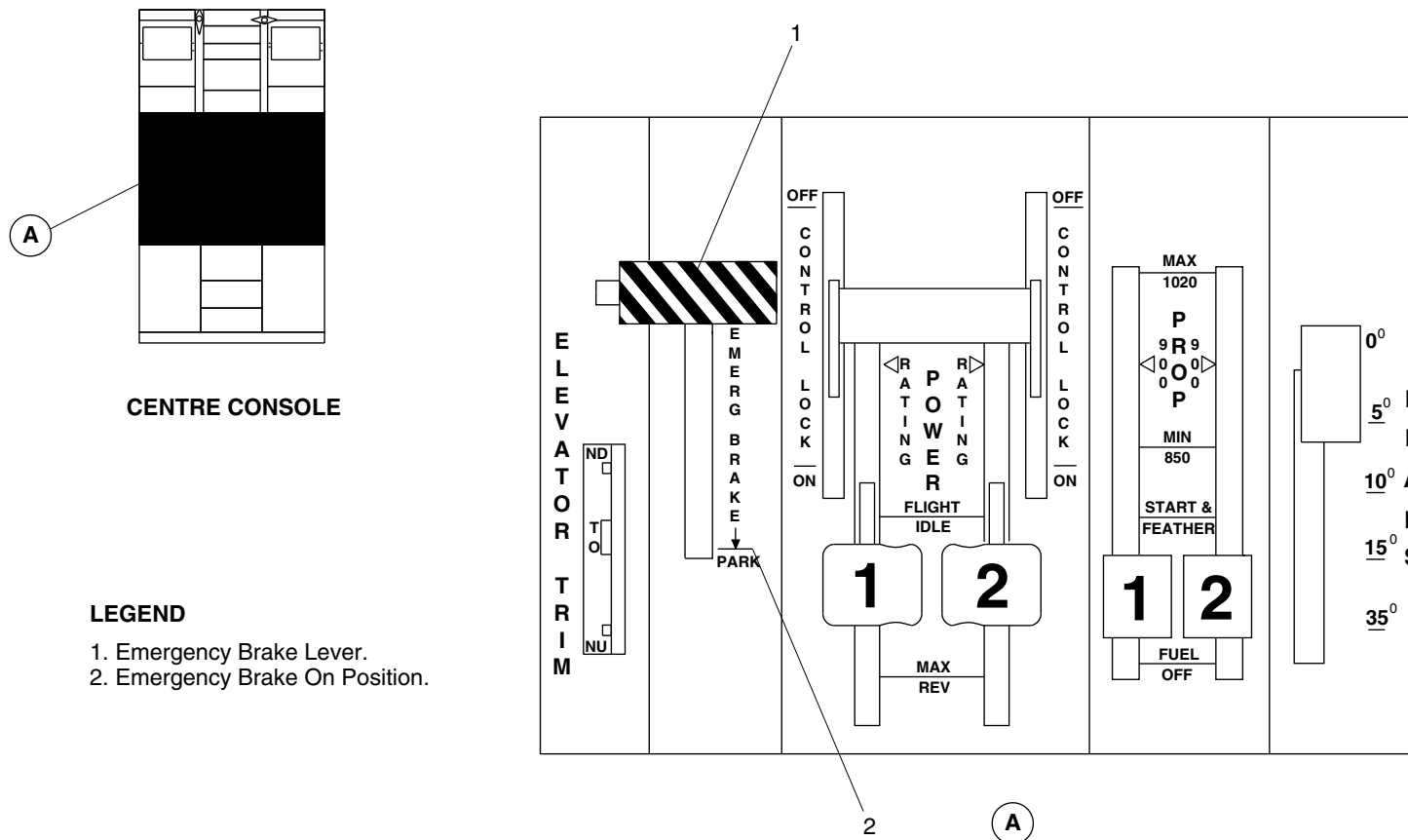
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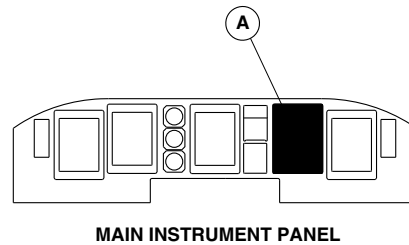
Parking Brake Control Detail
Figure 8

PSM 1-84-2A
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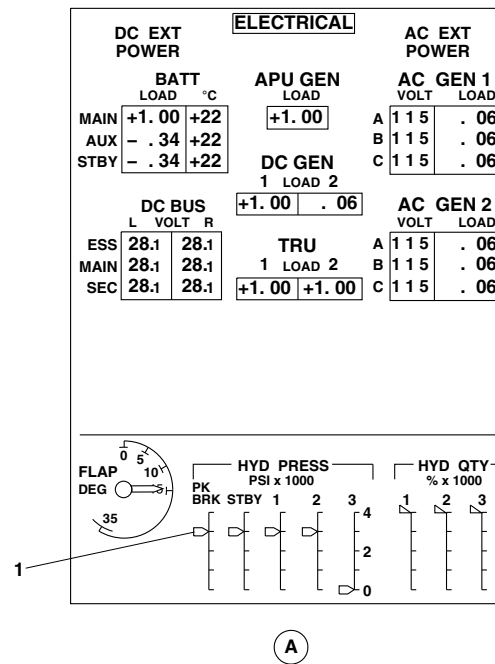
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Parking Brake Hydraulic Pressure Indicator.



Parking Brake Pressure Indication
Figure 9

PSM 1-84-2A
EFFECTIVITY:
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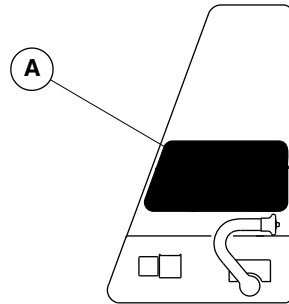
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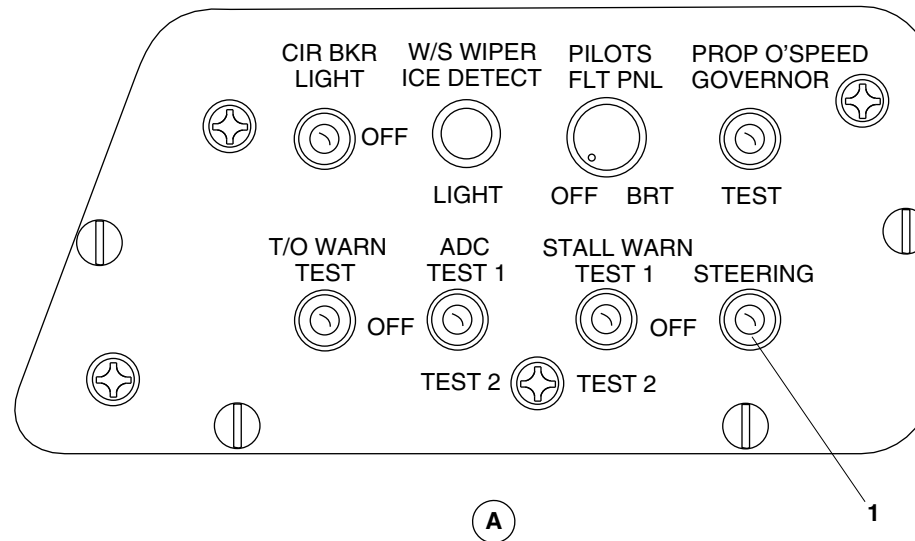
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEFT SIDE CONSOLE

LEGEND

1. Nosewheel Steering Toggle Switch.



fsm08a01.cgm

Nosewheel Steering Switch Detail
Figure 10

PSM 1-84-2A
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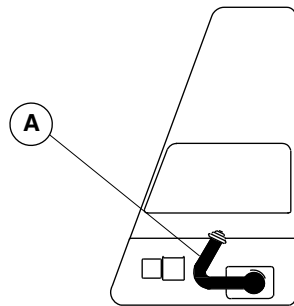
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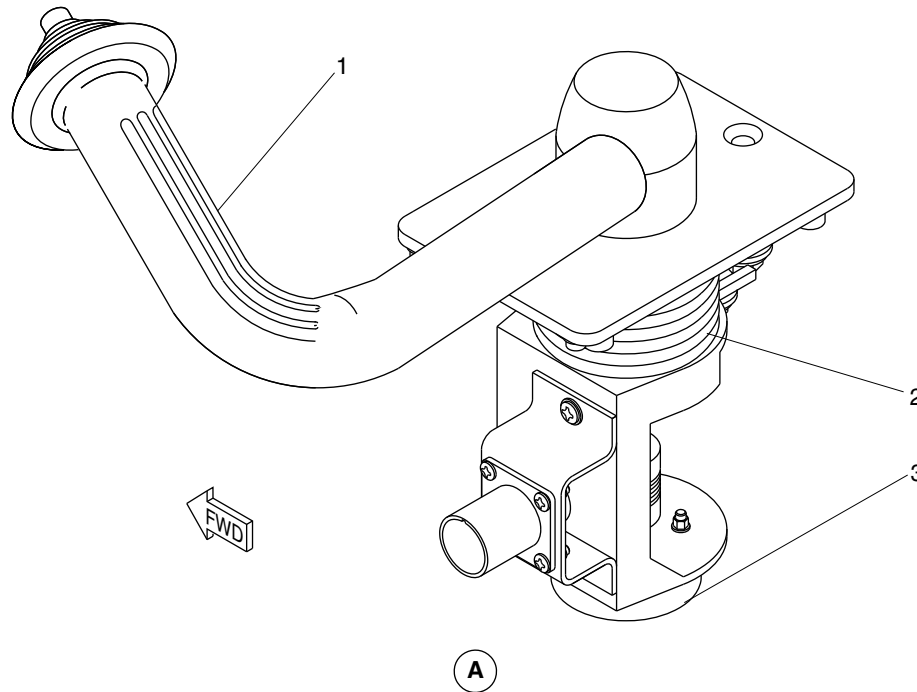
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEFT SIDE CONSOLE

LEGEND

- 1. Pilot Hand Control.
- 2. Centering Spring.
- 3. Potentiometer.



fs494a01.cgm

Nosewheel Steering Hand Control Locator
Figure 11

PSM 1-84-2A
EFFECTIVITY:
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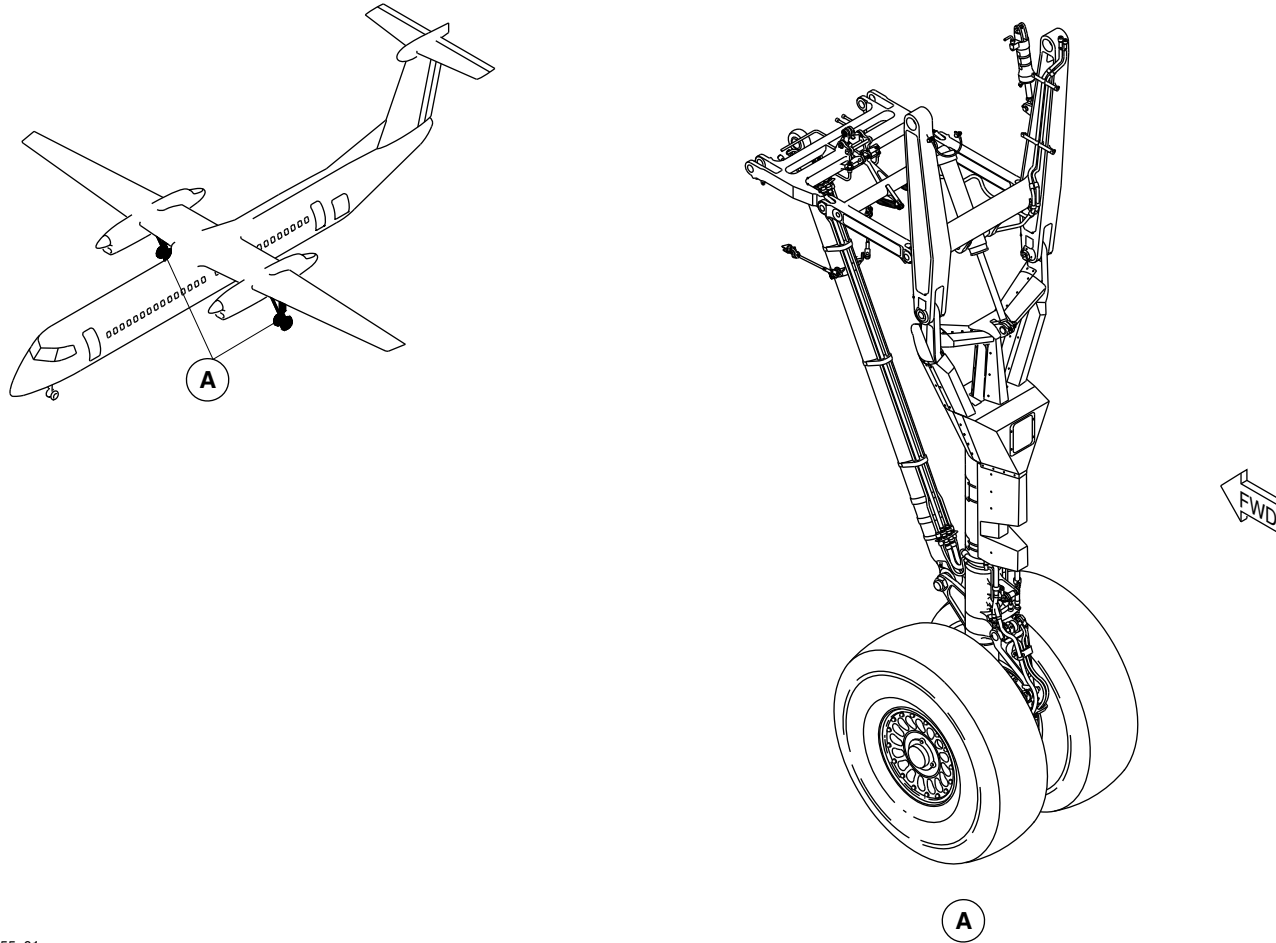
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fs555a01.cgm

Main Landing Gear Assembly Locator
Figure 12

PSM 1-84-2A
EFFECTIVITY:
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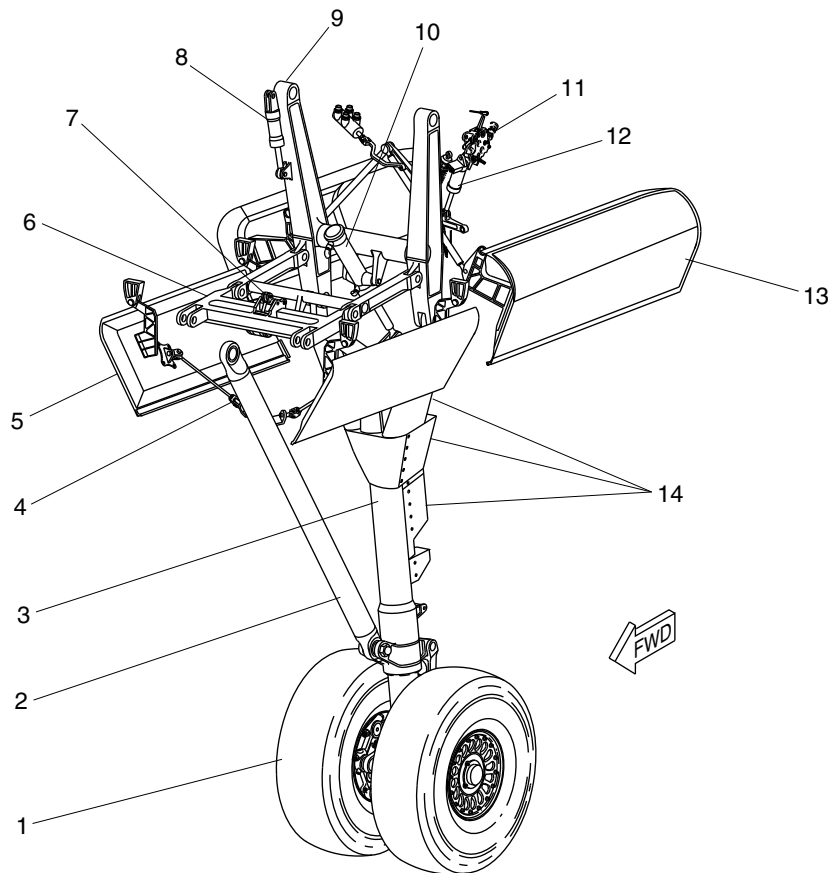
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Wheel.
2. Drag Strut.
3. Shock Strut.
4. Forward Door Linkage.
5. Forward Door.
6. Stabilizer Brace.
7. Stabilizer Lock.
8. Auxiliary Extension Actuator.
9. Yoke.
10. Retraction Actuator.
11. Uplock Assembly.
12. Aft Doors Mechanism.
13. Aft Door.
14. Fairings.

fs250a01.cgm

Main Landing Gear and Doors Detail
Figure 13

PSM 1-84-2A
EFFECTIVITY:
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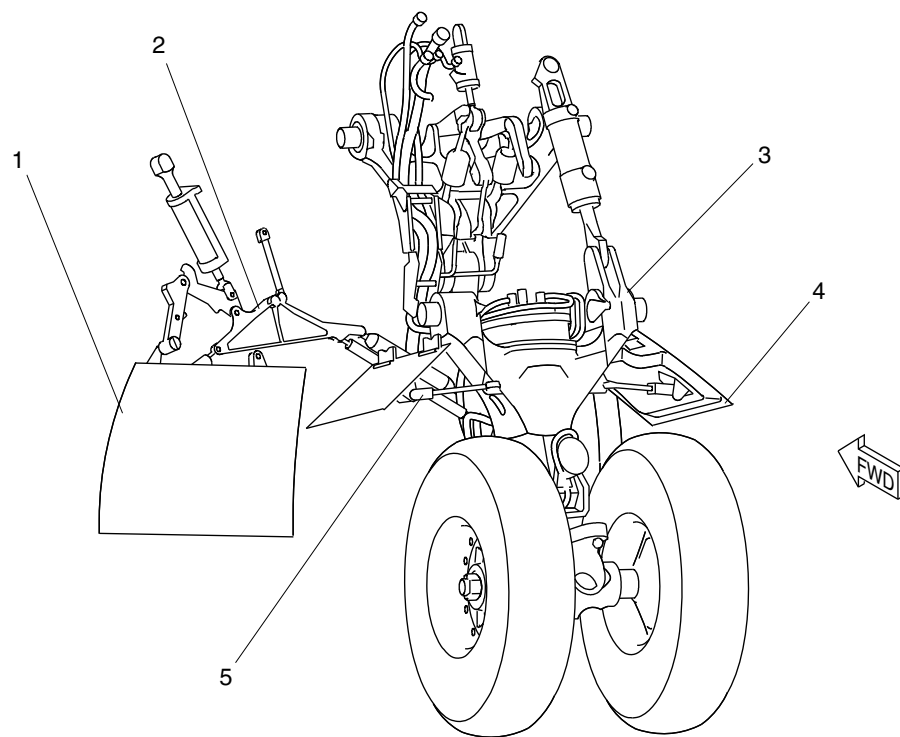
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LEGEND

1. Forward Door.
2. Forward Door Operating Mechanism.
3. Shock Strut Assembly.
4. Aft Door.
5. Aft Door Actuating Linkage.

fs574a01.cgm

Nose Landing Gear and Doors Detail
Figure 14

PSM 1-84-2A
EFFECTIVITY:
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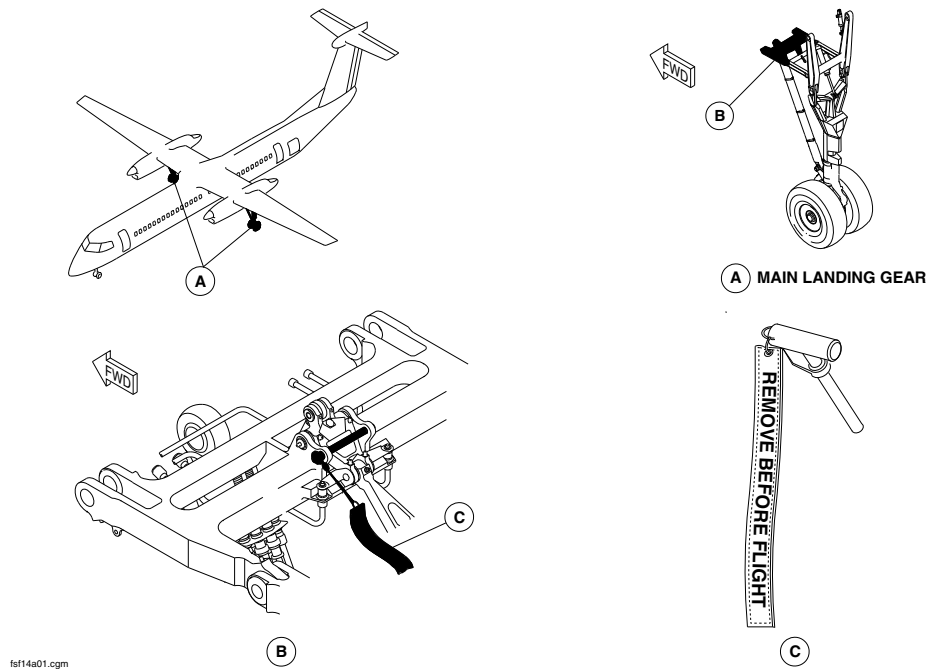
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Main Landing Gear Ground Lock Pin Locator
Figure 15

PSM 1-84-2A
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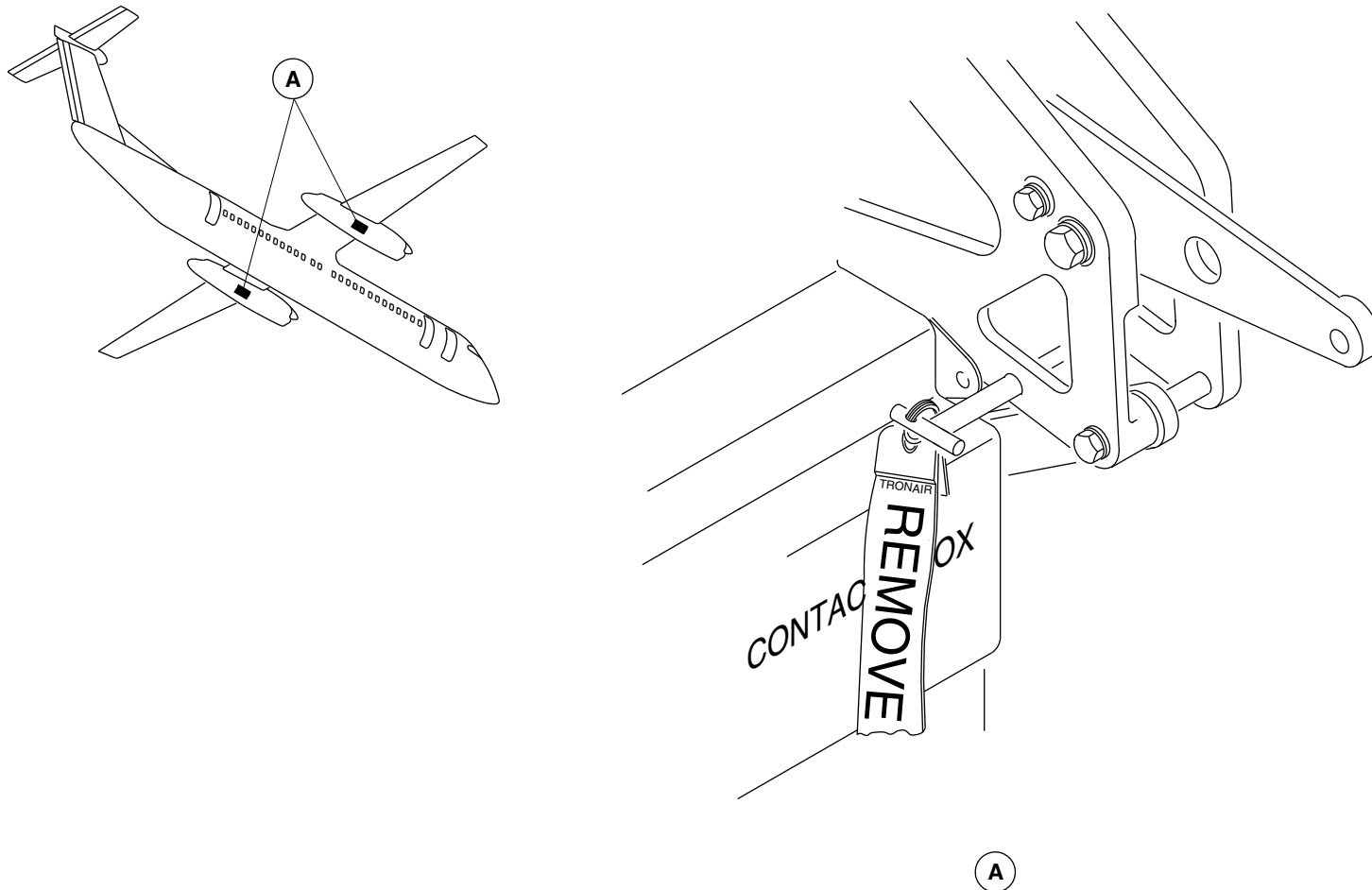
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Main Gear Hydraulic Doors Lock Pin Locator
Figure 16

PSM 1-84-2A
EFFECTIVITY:
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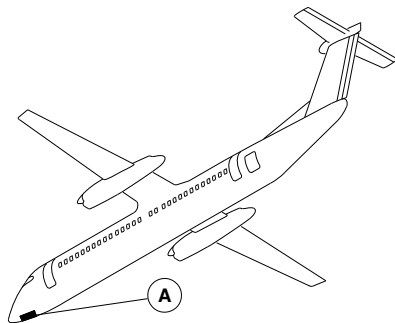
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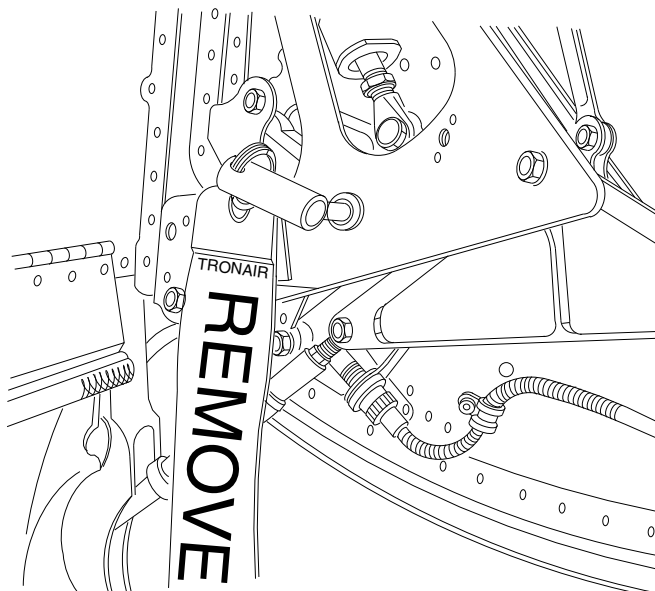
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NOTE

View shown with
debris guard removed.



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fsf13a01.cgm

Nose Landing Gear Hydraulic Doors Ground Lock Pin Locator
Figure 17

PSM 1-84-2A
EFFECTIVITY:
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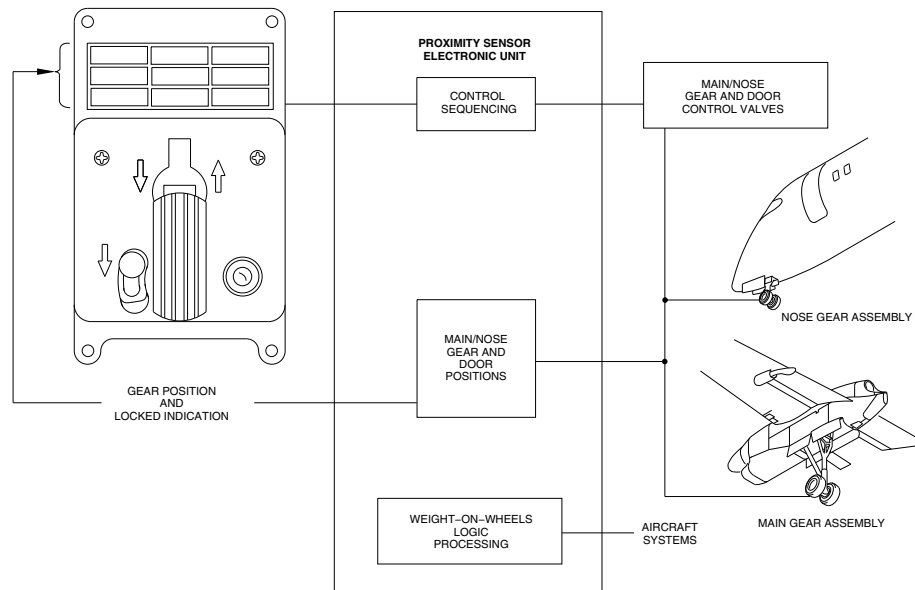
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Landing Gear Schematic
Figure 18

PSM 1-84-2A
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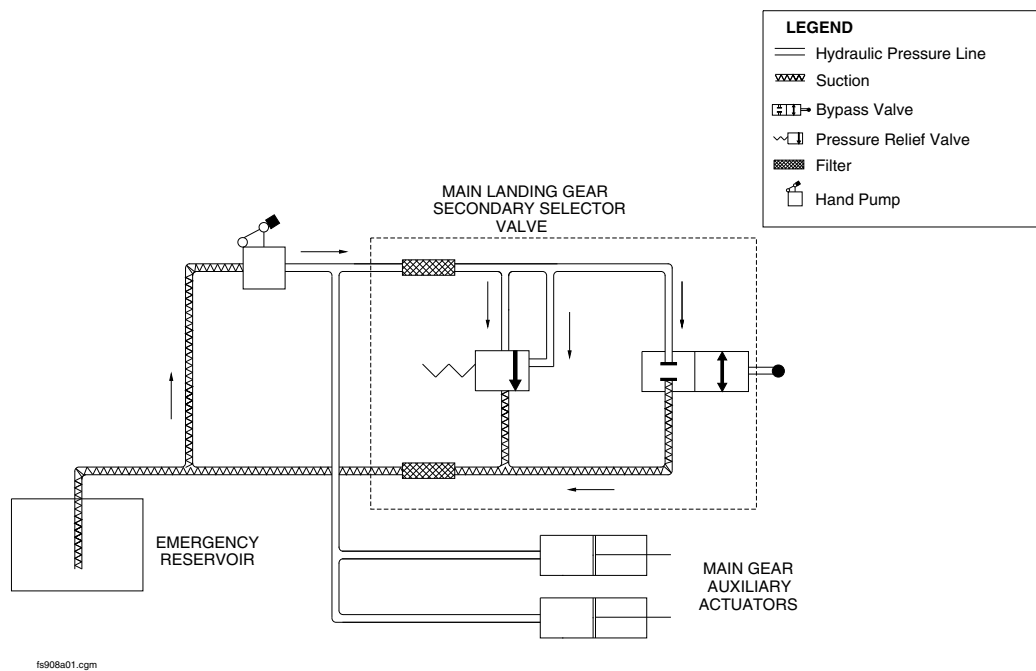
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Main Landing Gear Alternate Extension Schematic
Figure 19

PSM 1-84-2A
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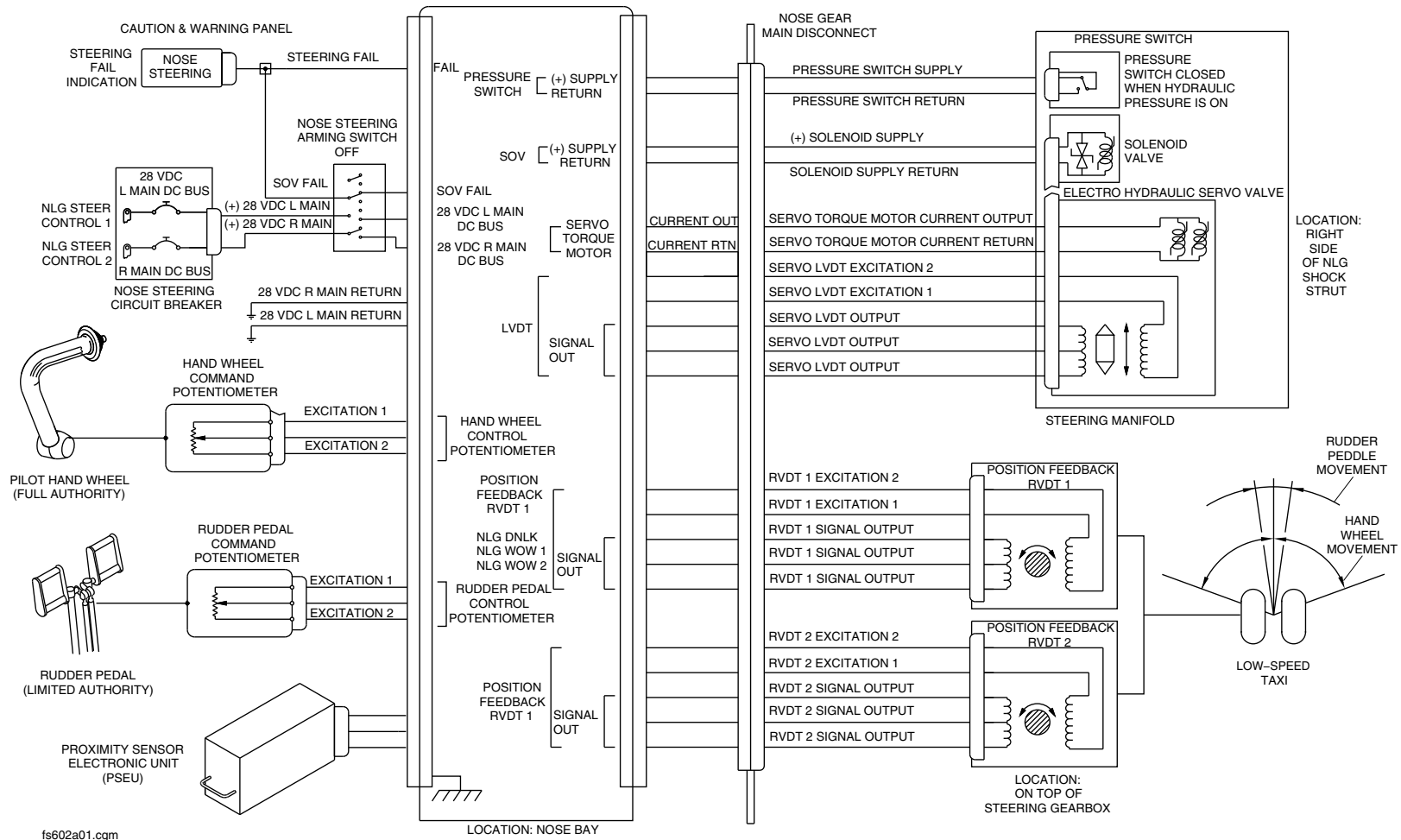
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Nosewheel Steering Electrical Schematic
Figure 20

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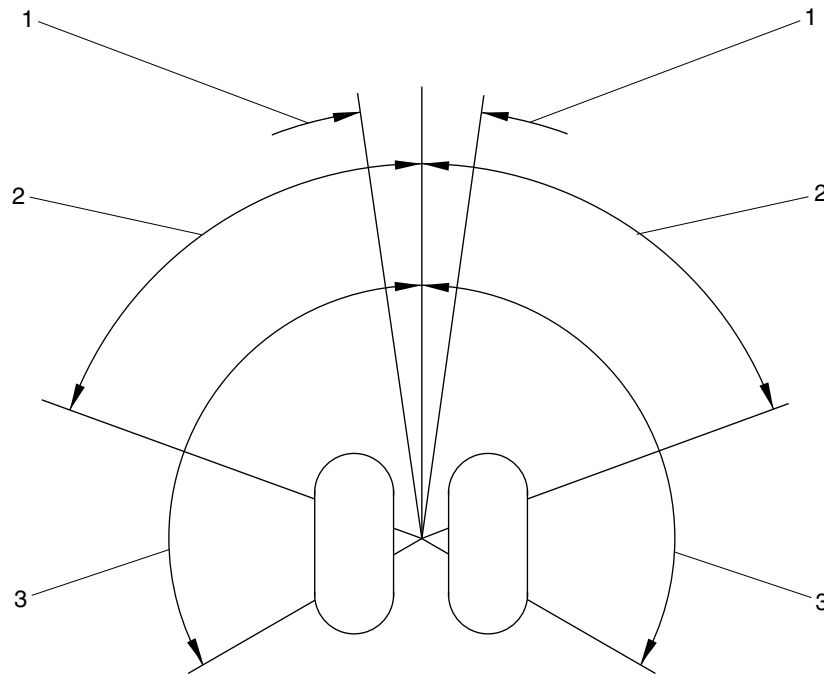
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. High-Speed Taxi Mode 8 +/- 1 Degrees Left and Right from Rudder Pedals.
2. Low-Speed Taxi Mode 70 Degrees Left and Right from Steering Control Handle.
3. Unpowered (Caster) Mode 120 Degrees Left and Right.

fs978a01.cgm

Nosewheel Steering Limits
Figure 21

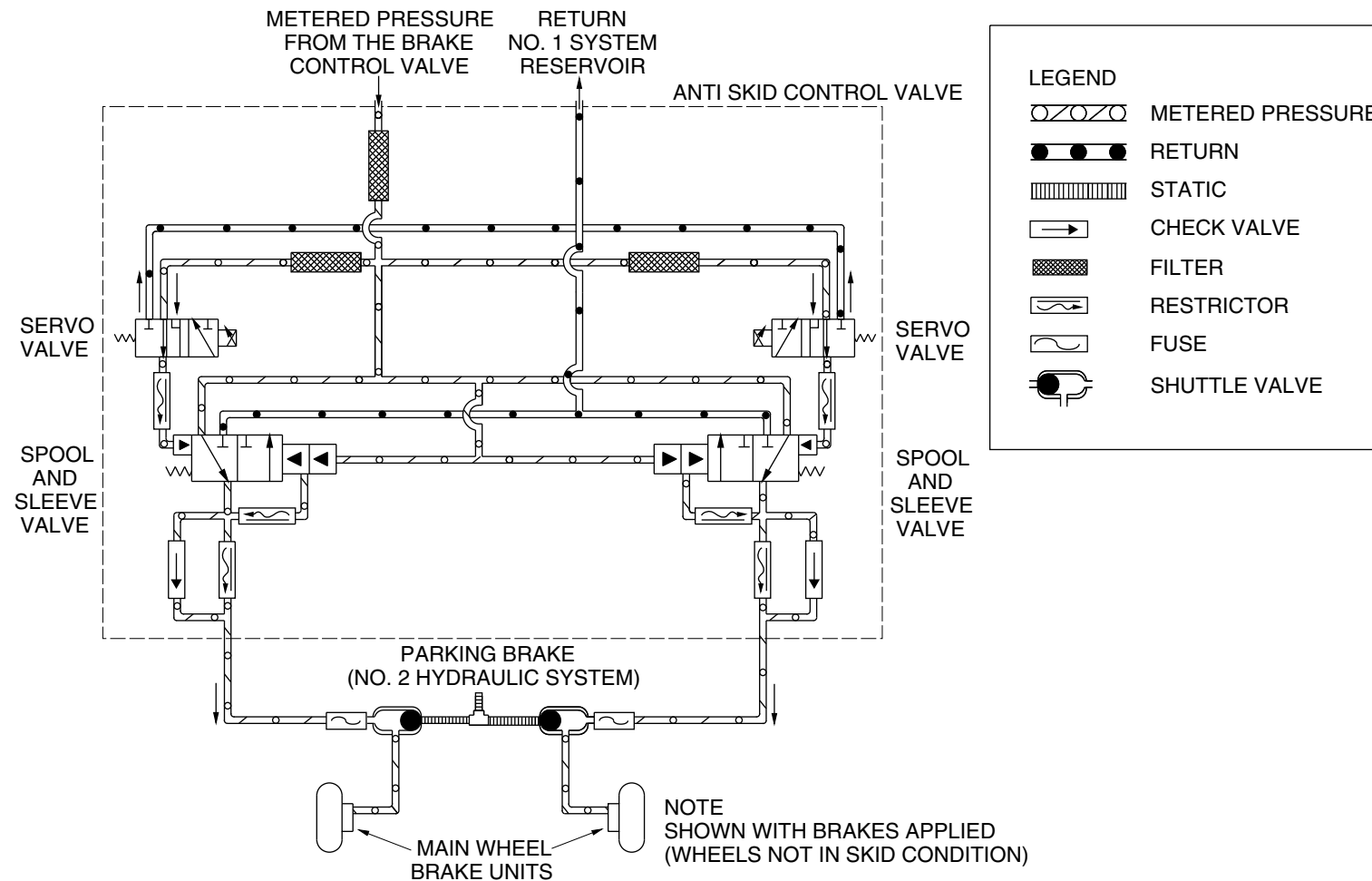
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fs667a01.cgm

Normal Braking With Anti-Skid Schematic
Figure 22

PSM 1-84-2A
EFFECTIVITY:
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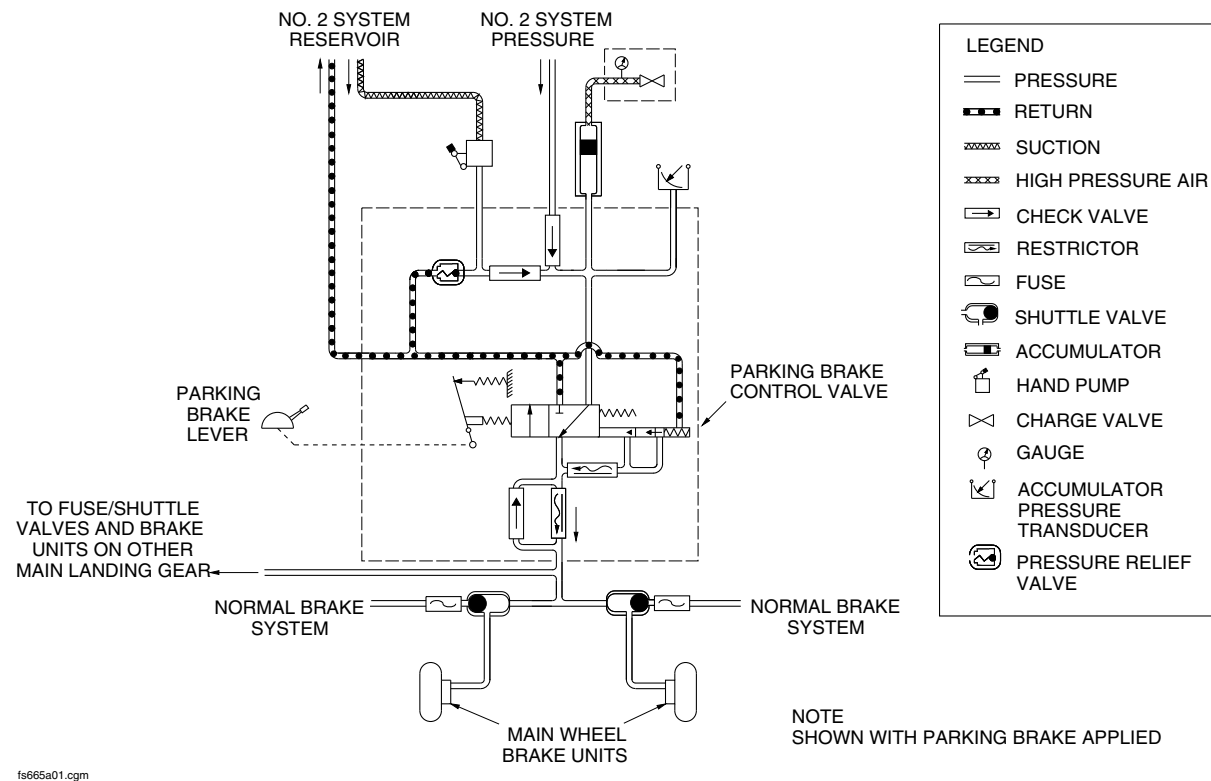
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Parking Brake Schematic
Figure 23

PSM 1-84-2A
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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MAIN LANDING GEAR AND DOORS

Introduction

The Main Landing Gear (MLG) absorbs and distributes the landing loads. The MLG and wheels support the aircraft on the ground. The MLG doors supply the fairing for the engine nacelle with the MLG in the retracted position.

General Description

Refer to Figures 1 and 2.

Operation of the MLG is controlled by the sequencing of components of the retraction and extension system. Hydraulic pressure is supplied to the lock actuator on the stabilizer brace to retract and break the overcenter lock, to allow the stabilizer brace to fold. The aft MLG doors actuator extends to open the doors. The aft doors mechanism operates the sequence valve assembly to supply hydraulic pressure to retract the MLG retraction actuator. The forward doors are attached to and move with the landing gear drag strut.

The shock absorbing unit is located inside the outer cylinder of the shock strut. The unit moves vertically in the shock strut cylinder to absorb shocks. The piston of the shock absorbing unit is attached to the shock strut by torque links. Proximity sensors on the shock strut use the upper torque link as targets to give a weight-on-wheels or a

weight-off-wheels indication to the Proximity Sensor Electronic Unit (PSEU).

Detailed Description

Refer to Figures 3 and 4.

The retraction actuator pulls the shock strut up and rearwards into the aft section of the MLG wheelwell. The stabilizer brace folds upwards and pulls the yoke forward. The shock strut hinges on the bottom of the yoke and on the bottom of the drag strut. The MLG uplock roller engages the uplock assembly and locks the MLG in the retracted position. The aft door actuator extends to close the aft wheelwell doors. The forward doors of the MLG wheelwell are closed by the upward movement of the drag strut when the shock strut moves backwards and up.

Proximity sensors located on the outer cylinder indicate a Weight-Off-Wheels condition to let the MLG retract. A proximity sensor on the aft doors mechanism gives an indication when the doors move and when the doors are closed and locked. Proximity sensors on the stabilizer brace overcenter lock mechanism give an indication that the MLG downlock is not in the downlock position. A proximity sensor on the uplock assembly gives an indication that the MLG is locked in the retracted position.

Extension of the MLG is controlled by the components operation of the retraction and extension system. Hydraulic pressure is supplied to the aft MLG doors actuator to extend and open the doors. The uplock assembly actuator extends to unlock and release the MLG uplock roller. The retraction actuator extends to lower the shock strut.

The lock actuator on the stabilizer brace unfolds the stabilizer brace. The bottom of the yoke moves backwards and the shock strut pivots around the yoke bottom hinge points and the drag strut hinge points.

PSM 1-84-2A

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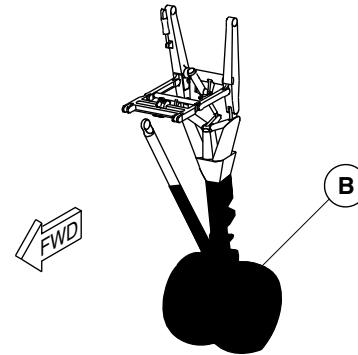
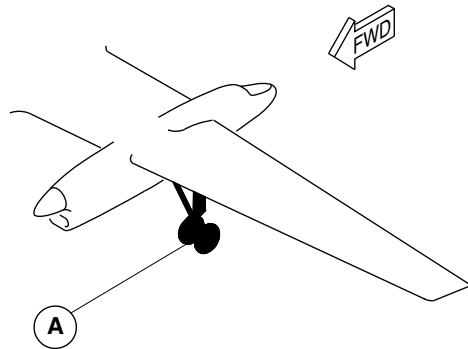
When the stabilizer brace has unfolded, the lock actuator pushes the stabilizer brace into an overcenter locked position. The actuator of the MLG wheelwell aft doors is sequenced to extend and close the doors. The forward doors of the MLG wheelwell are opened by the downward movement of the drag strut when the extension sequence occurs.

A proximity sensor on the aft doors mechanism gives an indication when the doors move and when the doors are closed and locked. A proximity sensor on the uplock assembly gives an indication that the MLG is unlocked from the uplock assembly. Proximity sensors on the stabilizer brace overcenter lock mechanism give an indication that the MLG is in a down and locked condition. Proximity sensors on the outer cylinder indicate a Weight-On-Wheels condition and will stop the MLG from retracting.

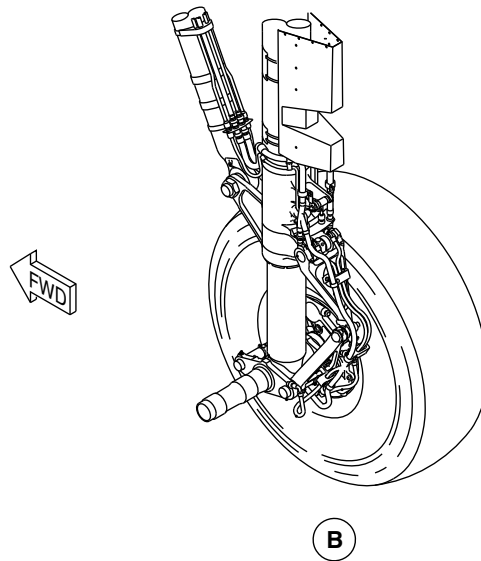


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A LOWER MLG AND WHEEL ASSEMBLY



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Main Landing Gear Locator
Figure 1

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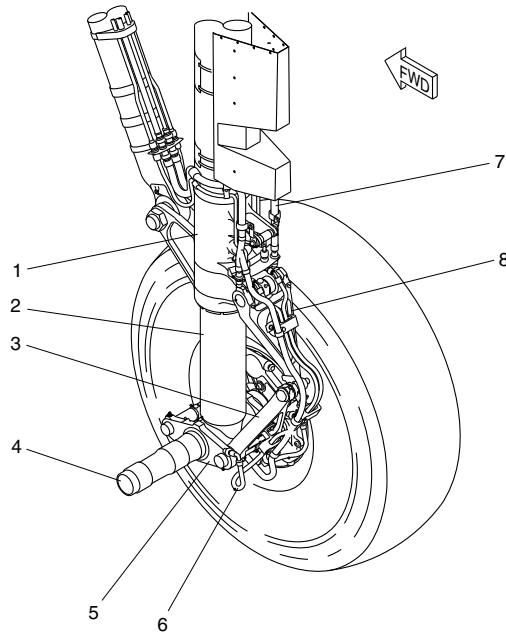
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LEGEND

1. Outer Cylinder.
2. Piston.
3. Lower Torque Link.
4. Axle.
5. Jack Plate.
6. Hydraulic Brake Hose.
7. Electrical Harness.
8. Upper Torque Link.

Main Landing Gear Detail
Figure 2

PSM 1-84-2A
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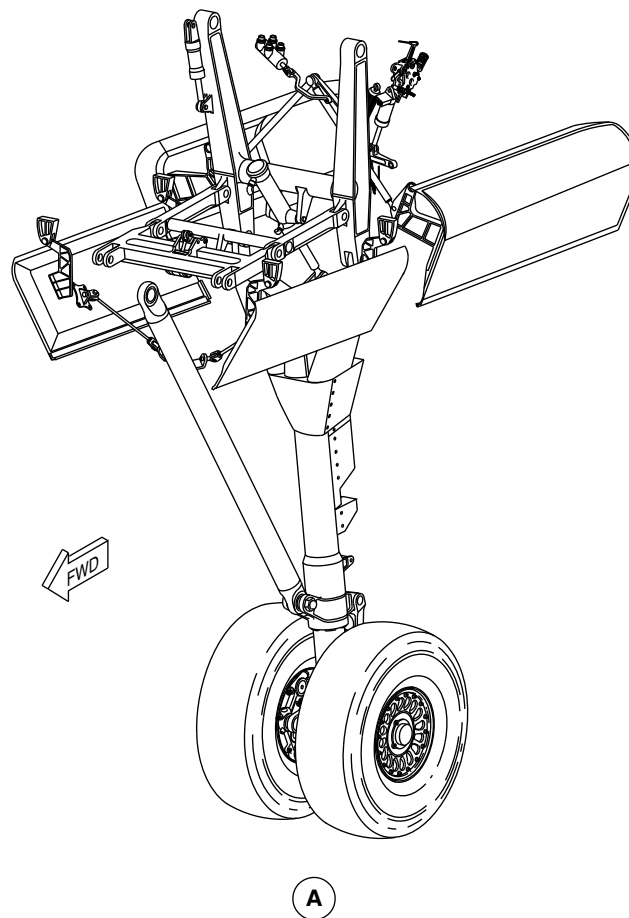
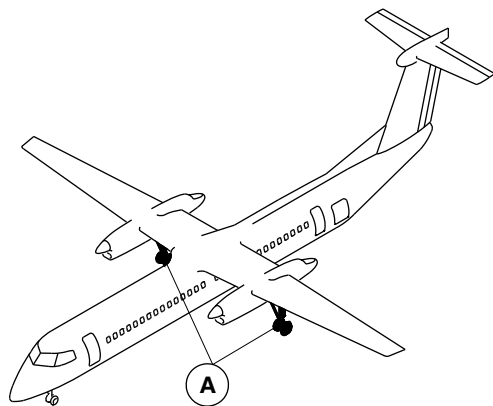
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NOTE

Left gear shown.
Right gear similar.

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Main Landing Gear and Doors Locator
Figure 3

PSM 1-84-2A
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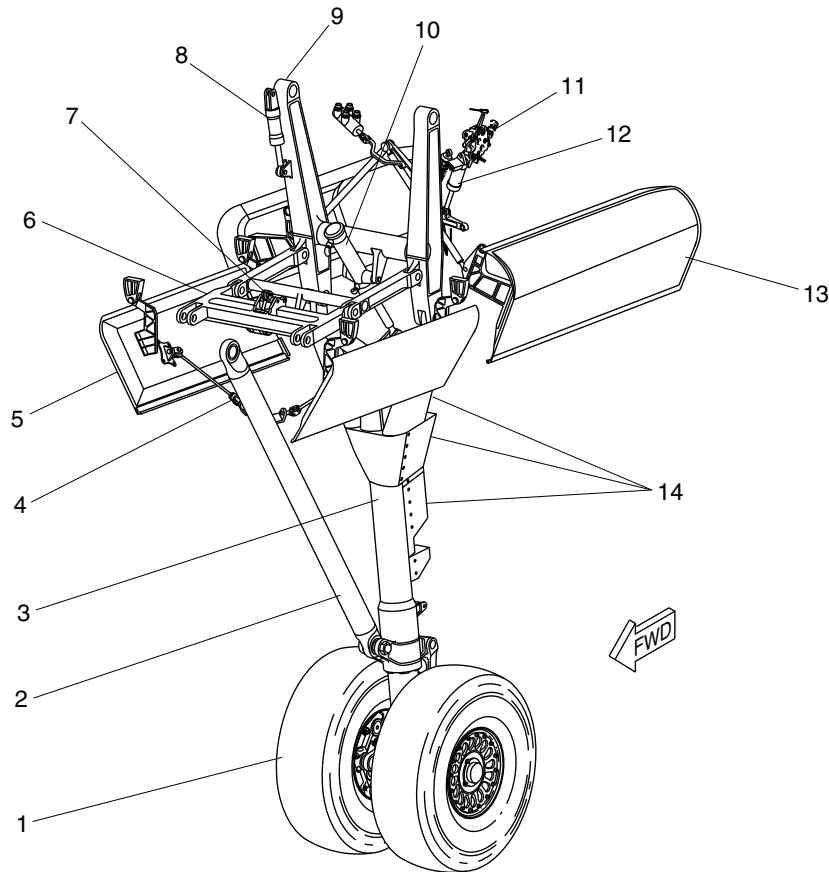
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LEGEND

1. Wheel.
2. Drag Strut.
3. Shock Strut.
4. Forward Door Linkage.
5. Forward Door.
6. Stabilizer Brace.
7. Stabilizer Lock.
8. Auxiliary Extension Actuator.
9. Yoke.
10. Retraction Actuator.
11. Uplock Assembly.
12. Aft Doors Mechanism.
13. Aft Door.
14. Fairings.

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Main Landing Gear and Doors Detail
Figure 4

PSM 1-84-2A
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MAIN LANDING GEAR

Introduction

The Main Landing Gear (MLG) absorbs and distributes the loads of landing. The MLG also supports the aircraft on the ground.

General Description

[Refer to Figure 1.](#)

Each MLG assembly is installed on the airframe structure in the wheelwell of the related engine nacelle. Each MLG has two wheel and tire assemblies and retracts rearwards into the aft section of the wheelwell.

A retraction actuator is attached to the yoke and to the shock strut to extend and retract the MLG. A lock actuator on the stabilizer brace locks the MLG in the down position. An auxiliary extension actuator is attached to the airframe structure in the aft section of the wheelwell and to the arm of the yoke. The auxiliary extension actuator extends the MLG during an alternate extension sequence. An uplock assembly is attached to the top of the aft section of the wheelwell and locks the MLG in the retracted position.

A proximity sensor on the uplock assembly gives a signal to the Proximity Sensor Electronic Unit (PSEU) of the status of the MLG. Proximity sensors on the stabilizer brace send signals to the PSEU when the MLG is down and locked. Proximity sensors on the torque

links give an indication to the PSEU of a Weight-On-wheels or a Weight-Off-Wheels condition.

The Main Landing Gear includes the components that follow:

- Yoke (32-11-21)
- Shock Strut (32-11-01)
- Stabilizer Brace (32-11-16)
- Drag Strut (32-11-11)
- Uplock Assembly (32-31-21)
- Downlock Release Actuator (32-31-16)
- Retraction Actuator (32-31-11)
- Auxiliary Extension Actuator (32-34-06).

Detailed Description

[Refer to Figures 2 and 3.](#)

The yoke is attached to the top of the aft section of the wheelwell. The shock strut is attached to the bottom of the yoke. A stabilizer brace keeps the yoke in position in the wheelwell. The stabilizer brace is attached to the front of the yoke and to the forward frame of the forward section of the wheelwell.

The shock strut is held in position, in the wheelwell, by the drag strut. The drag strut is attached to the bottom of the shock strut cylinder and to the forward frame of the forward section of the wheelwell. Fairings attached to the shock strut streamline the shock strut when the MLG is extended in flight. The shock strut has provision for the wheels, brake units, and anti-skid devices to attach to the axles. A

PSM 1-84-2A

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jack pad on the bottom of the shock strut is for the aircraft to be lifted on jacks to change wheels.

Yoke

The yoke assembly is an H-shaped structural component that connects the upper end of the shock strut to the airframe structure in the aft section of the wheelwell. The top of the yoke is attached to the airframe by two lubricated hinge points. The bottom of the yoke is attached to the top of the shock strut by two lubricated hinge points. The yoke attaches to the stabilizer brace at the cross beam with two lubricated hinge points. The retraction actuator cylinder attaches to the cross beam of the yoke with two lubricated hinge points. The rod end of the auxiliary retraction actuator is attached to lugs at the top of the yoke with lubricated hinge points. The yoke rotates about the top hinge points and the bottom of the yoke moves forward in the wheelwell during the retraction sequence.

Shock Strut

The shock strut outer cylinder is a Y-shaped component with a single stage nitrogen and oil filled shock absorbing piston fitted inside the outer cylinder. Fixed fairings are attached to the forward and aft sides of the shock strut Y-shaped top section and to the outer cylinder. The fairings have an aerodynamic shape and are made from composite material. The anti-skid devices, brake units, and wheel assemblies are attached to a removable and interchangeable axle at the bottom of the shock strut piston. An uplock roller on the back of the outer cylinder engages the uplock assembly in the aft section of the wheelwell to lock the MLG in the retracted position.

Rotation of the shock strut piston and axle is prevented by two connected torque links. The lower torque link is attached to two lugs

on the rear of the shock strut piston. The upper torque link is attached to two lugs on the bottom of the outer cylinder.

The top end of the shock strut is attached to the yoke with lubricated hinge points. The lower end of the shock strut outer cylinder is attached to the lower end of the drag strut with a lubricated hinge point. The rod end of the retraction actuator is attached to the top of the shock strut with a lubricated hinge point.

Proximity sensor targets on the torque links move as the piston moves to give a Weight-On-Wheels or a Weight-Off-Wheels indication to the PSEU. The proximity sensor and anti-skid wheel speed transducer electrical harnesses and brake hydraulic lines are fitted to the bottom section of the shock strut cylinder. Nitrogen and oil charging and bleeding valves are located on the outer cylinder.

Stabilizer Brace

The stabilizer brace is a two piece folding structural component. The stabilizer brace keeps the yoke and the shock strut in position when the MLG is in the extended or retracted position. The forward section of the stabilizer brace is attached to the airframe structure in the forward section of the wheelwell with two lubricated hinge points. The aft section of the stabilizer brace is attached to the yoke with two lubricated hinge points.

The stabilizer brace has an overcenter link subassembly attached between the two sections of the stabilizer brace. The overcenter link subassembly supplies a mechanical lock for the MLG in the down position. The mechanical lock is broken by the lock actuator. The two sections of the link subassembly fold upward to move the bottom of the yoke forward during the MLG retraction sequence. The links are moved into a mechanical lock position by the lock actuator during the



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

MLG extension sequence. Two lock springs keep the links in the mechanical lock position.

The lock springs lock the stabilizer brace in the down position when the alternate extension system is in operation. The stabilizer brace has provision for the installation of a ground lock pin when the MLG is extended.

Drag Strut

The drag strut is a fixed length tubular component. The drag strut is attached at the lower end to the shock strut outer cylinder with a lubricated hinge point. The drag strut is attached at the upper end to the airframe structure of the forward section of the wheelwell with a lubricated hinge point. A bracket assembly near the top end of the drag strut is to attach to the linkages of the forward doors of the wheelwell.

Brackets attach the electrical harnesses for the proximity sensors and wheel speed transducers, and hydraulic pipes for the brake units along the rear of the drag strut. A grounding point to ground the aircraft during maintenance procedures is fitted to the top of the drag strut.

The landing gear pivots about the drag strut upper end hinge during MLG retraction and extension.

Uplock Assembly

[Refer to Figures 7 and 8.](#)

The uplock assembly is made up of two side plates to form the body of the uplock assembly. The uplock assembly is made up of a latch

assembly, a lock link, an actuator assembly, an alternate release arm, internal extension springs, and a torsion spring.

The uplock assembly is attached to a frame in the top of the aft section of the wheelwell. The latch assembly and the lock link are attached to the body with lubricated hinge points. The latch assembly and the lock link pivot in the body of the uplock assembly. This action locks the uplock roller of the MLG in the retracted position.

The uplock roller on the MLG engages the inner face of the latch assembly and the latch assembly pivots into the locked position. The force of the two extension springs keeps the latch assembly fully rotated to lock the MLG in the retracted position. The lock link prevents the latch assembly from rotating to the unlocked position.

The lock link is rotated to the unlocked position when the piston of the uplock actuator extends under pressure from the No. 2 hydraulic system. The latch assembly returns to the locked position when hydraulic pressure is removed from the uplock actuator. The external compression spring on the uplock actuator returns the piston to the retracted position when hydraulic pressure is removed from the uplock actuator.

When the landing gear alternate release handle in the flight compartment is pulled, a cable rotates the alternate release arm. The lower portion of the alternate release arm touches the latch assembly and rotates the latch assembly to release the MLG uplock roller. The MLG is partially lowered by gravity. The torsion spring returns the alternate release arm to the normal position when the alternate release handle in the flight compartment is reset.

A proximity sensor mounted on the uplock assembly gives an indication to the PSEU on the status of the uplock assembly. The target on the lock link moves away from the proximity sensor when the lock link is rotated during the MLG extension sequence. This



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gives an indication to the PSEU that the uplock assembly is in the unlocked position. The target on the lock link moves to the proximity sensor when the lock link is rotated during the MLG retraction sequence. This gives an indication to the PSEU that the uplock assembly is in the locked position.

Downlock Release Actuator

[Refer to Figure 5.](#)

[Refer to Figure 6.](#)

The downlock release actuator is of a simple two port design with hydraulic pressure supplied by the No.2 hydraulic system. The downlock release actuator is attached to the stabilizer brace forward section with lubricated hinge points. The rod end of the downlock release actuator is attached to the forward lock link of the stabilizer brace assembly with a lubricated hinge point.

The piston of the downlock release actuator retracts to break the lock links overcenter position. Further movement folds the stabilizer brace and pulls the yoke forward during the retraction sequence. The piston of the downlock release actuator extends to bring the lock links into an overcenter position to lock the stabilizer brace.

Retraction Actuator

[Refer to Figure 4.](#)

The retraction actuator is of a simple two port design with hydraulic pressure supplied by the No.2 hydraulic system. The retraction actuator is attached to the yoke cross beam lugs with lubricated hinge points. The rod end of the retraction actuator is attached to the top of the shock strut with a lubricated hinge point.

The piston of the retraction actuator retracts to retract the MLG. The piston extends to lower the MLG.

Auxiliary Extension Actuator

[Refer to Figure 9.](#)

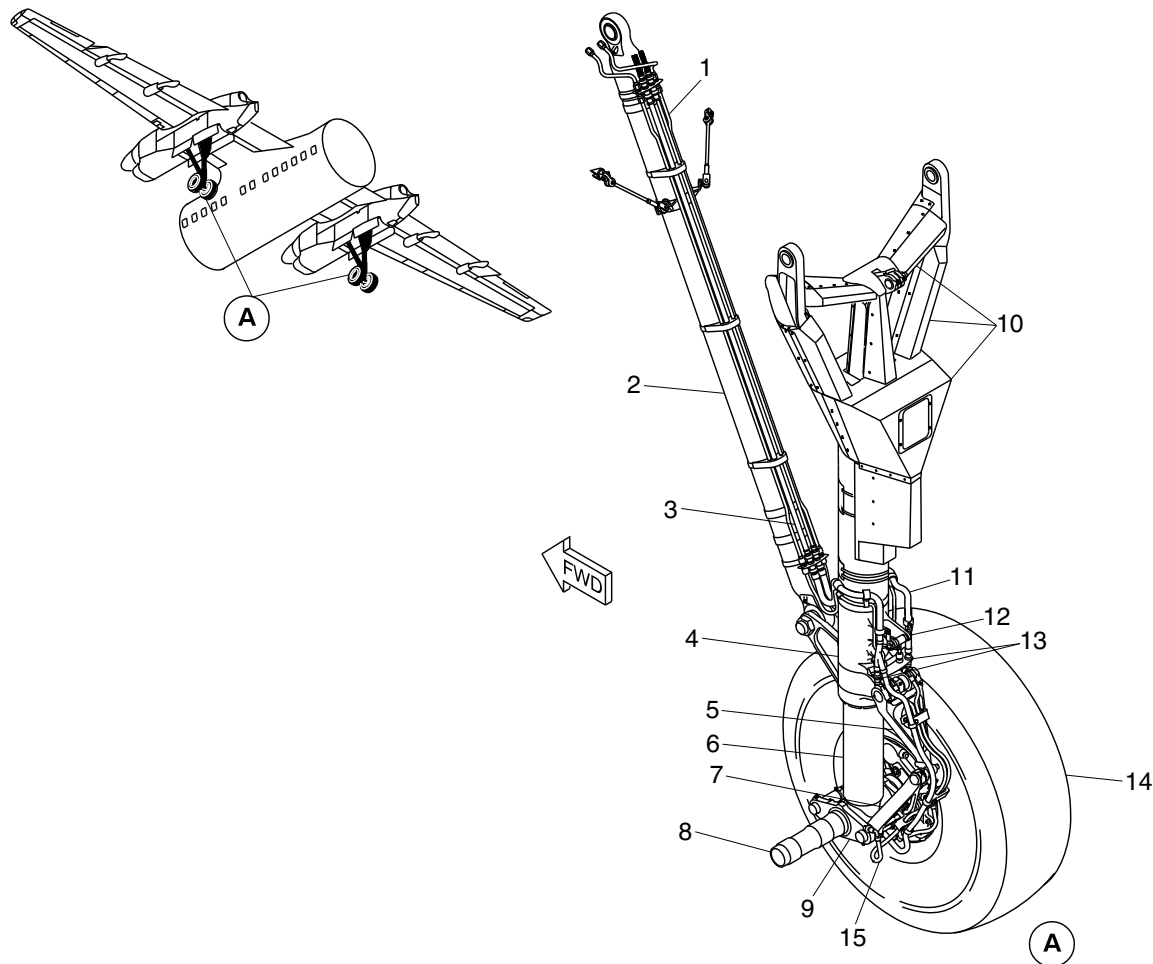
The auxiliary extension actuator is of a one–port design with hydraulic pressure supplied by the handpump hydraulic system. The auxiliary extension actuator attaches to the airframe structure at the top of the wheelwell. The rod end of the auxiliary extension actuator attaches to the forward side of the yoke with a lubricated hinge point.

The piston of the auxiliary extension actuator extends to lower the MLG during the alternate extension sequence.



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LEGEND

1. Electrical harness.
2. Drag strut.
3. Hydraulic brake line.
4. Shock strut.
5. Upper torque link.
6. Shock strut piston.
7. Lower torque link.
8. Axle.
9. Jack pad.
10. Fairing.
11. Electrical harness.
12. Uplock roller.
13. Proximity sensor.
14. Wheel and brake assembly.
15. Hydraulic brake line.

fs572a01.dg, ck/gv, jul14/2011

Main Landing Gear – Component Details
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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Config 001

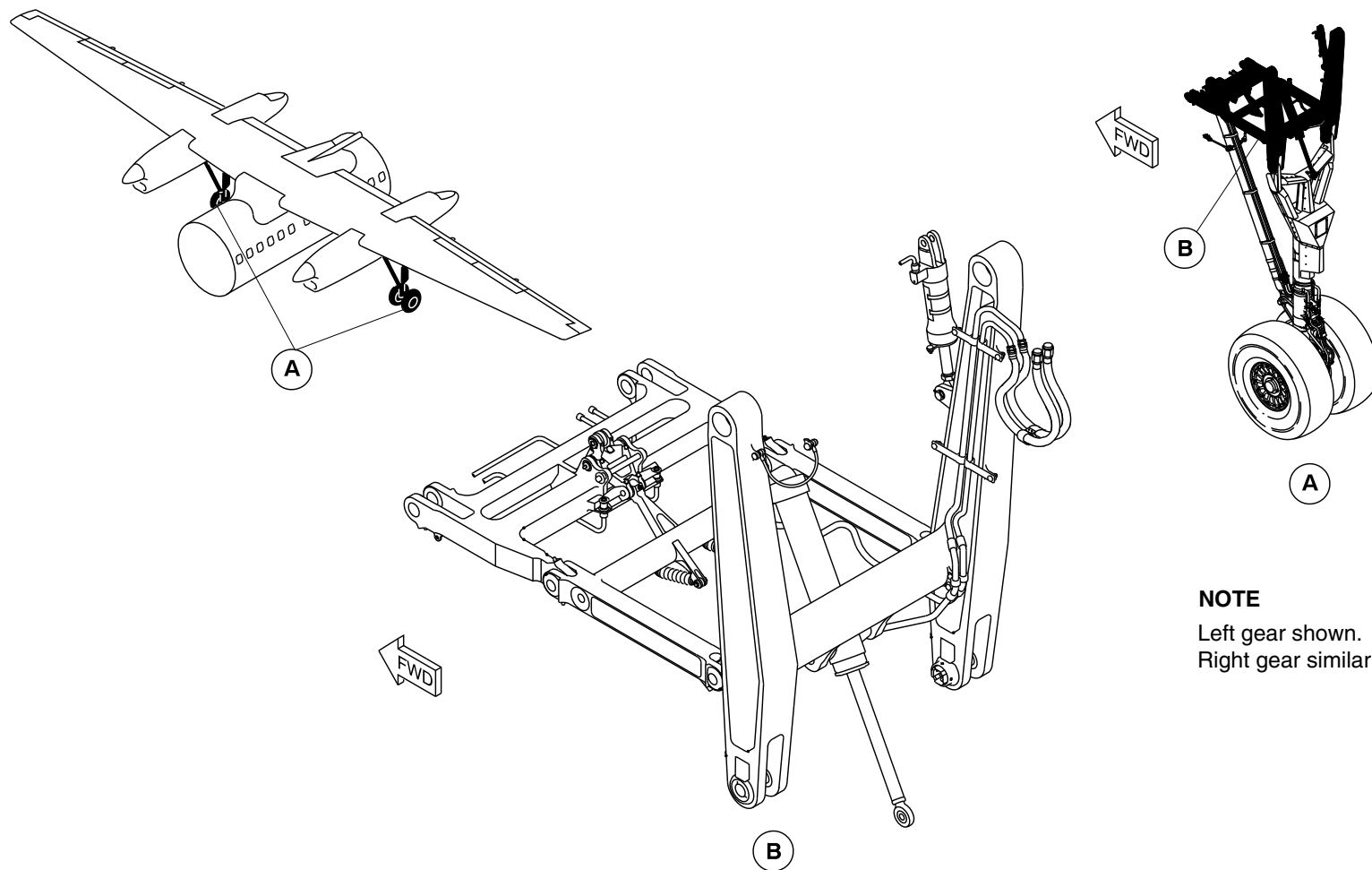
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NOTE

Left gear shown.
Right gear similar.

fs567a01.dg, ck, jun23/2011

MLG Stabilizer Brace Locator
Figure 2

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 32-11-00
Config 001

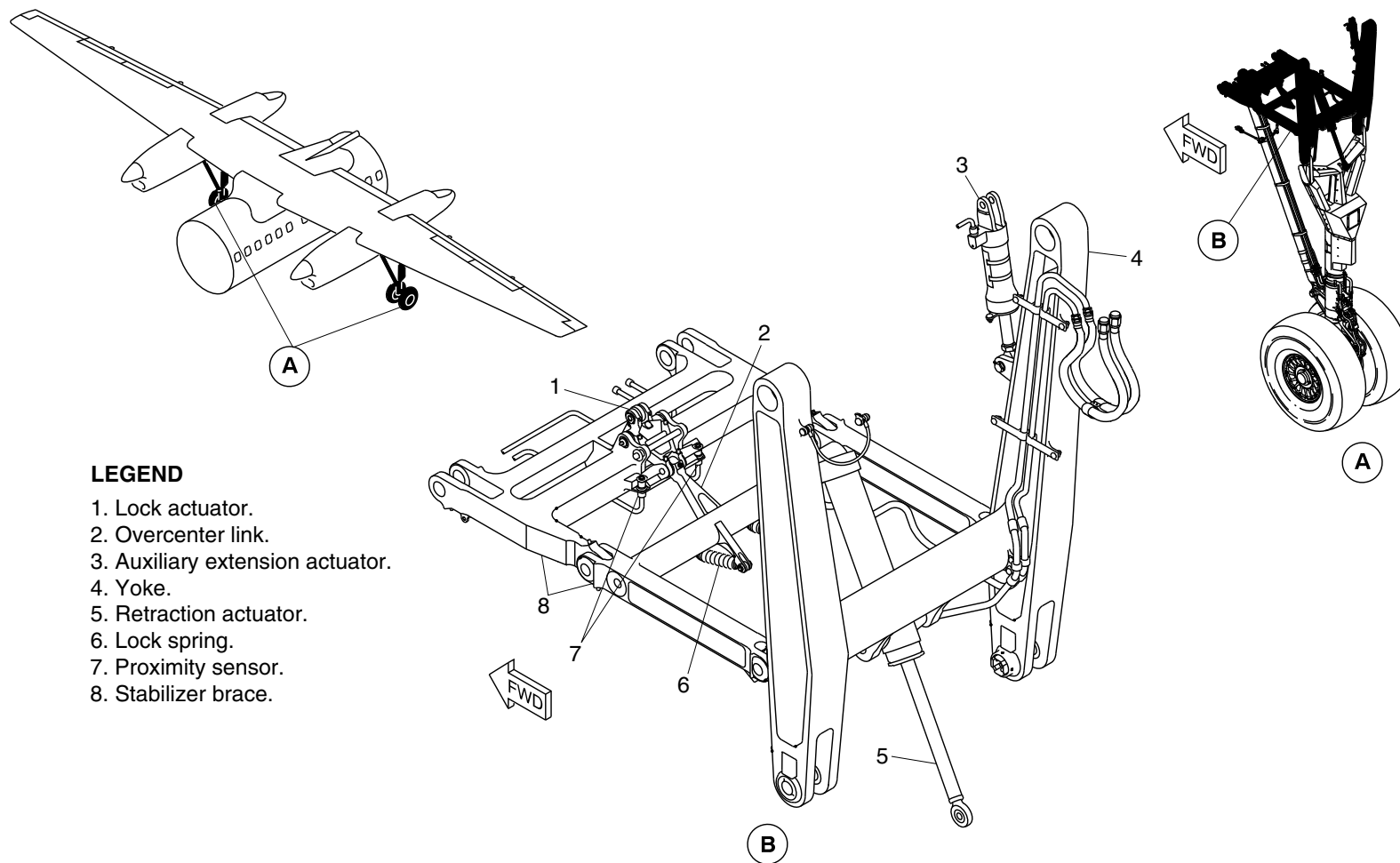
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MLG Stabilizer Brace Detail
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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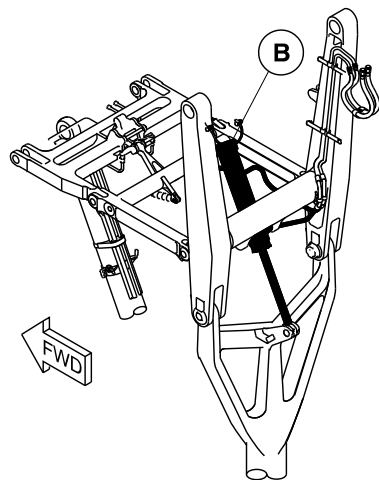
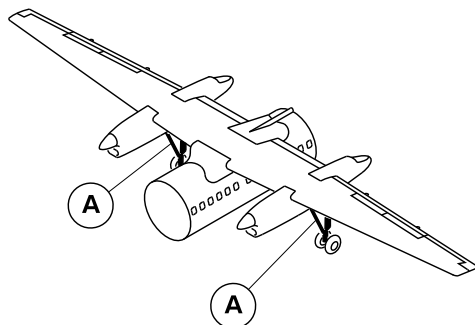


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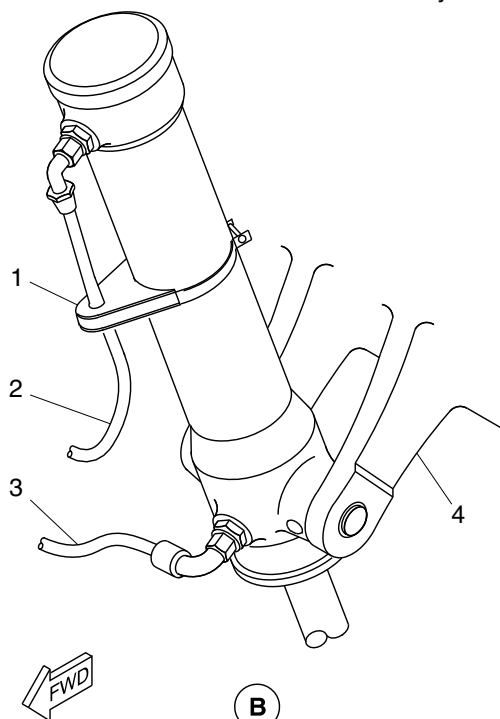
LEGEND

1. Insert.
2. Hydraulic line extension.
3. Hydraulic line retraction.
4. Yoke assembly.

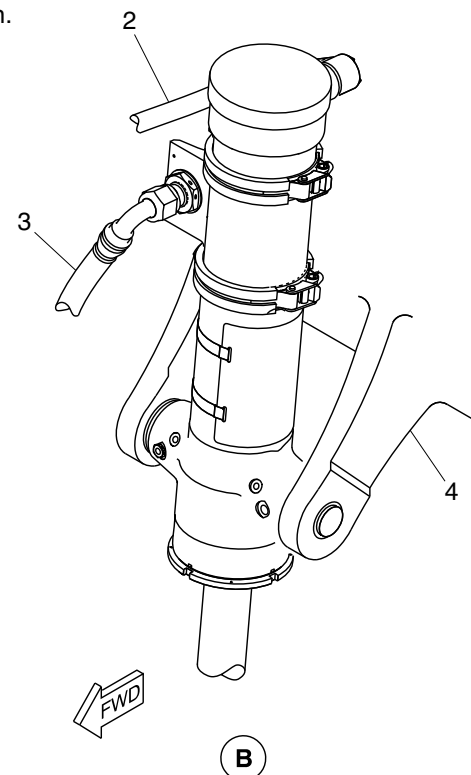


MAIN LANDING GEAR (MLG)

fs105a01.dg, ck/cm, jan10/2013



**PRE MOD 4-902327
PRE SB84-32-105**



**POST MOD 4-902327
POST SB84-32-105**

MLG Retraction Actuator Locator
Figure 4

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 32-11-00
Config 001

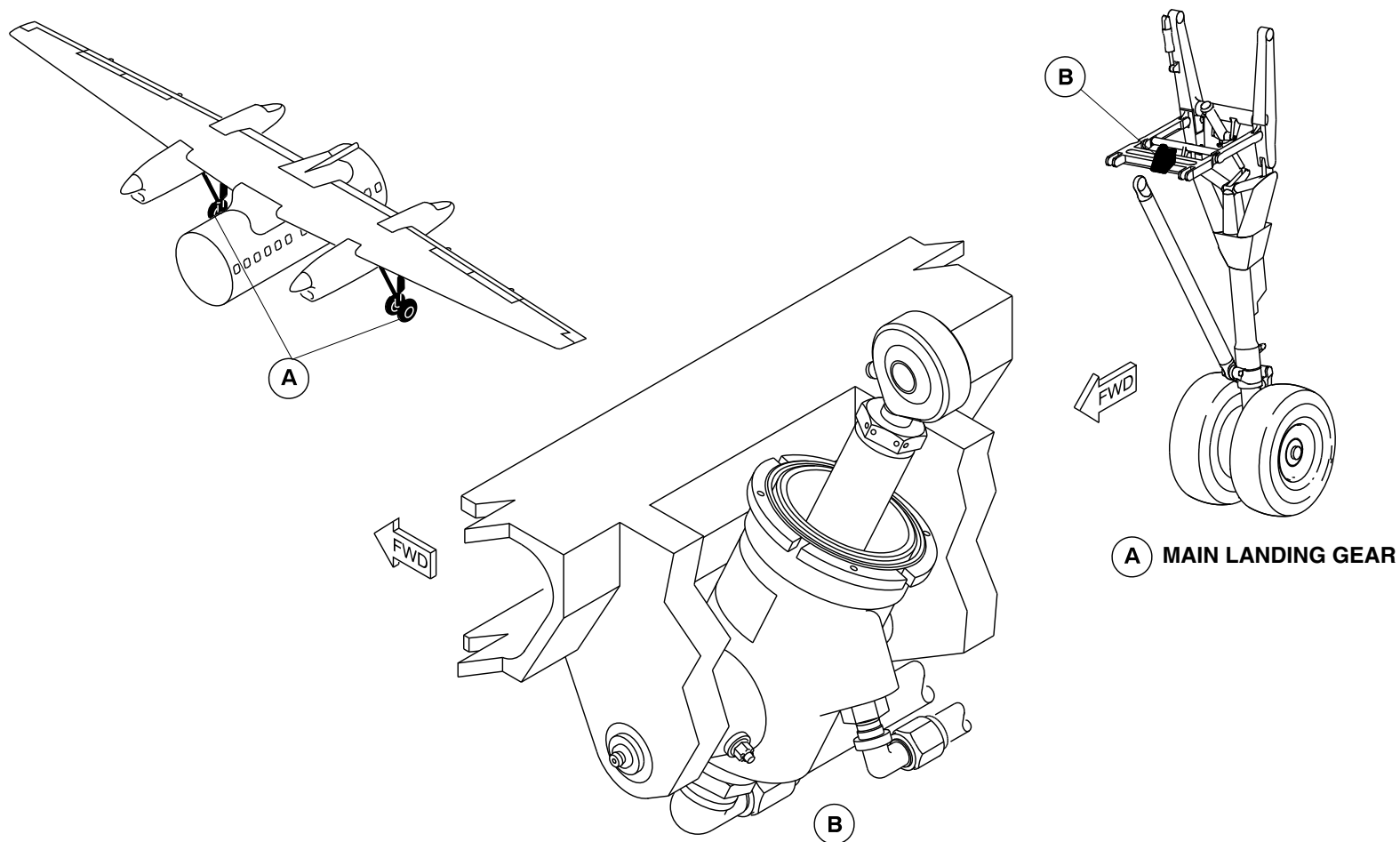
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fs306a01.dg, ck, jun21/2011

MLG Downlock Actuator Locator
Figure 5

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 32-11-00
Config 001

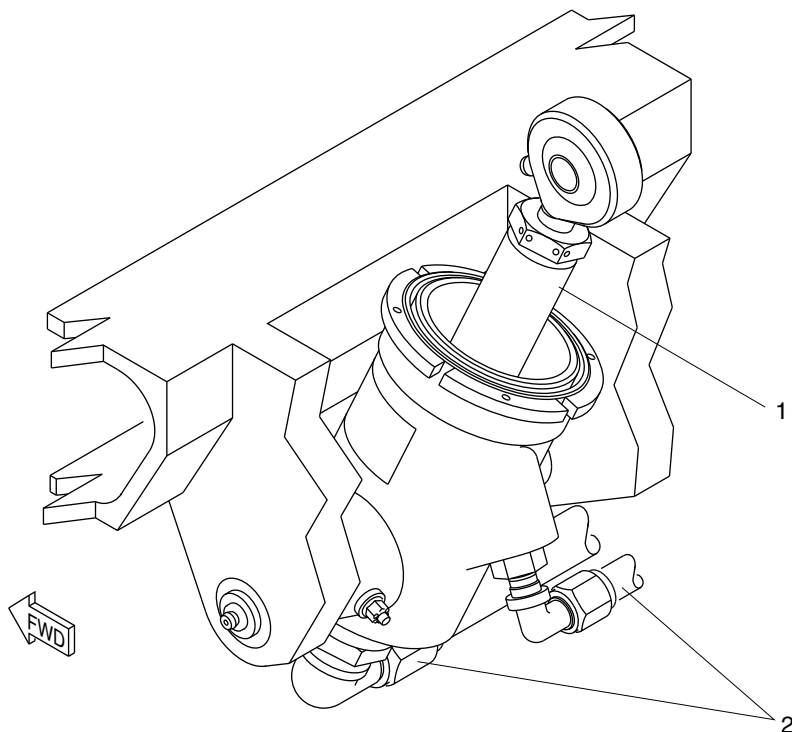
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LEGEND

- 1. MLG Downlock Actuator.
- 2. Hydraulic Hose.

fs307a01.cgm

MLG Downlock Actuator Detail
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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Config 001

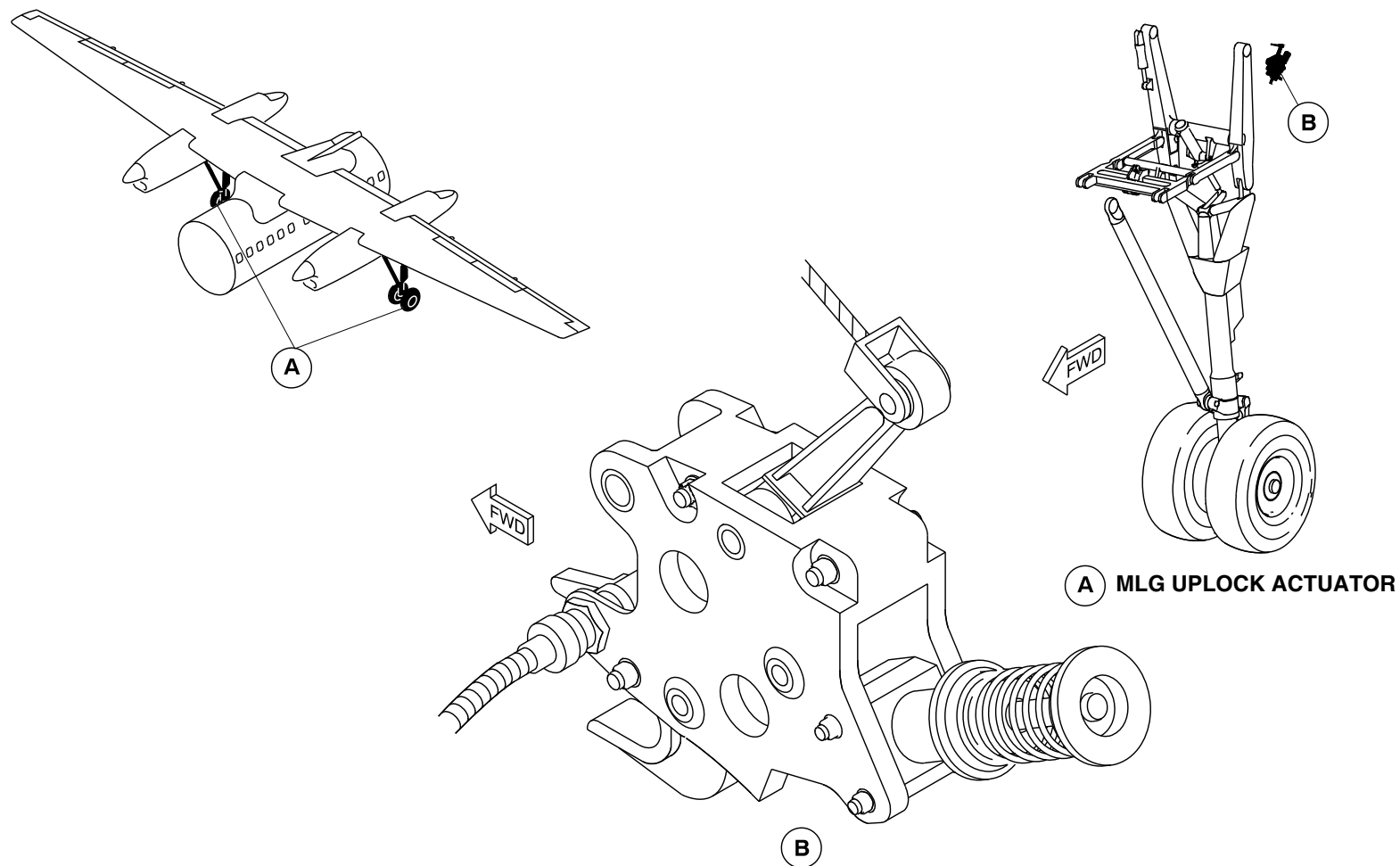
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fs134a01.dg, ck, jun21/2011

MLG Uplock Actuator Locator
Figure 7

PSM 1-84-2A
EFFECTIVITY:
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Config 001

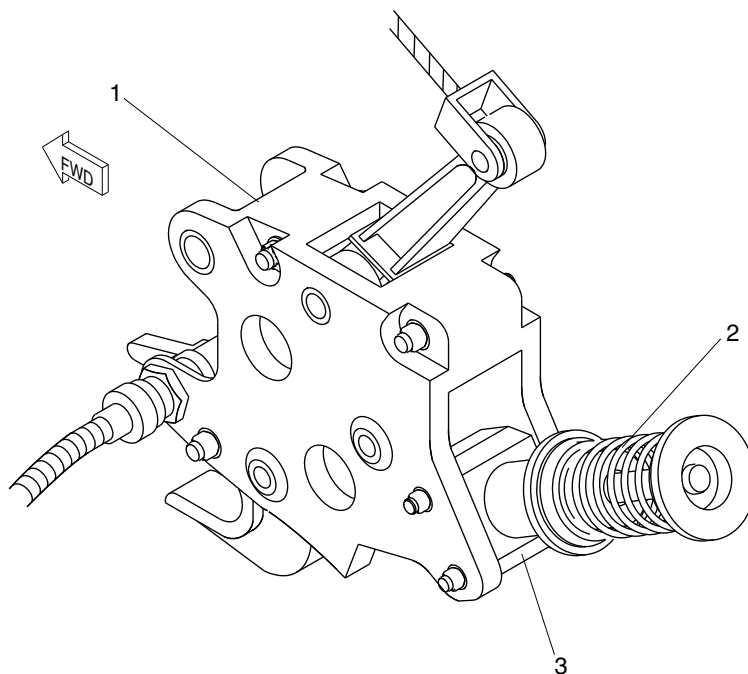
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LEGEND

- 1. MLG Uplock Assembly.
- 2. External Spring.
- 3. MLG Uplock Actuator.

fs138a01.cgm

MLG Uplock Actuator Detail
Figure 8

PSM 1-84-2A
EFFECTIVITY:
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Config 001

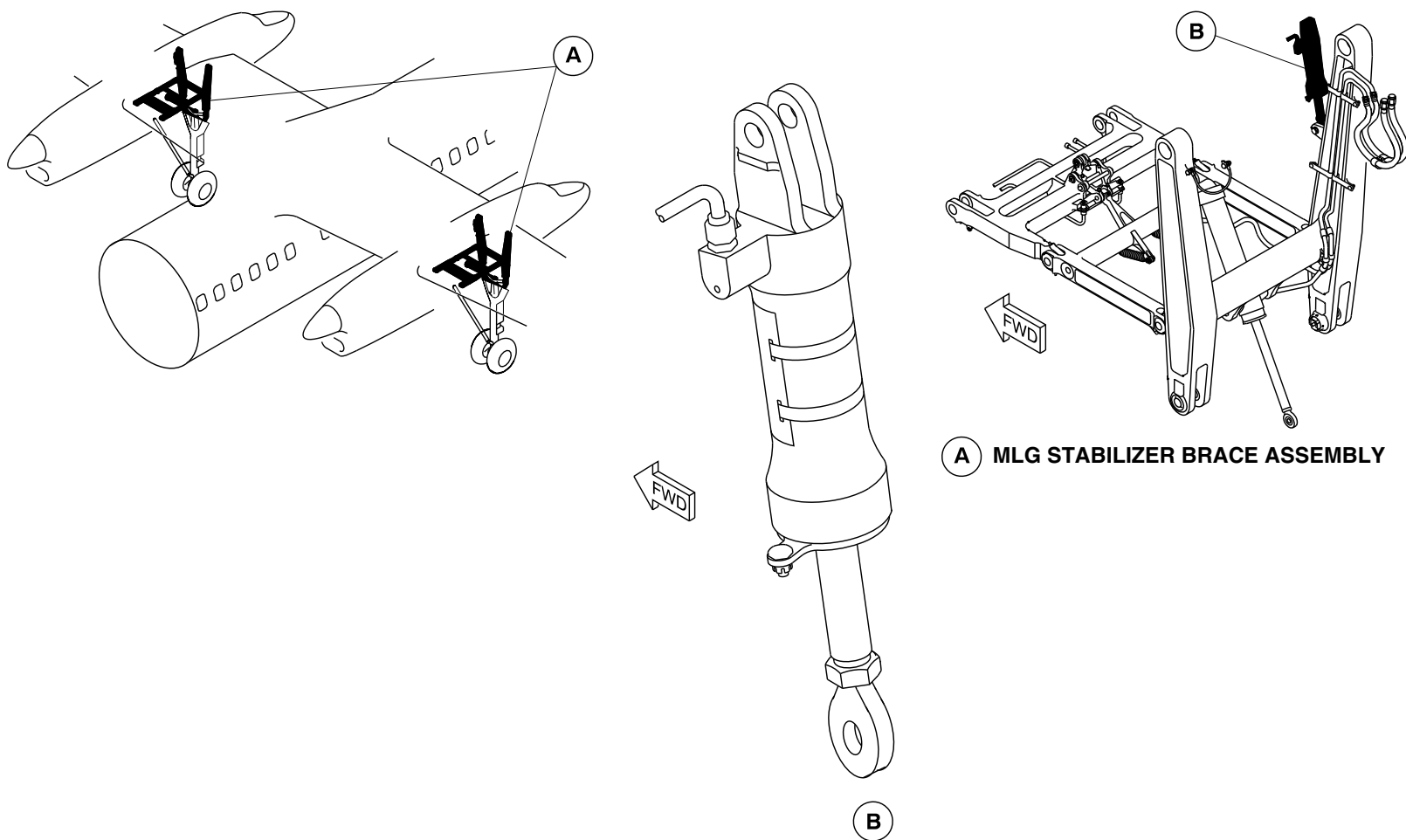
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fs146a01.dg, ck, jun23/2011

Alternate Extension Actuator
Figure 9

PSM 1-84-2A
EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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MAIN LANDING GEAR DOORS

Introduction

The Main Landing Gear (MLG) doors enclose the retracted MLG in the engine nacelle. The four doors on each engine nacelle make up the fairings for the MLG in the retracted position.

General Description

[Refer to Figures 1 , 2 and 3.](#)

Two doors close off the forward section of the MLG wheelwell. The forward doors are open when the MLG is extended and closed when the MLG is retracted. The doors are opened and closed by adjustable links connected to the the drag strut.

Two doors close off the aft section of the MLG wheelwell. The aft doors are closed when the MLG is retracted and closed when the MLG is extended.

An alternate release arm is used to open the doors during alternate extension of the MLG.

The status of the aft doors is shown on the LANDING CONTROL panel in the flight compartment.

The MLG include the components that follow:

- Forward Doors (32-12-00)
- Aft Doors (32-12-01)

- Forward Doors Linkages (32-12-16)
- Aft Doors Linkage Mechanism (32-12-06)
- Aft Doors Actuator (32-31-26).

Detailed Description

[Refer to Figure 2.](#)

The forward doors are operated by mechanical links attached to the drag strut of the MLG. When the MLG is extended the forward doors are opened mechanically. When the MLG is retracted the doors are closed mechanically

The aft doors are opened and closed by a single hydraulic actuator and a linkage mechanism. The linkage mechanism has a mechanical sequence valve that is connected to the pivot arm by a rod. The door pushrods, lock links, and the tension spring are attached to the pivot arm. The pushrods connect the doors to the pivot arm. The lock links hinge point supplies an overcenter lock for the doors in the closed position.

An alternate release arm breaks the overcenter lock of the lock links hinge point to open the doors during alternate extension of the MLG.

A proximity sensor sends a signal to the Proximity Sensor Electronic Unit (PSEU). The PSEU sends a signal to the LANDING GEAR Control Panel in the flight compartment, to show the status of the aft doors.

PSM 1-84-2A
EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Forward doors

Refer to Figure 4.

The forward doors are double skinned and are made in the same shape as the bottom contour of the center section of the engine nacelle. The doors are connected to the bottom of the center section of the engine nacelle by hinges. Each door has a hinge on the forward and aft corner. The doors are symmetrical and meet at the engine nacelle centerline when closed. The door operating links, are attached to fittings on the forward section of the doors and to the MLG drag struts.

Forward Doors Linkages

Refer to Figure 5.

The forward door linkages have articulated, adjustable links. The links are attached to a bracket assembly on the upper part of the drag strut and to fittings on the forward section of the doors. The links open the doors as the drag strut moves when the MLG extends. The links close the doors as the drag strut moves when the MLG retracts.

Aft Doors

Refer to Figure 6.

The aft doors are double skinned and are made in the same shape as the bottom contour of the aft section of the engine nacelle. The doors are connected to the bottom of the aft section of the engine nacelle by hinges. Each door has a hinge on the forward corner and a hinge on the aft corner. The doors are symmetrical and meet at the

engine nacelle centerline when closed. The door operating linkage mechanism is attached to the forward hinges of the doors.

Aft Doors Linkage Mechanism

Refer to Figure 7.

The door linkage has two push rods that are attached to the forward hinges of the aft doors and to the pivot arm. The pivot arm is attached to a frame assembly on the left side of the aft section of the wheelwell. The pivot arm has hinge points to form a bellcrank between the push rods and the lock links. The lock links are attached to the bottom of the pivot arm and to the side of the aft section of the wheelwell. The lock links hinge point is pushed into an overcenter position by the actuator to lock the doors in the closed position.

The lock links hinge point is pulled from an overcenter position by the actuator to open the doors. The lock links fold and rotate the pivot arm to push open the doors with the push rods. During the alternate extension of the MLG, the alternate release arm rotates to move the lock links from an overcenter position. A tension spring is attached to the pivot arm and to the side of the aft section of the wheelwell. The tension spring turns the pivot arm for the push rods to push open the doors.

Refer to Figure 8.

A proximity sensor is installed on the left side of the aft section of the wheelwell. The proximity sensor sends a signal to the Proximity Sensor Electronic Unit (PSEU). The PSEU sends a signal to the LANDING GEAR control panel, located in the flight compartment, to show the status of the aft doors. When the doors are open the amber L.DOOR and R.DOOR lights come on. The lock links in an overcenter position will show that the doors are closed and locked, and the lights will go off. When the lock links are moved from the



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

overcenter position, the sensor target on the lock links is moved away from the proximity sensor. This will give an indication that the doors are not closed and locked.

Aft Doors Actuator

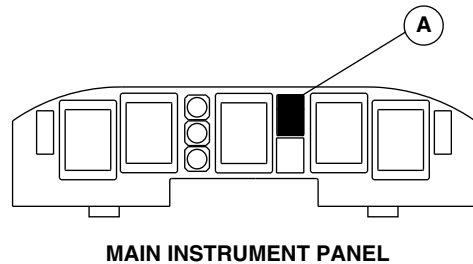
[Refer to Figure 7.](#)

The hydraulic actuator is a two port design connected to the No.2 hydraulic system of the aircraft. The actuator is located on the left side of the aft section of the wheelwell. Lubricated hinge points attach the body of the actuator to the side of the wheelwell and the rod end of the piston to the pivot arm. The hydraulic actuator is sequenced by a solenoid sequence valve. The hydraulic actuator retracts to open the doors and extends to close the doors.



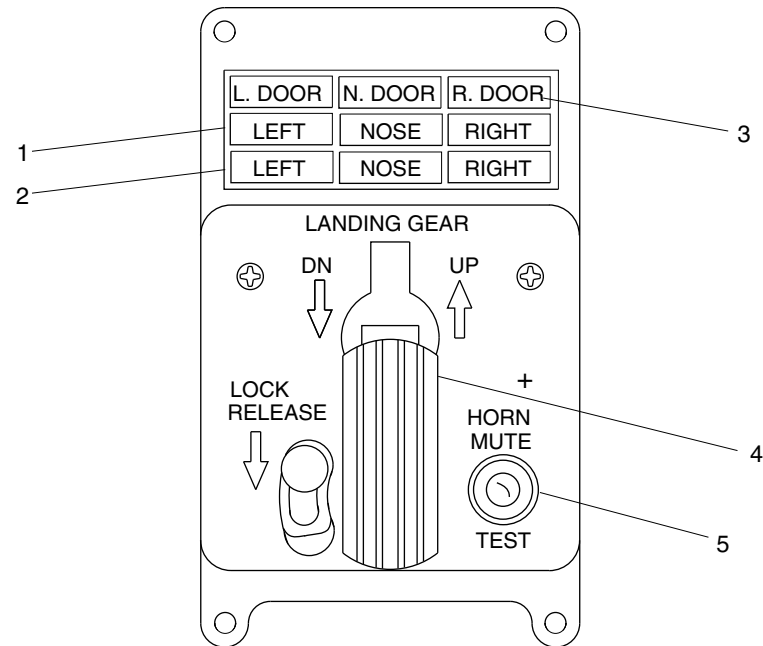
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. "Green" Gear Downlocked.
2. "Red" Gear Not Uplocked Or Downlocked.
3. "Amber" Door Open.
4. Handle "Amber" (Concurrent W/"Red" Unsafe Lights) Gear In-Transit.
5. Warning Horn Mute/Test Switch.



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Main Landing Gear Control
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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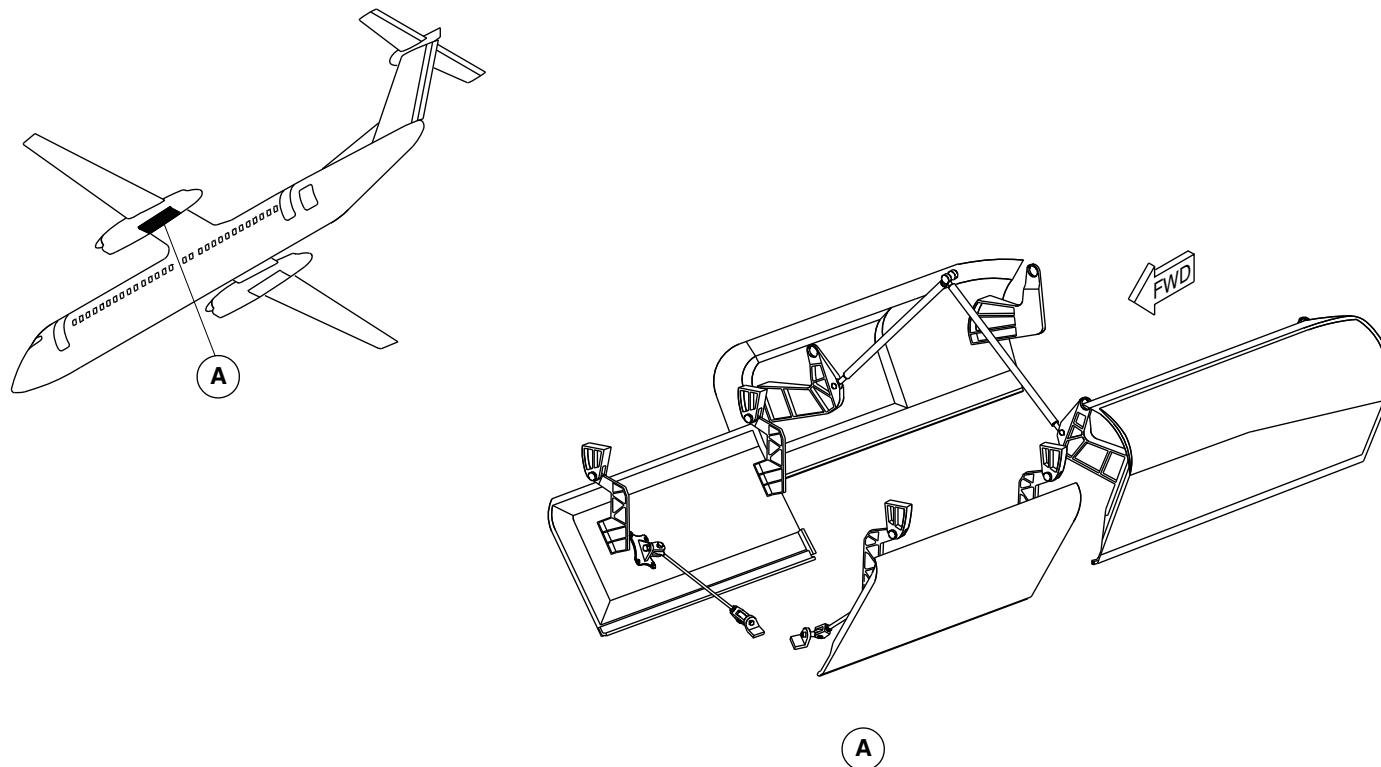
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Main Gear Doors Locator
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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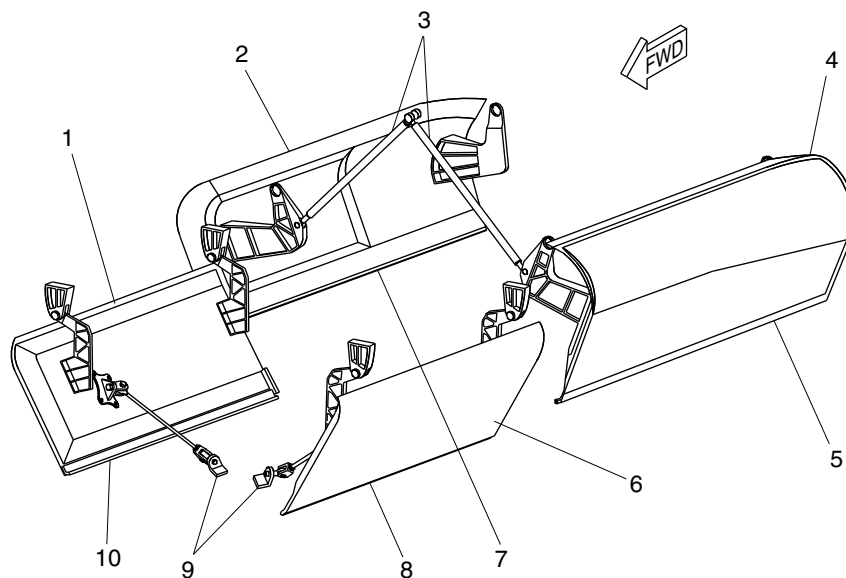


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LEGEND

1. Forward Door.
2. Aft Door.
3. Push Rods.
4. Aft Door.
5. Door Seal.
6. Forward Door.
7. Door Seal.
8. Door Seal.
9. Forward Door Connecting Rods.
10. Door Seal.



fs157a01.cgm

Main Gear Doors Detail
Figure 3

PSM 1-84-2A
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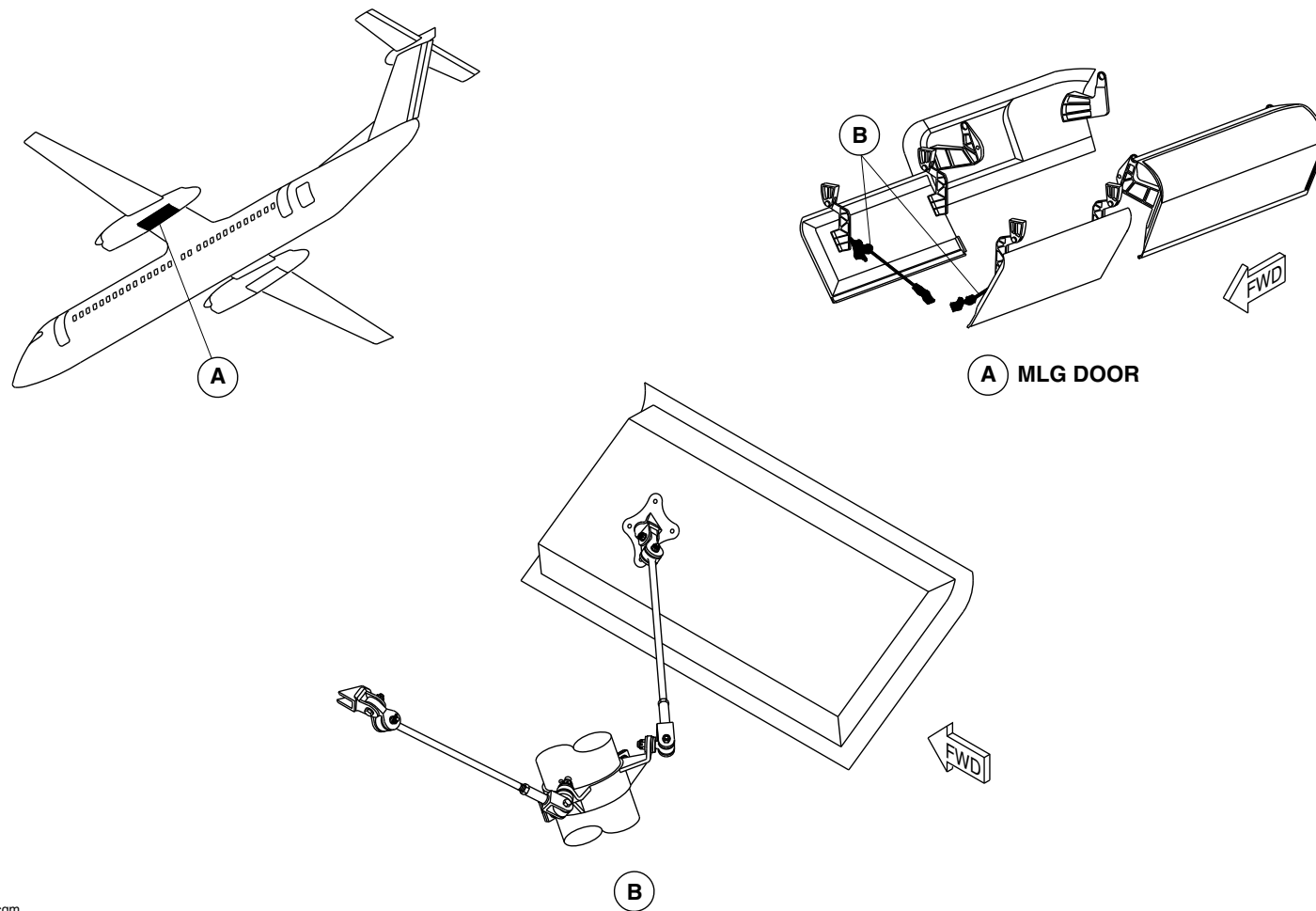
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fs158a01.cgm

Main Gear Doors Fwd Mechanism Locator
Figure 4

PSM 1-84-2A
EFFECTIVITY:
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Config 001

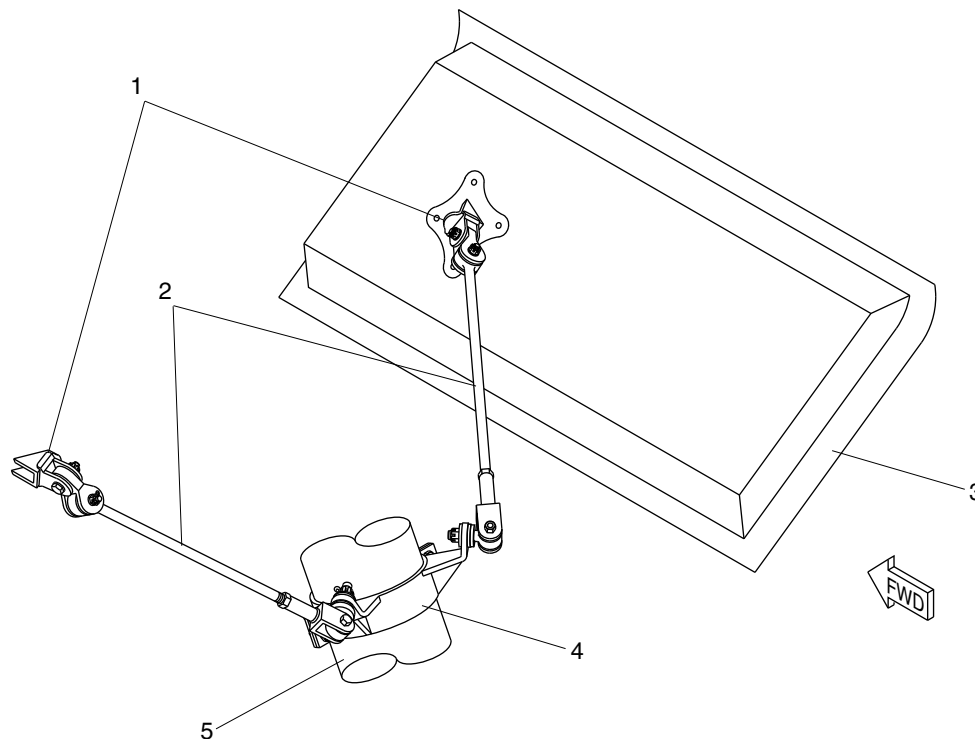
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LEGEND

- 1. Door Fitting.
- 2. Push Rods.
- 3. Forward Door.
- 4. Bracket Assembly.
- 5. Drag Strut.

fs159a01.cgm

Main Gear Doors Fwd Mechanism Detail
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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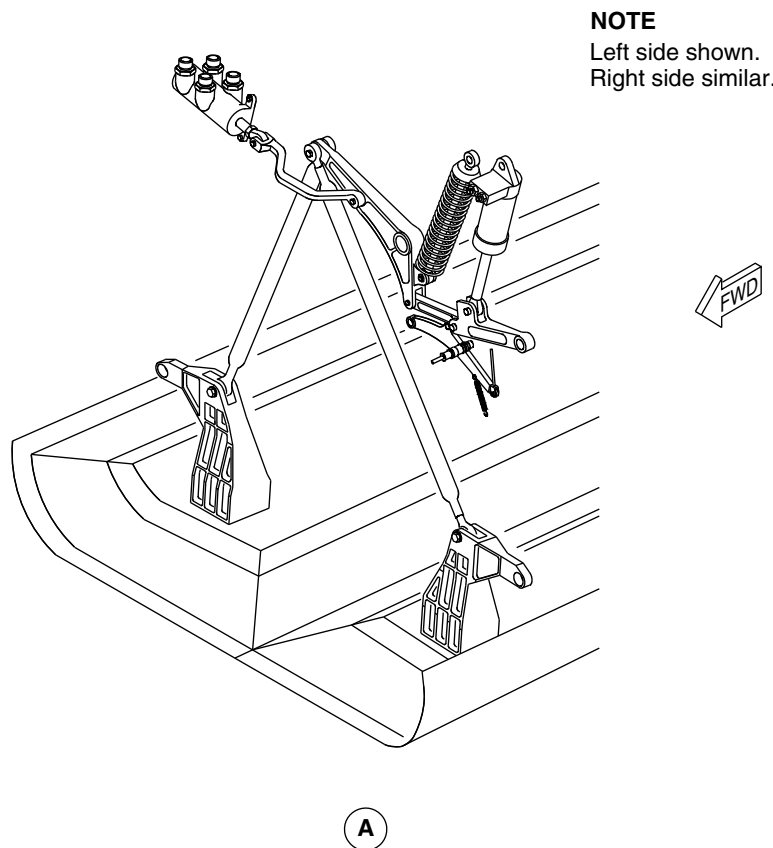
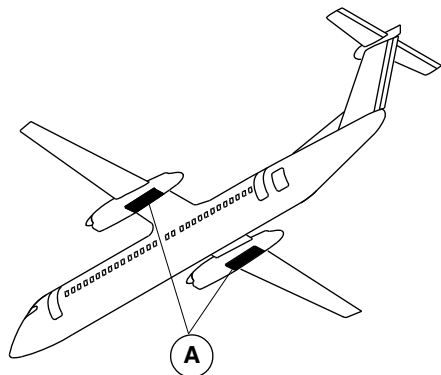
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fs660a01.cgm

MLG Aft Doors Linkage Mechanism Locator
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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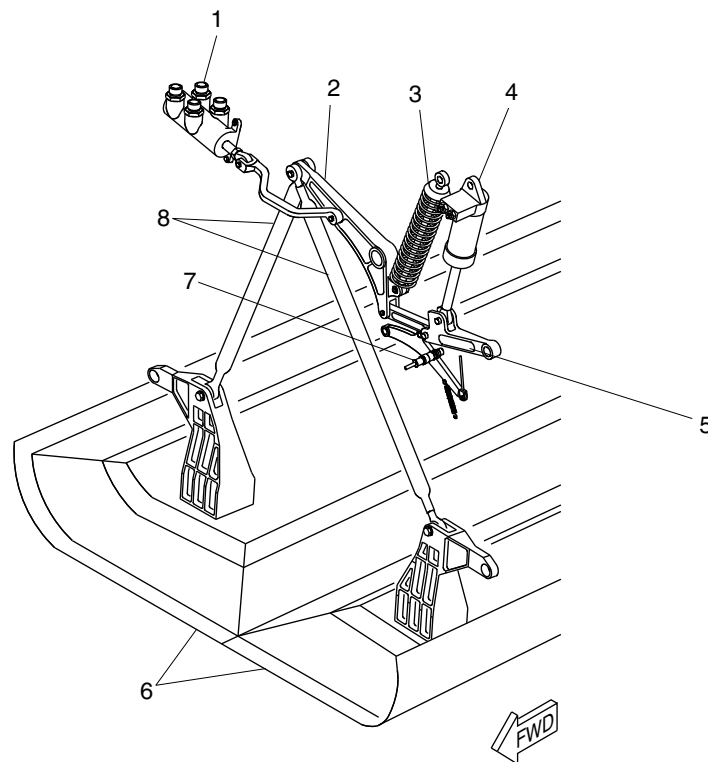
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LEGEND

- 1. Mechanical Sequence Valve.
- 2. Pivot Arm.
- 3. Tension Spring.
- 4. Actuator.
- 5. Lock Link.
- 6. Aft Door.
- 7. Proximity Sensor.
- 8. Push Rods.

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MLG Aft Doors Linkage Mechanism Detail
Figure 7

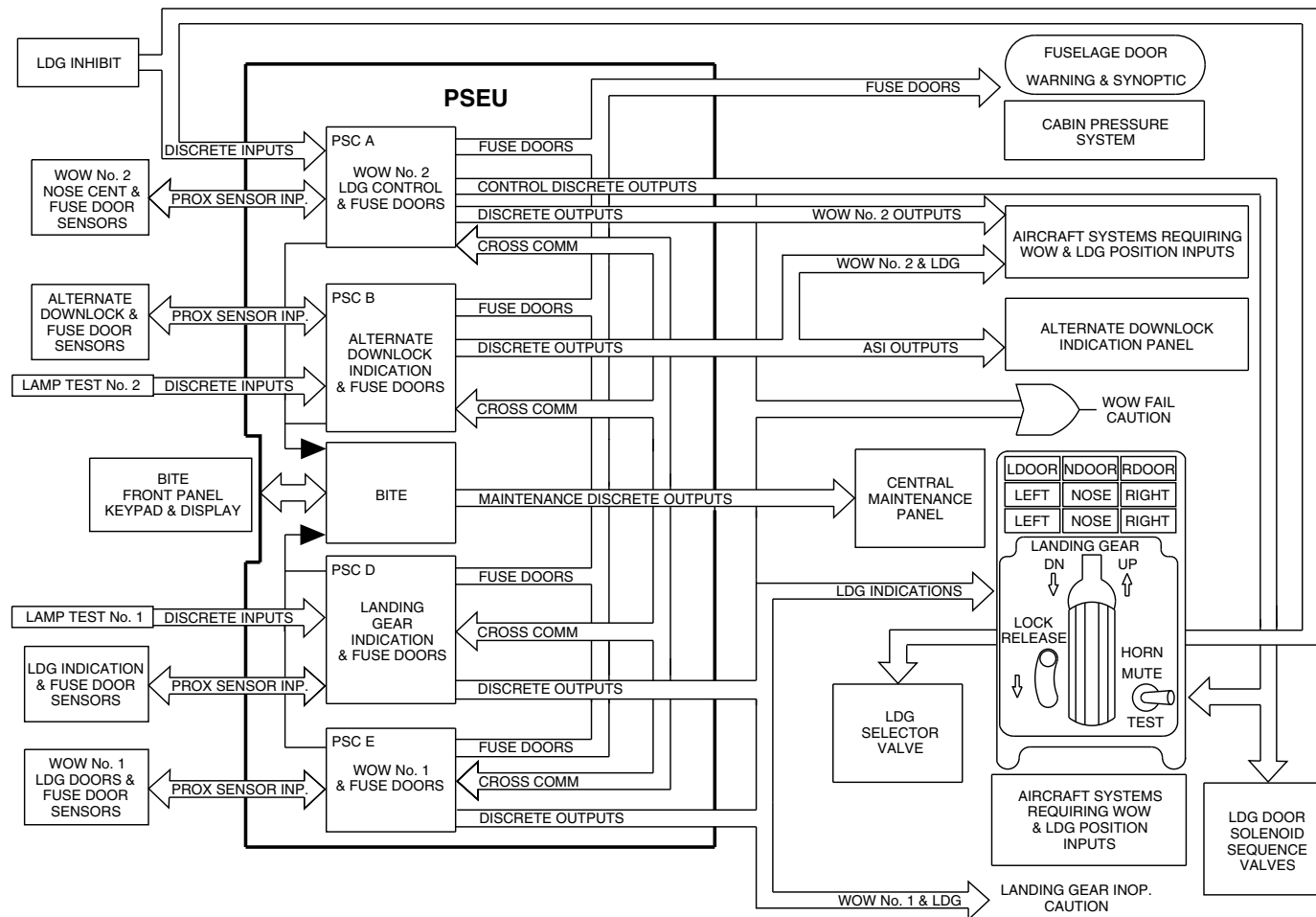
PSM 1-84-2A
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fs582a01, bm, jun06/2011

PSEU Inputs/Outputs Block Diagram
Figure 8

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**ON A/C ALL

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NOSE LANDING GEAR

Introduction

The nose landing gear (NLG) lets the forward fuselage absorb the shock of landing and gives stability and direction during aircraft taxi. The NLG doors enclose the retracted NLG in the forward fuselage and add to the aerodynamic efficiency of the aircraft

General Description

Refer to Figures 1 and 2.

The NLG is installed in the wheelwell in the nose fuselage, located forward of the forward pressure bulkhead. When the NLG retracts the forward and aft doors enclose it in the wheelwell.

A taxi light is installed on the lower right side of the outer cylinder of the shock strut.

The GROUND LOCK control handle operates a cable system to engage the downlock safety lever.

The nose landing gear assembly has the components that follow:

- Shock Strut Assembly (32-21-01)
- Drag Strut Assembly (32-21-06)
- Harness Electrical (32-21-11).

Detailed Description

Refer to Figures 5 and 6.

The NLG has two hydraulically operated forward doors and two mechanically operated aft doors, actuated by the NLG when the landing gear extends and retracts. When the NLG retracts into the wheelwell, the two sets of doors close around it.

A taxi light is installed on the lower right side of the outer cylinder of the shock strut.

The ground lock control handle operates a cable to engage the downlock safety lever. The lever is engaged when the downlock arm of the pivot tube assembly is on the drag strut of the NLG.

Shock Strut Assembly

The one stage, nitrogen/oil filled shock strut assembly uses a lever suspension geometry. The shock strut has an outer cylinder, an inner cylinder, a piston assembly, a link, and a trailing arm. The trailing arm has a jacking point to retract and extend the NLG.

Two trunnion arms are installed at the upper rear portion of the outer cylinder, where the trunnion bearings are attached and the NLG is installed. The trunnion bearings are the NLG pivot points during extension and retraction. The retraction actuator is attached to the mounting lugs on the top of the right trunnion arm. At the top forward side of the outer cylinder are two mounting lugs. The lower drag strut is attached to the two mounting lugs. A taxi light is installed on the bottom right side of the outer cylinder.

The power steering hydraulic motor and gearbox assembly are installed between the trunnion arms. The NLG steering centering



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actuator assembly, the steering sensor, and the centering cam are installed on the aft lower side of the outer cylinder.

The outer cylinder has attachment points for the electrical harnesses. The Weight On Wheels (WOW) proximity sensors are installed on the inner cylinder above the tow fitting assembly.

The piston assembly in the inner cylinder is attached to a link. The link is installed between two lugs on the trailing arm. The link transmits ground loads from the trailing arm to the piston assembly.

A low pressure valve is on the top surface of the inner cylinder. The valve is used to service the piston assembly with hydraulic fluid and dry nitrogen.

Drag Strut Assembly

Refer to Figures 5 and 6.

The drag strut assembly stabilizes the shock strut in either the retracted or the extended position.

The drag strut assembly has the components that follow:

- An upper drag strut;
- A lower drag strut
- A pivot tube assembly
- Two spring assemblies
- A link assembly
- An upper lock link
- A lower lock link.

The two spring assemblies and the pivot tube assembly are installed on the upper drag strut. The pivot tube assembly has six arms:

- Two for the spring assemblies
- One for the downlock actuator
- One for the link assembly
- One for the emergency uplock release mechanism
- One for the ground lock mechanism.

The lower lock link is installed on the lower drag strut and is connected to the upper lock link. The upper lock link is installed on the upper drag strut and is connected to the lock link assembly. The over centering action of the upper and lower lock links makes a mechanical lock when the gear is in the retracted and the extended positions.

NLG Electrical Harness

Four electrical harnesses are attached at different places on the shock strut. Two additional electrical harnesses are attached to the drag strut.

The four harnesses on the shock strut give power/signal input for the components that follow:

- The taxi light
- The WOW proximity sensors
- The steering proximity sensor
- The steering manifold pressure switches
- The Rotary Variable Differential Transformers (RVDTs).



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The two harnesses on the drag strut supply electrical power/signal input for the primary and alternate lock proximity sensors.

Training Information Points

When you tow the aircraft or move the NLG you must obey the turn limits. Do not move the NLG more than 120° right and left of center. If you do not do this, you can cause damage to the equipment. There are labels on the two aft NLG doors that show these limits.

When you do maintenance on the nose landing gear make sure that the nose landing gear ground lock is engaged, the door lock pin is installed and that the hydraulic pressure is shut off. If you do not do this, you can cause injuries to persons and/or damage to the equipment.

Make sure that the aircraft is well clear of the hangar structure and ground equipment before you release the pressure from the landing gear shock struts. If you do not do this, damage to the aircraft and/or equipment can occur.

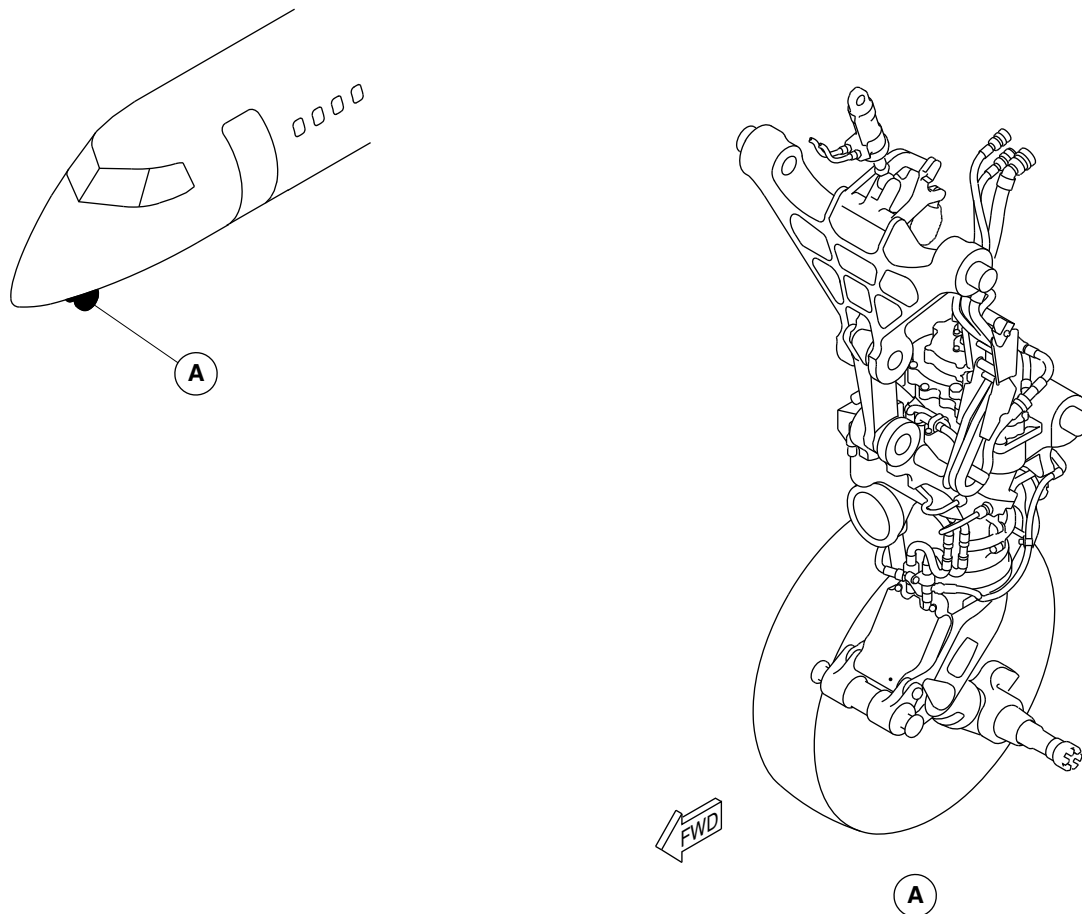
Use only nitrogen when you charge the landing gear shock struts. Do not use compressed air. If you use compressed air, damage to the shock struts can occur.

Use the approved hydraulic fluid when you service the landing gear. The use of phosphate ester–base fluids is strictly prohibited. If you use unapproved fluids, severe deterioration of the shock strut seals will occur.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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Nose Landing Gear Aft View Locator
Figure 1

PSM 1-84-2A
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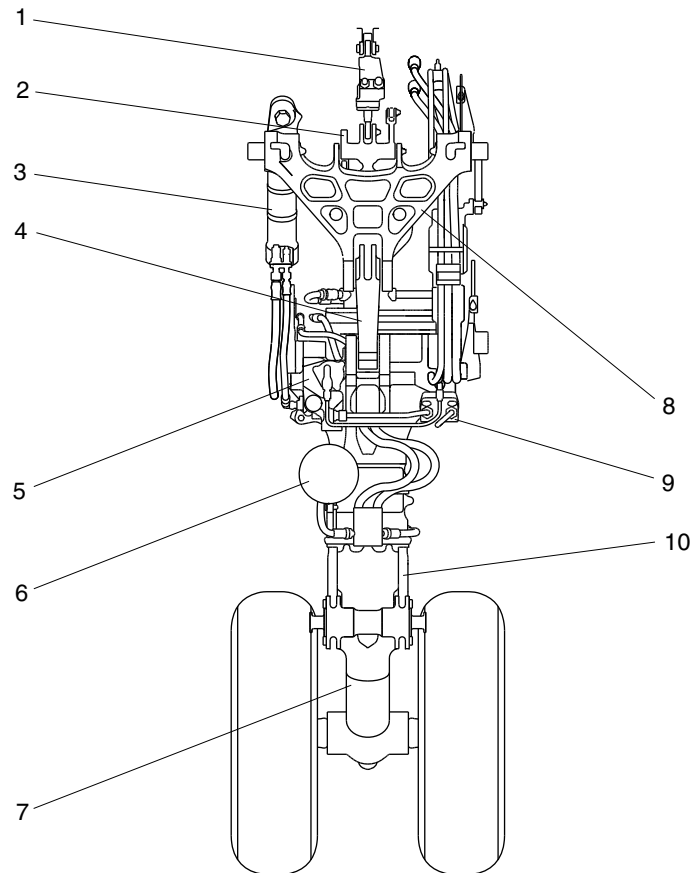
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LEGEND

- 1. Downlock Actuator.
- 2. Pivot Tube Assembly.
- 3. Retraction Actuator.
- 4. Lower Drag Strut.
- 5. Nosewheel Steering Manifold.
- 6. Taxi Light.
- 7. Trailing Arm.
- 8. Upper Drag Strut.
- 9. Steering Hydraulic Motor.
- 10. Inner Cylinder.

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Nose Landing Gear Aft View Detail
Figure 2

PSM 1-84-2A
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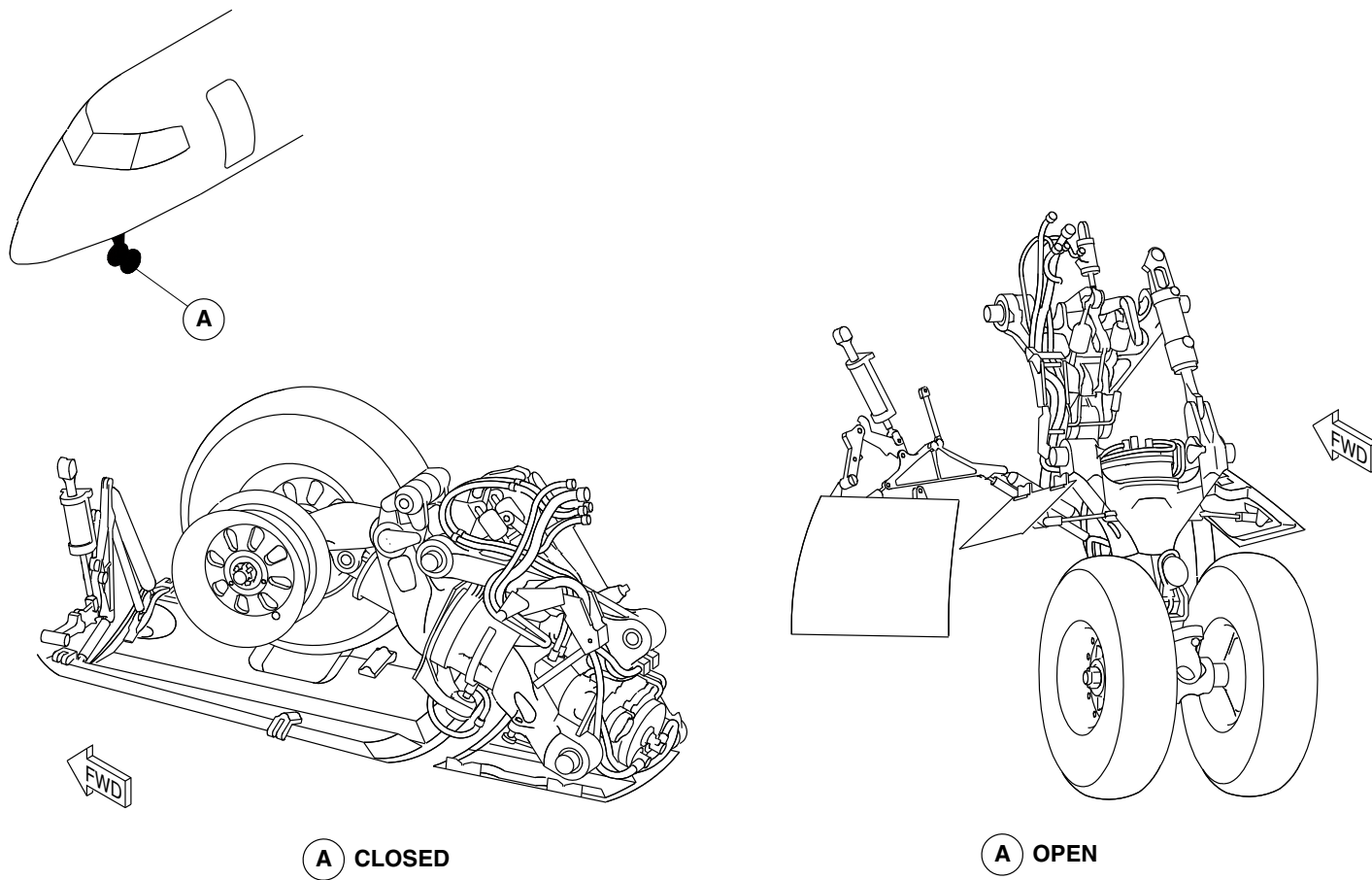
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Nose Landing Gear and Doors – Locator
Figure 3

PSM 1-84-2A
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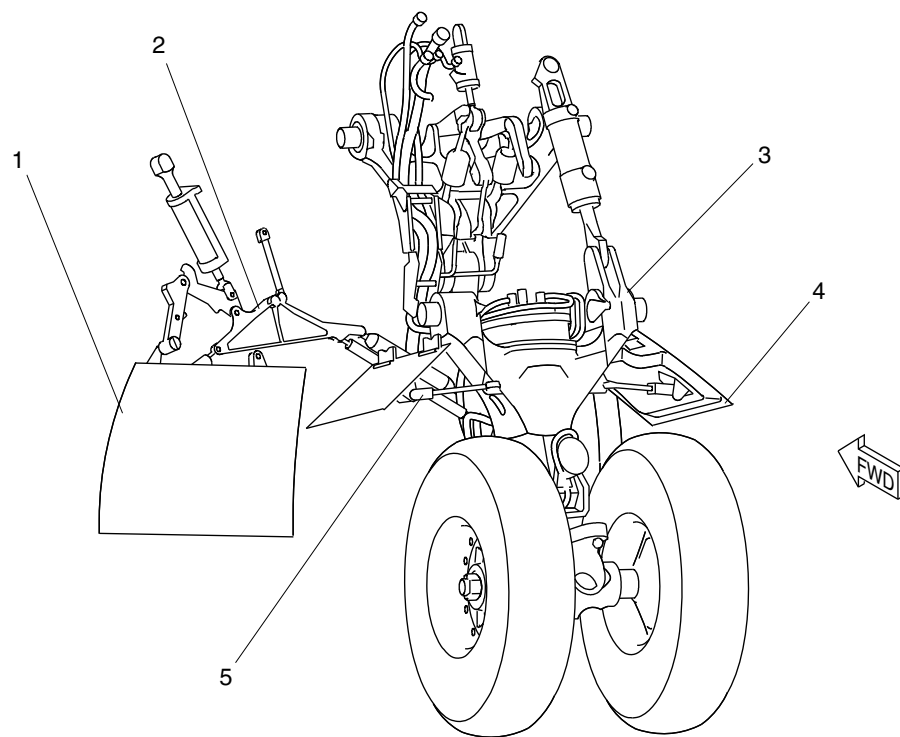
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Forward Door.
2. Forward Door Operating Mechanism.
3. Shock Strut Assembly.
4. Aft Door.
5. Aft Door Actuating Linkage.

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Nose Landing Gear and Doors – Details
Figure 4

PSM 1-84-2A
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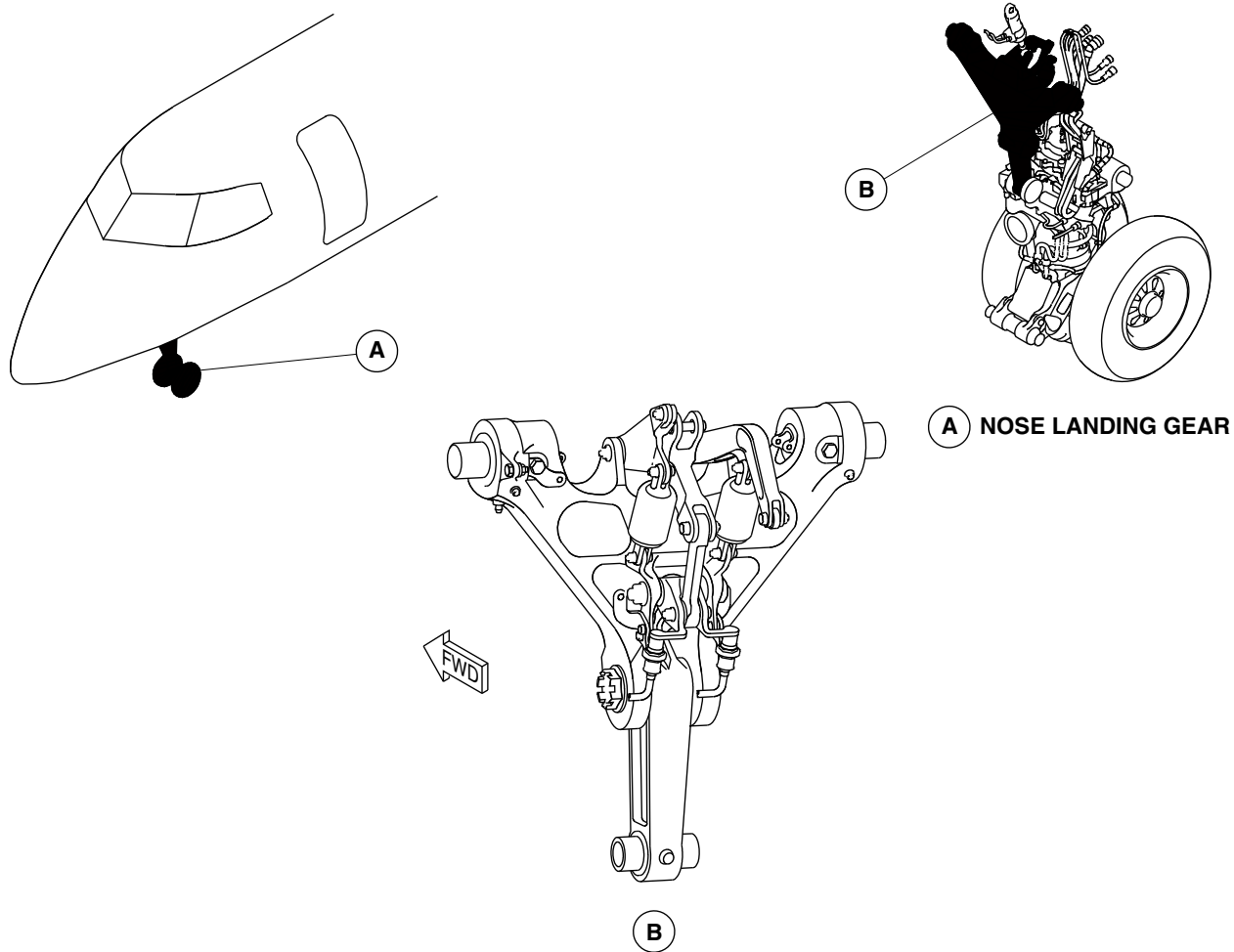
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Nose Gear Strut/Steering Locator
Figure 5

PSM 1-84-2A
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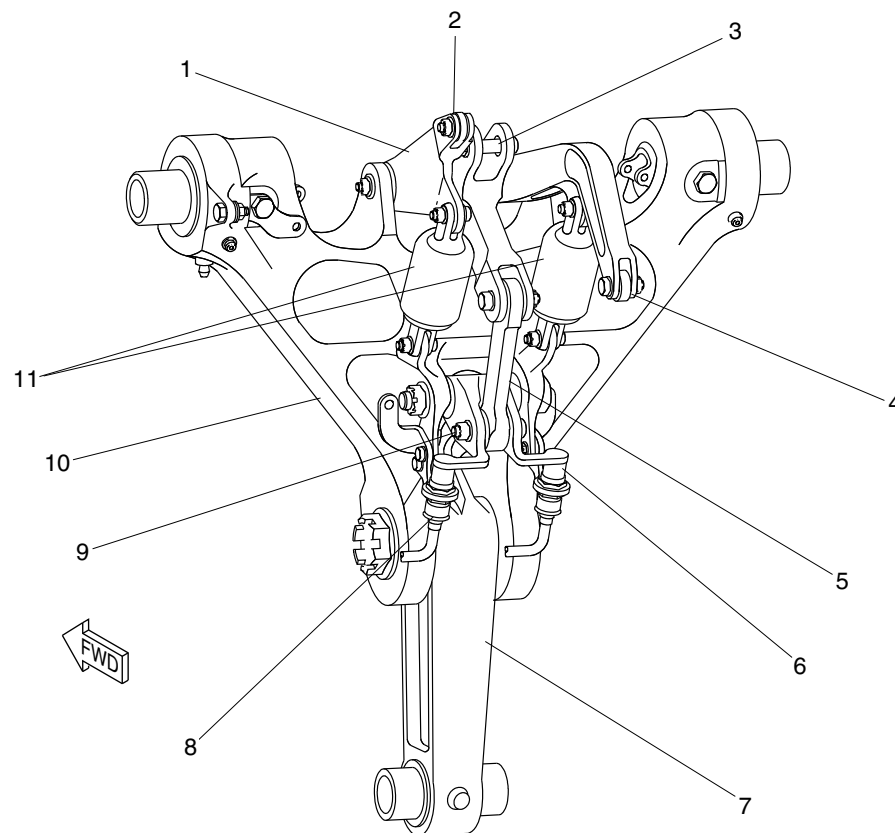
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LEGEND

1. Pivot Tube Assembly.
2. Pivot Tube Arm Downlock Actuator for Ground Lock Attachment Point.
3. Pivot Tube Arm Downlock Actuator for Ground Lock Attachment Point.
4. Pivot Tube Arm for Emergency Uplock Release.
5. Lock Link Assembly.
6. NLG Alternate Lock Proximity Sensor.
7. Lower Drag Strut.
8. NLG Primary-Lock Proximity Sensor.
9. Upper Lock Link.
10. Upper Drag Strut.
11. Spring Assembly.

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Nose Gear Strut/Steering Detail
Figure 6

PSM 1-84-2A
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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NOSE LANDING GEAR DOORS

Introduction

The Nose Landing Gear (NLG) doors enclose the retracted NLG in the forward fuselage wheelwell and add to the aerodynamic efficiency of the aircraft.

General Description

Refer to Figure 1.

The four NLG doors consist of two hydraulically operated forward doors, and two mechanically operated aft doors. Refer to Figure 2.

The NLG doors have the components that follow:

- The NLG Forward Doors (32-22-11)
- The NLG Forward Doors Connecting Rods (32-22-16)
- The NLG Aft Doors (32-22-01)
- The NLG Aft Doors Connecting Rods (32-22-06).

Detailed Description

The two NLG forward doors are hydraulically operated to open before the NLG extends or retracts, and close after the NLG has extended or retracted. The two NLG aft doors are connected to the NLG shock strut and are actuated by the NLG.

NLG Aft Doors

The two NLG aft doors are connected to the NLG shock strut by two adjustable door linkage assemblies. These doors are open when the NLG is extended and close when the NLG is retracted. Each NLG aft door is hinged to the fuselage at two places.

NLG Aft Doors Linkage Assemblies

The two adjustable connecting rods, connect the NLG shock strut to the doors for mechanical operation and let the doors be rigged correctly, in the closed position.

NLG Forward Doors

The two NLG forward doors are hinged to the fuselage at two places. The forward hinges form the attachment brackets for the operating mechanism. The operating mechanism has two adjustable connecting rods, an extension spring, a bellcrank, a toggle link assembly, a link assembly, a mechanical sequence valve, and a hydraulic actuator. This operating mechanism is attached to the forward wall of the nosewheel bay and is protected by a removable debris guard. Each door is made up of an inner skin and an outer skin. The inner skin is contoured to give clearance for the nosewheels in the retracted position.

NLG Forward Doors Connecting Rods

The two adjustable connecting rods connect the bellcrank to the doors and let the doors be rigged correctly in the closed position.

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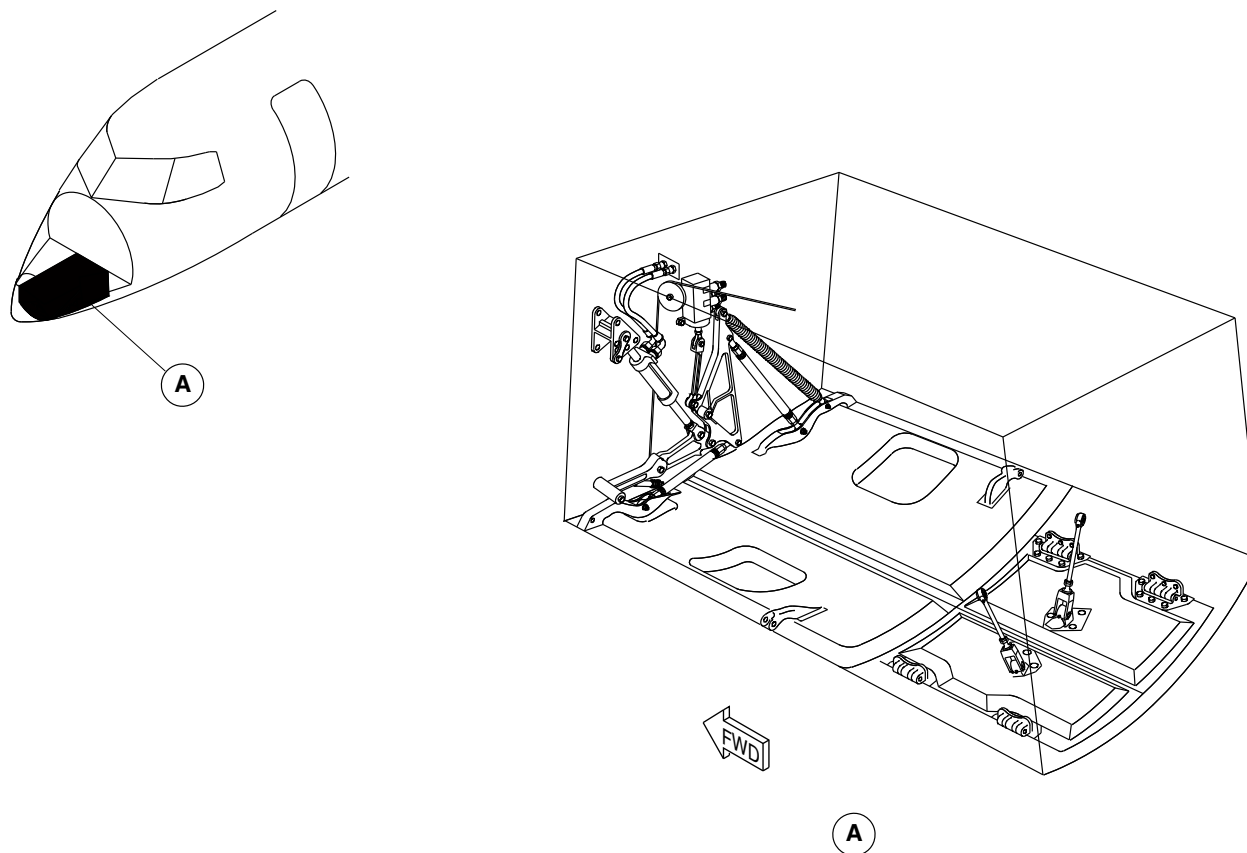
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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Nose Landing Gear Compartment Locator
Figure 1

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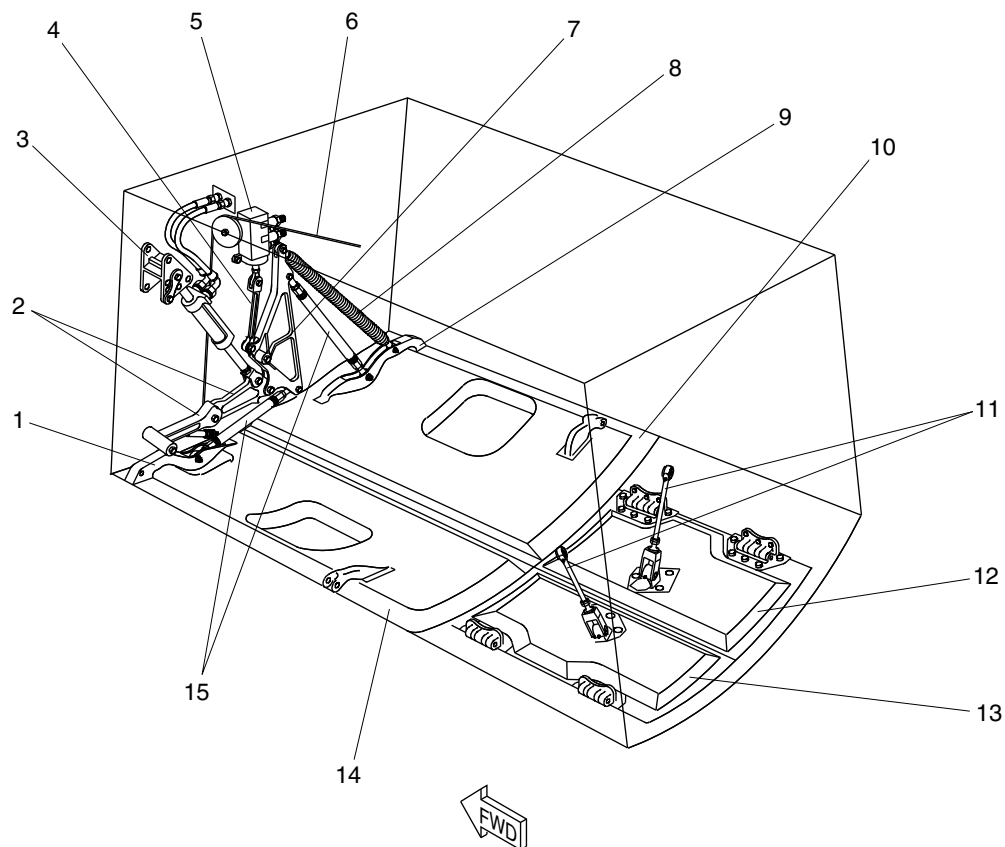


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LEGEND

1. Left Door Attachment Bracket.
2. Toggle Link Assembly.
3. Forward Door Actuator Assembly.
4. Link Assembly.
5. Mechanical Sequence Valve.
6. Alternate Release Cable.
7. Bell Crank.
8. Spring Assembly.
9. Right Door Attachment Bracket.
10. Right Forward Door.
11. NLG Aft Door Linkage Assembly.
12. Right Aft Door.
13. Left Aft Door.
14. Left Forward Door.
15. Push Rods.



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Nose Landing Gear Compartment Detail
Figure 2

PSM 1-84-2A
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

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LANDING GEAR EXTENSION AND RETRACTION

Introduction

The landing gear Retraction Control System (RCS) controls the hydraulic retraction and extension of the landing gear. It also controls the alternative manual extension of the main landing gear (MLG) and the nose landing gear (NLG). The RCS shows the landing gear and landing gear door status to the flight crew in the flight compartment.

General Description

[Refer to Figures 1 and 2.](#)

Hydraulic retraction and extension starts when the landing gear selector lever is unlocked and moved to the desired command position. The Proximity Sensor Electronic Unit (PSEU) checks the status of the landing gear and the landing gear doors, and compares it to the command position selected. The PSEU then controls the hydraulic sequences to either fully extend or fully retract the landing gear. The indicator lights installed on the landing gear control panel assembly show the landing gear and landing gear door status. A LDG GEAR INOP caution light is shown on the flight compartment Caution and Warning panel in the event of a system malfunction.

Each MLG wheelwell has four doors. The two aft doors are hydraulically operated and the two forward doors are mechanically operated.

The NLG wheelwell has four doors. The two forward doors are hydraulically operated and the two aft doors are mechanically operated.

[Refer to Figures 3 and 4.](#)

The alternate extension system is an alternate means to manually extend the landing gear when the No. 2 hydraulic system or the PSEU itself fails.

Detailed Description

Normal landing gear operation is controlled by the PSEU as a result of input signals from the landing gear selector lever. The No. 2 hydraulic system supplies the power to move the landing gear.

[Refer to Figure 1.](#)

The PSEU does sequencing logic for the RCS and gives indications for the landing gear and the landing gear door position. The PSEU interfaces with the landing gear selector lever to turn on the indicator lights on the landing gear control panel.

The Landing Gear Control Panel indicator lights are as follows:

- LEFT, NOSE, RIGHT green lights indicate the gear is down and locked.
- LEFT, NOSE, RIGHT red lights indicate the gear is not locked up, or down
- L DOOR, N DOOR, R DOOR amber lights indicate landing gear doors are open.

[Refer to Figure 5.](#)

The PSEU also interfaces with the caution and warning panel to turn on the LND GEAR INOP light, when a malfunction is detected.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Refer to Figure 4.

If the primary downlock indication fails, an alternate downlock indication system is available. There are individual green indicator lights for each landing gear, to indicate a down and locked condition. These indicator lights are installed in the flight compartment floor, under the ALTERNATE EXTENSION door.

The MLG is designed to retract rearward into the wheelwell and extends toward the nose of the aircraft. The NLG retracts forward into the nose wheelwell and extends toward the aft of the aircraft. The four doors of each landing gear hydraulically and mechanically open in sequence with the landing gear movement. The two aft MLG doors and the two forward NLG doors are operated hydraulically by the PSEU.

Mechanical sequence valves check the time of landing gear extension and retraction in relation to the hydraulic landing gear doors movement. The two forward MLG and the two aft NLG doors are mechanically linked to the landing gear. The doors are opened and closed by the movement of the landing gear. The mechanically actuated landing gear doors stay open while the landing gear are in the extended position. Hydraulically actuated doors are closed at the end of the extension and retraction cycle.

There are three conditions for the PSEU to activate the selector valve. These conditions are as follows:

- The landing gear selector lever is moved to the UP position
- The PSEU receives a signal that the aircraft is airborne
- The NLG is centered.

The PSEU electrically signals the selector valve to supply hydraulic power to the retract circuit of the landing gear hydraulic system. When the landing gear selector lever is moved to the DOWN position, the PSEU signals the selector valve to supply hydraulic power to the extend circuit of the landing gear hydraulic system.

The alternate extension system is a self resetting, cable actuated design. The system is accessible in the flight compartment. The alternate extension system has a bypass valve and a manual handpump hydraulic system. The bypass valve isolates the landing gear hydraulics from the No.2 hydraulic system. The manual handpump hydraulic system helps move the MLG into the down and locked position.

For an alternate extension of the landing gear to occur, the NLG forward doors, the MLG aft doors, and the landing gear uplocks open mechanically. The NLG free falls to the down and locked position with the help of the airflow over the fuselage. The MLGs free fall and are moved to the down and locked position by the alternate extension actuators. A LANDING GEAR SELECTOR INHIBIT switch installed in the flight compartment ceiling is used to isolate the normal landing gear selector valve during flight. The LANDING GEAR SELECTOR INHIBIT switch must be set from NORMAL to INHIBIT before the alternate extension system can be used.

Training Information Points

Make sure that all hydraulic power and electrical power is off before any maintenance is performed on the landing gear. The landing gear can accidentally retract. This can cause injuries or death to persons and/or damage to equipment.



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Make sure that all persons and equipment are clear of the landing gear and door movement area otherwise injuries to persons and/or damage to equipment can occur.

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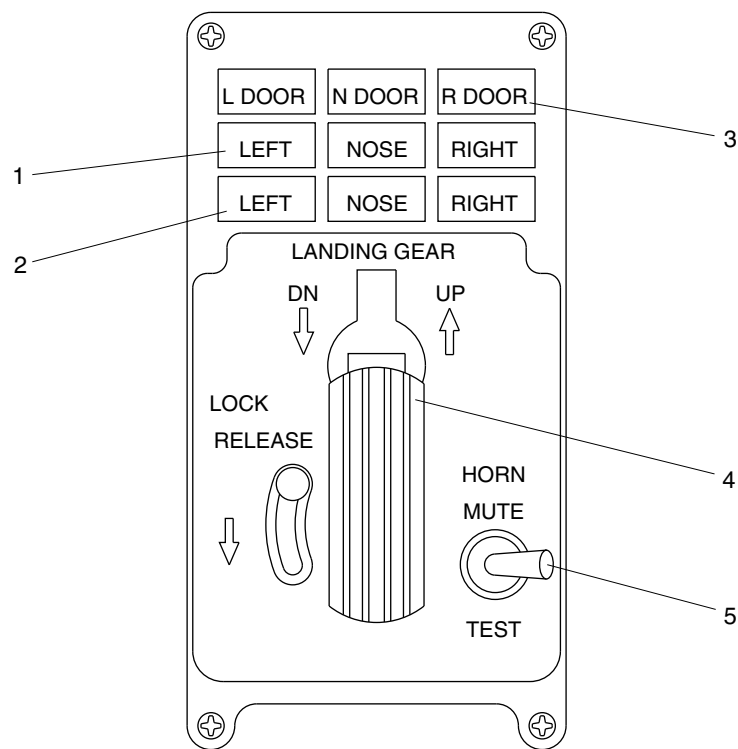
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LEGEND

1. "Green" Gear Downlocked.
2. "Red" Gear Not Uplocked Or Downlocked.
3. "Amber" Door Open.
4. Handle "Amber" (Concurrent W/"Red" Unsafe Lights) Gear In-Transit.
5. Warning Horn Mute/Test Switch.

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Landing Gear Selector
Figure 1

PSM 1-84-2A
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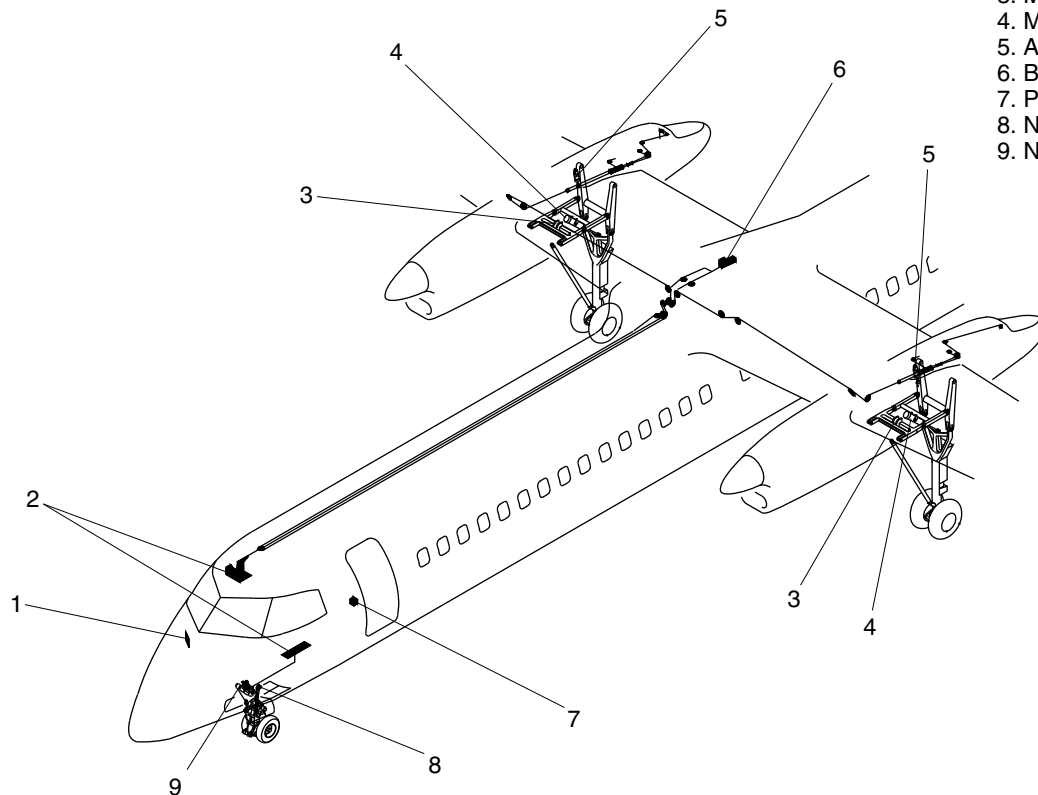


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LEGEND

1. Landing Gear Selector Lever and Control Panel.
2. Alternate Release Handles.
3. MLG Lock Actuator.
4. MLG Retraction Actuator.
5. Alternate Extension Actuator.
6. Bypass Valve.
7. Proximity Sensor Electronic Unit.
8. NLG Lock Actuator.
9. NLG Retraction Actuator.



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Landing Gear Extension and Retraction – Component Details
Figure 2

PSM 1-84-2A
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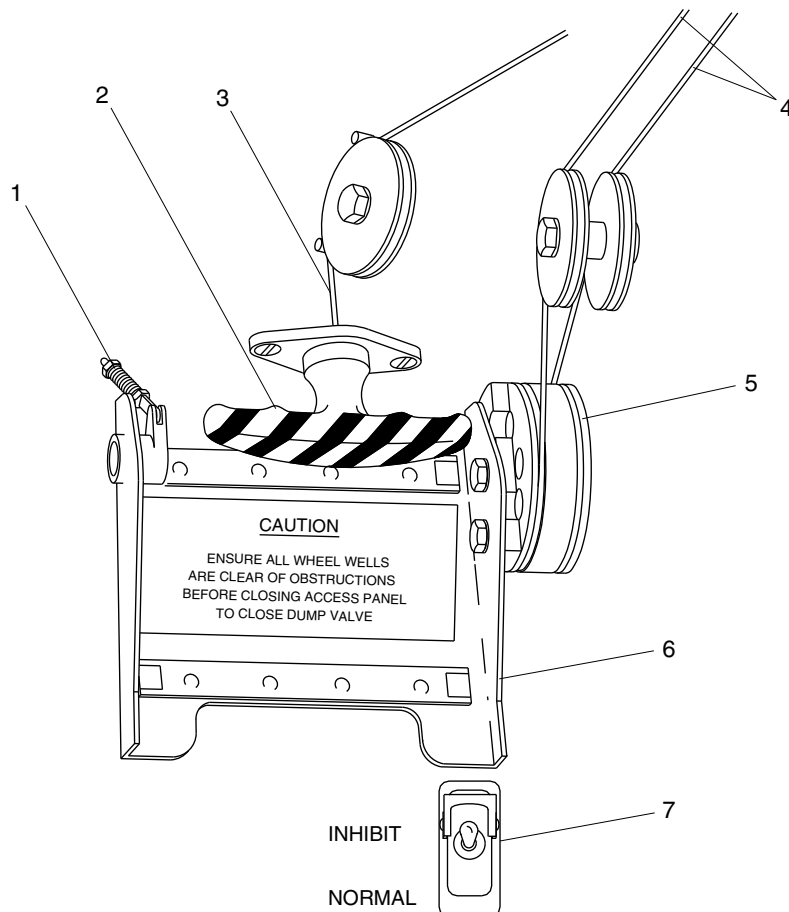
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LEGEND

1. Door Compression Spring.
2. Main Gear Alternate Release Handle.
3. Cables to Slide Assembly.
4. Cables to Bypass Valve.
5. Door Pulley.
6. Landing Gear Alternate Release Door.
7. Landing Gear Inhibit Toggle Switch.

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Landing Gear Alternate Extension Controls Page 1
Figure 3

PSM 1-84-2A
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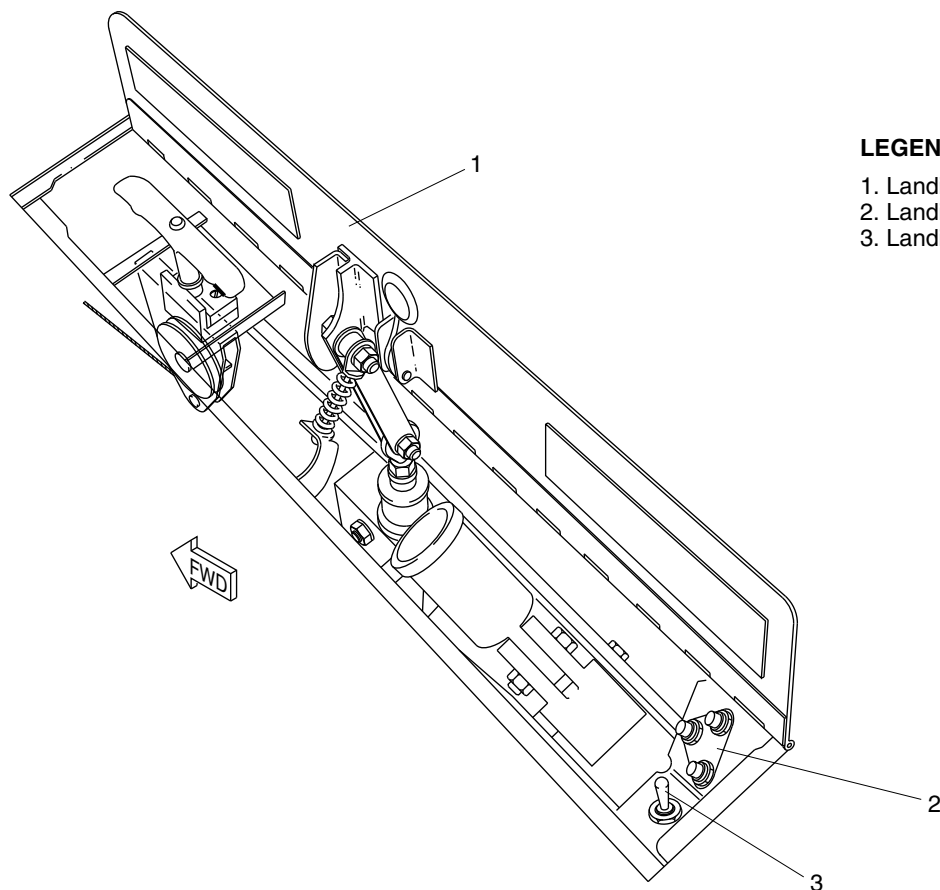
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LEGEND

1. Landing Gear Alternate Extension Door.
2. Landing Gear Downlock Verification Lights.
3. Landing Gear Downlock Verification Switch.

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Landing Gear Alternate Extension Controls Page 2
Figure 4

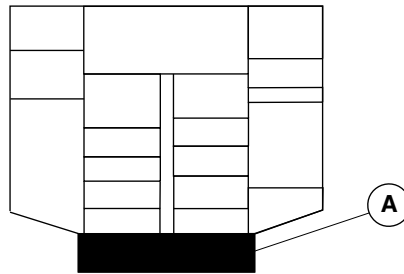
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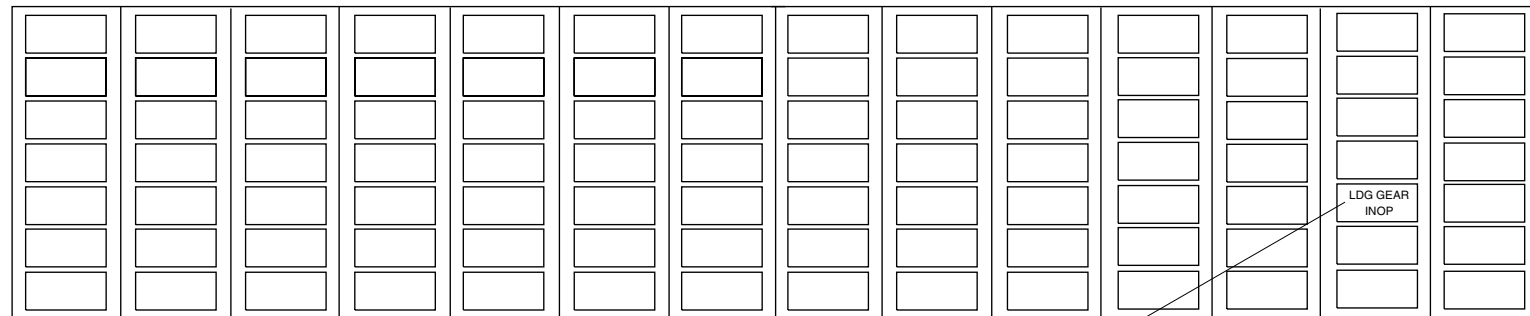
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OVERHEAD CONSOLE

LEGEND

1. Landing Gear Inoperative (Amber).



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Landing Gear Inop. Caution Light
Figure 5

PSM 1-84-2A
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**ON A/C ALL

32-31-00-001

PRIMARY EXTENSION AND RETRACTION

Introduction

The Primary Extension and Retraction system controls the extension and retraction of the landing gear and the landing gear doors operation.

General Description

[Refer to Figure 1.](#)

The LANDING GEAR selector lever and the Proximity Sensor Electronic Unit (PSEU) control the operation of the landing gear. The No. 2 hydraulic system supplies the power to move the landing gear.

The primary extension and retraction system has the components that follow:

- Panel Control, Landing Gear (32-31-01)
- Valve Selector, Landing Gear (32-31-06)
- Actuator Retraction MLG (32-31-11)
- Actuators Downlock MLG (32-31-16)
- Actuators Uplock MLG (32-31-21)
- Actuator Doors MLG (32-21-26)
- Mechanism Doors MLG (32-31-31)

- Valves Mechanical Sequence MLG Doors (32-31-36)
- Restrictors Inline MLG Doors (32-31-38)
- Valve Solenoid Sequence MLG Doors (32-31-41)
- Actuator Retraction NLG (32-31-46)
- Actuator NLG Lock (32-31-51)
- Actuator NLG Doors (32-31-56)
- Mechanism NLG Doors (32-31-61)
- Valve Mechanical Sequence NLG Doors (32-31-66)
- Restrictor Inline NLG (32-31-68)
- Valve Solenoid Sequence NLG Doors (32-31-71)
- A Power Transfer Unit (29-11-61).

Detailed Description

[Refer to Figure 2.](#)

Hydraulic retraction and extension starts when the LANDING GEAR selector lever is moved to the desired position. The PSEU checks the status of the landing gear and the landing gear doors, and compares it to the command selected. The PSEU controls the hydraulic sequences to either fully extend or fully retract the landing gear. The status of the landing gear and the landing gear doors is shown by the indicator lights on the LANDING GEAR control panel. A LDG GEAR INOP caution light on the Caution and Warning panel shows a fault in the landing gear retraction and extension system.

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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The following is an in depth description of the Extension and Retraction systems in the order that follows:

- Retraction sequence
- Retraction sequence configuration
- Main landing gear retraction
- Nose landing gear retraction
- Retraction sequence completion
- Extension sequence configuration
- Main landing gear extension
- Nose landing gear extension.

Retraction Sequence

[Refer to Figure 1.](#)

When the landing gear is down and locked, the solenoid sequence valves and the down solenoid of the selector valve are kept in an energized condition. This keeps hydraulic pressure on the down side of the retraction actuators. The lock actuators are also used to downlock the landing gear and to keep the hydraulically actuated wheelwell doors closed. Retraction of the landing gear starts when the LANDING GEAR selector lever is unlocked and moved to the UP position.

Retraction Sequence Configuration

[Refer to Figure 3.](#)

When the LANDING GEAR selector lever is moved to the UP position, the down solenoid of the landing gear selector valve is de-energized. The PSEU makes sure that the aircraft is airborne and that the Nose Landing Gear (NLG) is centered. The up solenoid of the landing gear selector valve, now receives electrical power. When the landing gear selector valve receives electrical power, the up solenoid configures the selector valve. This lets hydraulic system pressure flow into the retract side of the landing gear hydraulic system.

Main Landing Gear Retraction

Hydraulic pressure is supplied to each Main Landing Gear (MLG) downlock release actuator to release the downlock. Hydraulic pressure is supplied through an energized solenoid sequence valve to the open side of the MLG aft door actuators.

NOTE

The solenoid sequence valves stay energized from the time of closing the doors during the landing gear extension sequence. They will not change the state.

This causes the MLG aft doors to open. The operation of the downlock release actuator and the MLG aft door actuators is monitored by the proximity sensors.

The MLG door mechanism operates a mechanical sequence valve to interlock the retraction/extension part of the hydraulic system. This continues until the doors are open enough for the landing gear to retract so that it does not touch the doors. At approximately 93 percent travel of the MLG aft doors, the MLG aft doors linkage



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

operates the mechanical sequence valve. When the mechanical sequence valve is in operation, hydraulic pressure is supplied to the up side of the MLG retraction actuator.

The MLG starts to travel to the fully retracted position where it is locked in place by the mechanical uplock. The proximity sensors monitor the gear and door positions. When the PSEU receives the input signals that the MLG is up and locked, the PSEU de-energizes the solenoid sequence valves. This causes the solenoid sequence valves to supply pressure to the close side of the MLG door actuators and close the doors.

At approximately 7 percent reverse travel of the MLG doors, the mechanical sequence valves stop their operation. This action removes pressure from the up side of the MLG retraction actuators. Inline restrictors bypass the mechanical sequence valves and keep the MLG retraction actuators pressurized to 3000 psi (20684 kPa). Pressure is kept at 3000 psi (20684 kPa) until the landing gear hydraulic system is depressurized upon completion of the retraction cycle.

Nose Landing Gear Retraction

Hydraulic pressure is supplied to the NLG uplock/downlock release actuator to release the downlock. Hydraulic pressure is supplied through an energized solenoid sequence valve to the open side of the NLG forward doors actuator.

NOTE

The solenoid sequence valve stays energized from the time of closing the doors during the landing gear extension sequence. It will not change the state.

This causes the NLG forward doors to start to open. The operation of the NLG uplock/downlock release actuator and the NLG forward doors actuator is checked by the proximity sensors.

At approximately 92 percent travel of the NLG forward doors, the door linkage operates the mechanical sequence valve. When the mechanical sequence valve is in operation, it stops to interlock the retraction/extension part of the NLG hydraulic system. Hydraulic pressure is supplied to the up side of the NLG retraction actuator. The NLG starts to travel to the fully retracted position where it is locked in place by the mechanical uplock.

The proximity sensors make sure that these procedures have been completed. When the PSEU receives the input signals that the NLG is up and locked, the PSEU de-energizes the solenoid sequence valves. This causes the solenoid sequence valves to pressurize the NLG forward doors actuator to close the doors.

At approximately 8 percent reverse travel of the NLG forward doors, the mechanical sequence valve stops its operation. This action removes pressure from the up side of the NLG retraction actuator. Inline restrictors bypass the mechanical sequence valve and keep the NLG retraction actuator pressurized to 3000 psi (20684 kPa). Hydraulic pressure is kept at 3000 psi (20684 kPa) until the landing gear hydraulic system is depressurized upon completion of the retraction cycle. A proximity sensor senses that the NLG forward doors are closed and sends an input signal to the PSEU.

Retraction Sequence Completion

[Refer to Figure 3.](#)

The MLG and NLG proximity sensors sense that the gear is up and locked and that the MLG and NLG doors are closed. The PSEU gives a three second interval before it de-energizes the up and the



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

down solenoid in the selector valve. The PSEU then depressurizes the landing gear hydraulic circuit.

Extension Sequence Configuration

Refer to Figure 4.

The landing gear starts to extend when the LANDING GEAR selector lever is unlocked and moved to the DN position. The down solenoid of the selector valve receives electrical power. The selector valve now supplies aircraft hydraulic system pressure and flow into the extend side of the landing gear hydraulic system.

Main Landing Gear Extension

When the LANDING GEAR selector lever is moved to the DN position, the solenoid sequence valves stay de-energized. The de-energized solenoid sequence valves supply hydraulic pressure to the open side of the MLG aft doors actuators to open the MLG aft doors.

NOTE

The solenoid sequence valves stay de-energized from the time of closing the doors during the landing gear retraction sequence. They will not change the state.

At approximately 93 percent travel of the MLG aft doors, the MLG aft doors linkage operates the mechanical sequence valve. The valve supplies hydraulic pressure to the uplock release actuators and to the down side of the MLG retraction actuators. The MLG starts to travel to the down and locked position.

There are three proximity sensors used to monitor the MLG extension sequence. Each MLG has two down and locked sensors

and one MLG aft doors, closed sensor. When the PSEU receives input signals that the MLG is down and locked, the PSEU energizes the solenoid sequence valves.

Pressure is supplied to the MLG aft doors actuators to close the MLG aft doors. At approximately 7 percent reverse travel of the MLG doors, the mechanical sequence valves stop their operation. This action removes pressure from the up side of the MLG retraction actuators. Inline restrictors keep the MLG retraction actuators pressurized to 3000 psi (20684 kPa) at the end of the extension sequence.

Nose Landing Gear Extension

When the LANDING GEAR selector lever is moved to the DN position, hydraulic pressure is supplied to the open side of the NLG doors actuator through the de-energized solenoid valve. This causes the NLG forward doors to open.

NOTE

The solenoid sequence valve stays de-energized from the time of closing the doors during the landing gear retraction sequence. It will not change the state.

At approximately 92 percent travel of the NLG forward doors, a mechanical sequence valve is actuated by the door linkage. When the mechanical sequence valve operates, hydraulic pressure is sent to the lock actuator and the down side of the NLG retraction actuator. The NLG then starts to travel to the down and locked position.

There are five proximity sensors that monitor the NLG extension sequence. The NLG has two down sensors on the NLG trunnion arms, two down and locked sensors on the NLG drag strut, and one NLG doors closed sensor. When the PSEU receives the input signals that the NLG is down and locked, the solenoid sequence valve is



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energized by the PSEU. Hydraulic pressure is supplied to the NLG forward doors actuator to close the doors.

At approximately 8 percent reverse travel of the NLG forward doors, the mechanical sequence valve stops its operation. This action removes pressure from the uplock/downlock actuator and the NLG retraction actuator. An inline restrictor keeps the NLG retraction actuator and the lock actuator pressurized to 3000 psi (20684 kPa) at the end of the extension cycle. A mechanical downlock, locks the NLG in the fully extended position. The PSEU keeps power on the down solenoid of the selector valve as long as the LANDING GEAR selector lever is in the DN position.

Landing Gear Control Panel

Refer to Figures 5 and 6.

The landing gear is controlled and monitored from the LANDING GEAR control panel located on the right side of the Engine Display (ED). The panel has a LANDING GEAR selector lever, a LOCK RELEASE selector lever, landing gear and landing gear door advisory lights, and a landing gear warning HORN MUTE/TEST switch. The landing gear is commanded to the UP or DN position with the LANDING GEAR selector lever. The LANDING GEAR selector lever LOCK RELEASE must be held down before the selector lever can be moved to either the UP or the DN position. An amber light in the LANDING GEAR selector lever comes on to show that the landing gear position does not agree with the LANDING GEAR selector handle position.

Nine advisory lights are located at the top of the landing gear control panel. Three amber lights L DOOR, N DOOR, and R DOOR show that the landing gear doors are open during extension and retraction, or after the sequence is completed. Three green lights show that the

LEFT, NOSE, and RIGHT landing gear are down and locked. Three red lights show that the LEFT, NOSE, and RIGHT landing gear are unlocked.

When the LANDING GEAR selector lever is unlocked and moved to the UP position, the landing gear downlocks are released. Hydraulic pressure is supplied to the landing gear actuators and the landing gear is retracted. When the LANDING GEAR selector lever is unlocked and moved to the DN position, the landing gear uplocks are released. Hydraulic pressure is supplied to the landing gear actuators and the landing gear is extended.

A landing gear configuration warning tone sounds when the aircraft is in the landing configuration and not all the landing gear are down and locked. The two position landing gear warning HORN MUTE/TEST switch is a momentary type toggle switch. When the switch is set to the TEST position, the landing gear configuration warning tone will sound at a constant 800 Hz. The MUTE position is used for training purposes (for information on the landing gear configuration warning system, refer to SDS 32–62–00).

Landing Gear Selector Valve

Refer to Figures 7 and 8.

The landing gear selector valve is a self contained assembly which has two solenoid valves. The landing gear selector valve controls hydraulic pressure to position a directional control valve that is spring centered. The position of the valve controls the supply of hydraulic pressure to either the UP or DOWN hydraulic circuits of the landing gear system. The landing gear system will be configured for either normal retraction or extension.



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Main Landing Gear Retraction Actuator

Refer to Figure 9.

The main landing gear retraction actuator is a hydraulic device that has two ports. There are restrictors in the retract and extend ports. The rod end of the actuator piston has a ball and race with a lubrication fitting. The MLG retraction actuator cylinder is attached to the lower front of the MLG yoke cross beam. The rod end is attached to the center top of the MLG shock strut.

MLG Downlock Actuator

Refer to Figures 10 and 11.

The MLG downlock actuator has two ports. The rod end of the actuator has a ball and race with a lubrication fitting. Hydraulic lines are attached to the two ports and are sealed with O-rings. The MLG downlock actuator is attached to the MLG stabilizer brace assembly.

MLG Uplock Actuator

Refer to Figures 12 and 13.

The MLG uplock actuator is a part of the MLG uplock assembly. The MLG uplock actuator has an external spring that helps to return it to its retracted position when hydraulic pressure is removed.

MLG Doors Actuator

Refer to Figure 14.

The MLG doors actuator is a hydraulic device that has two ports. A restrictor valve is installed at the extension port and an internal restrictor is located under the retraction port in the cylinder wall.

These restrictors are used to monitor the time for extension and retraction. Lubrication fittings are located at the attachment point of the actuator cylinder and on the piston rod end.

MLG Doors Mechanism

Refer to Figure 15.

The MLG door actuator powers the MLG doors mechanism to open or close the aft MLG doors. When the MLG doors actuator retracts, it unlocks the over-center lock link. As the actuator continues to retract, it folds the lock link and turns the bellcrank clockwise. The bellcrank acting on the two push rods, pushes the aft doors to the open position.

When the MLG doors actuator extends, it unfolds the lock link to the over-center position. As the actuator continues to extend, it unfolds the lock link and turns the bellcrank anti-clockwise. The bellcrank, acting on the two push rods, pulls the aft doors to the closed position. The aft doors are closed when the MLG is in the extended or the retracted position. The forward doors are connected to and operated by the upper part of the shock strut of the MLG through two links. The forward doors are open when the MLG is extended and closed when the MLG is retracted.

MLG Doors Mechanical Sequence Valve

Refer to Figure 15.

The MLG mechanical sequence valve is a bolt-on assembly. During landing gear retraction and extension, the mechanical sequence valve operates to correctly sequence the MLG uplock release and the MLG retraction actuators. This prevents the landing gear from traveling through partially opened MLG landing gear doors.



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MLG Doors Inline Restrictor

Refer to Figure 15.

The MLG doors inline restrictors are unions that give a specific hydraulic restriction, to help in the door timing sequence. The MLG doors inline restrictors bypass the MLG mechanical sequence valves to keep the MLG retraction actuators pressurized to 3000 psi (20684 kPa). The pressure is kept at 3000 psi (20684 kPa) until the retraction cycle is completed.

MLG and NLG Doors Solenoid Sequence Valve

Refer to Figure 16.

The MLG and NLG doors solenoid sequence valves are located in each landing gear wheelwell. The doors solenoid sequence valve is a self contained, bolt-on assembly. This valve lets the hydraulically operated wheelwell doors open and close.

Nose Landing Gear Retraction Actuator

Refer to Figure 17.

The nose landing gear retraction actuator has two ports. The rod end of the piston is installed with bushings and a lubrication fitting. The cylinder end of the actuator has a ball and race with a lubrication fitting. Hydraulic lines are attached to the two ports and are sealed with O-rings. The NLG retraction actuator piston is attached at the top right hand of the NLG shock strut and the cylinder is attached to the aircraft structure.

NLG Lock Actuator

The NLG lock actuator has two ports. The piston is held in the cylinder by a gland nut and a cup washer. Both ends of the actuator have a ball and race with a lubrication fitting. The NLG lock actuator is attached to the NLG drag strut near the lock links.

NLG Doors Actuator

Refer to Figure 18.

The NLG door actuator has two ports. The piston is held in the cylinder by a gland nut. The NLG door actuator extends to break the linkage over-center and open the doors. The NLG door actuator retracts to break the linkage over-center and close the doors.

NLG Doors Mechanism

Refer to Figure 19.

The NLG doors mechanism is actuated by the NLG door actuator to open or close the forward NLG doors. When the NLG doors actuator retracts, it unlocks the over centered lock link. As the actuator continues to retract, it folds the lock link and turns the bellcrank clockwise. The bellcrank acting on the two push rods pushes the forward doors to the open position.

When the NLG doors actuator extends, it unfolds the lock link to the over-center position. As the actuator continues to extend, it unfolds the lock link and turns the bellcrank anti-clockwise. The bellcrank acting on the two push rods, pulls the forward doors to the closed position. The forward doors are closed when the NLG is in the extended or the retracted position. The aft doors are connected to and operated by the upper part of the shock strut of the NLG through



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two articulated links. The aft doors are open when the NLG is extended and closed when the NLG is retracted.

NLG Doors Mechanical Sequence Valve

[Refer to Figure 20.](#)

The NLG mechanical sequence valve is a bolt-on assembly. It correctly sequences the NLG uplock/downlock release and the NLG retraction actuators. This prevents the landing gear from traveling through partially opened NLG landing gear doors. The NLG mechanical sequence valve is open when the landing gear is moving between either the fully extended or the fully retracted position. The NLG mechanical sequence valve is closed when the landing gear is locked in either the fully extended or the fully retracted position.

NLG Inline Restrictors

The NLG inline restrictors are unions that provide a specific hydraulic restriction. The NLG inline restrictors bypass the NLG mechanical sequence valves to keep the NLG retraction actuators pressurized to 3000 psi (20684 kPa). Hydraulic pressure is kept at 3000 psi (20684 kPa) upon completion of the retraction cycle.

NLG Doors Solenoid Sequence Valve

[Refer to Figure 16.](#)

The NLG doors solenoid sequence valve is a bolt-on assembly with four ports. The valve is directly operated by a single coil solenoid. This valve lets the hydraulically operated NLG wheelwell doors open and close.

Power Transfer Unit

[Refer to Figure 21.](#)

The Power Transfer Unit (PTU) includes a hydraulic motor operated by the No. 1 hydraulic system, which operates a hydraulic pump in the No. 2 hydraulic system. The PTU supplies hydraulic pressure to the No. 2 hydraulic system when the following occurs:

- The No. 1 hydraulic system is in operation
- There is no fluid loss in the No. 2 hydraulic system
- The PTU is activated either manually or automatically.

Automatic operation occurs when the No. 2 engine fails during take-off and flap lever is selected to greater than 0 degrees.

The PTU can be selected manually at any time to extend or retract the landing gear. To select the PTU manually, press the PTU CNTRL switchlight on the HYDRAULIC CONTROL panel. When the switchlight is pressed, a green light, located in the switchlight, comes on to show that the PTU is supplying hydraulic pressure. The PTU CNTRL switchlight must be selected to the "OFF" position after the landing gear is down and locked.



LEGEND

- Pressure No.2 HYD
- Extend Pressure
- Return Pressure
- Check Valve
- Restrictor

RIGHT MAIN GEAR

The diagram illustrates the hydraulic system for the right main gear. It shows the flow of pressure No.2 HYD, extend pressure, and return pressure through various valves and actuators. Key components include:

- SOLENOID A (DOWN)** and **SOLENOID B (UP)**: Control valves for the landing gear selector valve.
- LANDING GEAR SELECTOR VALVE (UP SOLENOID ENERGIZED)**: Selects between up and down gear positions.
- LANDING GEAR BYPASS VALVE**: Allows for manual bypass of the selector valve.
- MECHANICAL SEQUENCE VALVE**: Ensures proper sequencing of gear operations.
- SOLENOID SEQUENCE VALVE**: Controls the sequence of gear extension and retraction.
- REAR DOOR ACTUATOR**: Controls the rear door of the gear.
- NOSE DOOR ACTUATOR**: Controls the nose door of the gear.
- NOSE GEAR RETRACTION ACTUATOR**: Controls the retraction of the nose gear.
- MAIN GEAR RETRACTION ACTUATOR**: Controls the retraction of the main gear.
- AUXILIARY ACTUATOR**: Controls auxiliary functions.
- LOCK ACTUATOR** and **UPLOCK ACTUATOR**: Control the locking and unlocking of the gear.
- DOWNLOCK ACTUATOR**: Controls the downlocking of the gear.

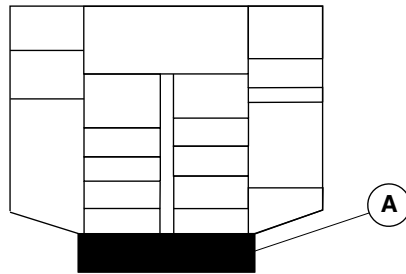
The system is designed to ensure safe and reliable gear operation under various conditions, including manual bypass and emergency procedures.

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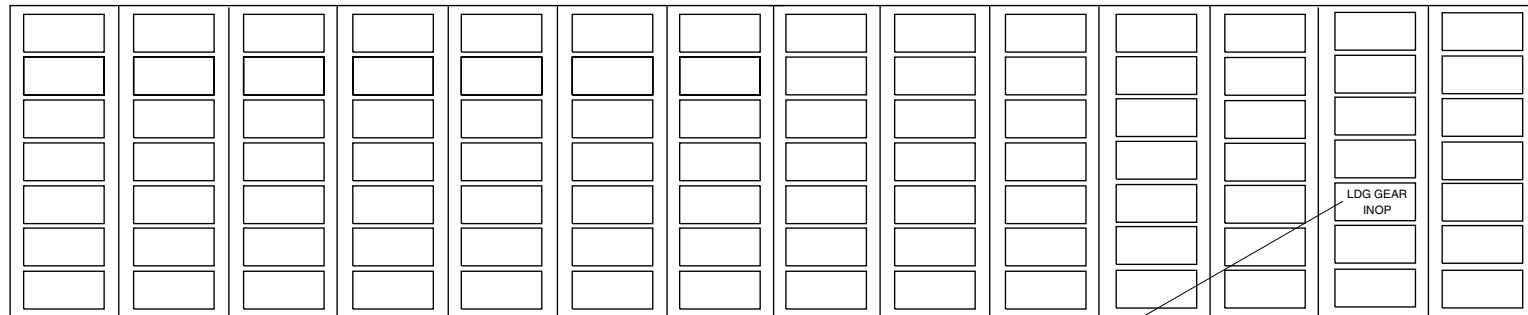
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OVERHEAD CONSOLE

LEGEND

1. Landing Gear Inoperative (Amber).



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LDG GEAR INOP caution light
Figure 2

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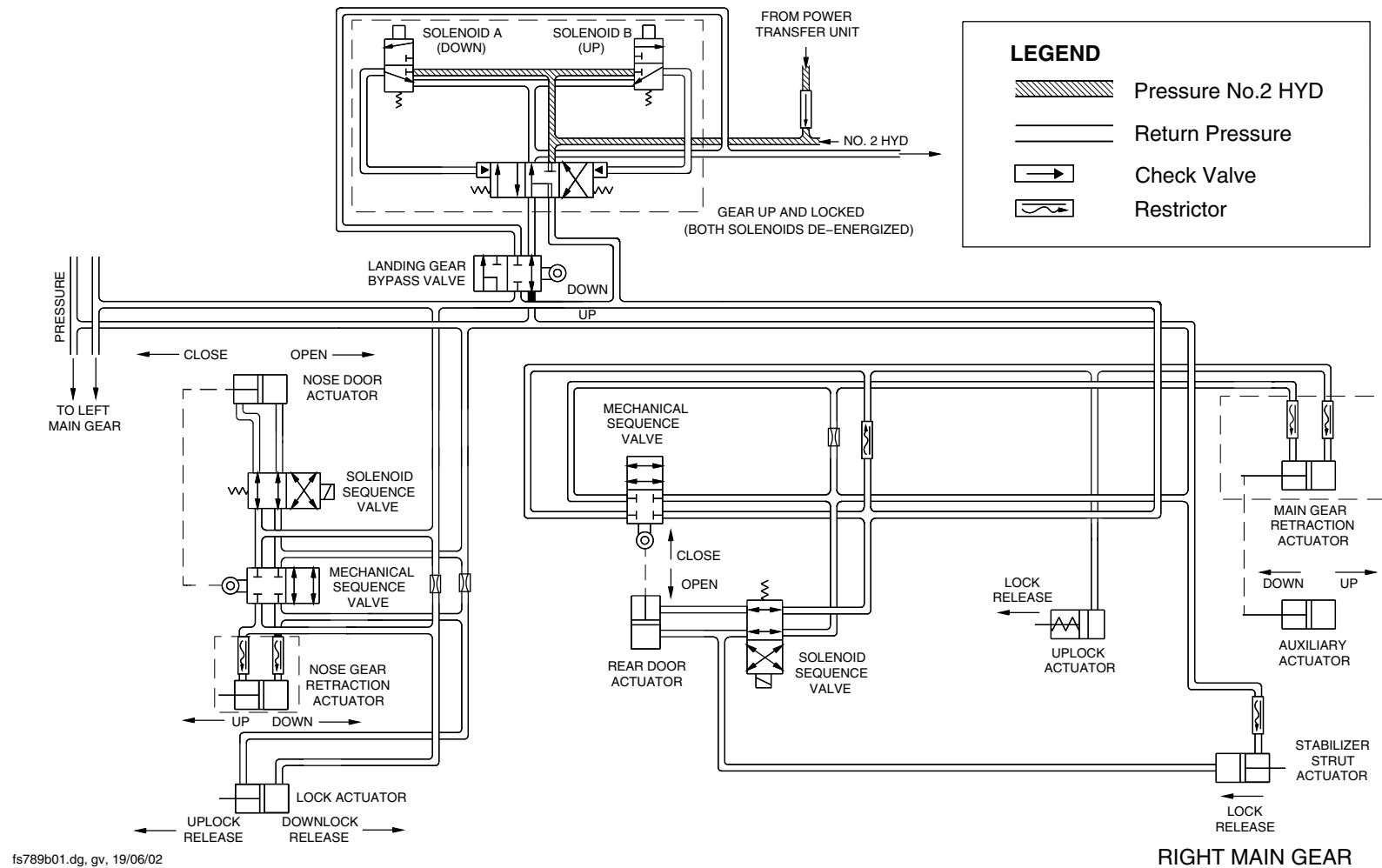
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Gear Up and Locked Doors Closed
Figure 3

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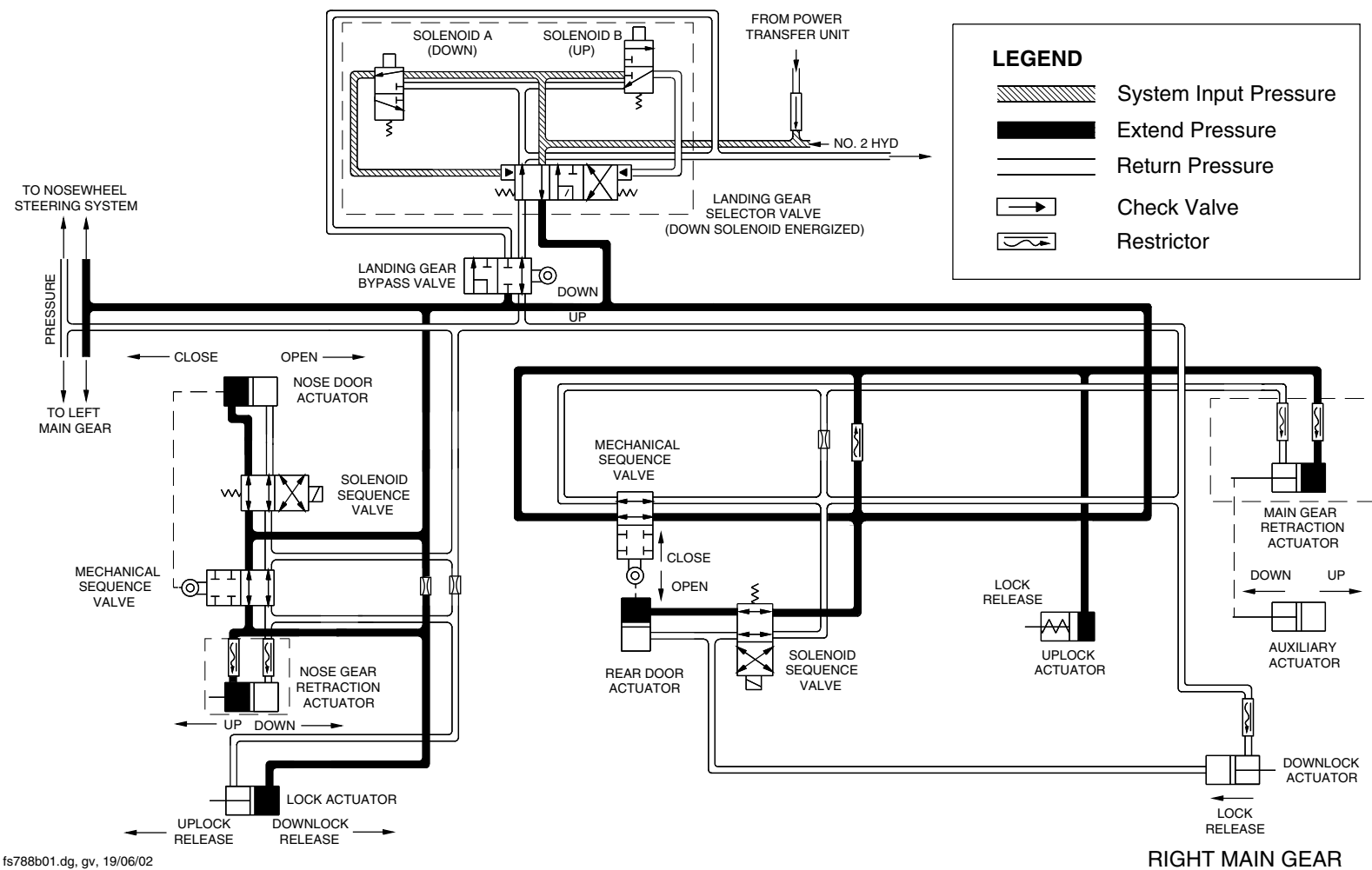
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Initial Down Selection Doors Open
Figure 4

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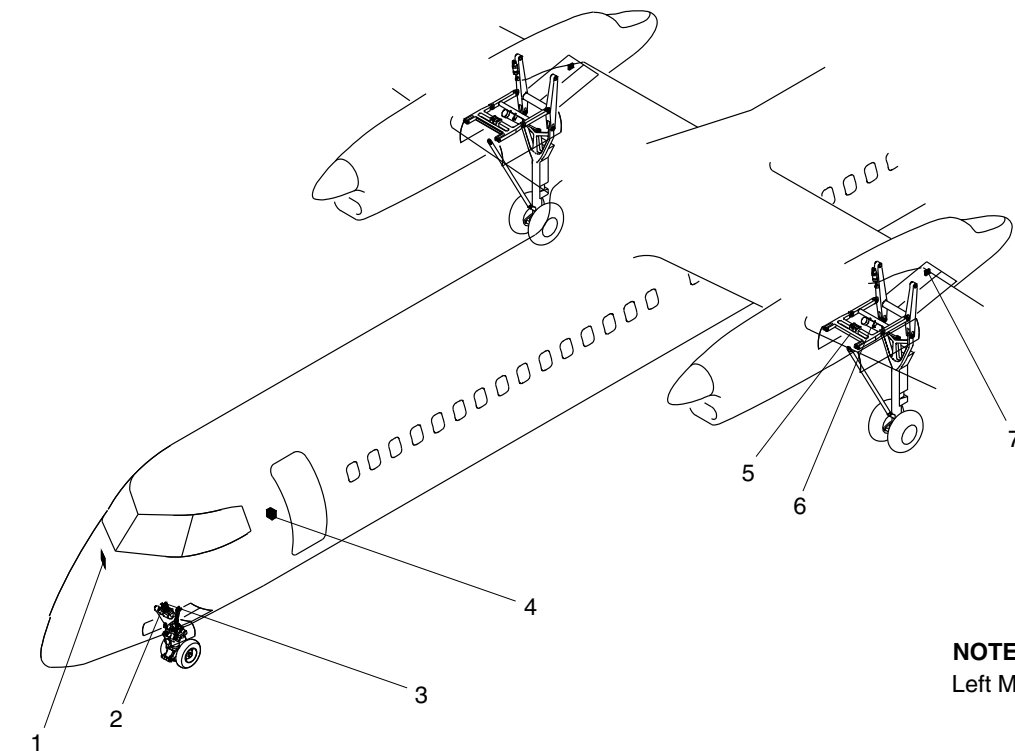


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LEGEND

1. Landing Gear Selector Level and Control Panel.
2. NLG Retraction Actuator.
3. NLG Lock Actuator.
4. Proximity Sensor Electronic Unit.
5. MLG Lock Actuator.
6. MLG Retraction Actuator.
7. Uplock Actuator.



fs312a01.cgm

NOTE

Left MLG similar to right MLG.

Landing Gear Primary Extension and Retraction – Component Location
Figure 5

PSM 1-84-2A

EFFECTIVITY:

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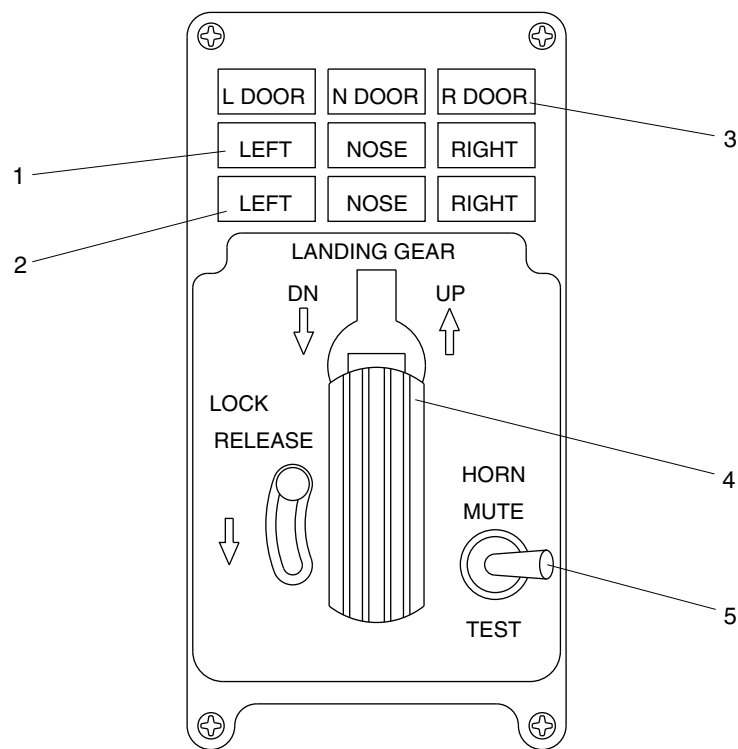
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LEGEND

1. "Green" Gear Downlocked.
2. "Red" Gear Not Uplocked Or Downlocked.
3. "Amber" Door Open.
4. Handle "Amber" (Concurrent W/"Red" Unsafe Lights) Gear In-Transit.
5. Warning Horn Mute/Test Switch.

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Primary Extension and Retraction, Landing Gear Selector Panel
Figure 6

PSM 1-84-2A
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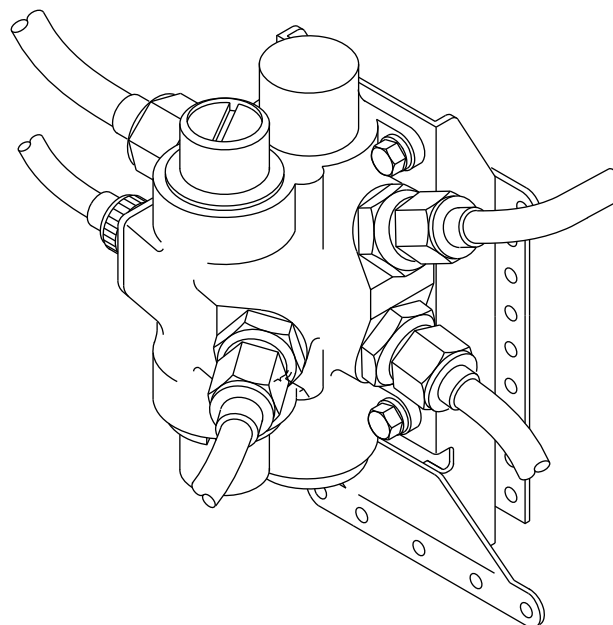
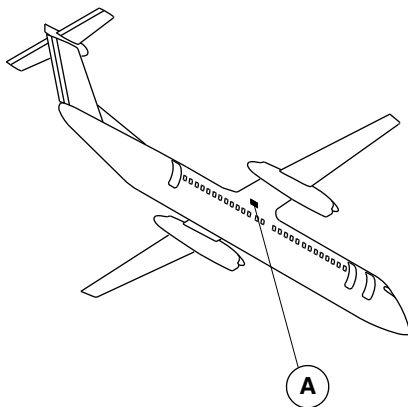
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Landing Gear Selector Valve – Locator
Figure 7

PSM 1-84-2A
EFFECTIVITY:
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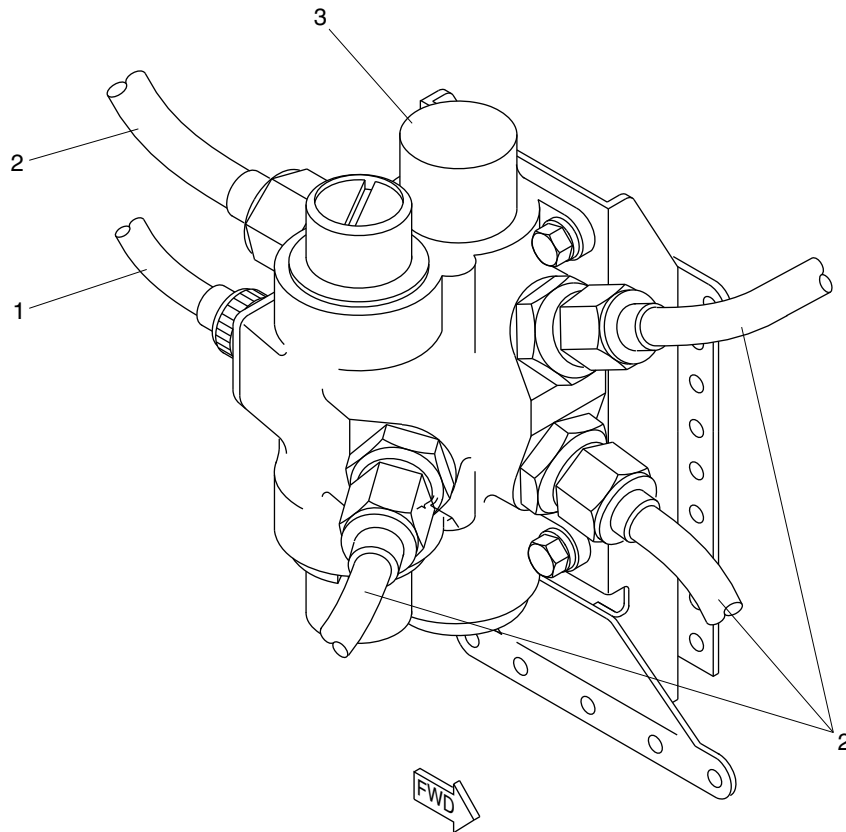
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LEGEND

- 1. Electrical Connector.
- 2. Hydraulic Line.
- 3. Selector Valve.

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Landing Gear Selector Valve Detail
Figure 8

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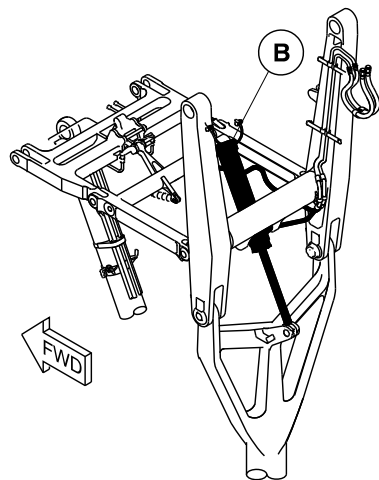
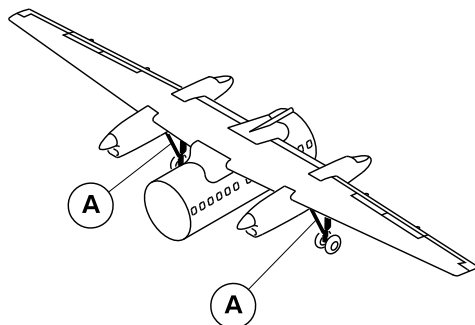


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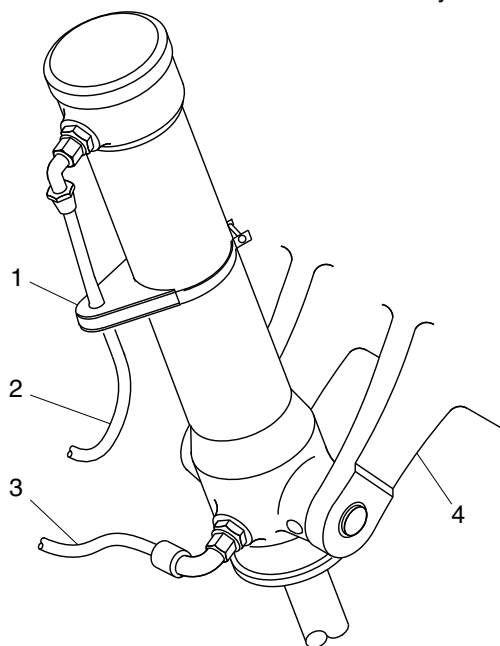
LEGEND

1. Insert.
2. Hydraulic line extension.
3. Hydraulic line retraction.
4. Yoke assembly.

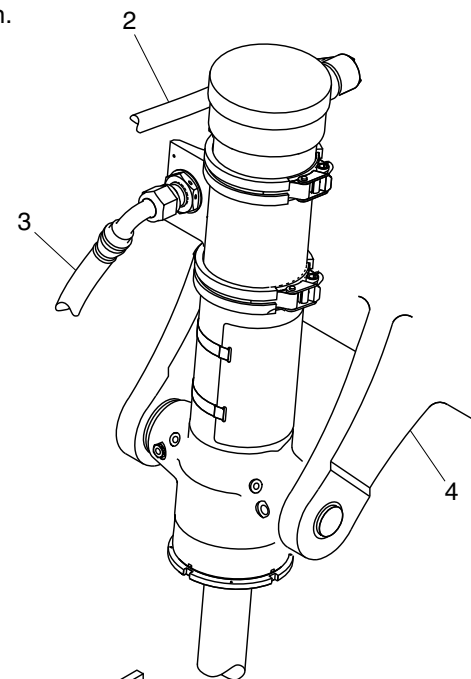


MAIN LANDING GEAR (MLG)

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**PRE MOD 4-902327
PRE SB84-32-105**



**POST MOD 4-902327
POST SB84-32-105**

MLG Retraction Actuator Locator
Figure 9

PSM 1-84-2A
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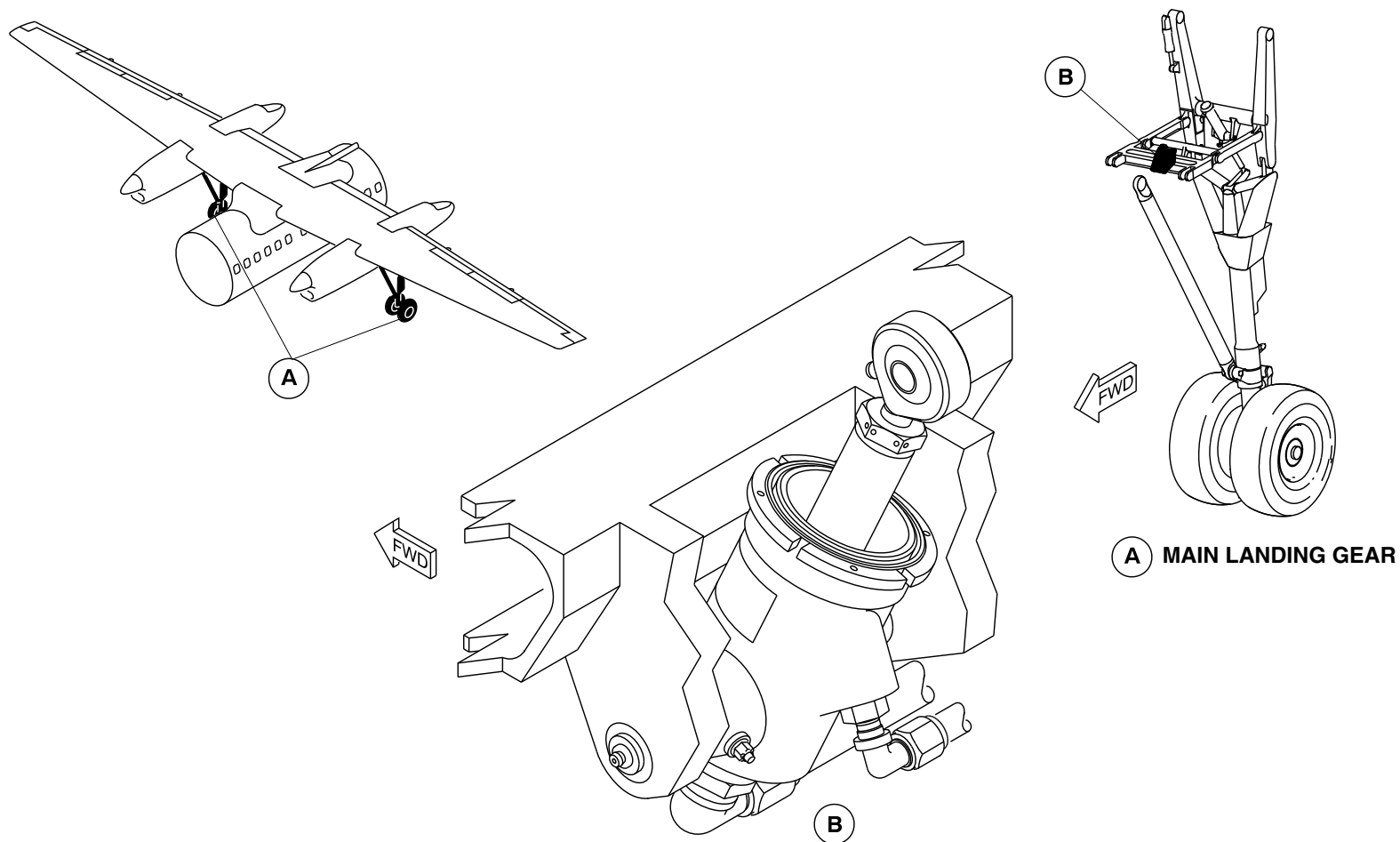
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MLG Downlock Actuator Locator
Figure 10

PSM 1-84-2A
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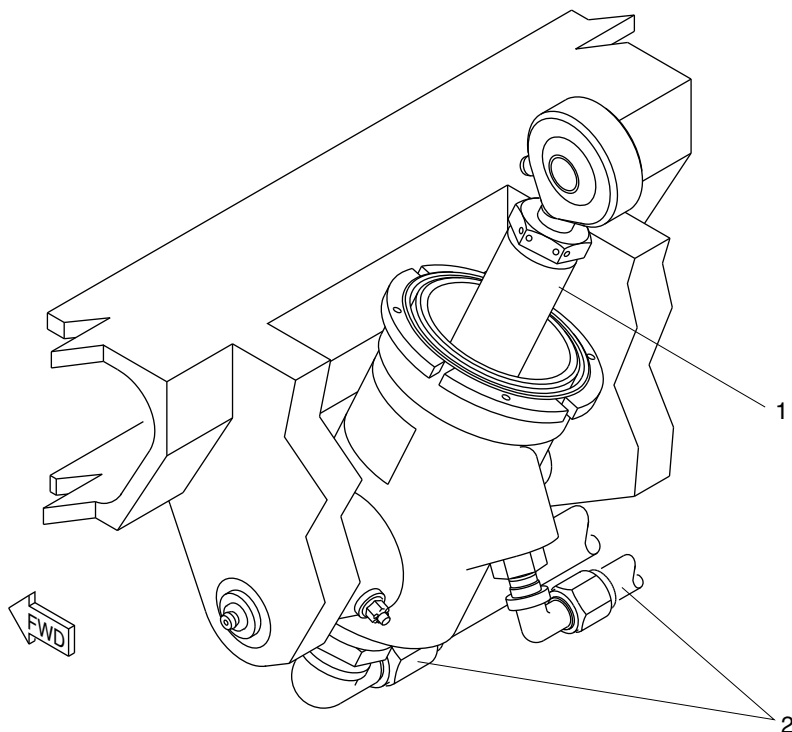
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LEGEND

- 1. MLG Downlock Actuator.
- 2. Hydraulic Hose.

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MLG Downlock Actuator Detail
Figure 11

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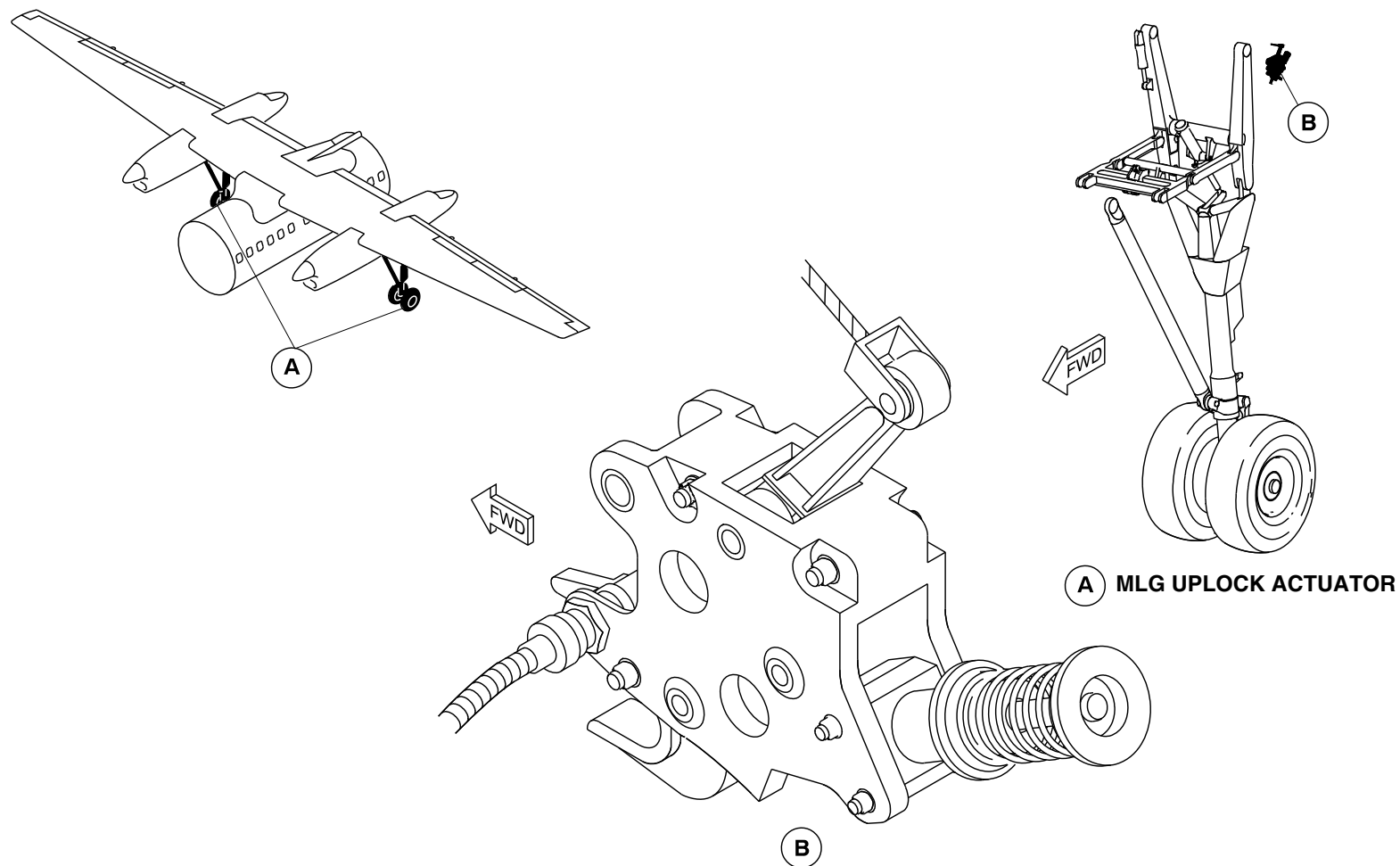
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MLG Uplock Actuator Locator
Figure 12

PSM 1-84-2A
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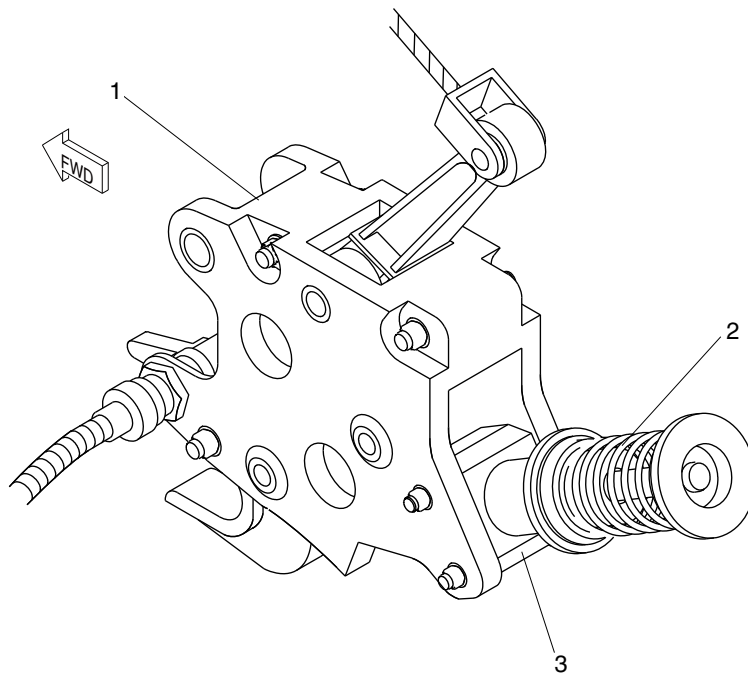
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LEGEND

- 1. MLG Uplock Assembly.
- 2. External Spring.
- 3. MLG Uplock Actuator.

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MLG Uplock Actuator Detail
Figure 13

PSM 1-84-2A
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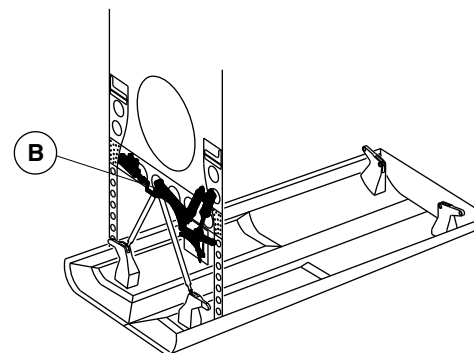
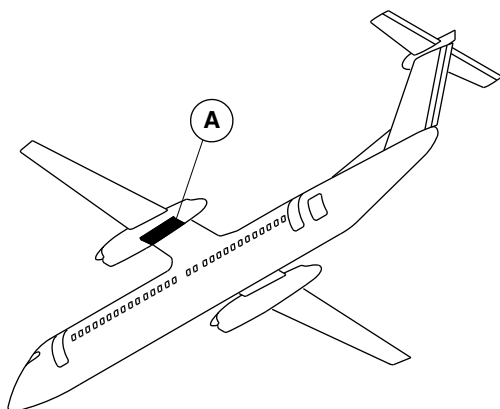
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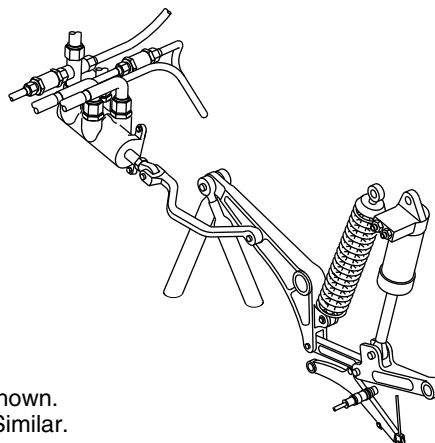


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A MLG DOORS



NOTE

Left Gear Shown.
Right Side Similar.

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MLG Aft Doors Mechanism Locator
Figure 14

PSM 1-84-2A
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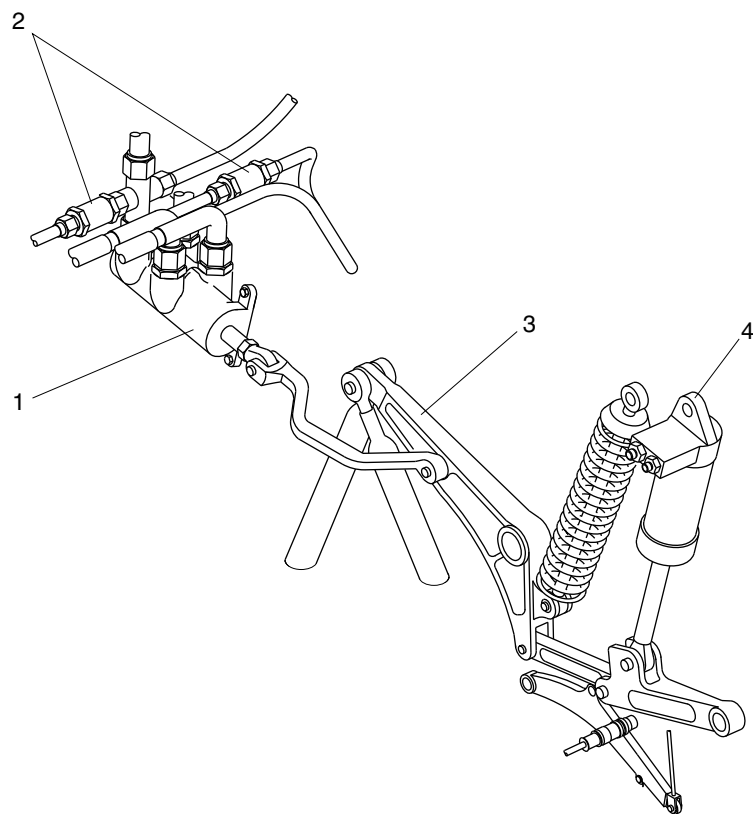
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LEGEND

- 1. MLG Doors Mechanical Sequence Valve.
- 2. MLG Doors Inline Restrictors.
- 3. MLG Doors Mechanism.
- 4. MLG Doors Actuator.

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MLG Aft Doors Mechanism Detail
Figure 15

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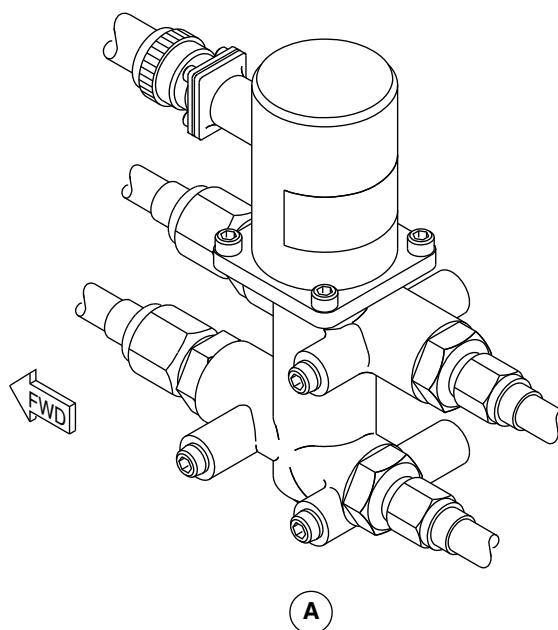
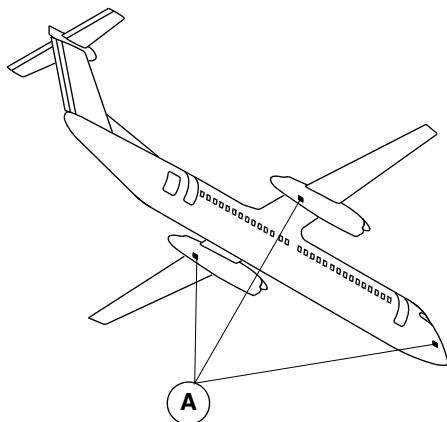
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Landing Gear Doors Solenoid Sequence Valve – Component Details
Figure 16

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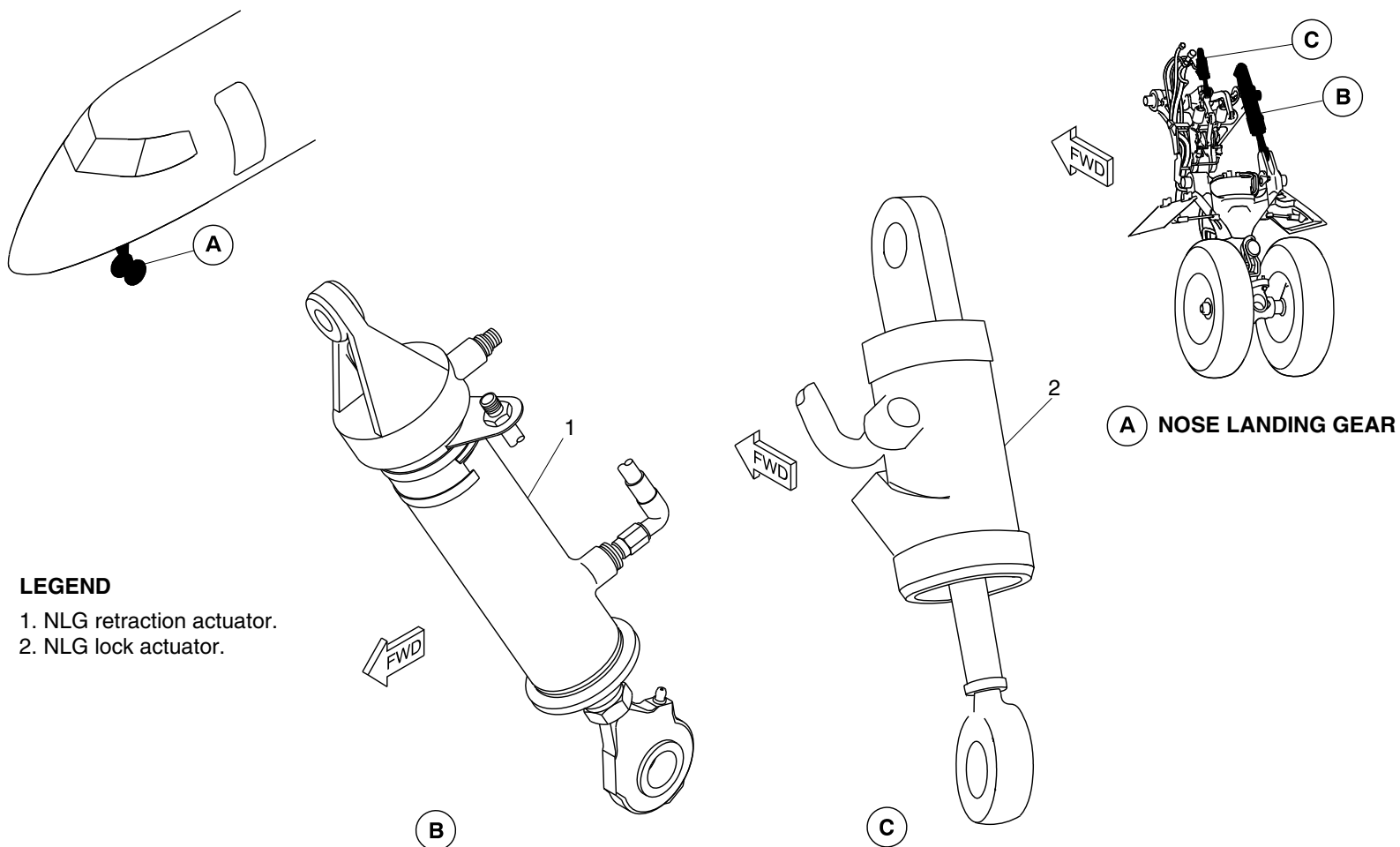
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NLG Actuators Locator
Figure 17

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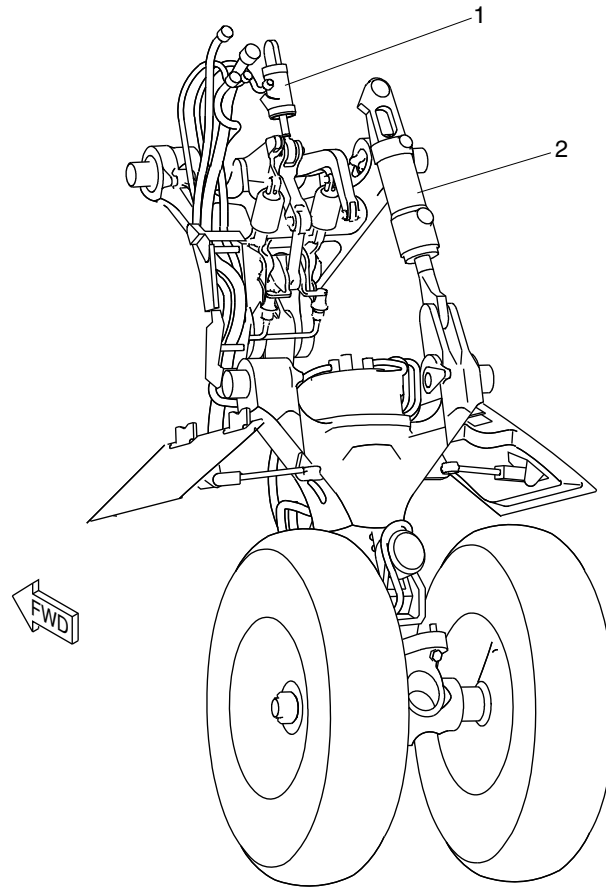
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LEGEND

- 1. NLG Lock Actuator.
- 2. NLG Retraction Actuator.

fs130a01.cgm

NLG Actuators Detail
Figure 18

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 32-31-00
Config 001

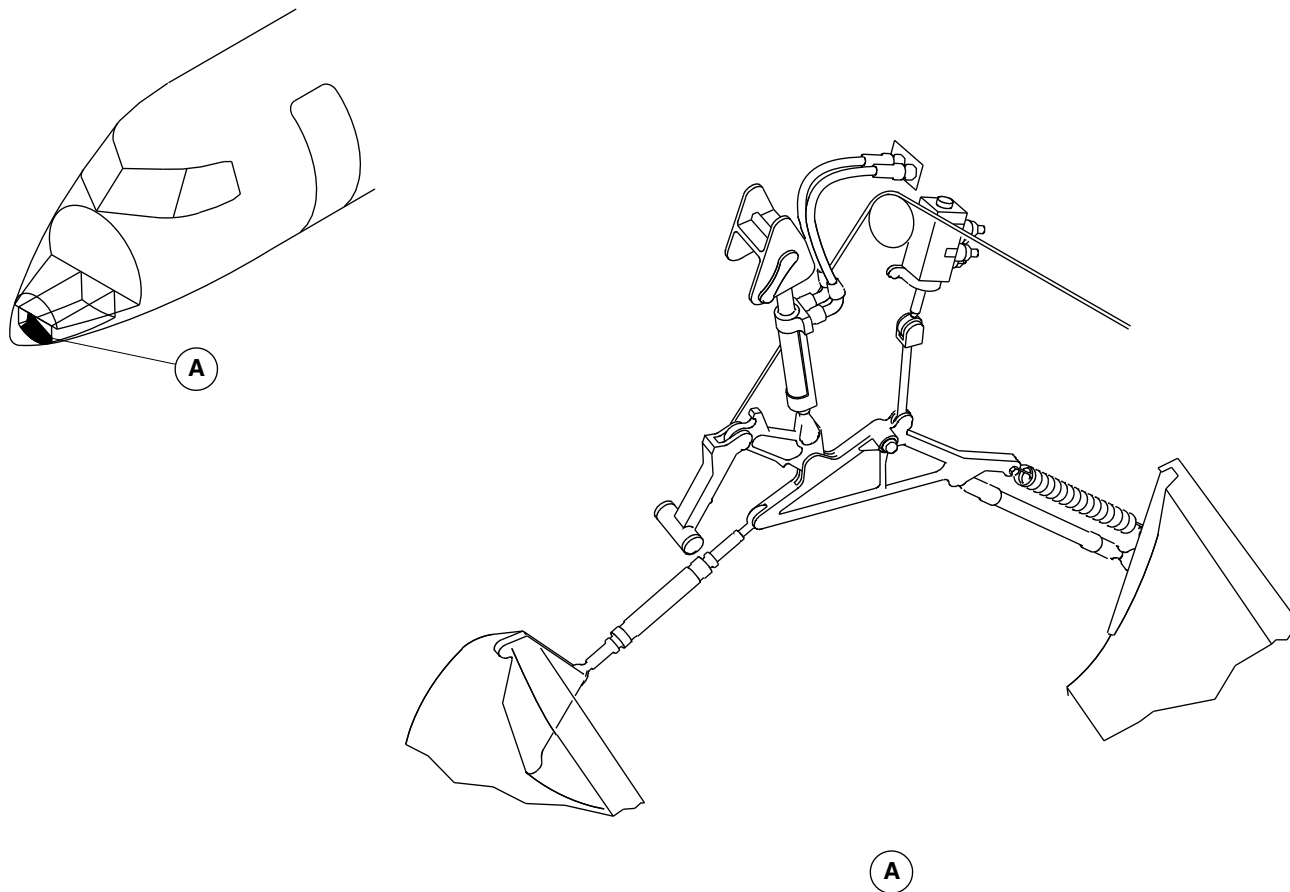
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



fs431a01.cgm

NLG Door Mechanism Locator
Figure 19

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 32-31-00
Config 001

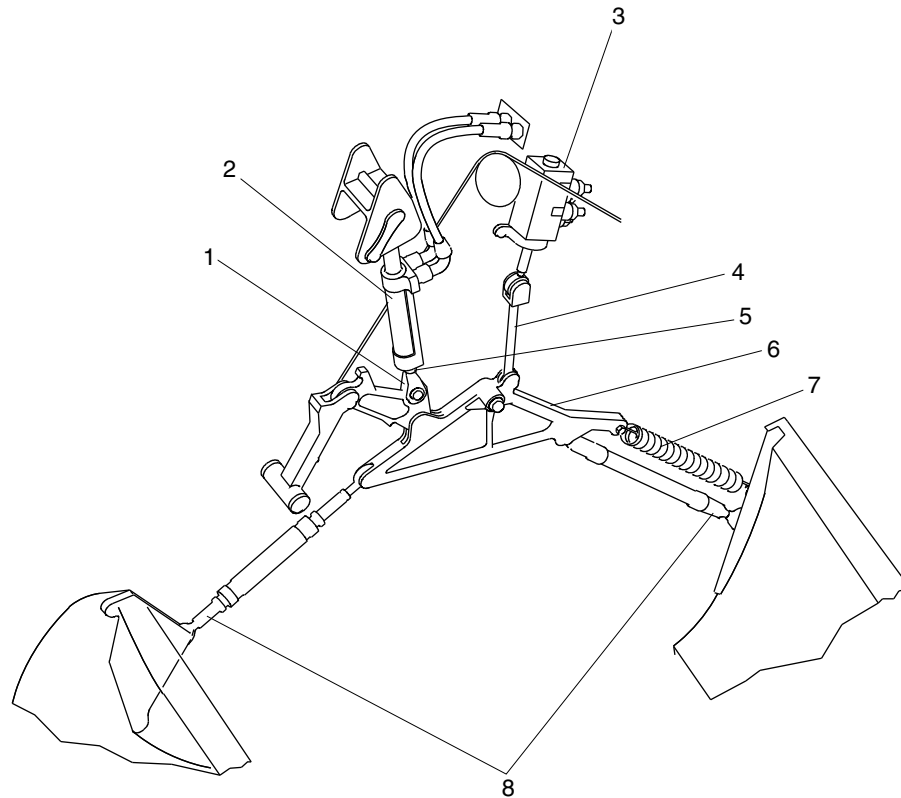
32-31-00

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LEGEND

1. Actuator Fork.
2. NLG Forward Door Actuator.
3. NLG Mechanical Sequence Valve.
4. Link.
5. NLG Doors Actuator.
6. NLG Door Mechanism.
7. Spring Assembly.
8. Push Rods.

fs430a01.cgm

NLG Door Mechanism Detail
Figure 20

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 32-31-00
Config 001

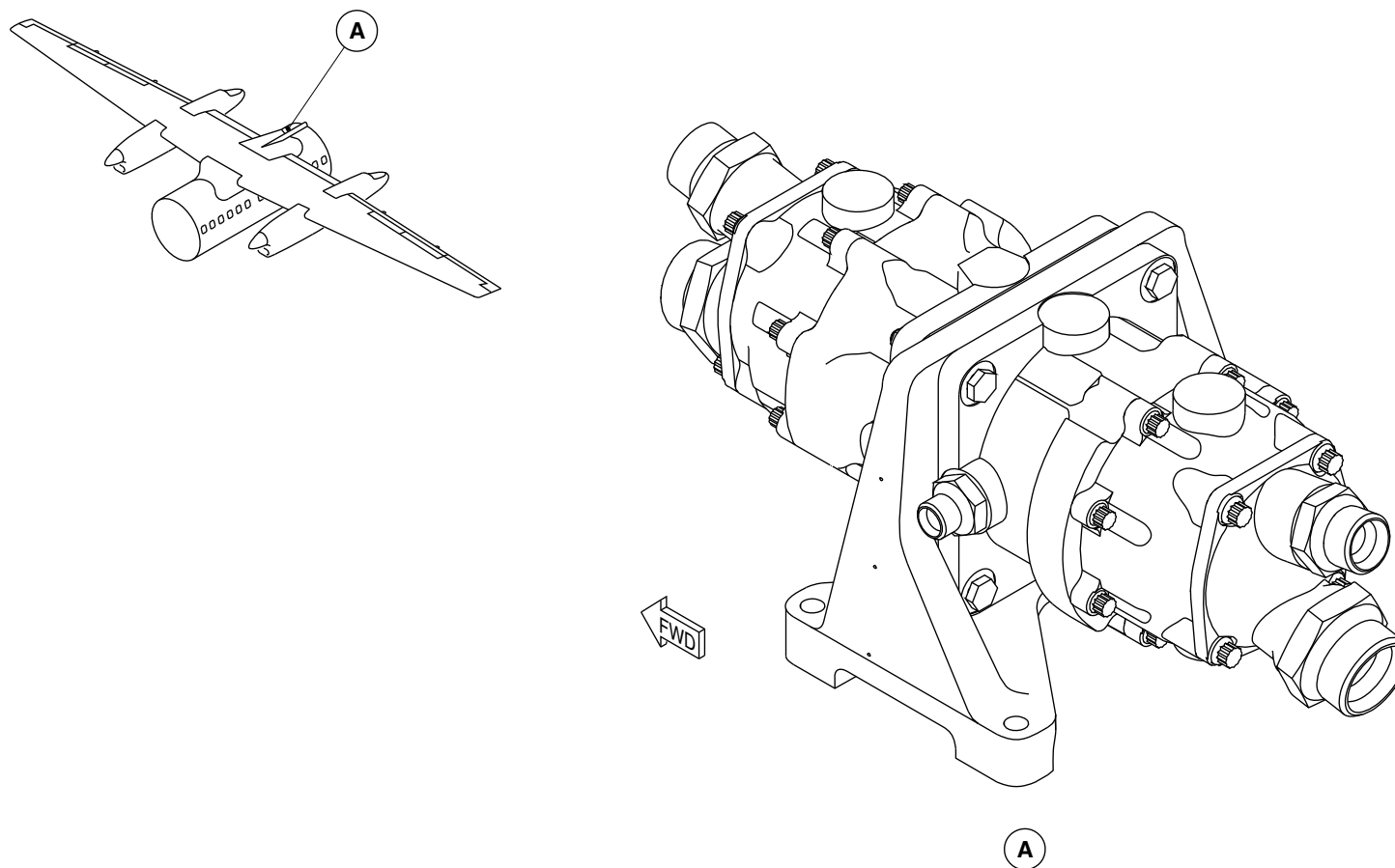
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PTU Locator
Figure 21

PSM 1-84-2A
EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

32–34–00–001

ALTERNATE EXTENSION SYSTEM

Introduction

The alternate extension system can be used to extend the landing gear when the No. 2 hydraulic system, or the Proximity Sensor Unit (PSEU) are not serviceable. The system can also be used, if the normal extension system fails to lock the landing gear in the down position.

General Description

The alternate extension system is a self resetting, cable actuated design. Access to the alternate extension system is through the flight compartment. The alternate extension system includes a bypass valve to isolate the landing gear hydraulics from the No. 2 hydraulic system. A manual handpump hydraulic system is available to assist in lowering the Main Landing Gear (MLG) into the down and locked position.

The alternate extension of the landing gear occurs when the Nose Landing Gear (NLG) forward doors, the MLG aft doors, and the landing gear uplocks are mechanically opened. The NLG free-falls to the down and locked position, assisted by the airflow. The MLGs free-fall and, if required, are pumped to the down and locked position by the alternate extension actuators.

To isolate the normal landing gear selector valve during flight, a LANDING GEAR INHIBIT switch is installed in the flight compartment

ceiling. To use the alternate extension system, the LANDING GEAR INHIBIT switch must be set from NORMAL to INHIBIT.

The alternate extension system has the components that follow:

- Two alternate release handles
- One alternate release door
- One alternate extension door
- Two alternate extension actuators
- A bypass valve
- Control cables.

Detailed Description

Alternate manual extension of the landing gear is accomplished in four stages.

The LANDING GEAR INHIBIT switch is set to the INHIBIT position to disable the PSEU.

[Refer to Figures 1 and 2.](#)

The LANDING GEAR ALTERNATE RELEASE door, located in the ceiling of the flight compartment, is opened. This actuates the bypass valve to isolate the No. 2 hydraulic system. The LANDING GEAR ALTERNATE RELEASE handle is pulled to open the MLG doors and release the MLG uplocks. Tension springs in the MLG door mechanism pull the doors open and the MLG free-falls into the down position.

[Refer to Figures 3 , 4 and 5.](#)

The LANDING GEAR ALTERNATE EXTENSION door located in the flight compartment floor is opened. The door operates the alternate

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landing gear selector valve and reveals the handpump socket. If the MLG does not reach the down and locked position, the handpump is used to fully extend the MLG. Springs installed on the MLG stabilizer brace move the lock links into the down and locked position. The LANDING GEAR ALTERNATE EXTENSION handle is pulled to unlock the NLG forward doors. Tension springs in the NLG door mechanism pull the doors open. When the doors are fully open, the NLG uplock is released by a further pull on the handle. The NLG free-falls to the fully extended position. Springs installed on the NLG drag strut move the lock links into the down and locked position.

[Refer to Figure 4.](#)

An alternate landing gear down and locked verification system is located in the in the flight compartment, in the aft section of the LANDING GEAR ALTERNATE EXTENSION compartment. The system has a two position, momentary type toggle switch, a light panel that contains three green lights, and four proximity sensors. Two of the proximity sensors are used in series and are attached to the NLG. One proximity sensor is installed on each of the MLG. The system elements on each landing gear are activated when the downlock verification switch is moved to the ON position. The system indication lights come on as each landing gear is locked down.

Alternate Release Handles

[Refer to Figures 6 and 7.](#)

The alternate extension of the NLG uses the NLG alternate release handle located in the floor of the flight compartment under the LANDING GEAR ALTERNATE EXTENSION door. The handle is connected by cables, turnbuckles and pulleys to the NLG uplock release arm, a pulley cam mechanism and to the NLG door release operating arm. The pulley cam mechanism makes sure that the NLG

forward doors open before the NLG uplock is opened to release the NLG.

[Refer to Figures 8 and 9.](#)

The alternate extension of the MLG uses the MLG alternate release handle, the handpump, and the auxiliary actuators. The alternate release handle is located in the ceiling of the flight compartment behind the LANDING GEAR ALTERNATE RELEASE door. To actuate the MLG bypass valve, open the LANDING GEAR ALTERNATE RELEASE door. The handle is connected by cables, turnbuckles, pulleys, a slide assembly, and a cam mechanism to the MLG uplocks and the MLG aft doors release mechanisms. The cam mechanism makes sure that the MLG aft doors open before the MLG uplocks are opened to release the MLGs.

Alternate Extension Actuator

[Refer to Figure 11.](#)

Alternate extension actuators are located on the right side of each MLG yoke. Hydraulic pressure required to operate the actuators is supplied by a handpump located in the flight compartment floor. The hydraulic fluid is supplied from the emergency hydraulic reservoir.

Bypass Valve

[Refer to Figure 10.](#)

The bypass valve is installed in the forward right wing root area, adjacent to the landing gear selector valve. The bypass valve isolates the No. 2 hydraulic supply lines from the main and nose gear retraction actuators, and connects the lines to the return lines. The bypass valve is actuated when the LANDING GEAR ALTERNATE RELEASE door is opened. Cables extend from the LANDING GEAR



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

ALTERNATE RELEASE door and attach to a lever at the bypass valve. The lever operates the bypass valve.

Control Cables

Refer to Figures 9 , 6 and 12.

The MLG ALTERNATE RELEASE door installed in the flight compartment ceiling has a closed loop cable system. When the door is opened, it actuates the bypass valve to the BYPASS position. This isolates the main landing gear hydraulic system and depressurizes the extension actuators. It also lets the MLG be pumped to the down and locked position. When the door is closed, the bypass valve returns to the NORMAL position.

Control cables are routed from the MLG alternate release handle through a slider mechanism located in the ceiling of the aircraft forward of the wing root. The slider mechanism lets the full force of the pull be concentrated first on the release of one MLG and then on the other. From here, the control cables are routed through to the engine nacelles to the cam mechanisms. The cam mechanism allows the force of the pull to be concentrated first on the release of the MLG aft doors and then on the MLG.

Control cables are routed from the NLG alternate release handle through a pulley cam mechanism located in the forward fuselage above the nose wheel well. The pulley cam mechanism lets the full force of the pull be concentrated first on the release of the NLG forward doors and then on the NLG.

Training Information Points

Make sure that all hydraulic power and electrical power is off before you do maintenance on the landing gear. The landing gear can

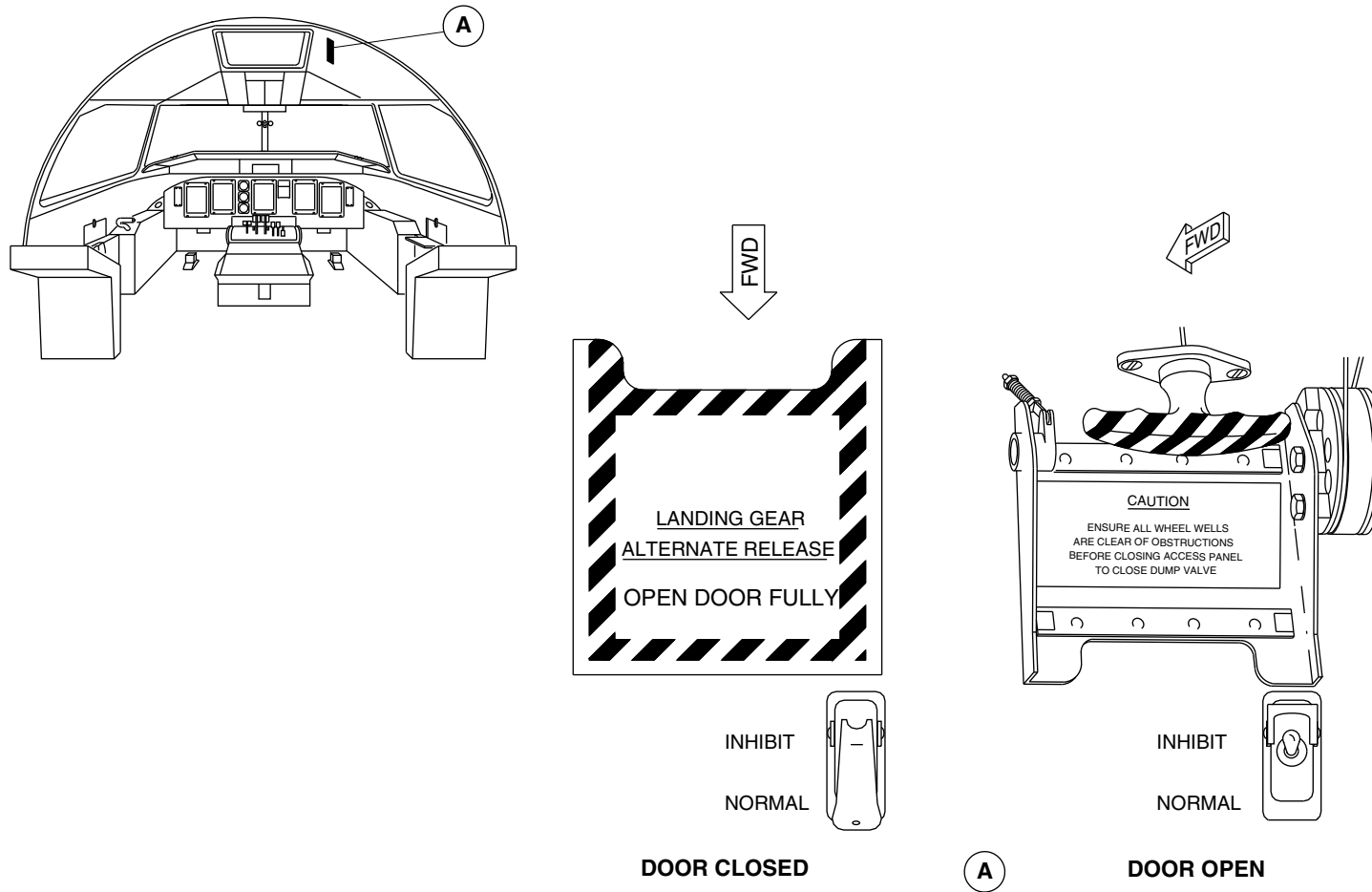
accidentally retract. This can cause injuries or death to persons and/or damage to equipment.

Make sure that all persons and equipment are clear of the landing gear and door movement area. If you do not do this, injuries to persons and/or damage to equipment can occur.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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Landing Gear Alternate Release Controls Locator
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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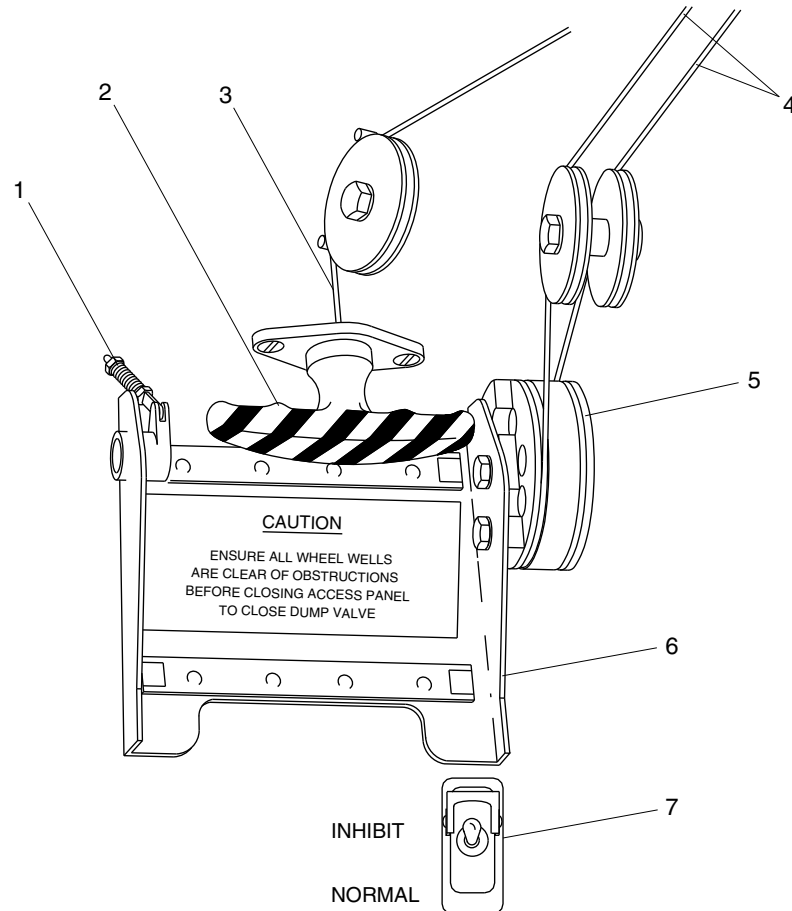
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LEGEND

1. Door Compression Spring.
2. Main Gear Alternate Release Handle.
3. Cables to Slide Assembly.
4. Cables to Bypass Valve.
5. Door Pulley.
6. Landing Gear Alternate Release Door.
7. Landing Gear Inhibit Toggle Switch.

fs434a01.cgm

Landing Gear Alternate Release Controls Detail
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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Config 001

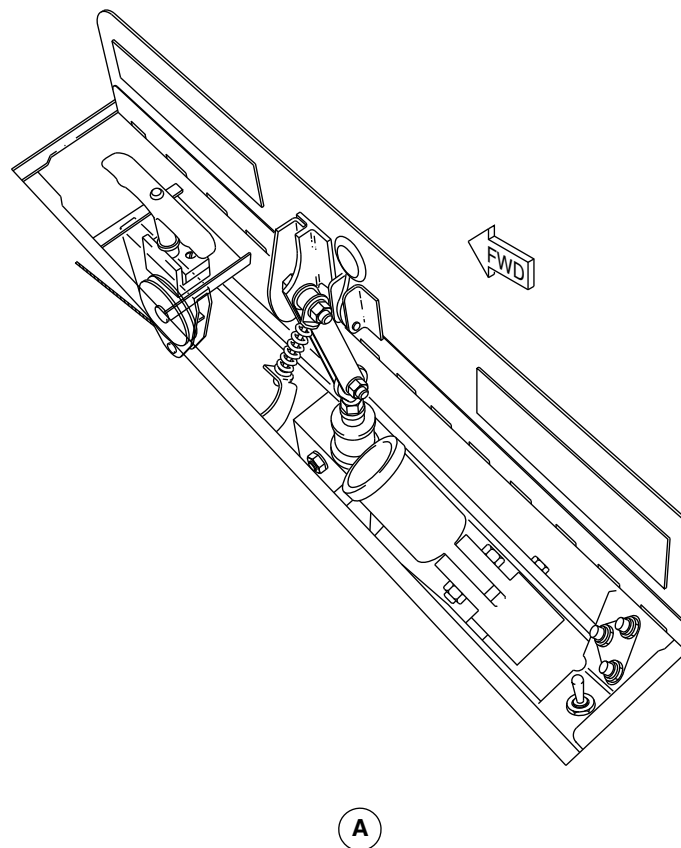
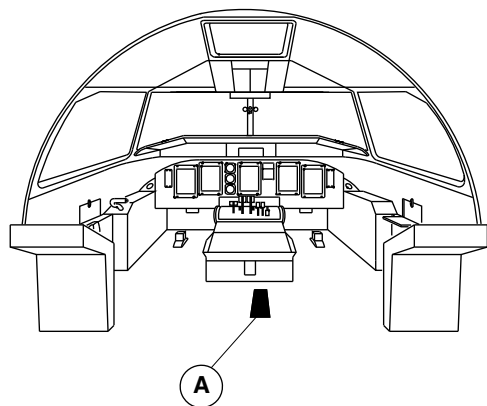
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fs215a01.cgm

Landing Gear Alt Extension Door Locator
Figure 3

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 32-34-00
Config 001

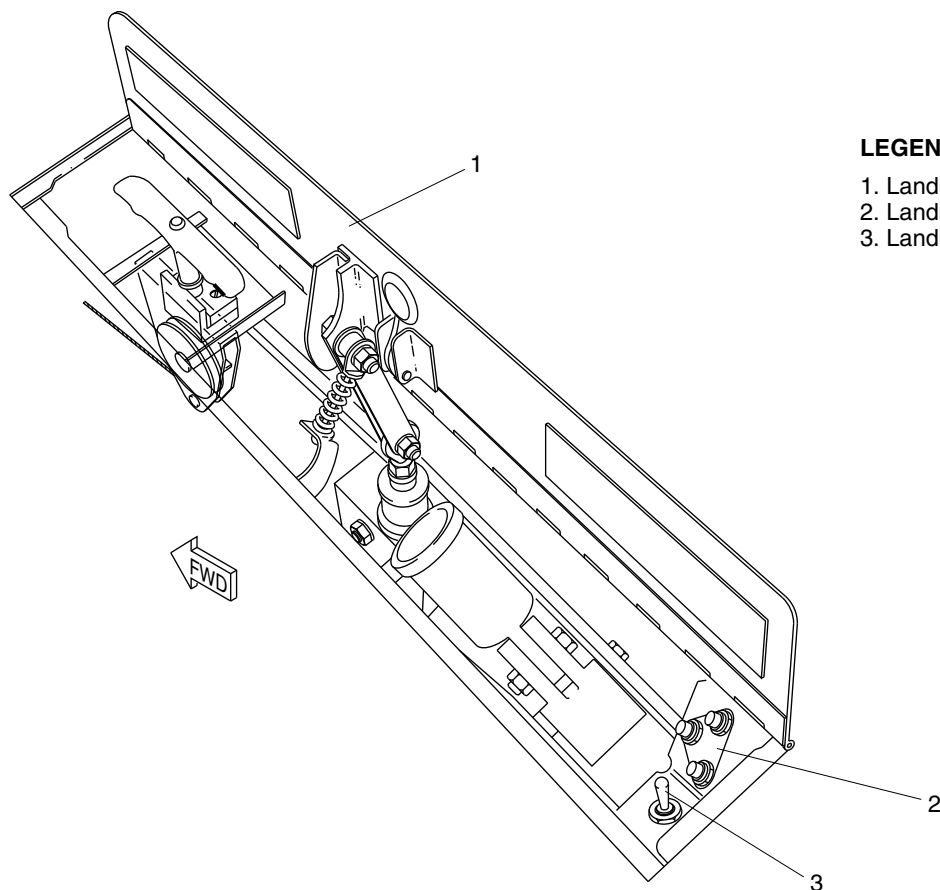
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LEGEND

1. Landing Gear Alternate Extension Door.
2. Landing Gear Downlock Verification Lights.
3. Landing Gear Downlock Verification Switch.

fs216a01.cgm

Landing Gear Alt Extension Door Detail
Figure 4

PSM 1-84-2A
EFFECTIVITY:
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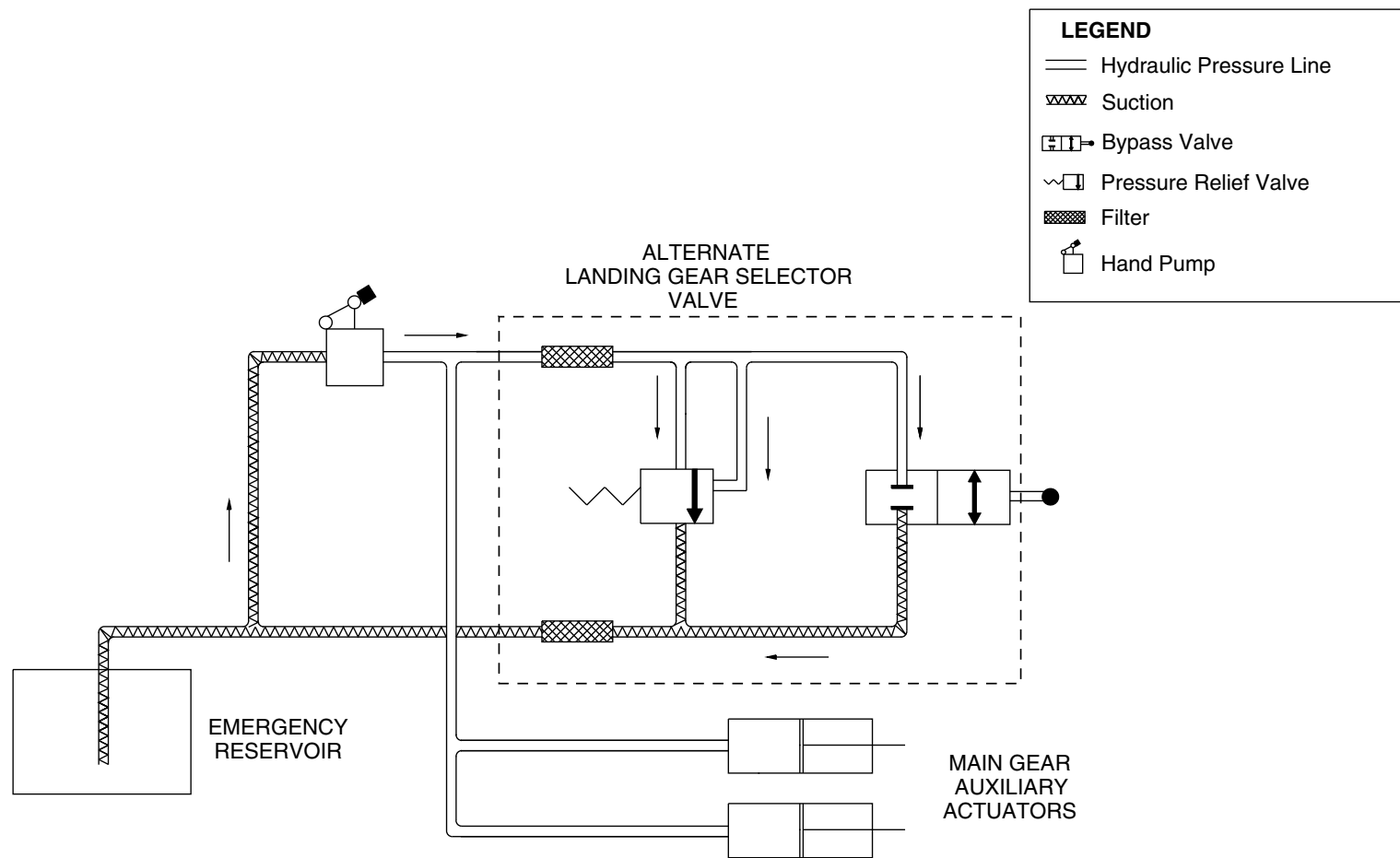
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Emergency Extension Hydraulic System
Figure 5

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 32-34-00
Config 001

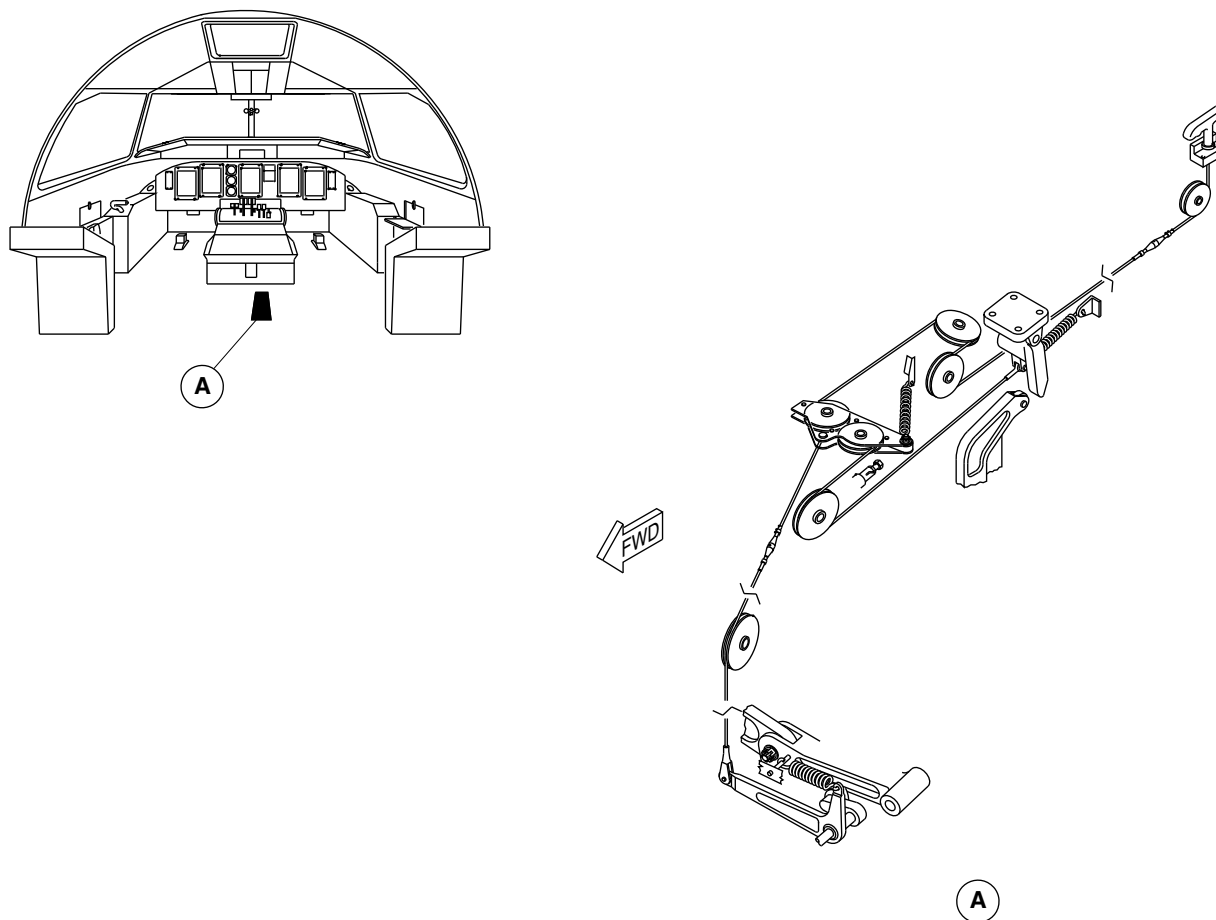
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fs269a01.cgm

Nose Gear Alternate Release Locator
Figure 6

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 32-34-00
Config 001

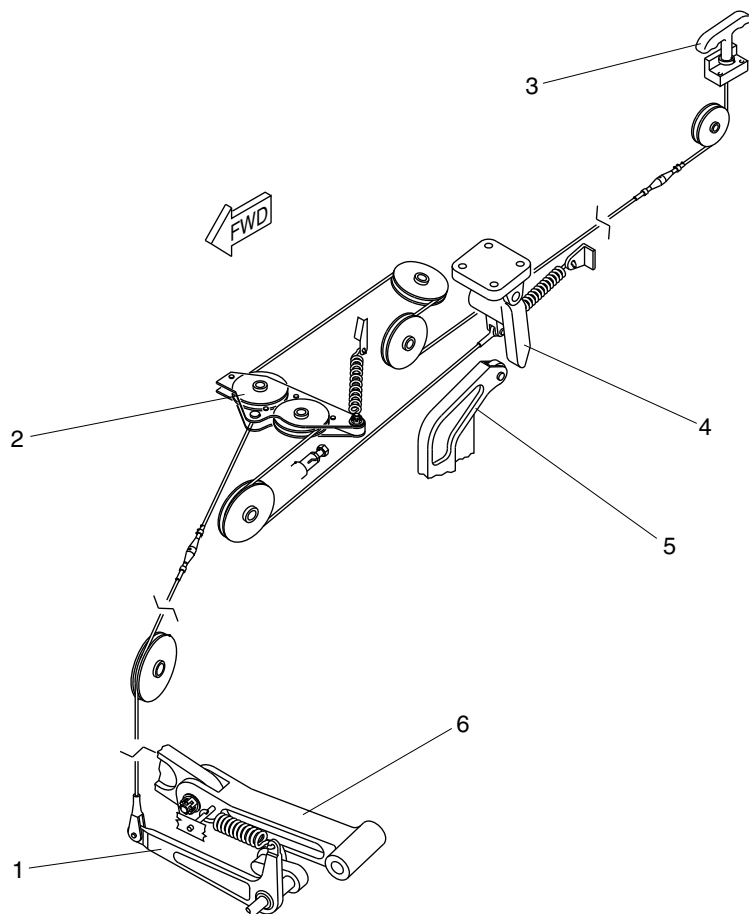
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fs270a01.cgm

LEGEND

1. Front Door Alternate Release Operating Arm and Roller Assembly.
2. Pulley Cam Mechanism.
3. Nose Gear Alternate Release Handle (Flight Compartment Floor).
4. NLG Uplock Release Arm.
5. Pivot Tube (Nose Gear Release).
6. Link Assembly.

Nose Gear Alternate Release Detail
Figure 7

PSM 1-84-2A
EFFECTIVITY:
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Config 001

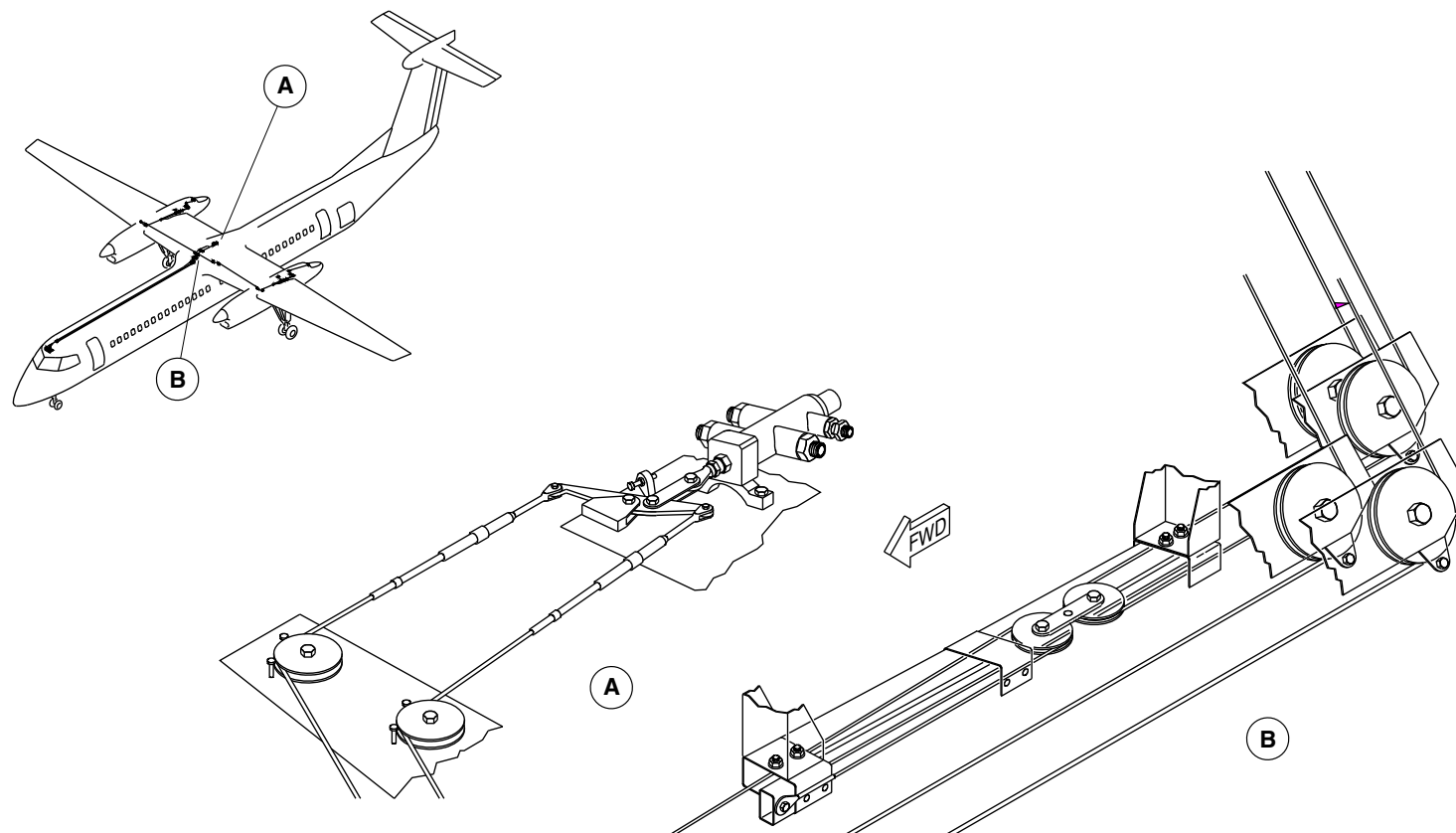
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Alternate Release Cables Locator
Figure 8

PSM 1-84-2A
EFFECTIVITY:
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Config 001

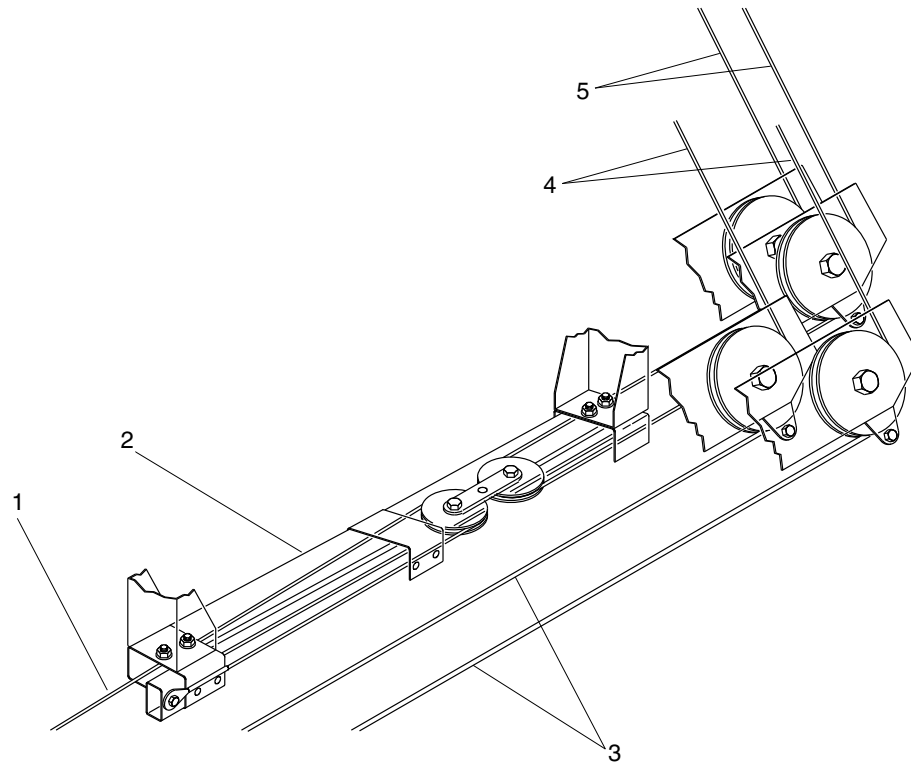
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fs402a01.cgm

LEGEND

1. Cable from Alternate Release Handle.
2. Slide Assembly.
3. Cables from Door Pulley.
4. Cables to Bypass Valve Operating Lever.
5. Cables to Left and Right Nacelles.

Alternate Release Cables Detail 1
Figure 9

PSM 1-84-2A
EFFECTIVITY:
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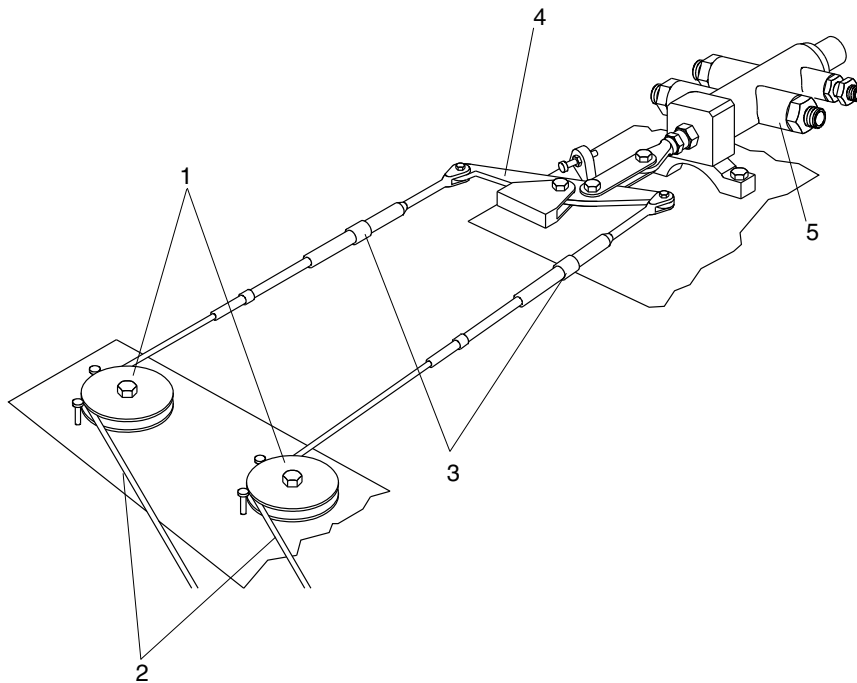


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LEGEND

- 1. Pulleys.
- 2. Cables from Door Pulley.
- 3. Turnbuckles.
- 4. Operating Lever.
- 5. Bypass Valve.



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Alternate Release Cables Detail 2
Figure 10

PSM 1-84-2A
EFFECTIVITY:
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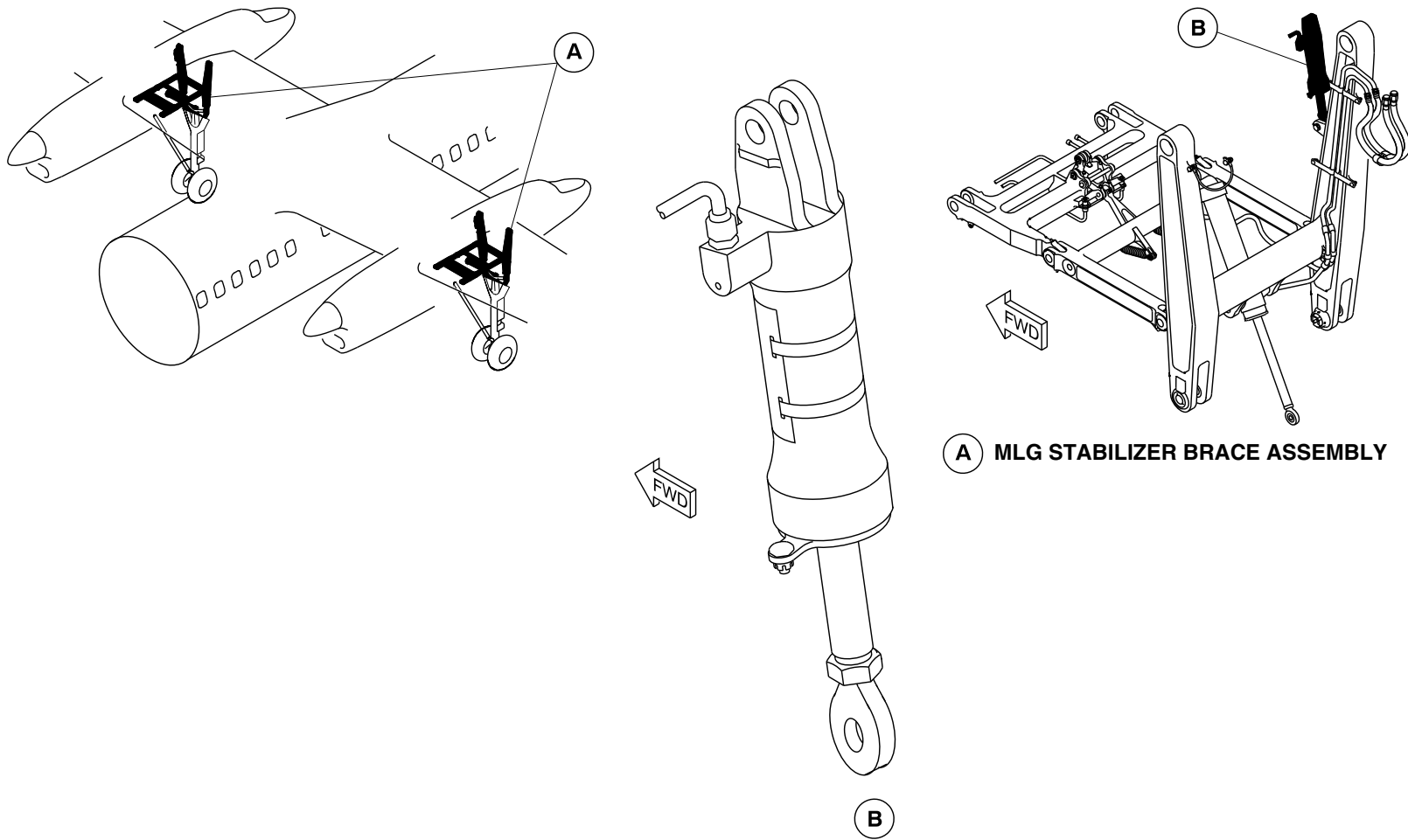
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Alternate Extension Actuator
Figure 11

PSM 1-84-2A
EFFECTIVITY:
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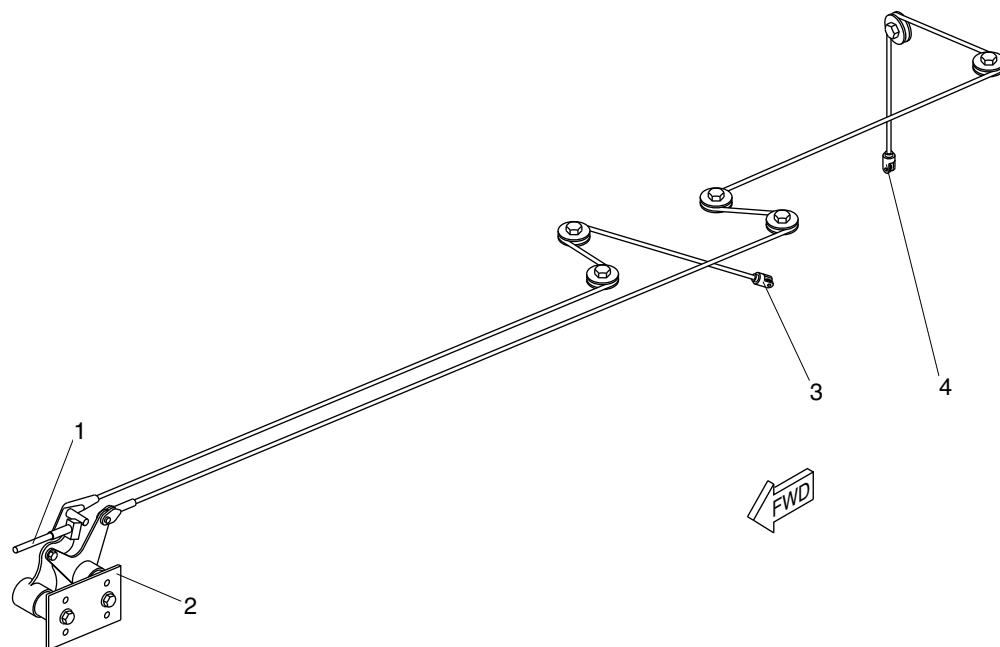
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LEGEND

- 1. Cable From Slide Assembly.
- 2. Cam Mechanism.
- 3. Cable to Main Gear Uplock Release.
- 4. Cable to Aft Door Release.

fs311a01.cgm

Landing Gear Alt Release Cables and Uplock Detail
Figure 12

PSM 1-84-2A
EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

32-40-00-001

WHEELS AND BRAKES

Introduction

The wheels support the weight of the aircraft and permit the aircraft to move on the ground. The brakes convert energy necessary to bring the aircraft to a stop.

General Description

[Refer to Figure 1.](#)

The aircraft has six wheels. Two wheels are installed on the nose landing gear and two are installed on each main landing gear. Two brake units are installed on each axle of the main landing gear.

The three systems that control the brakes are as follows:

- Normal brake system (32-42-00)
- Parking/emergency brake system (32-44-00)
- Anti-skid system. (32-46-00).

Detailed Description

[Refer to Figures 2 and 3.](#)

The normal brake system decreases the speed of the aircraft during ground operations. The normal brake system is controlled by the pilot's and copilot's brake pedals and is operated hydraulically from the No.1 hydraulic system.

The parking/emergency brake system is used to keep hydraulic pressure applied to the main wheel brake units when the aircraft is parked. The parking/emergency brake system can also be used to stop the aircraft when the normal brake system is not available. It is powered by the No. 2 system hydraulic system, and or the parking brake accumulator.

The anti-skid system controls hydraulic pressure to the four main wheel brake units to supply individual wheel anti-skid protection.

Normal Brake System

When the pilot's or copilot's left brake pedal is pushed, the No.1 hydraulic system pressure is supplied to the two left main wheel brake units. When the pilot's or copilot's right brake pedal is pushed, the No.1 hydraulic system pressure is supplied to the two right main wheel brake units. Hydraulic pressure is supplied to engage the brake units in proportion to the brake pedal travel.

Parking/Emergency Brake System

[Refer to Figure 4.](#)

The parking/emergency brake system is controlled by a lever on the center console in the flight compartment. This system is operated hydraulically from the No.2 hydraulic system.

[Refer to Figure 5.](#)

With the PARK/EMERG brake lever in the PARK position, hydraulic pressure is supplied to engage the brake units in proportion to the brake lever travel. Hydraulic pressure from the No.2 hydraulic system, or the parking brake accumulator, is supplied to engage all of the main wheel brake units. The parking brake accumulator is pressurized hydraulically by No.2 hydraulic system. When the aircraft



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

is on the ground the accumulator can be pressurized with a handpump installed in the right nacelle. The accumulator has a nitrogen pre-charge of 1000psi (6895 kPa).

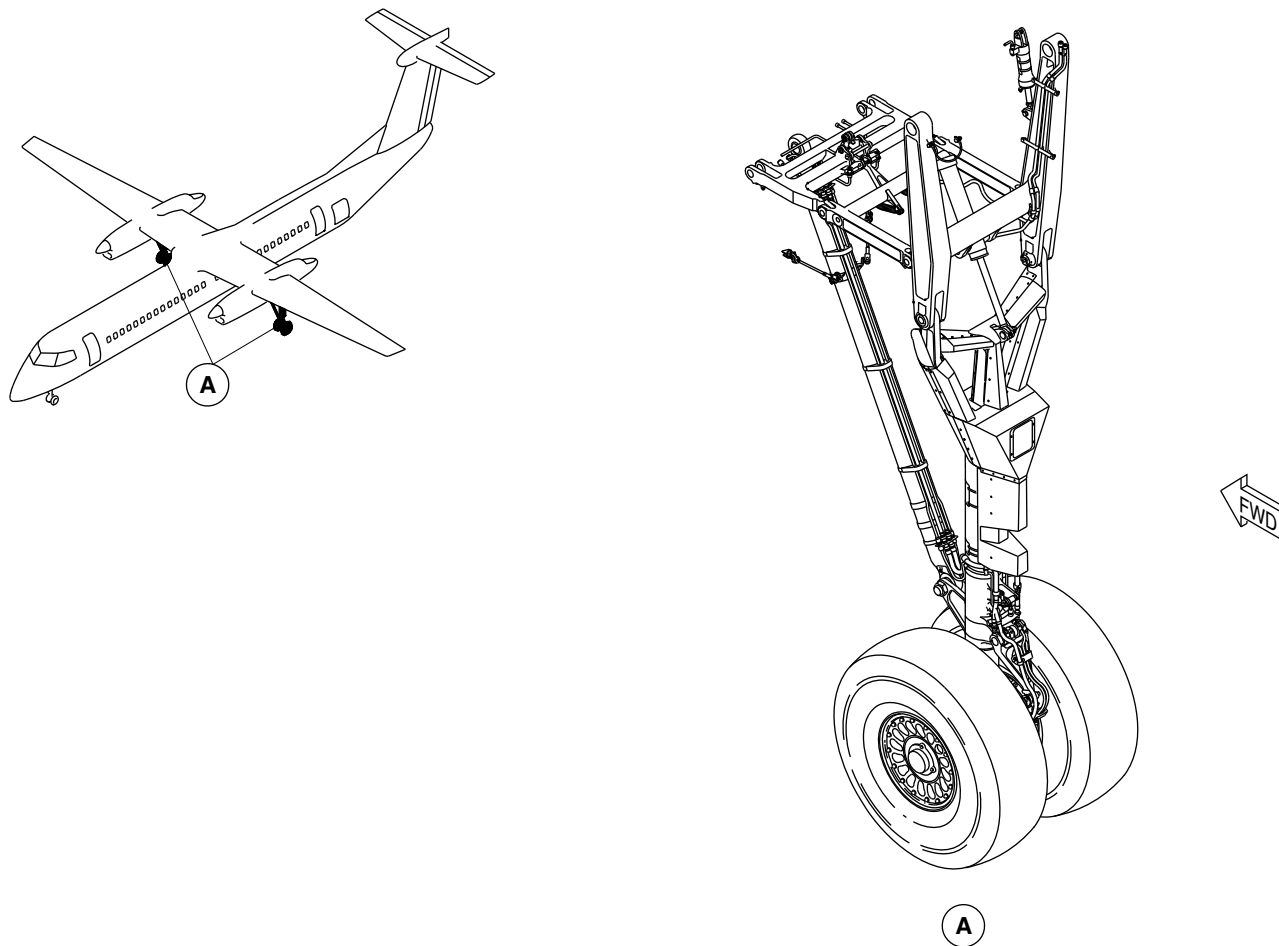
The Anti-Skid System

The anti-skid system controls the quantity of hydraulic pressure supplied to the main wheel brake units during the use of the normal brake system. Maximum brake pedal input will not cause the main wheels to skid when the anti-skid system is in operation. The anti-skid system is controlled by an ON/OFF/TEST switch located on the copilot's instrument panel. The anti-skid system operates with the normal brake system only and is not available with the parking/emergency brake system.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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Wheels and Brakes Locator
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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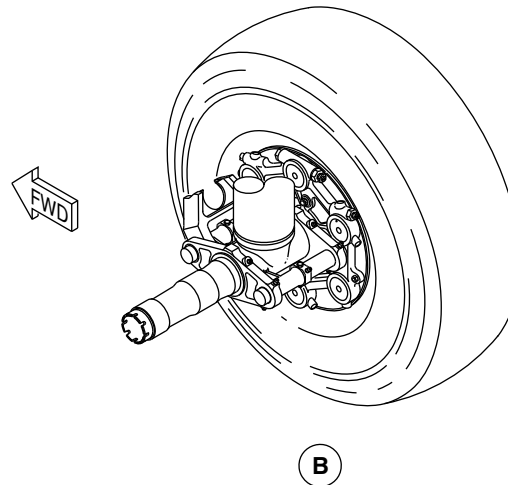
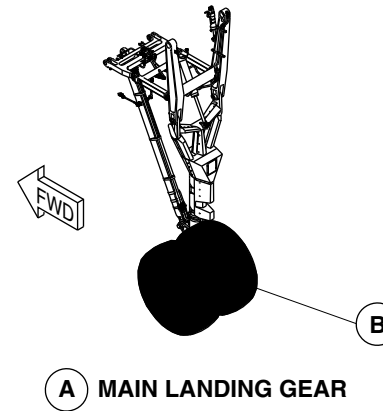
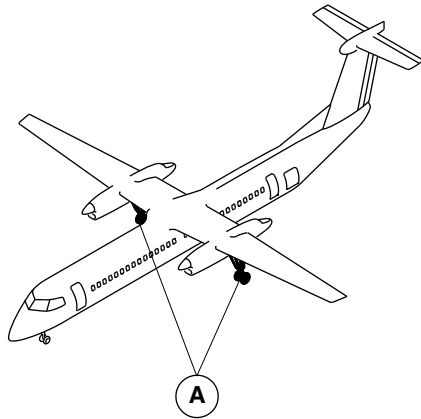
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



NOTE

Inboard Wheel and
Brake removed for clarity.

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Main Wheel Locator
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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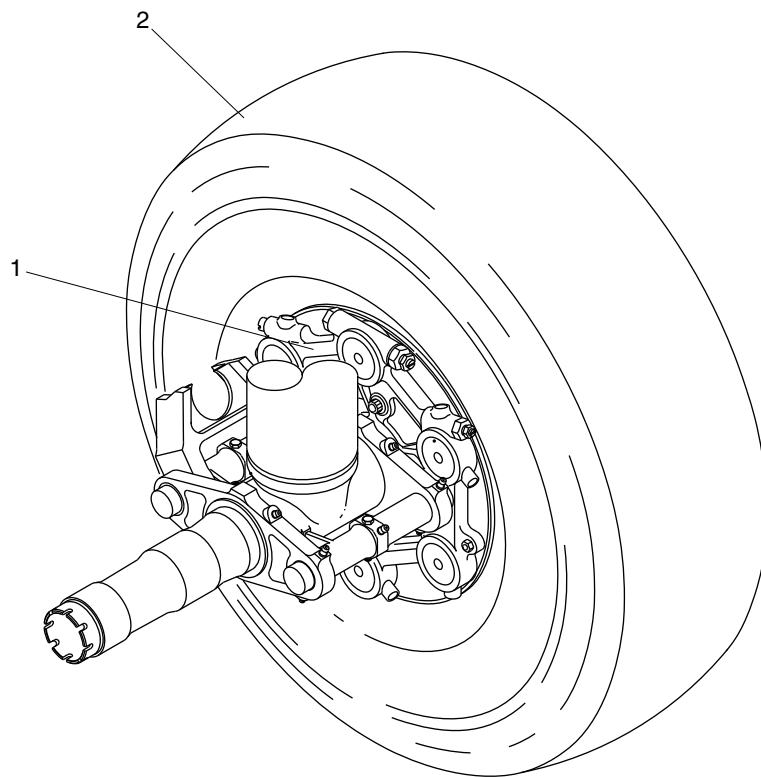
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DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Main Wheel Brake Unit.
- 2. Main Wheel and Tire Assembly.

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Main Wheel Detail
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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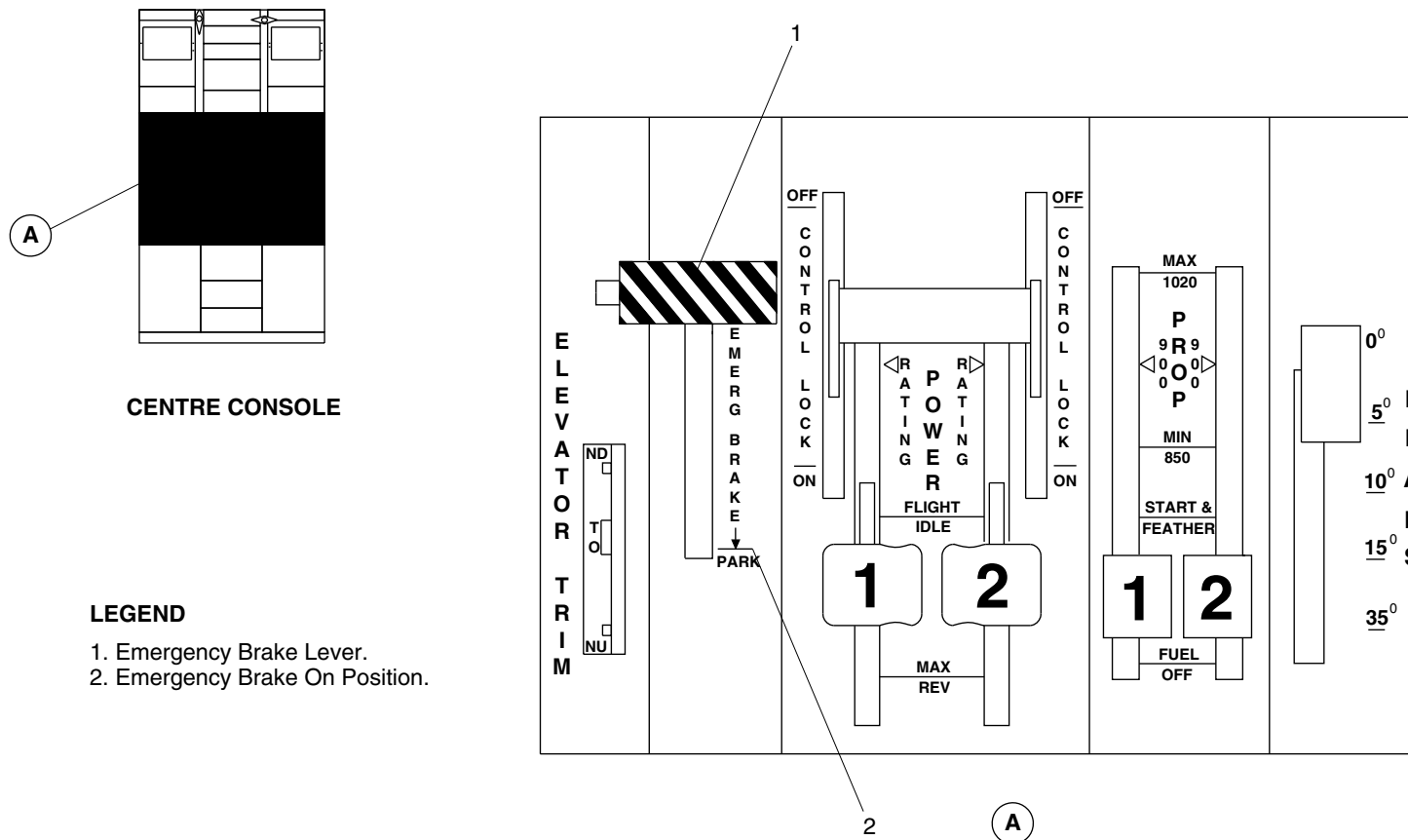
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LEGEND

1. Emergency Brake Lever.
2. Emergency Brake On Position.

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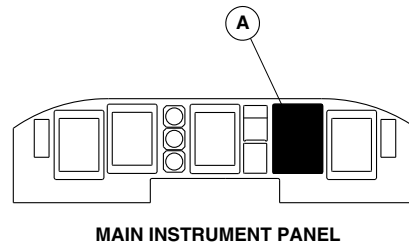
Emerg/ Park Handle
Figure 4

PSM 1-84-2A
EFFECTIVITY:
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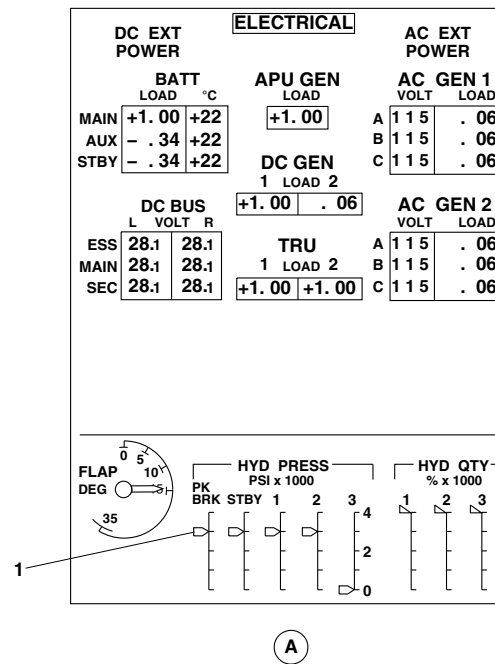
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Parking Brake Hydraulic Pressure Indicator.



Parking Brake Pressure Alternate Indication
Figure 5

PSM 1-84-2A
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

32-41-00-001

WHEELS

Introduction

The wheels support the weight of the aircraft and let the aircraft move while on the ground.

General Description

The aircraft has six wheels. Two wheels are installed on the nose landing gear axle and two wheels are installed on each main landing gear axle.

Detailed Description

[Refer to Figure 1.](#)

Two wheels are installed on the axle of each main landing gear. The main wheels are a split-web type assembly made of two forged aluminum wheel halves. There are two different wheel assemblies that can be used on the main landing gear. One main wheel assembly has a 32×8.8–16, 18 ply tubeless tire. The other main wheel assembly has a 34×10.75–16, 16 ply tubeless tire.

[Refer to Figure 2.](#)

The main wheels are tubeless, split-web type assemblies made of forged aluminum wheel halves. The two halves are held together with equally spaced tie bolts, washers, and nuts. A preformed

packing is installed on the inner wheel half assembly to prevent air leakage through the wheel mating surfaces.

Tapered wheel bearings and bearing cups are installed in the inner and outer wheel halves. The bearings and cups are installed in pairs and lubricated with grease. Bearing seals are installed at the outer surface of the bearings to prevent contamination. A retaining ring holds the outer bearing in place. Grease retainers are installed next to each bearing cup to prevent the flow of grease into the center of the wheel. A hubcap and gasket is installed on the outer wheel half to protect the bearing from dirt and moisture.

An inflation valve assembly is installed in the outer wheel half assembly. An over inflation plug is installed 180 degrees from the inflation valve to prevent over inflation of the tire. The over inflation plug contains a disk that ruptures, and deflates the tire to zero pressure. Four thermal relief plugs are installed in the web of the inner wheel half. Each thermal relief plug is made of fusible metal that melts at a specified temperature. This will release the pressure in the tire and prevent an explosion if the wheel gets too hot.

Wheel inserts are installed on the inner wheel. The inserts engage the drive slots in the brake disks of the main wheel brake units, which turn the disks when the wheel turns. A one piece heatshield is installed in the inner wheel half. The heatshield decreases the transfer of heat from the brake assembly to the wheel and tire assembly.

Nose Wheels

[Refer to Figure 3.](#)

Two wheels are installed on the axle of the nose landing gear. The nosewheels are a split-web type assembly made of two forged



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

aluminum wheel halves. Each nosewheel has a 22x6.50–10, 8 ply tubeless tire.

[Refer to Figure 4.](#)

The nose wheels are tubeless, split-web type assemblies made of forged aluminum wheel halves. The two halves are held together with equally spaced tie bolts, washers, and nuts. A preformed packing is installed on the inner wheel half assembly to prevent air leakage through the wheel mating surfaces.

Tapered wheel bearings and bearing cups are installed in the inner and outer wheel halves. The bearings and cups are installed in pairs and lubricated with grease. Bearing seals are installed at the outer surface of the bearings to prevent contamination. A water deflector shield is installed on the inner wheel half assembly to keep water spray away from the bearing seals and bearing. A hubcap and gasket are installed on the outer wheel half to protect the bearing from dirt and moisture.

An inflation valve assembly is installed in the outer wheel half assembly. An overinflation plug is installed in the outer wheel half assembly to prevent overinflation of the tire. The plug contains a disk that ruptures, and deflates the tire to zero pressure.

Training Information Points

Do not use compressed air to charge tires. Use only compressed dry nitrogen. If you use compressed air, you can cause damage to the tires.

Make sure the aircraft is correctly jacked and supported before you remove the wheels.

Fully deflate the tires before you disassemble the wheel.

The valve core can be forcefully ejected if removed before the tire is completely deflated. Use an approved deflating tool.

Lubrication of the wheel bearings by mechanical or pressure procedures is recommended. These methods are more efficient and decrease possible contamination.

Bearings and cones are matched sets and must be kept together. Do not mix bearing and cone sets.

Do not spin bearings with compressed air.

To prevent wheel bearing failure, be careful with wheel bearings.

Do not use impact or power wrenches to tighten or torque wheel tie bolts and nuts.

Do not inflate the tires until the tie bolts have been torqued. Inflate tires in a metal cage to prevent injury to personnel.

When you install the water shield on the axle, make sure the words THIS SIDE OUT point to the nose wheel strut.

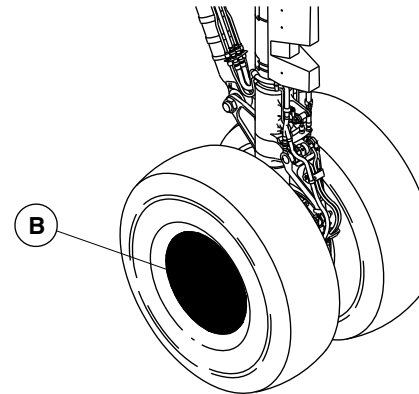
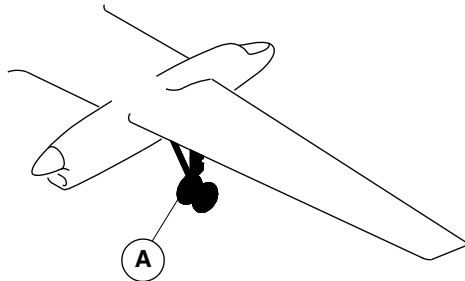


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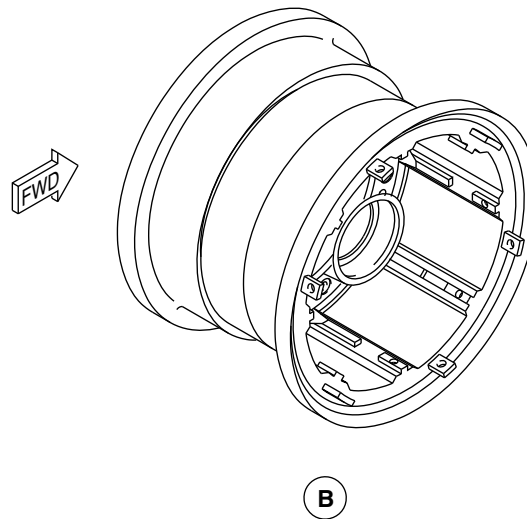
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NOTE

Left side shown.
Right side similar.



A MAIN LANDING GEAR



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Wheels, Main Wheel Locator
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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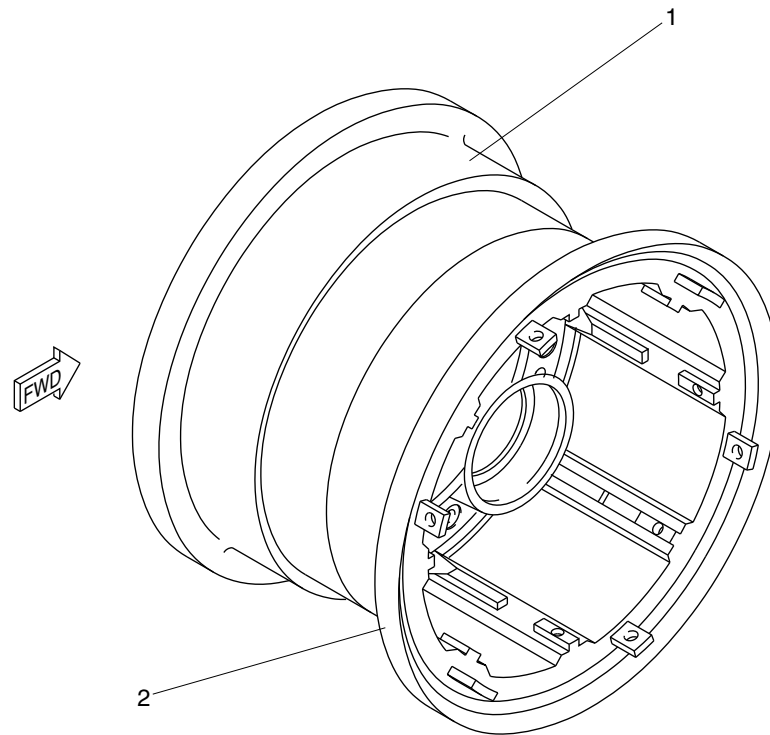
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LEGEND

- 1. Outer Wheel Half.
- 2. Inner Wheel Half.

fs145a01.cgm

Wheels, Main Wheel Detail
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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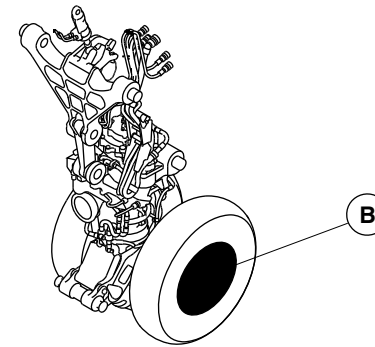
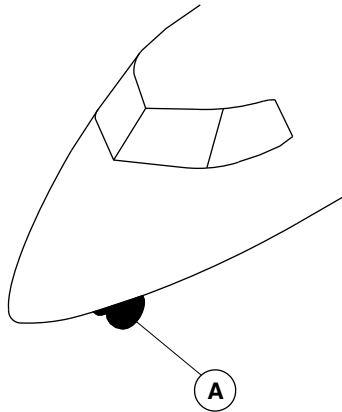
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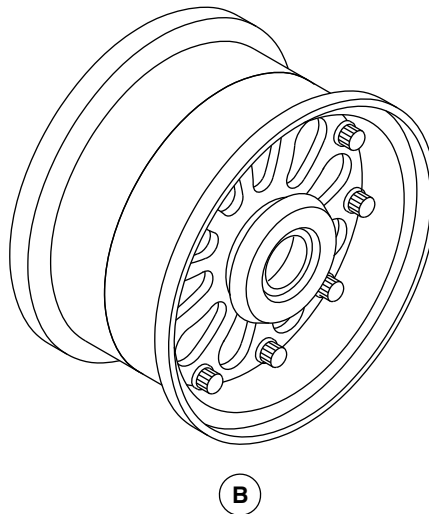


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A NOSE LANDING GEAR ASSEMBLY



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Nose Wheel Locator
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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Config 001

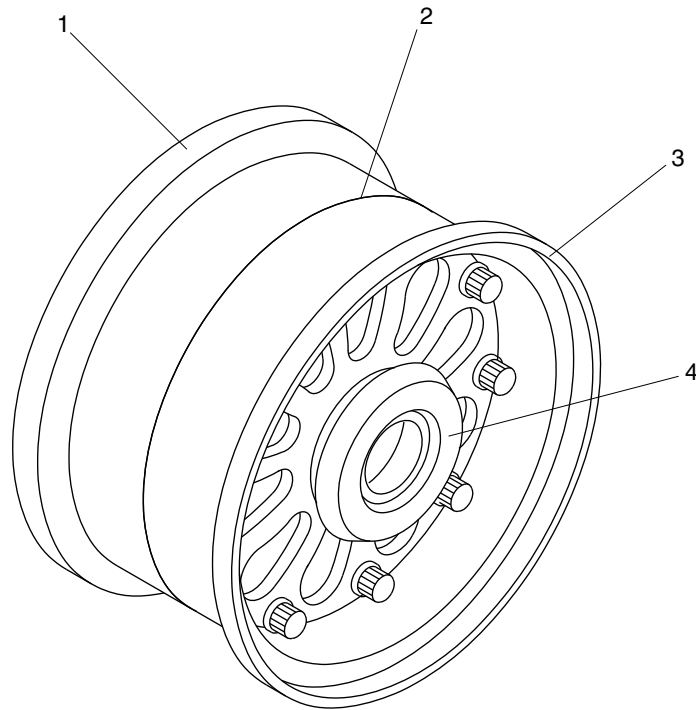
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LEGEND

- 1. Outer Wheel Half.
- 2. Preformed Packing.
- 3. Inner Wheel Half.
- 4. Water Deflector Shield.

fs743a01.cgm

Nose Wheel Detail
Figure 4

PSM 1-84-2A
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32-42-00-001

MAIN LANDING GEAR BRAKE SYSTEM

Introduction

The brake system for the Main Landing Gear (MLG) lets the aircraft brake normally to decrease its speed during ground operations.

General Description

The MLG brake system is controlled by the pilot's and copilot's brake pedals.

The brake system has the components that follow:

- Main wheel brake units (32-42-01)
- Brake control valve (32-42-06)
- Fuse/shuttle valves (32-42-11)
- Pilots' brake pedals (32-42-16)
- Brake control cables (32-42-21).

Detailed Description

[Refer to Figure 1.](#)

When a pilot's brake pedal is pushed, the attached tie rod moves a pushrod to operate the brake control lever. Movement of the brake control lever extends the lever spring and pulls the control cable to operate the applicable lever on the brake control valve. Hydraulic

pressure from the No. 1 system is supplied to the skid control valve when the brake control valve lever is operated. The brake control valve will supply hydraulic pressure in proportion to the amount of brake pedal travel. Hydraulic pressure is supplied from the skid control valve through the fuse/shuttle valves to the applicable MLG brake units.

When the brake pedal is released, the lever spring returns the brake pedal to the off position. A spring in the brake control valve returns the valve to the off position. The brake pressure is released to return and the valve inlet port is closed.

When the MLG brake unit is actuated, pressurized hydraulic fluid from the MLG brake system moves the pistons to push the pressure plate assembly. The pressure plate assembly pushes the rotor disks, the stator disks, and the end plate together. As the stator disks and the rotor disks are pushed against each other, the friction between the disks causes resistance. The resistance is transmitted from the MLG brake unit to the wheel and tire assemblies causing the wheel to slow down.

Main Wheel Brake Units

[Refer to Figures 2 , 3 and 4.](#)

Two brake units are installed on each axle of the main landing gear. The brake pins on the MLG shock strut engage with the bushings in the piston housing assembly to transfer the braking loads to the landing gear.

The main wheel brake unit is a piston operated, carbon heat sink, multiple carbon disk brake.

The brake assembly has three primary components:

- The piston housing assembly



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

- The carbon heat sink stack
- The torque plate assembly.

The piston housing is made of aluminum. It contains six adjuster assemblies, two hydraulic bleeder valves and adapters, and one bleeder plug. The adapters and bleeder plugs are interchangeable to let the brake assembly be installed on the left or right side of each main landing gear axle. Two wear indicator pins are attached to the pressure plate assembly and extend out of the piston housing through sleeves to give an indication of brake wear. When the wear pin is level with the wear indicator sleeve, replacement of the MLG brake unit is required. The piston housing with its integral parts performs the actuation, retraction and clearance adjustment functions of the brake. The piston housing is installed to the torque plate assembly with bolts.

The carbon heat sink stack includes the pressure plate, the three rotor disk assemblies, the two stator disk assemblies, and the end plate assembly that is attached to the torque plate assembly. All of these parts have friction surfaces that cause the brake assembly to stop the wheel when the brake is actuated. The pressure plate assembly has drive slots on the inner diameter that engage with the lugs on the torque plate assembly. The pressure plate does not turn, but slides as necessary on the torque plate assembly when the brake is operated.

The rotor disk assemblies have drive slots on the outer diameter that engage with the inserts on the inner wheel half. The rotor disk assemblies turn with the wheel, and slide as necessary on the inserts on the inner diameter of the wheel when the brake is operated.

The stator disks have drive slots on the inner diameter that engage with the lugs on the torque plate assembly. The stator disks do not

turn, but slide on the torque plate as necessary when the brake is actuated. The stator disks are installed between the three rotor disks.

The end plate assembly has nine torque buttons that engage with holes in the torque plate assembly. The end plate assembly does not turn or slide when the brake operates. The torque plate assembly ensures that the pressure plate assembly, the stator disks, and the end plate assembly maintain alignment. Lugs on the torque plate transfer brake torque from the pressure plate assembly and the stator disks to the torque plate and the piston housing.

Brake Control Valve

[Refer to Figures 5 and 6.](#)

The brake control valve is installed in the left wing root at Sta. X401.00. The pilot's and copilot's left brake pedals are connected to the left lever of the brake control valve. The pilot's and copilot's right brake pedals are connected to the right lever of the brake control valve. The brake control valve has two internal metering sections in a common housing. Each metering section is actuated progressively by an integral lever mechanism assembly.

The housing also supplies common system pressure and return ports. Hydraulic pressure and return lines connect the brake control valve to the No.1 hydraulic system. Hydraulic lines connect the brake control valve to the anti-skid valves.

Fuse/Shuttle Valve

[Refer to Figure 7.](#)

There are four fuse/shuttle valves installed on the left wall of the left and the right nacelles. Hydraulic pressure is supplied from the No.1



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hydraulic system to the brake units through the skid control valves and the fuse/shuttle valves.

The fuse component of the fuse/shuttle valve closes the hydraulic line to the brake unit, if the fluid flow is more than a specified quantity. This will limit the leakage of hydraulic fluid if a brake line on the main landing gear is cut or if a brake unit develops a leak. The shuttle component of the fuse/shuttle valve allows either normal brake pressure or parking/emergency brake pressure to be used to actuate the brake units.

Brake Pedals

Refer to Figure 8.

The pilot's and copilot's brake pedals are installed in the flight compartment and are connected to the brake control valve. The pilots' inputs are applied to the brake control levers through pushrods. Movement of the pilot's brake pedals does not operate the copilot's brake pedals. The copilot's inputs are transmitted by a cable system to the spring loaded lever arm. Movement of the copilot's brake pedals will move the pilot's brake pedals through the interconnect cables from the copilot's brake pedal.

The linkage attached to the base of the pedals is spring loaded, to return the pedals to the off position when the pedals are released. The springs in the cable system return the copilot's pedals to off. Two spring rods remove the tension in the braking loads.

Training Information Points

Before you do maintenance on or around the landing gear, make sure that the main gear lockpins are installed, the nose gear

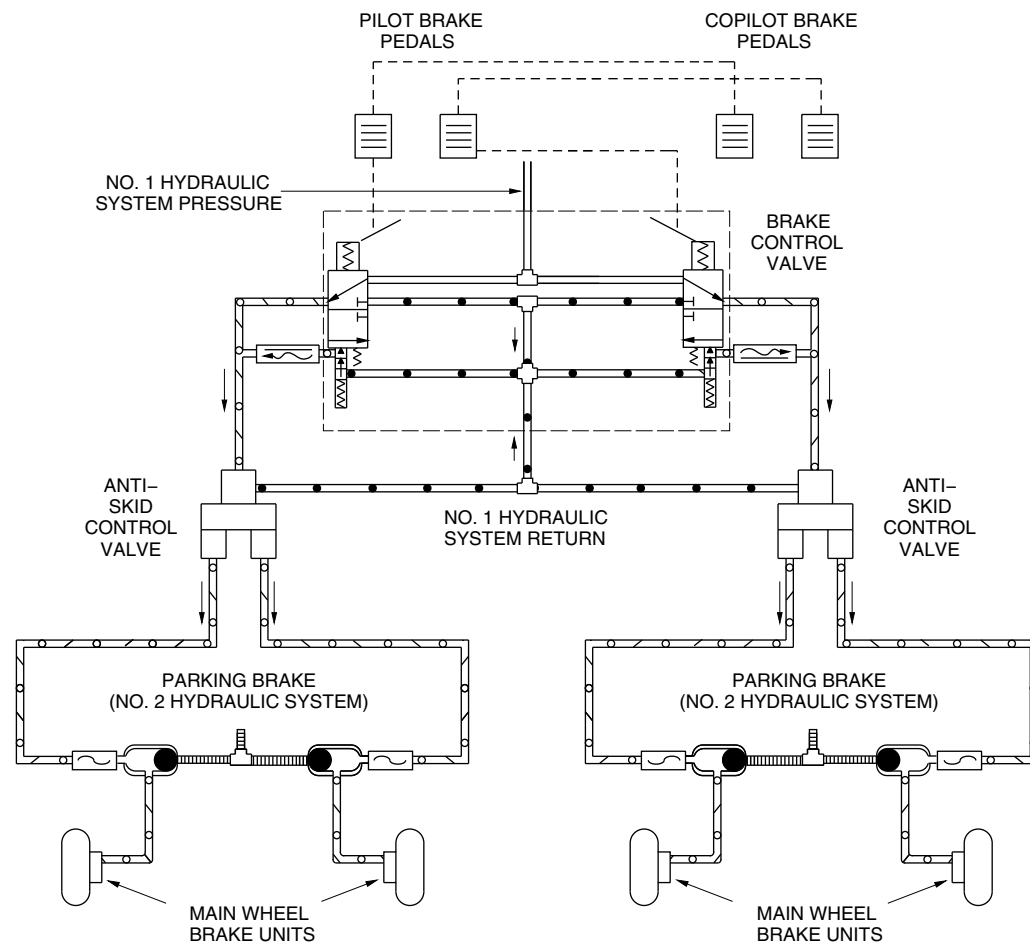
downlock is engaged and hydraulic and electric power are not connected.

Make sure the aircraft is correctly jacked and supported before you remove the wheels.

When the brake unit wear pin is level with the wear indicator sleeve, replace the brake assembly.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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LEGEND

	PRESSURE
	METERED PRESSURE
	RETURN
	STATIC
	FUSE
	SHUTTLE VALVE
	RESTRICTOR

NOTE

System shown with brakes applied.

Brakes System Hydraulic Synoptic
Figure 1

PSM 1-84-2A
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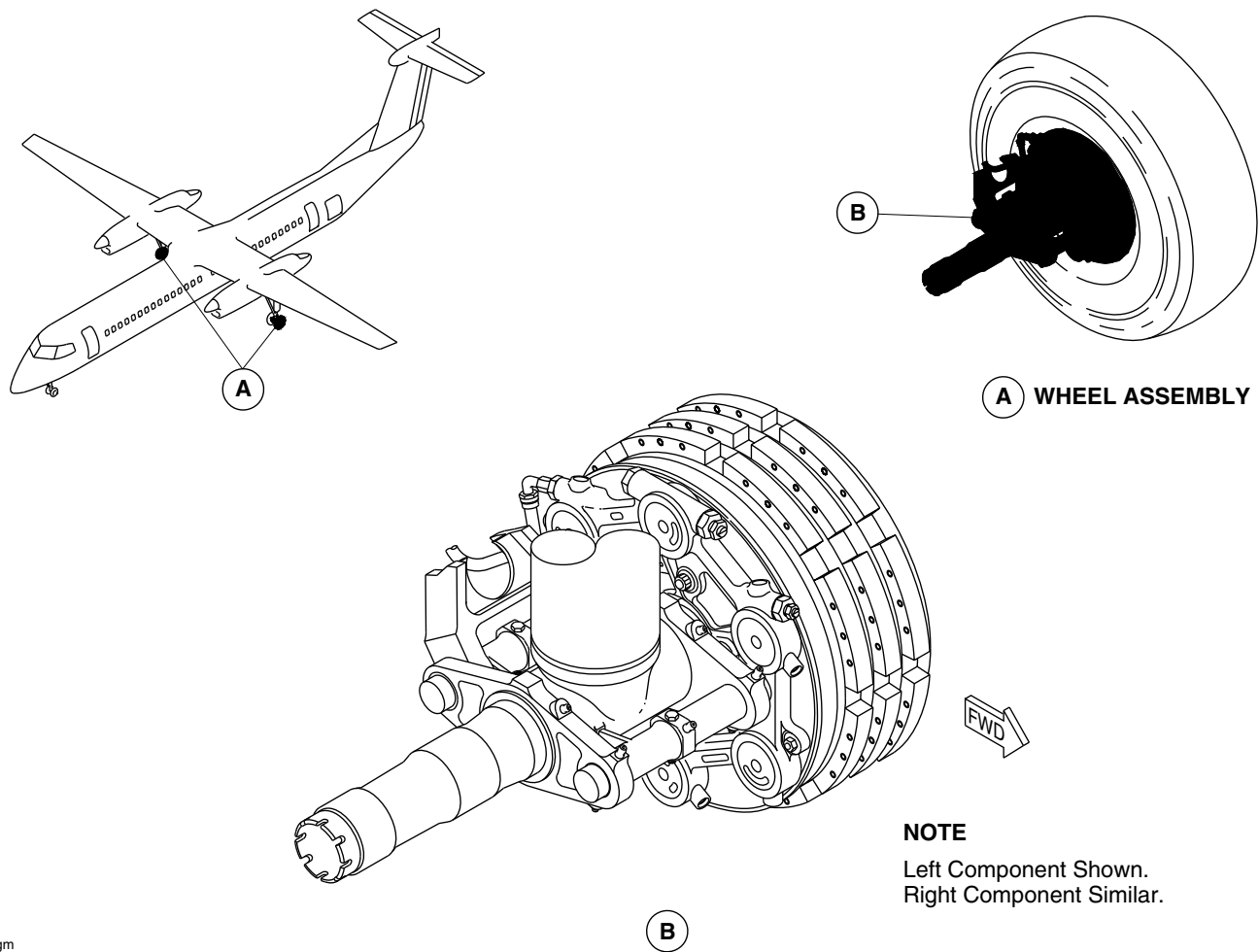
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Axle and Brake Assembly Locator
Figure 2

PSM 1-84-2A
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Config 001

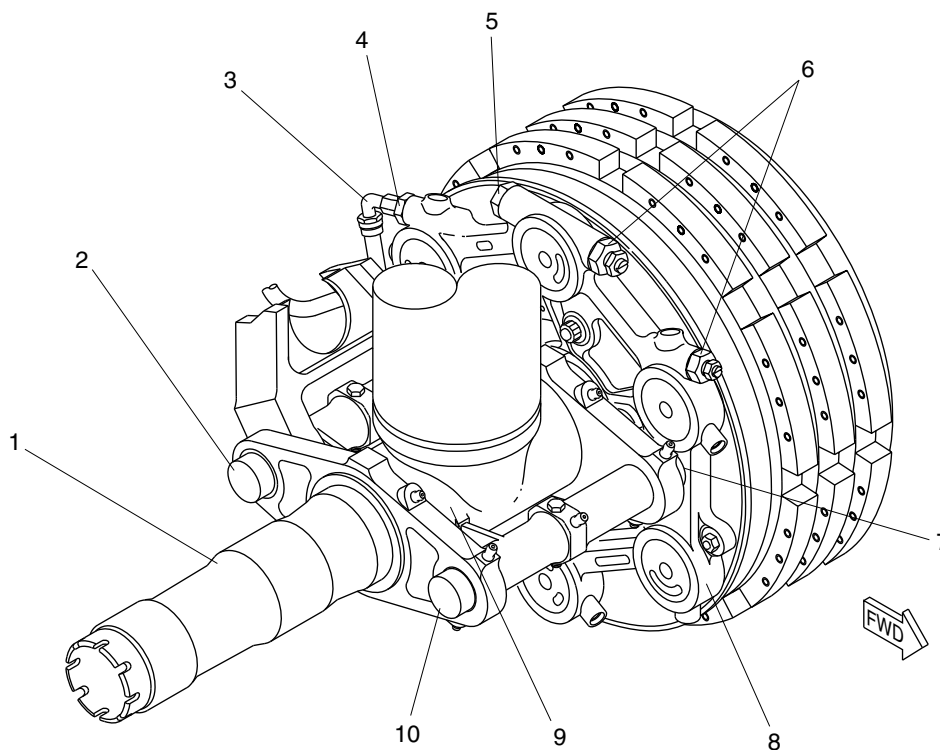
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LEGEND

1. Axle.
2. Brake Pin.
3. Hydraulic Line.
4. Reducer.
5. Bleeder Plug.
6. Bleeder Valve and Adapter.
7. Brake Pin Bushing.
8. Piston.
9. Main Landing Gear Shock Strut.
10. Brake Pin.

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Axle and Brake Assembly Detail
Figure 3

PSM 1-84-2A
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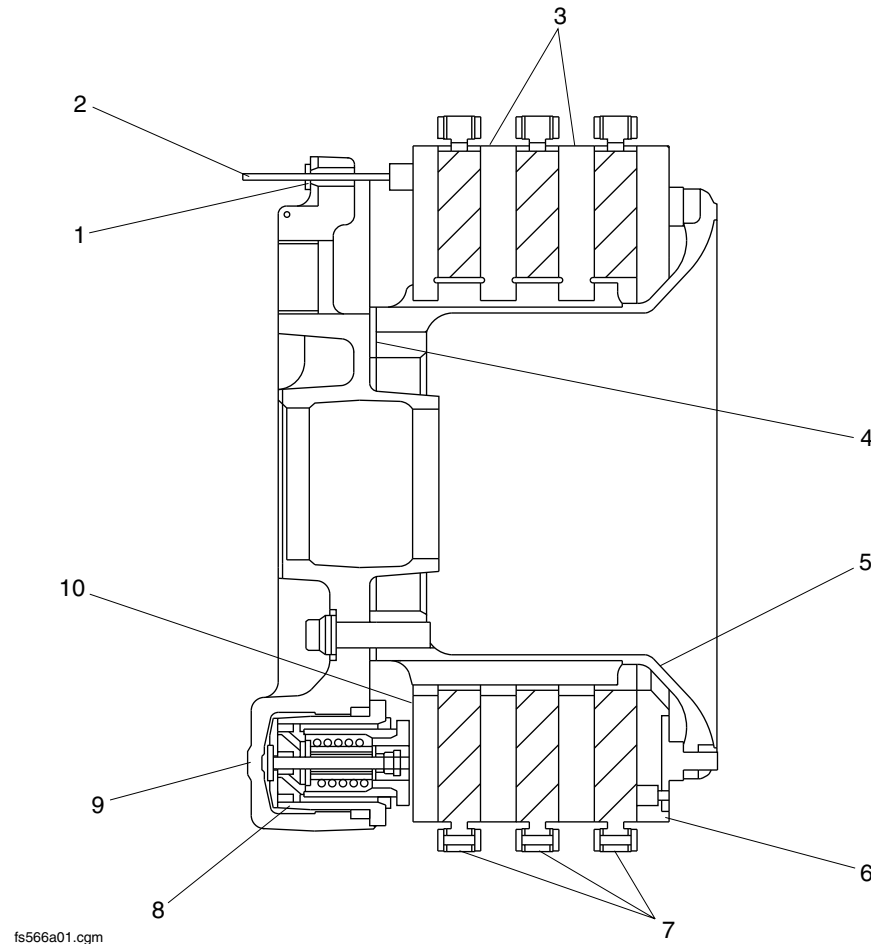
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LEGEND

1. Wear Indicator Sleeve.
2. Wear Pin Indicator.
3. Stator Disk.
4. Insulator.
5. Torque Plate Assembly.
6. End Plate Assembly.
7. Rotor Disk Assembly.
8. Adjuster Assembly.
9. Piston Housing Assembly.
10. Pressure Plate Assembly.

Brake Assembly Detail
Figure 4

PSM 1-84-2A
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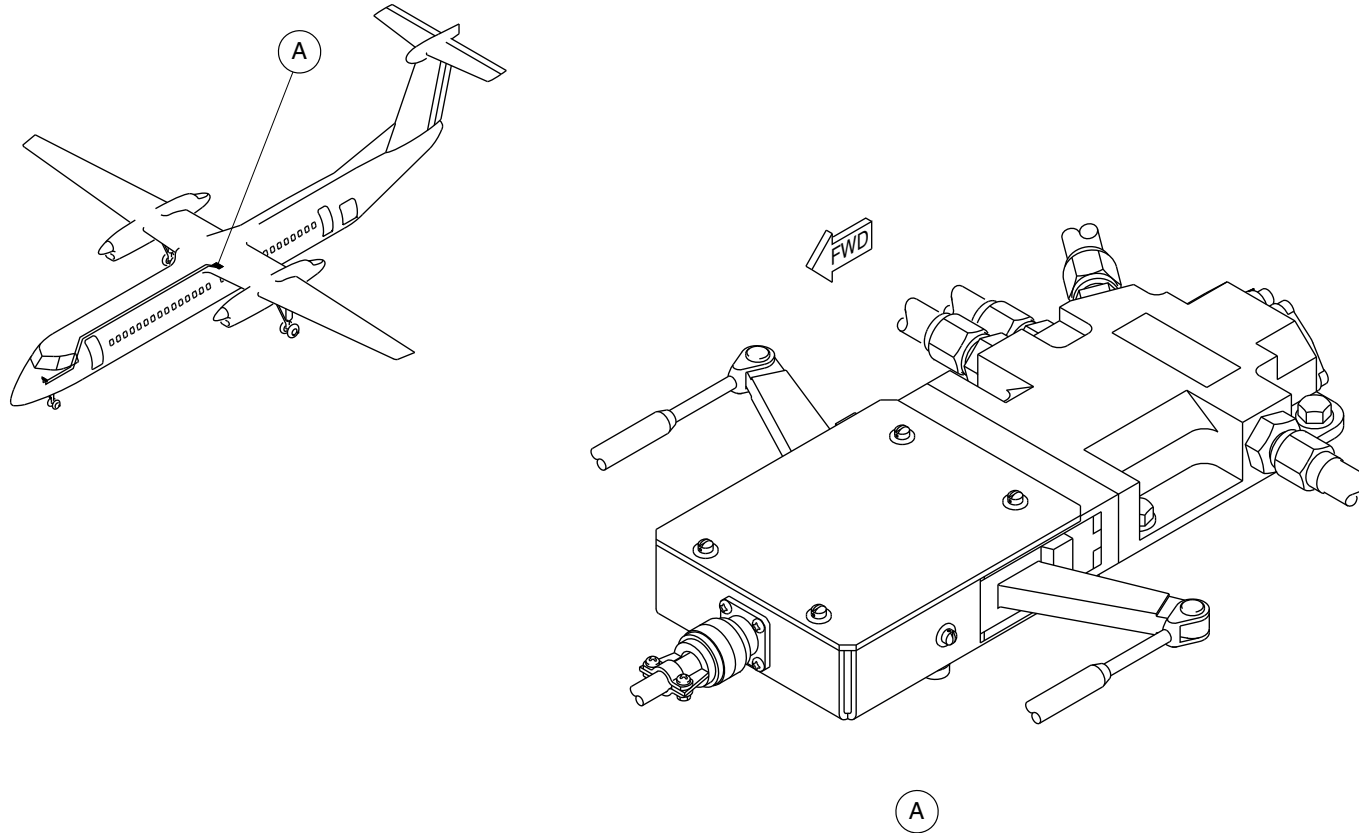
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Brake Control Valve Locator
Figure 5

PSM 1-84-2A
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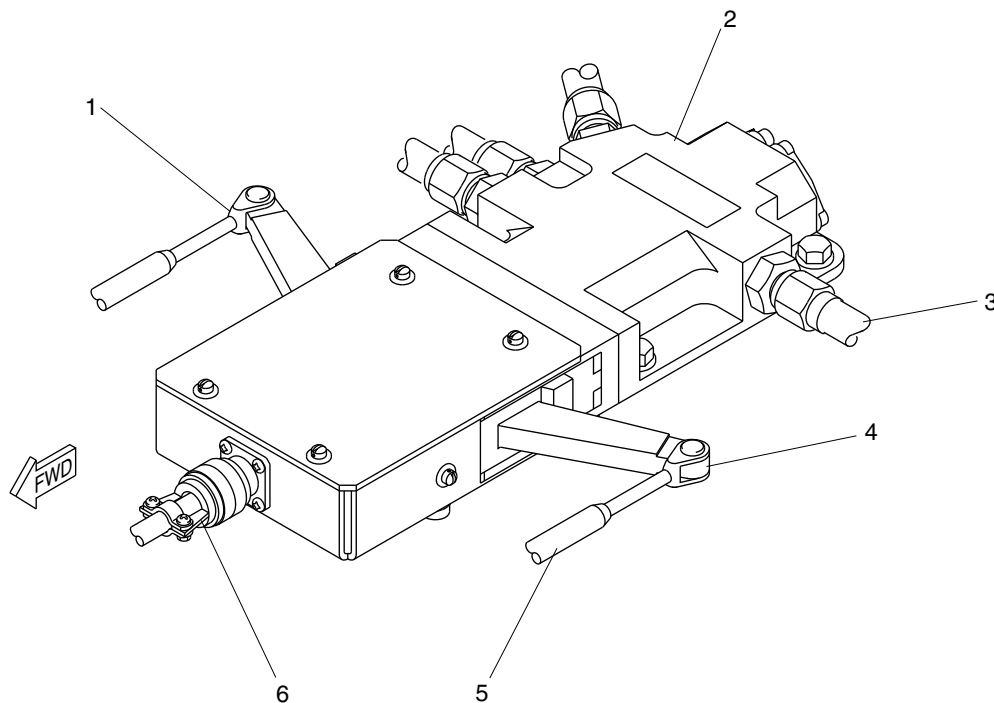


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LEGEND

1. Right Brake Lever.
2. Brake Control Valve.
3. Hydraulic Line.
4. Left Brake Lever.
5. Left Brake Cable Turnbuckle.
6. Electrical Plug.



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Brake Control Valve Detail
Figure 6

PSM 1-84-2A
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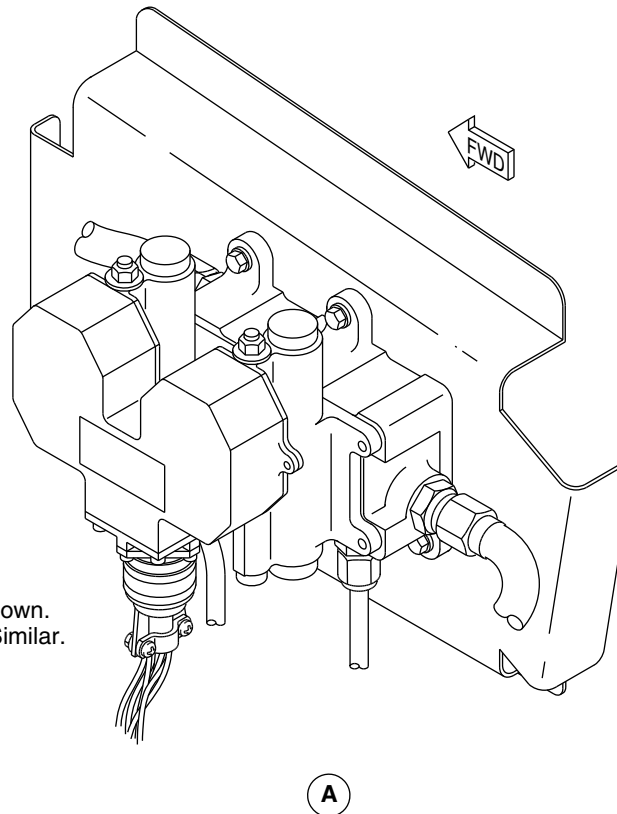
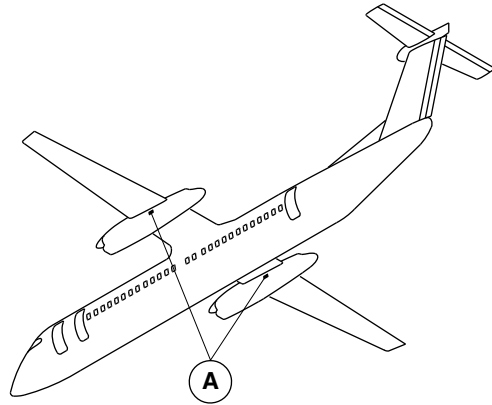
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NOTE

Left Component Shown.
Right Component Similar.

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Fuse/Shuttle Valve
Figure 7

PSM 1-84-2A
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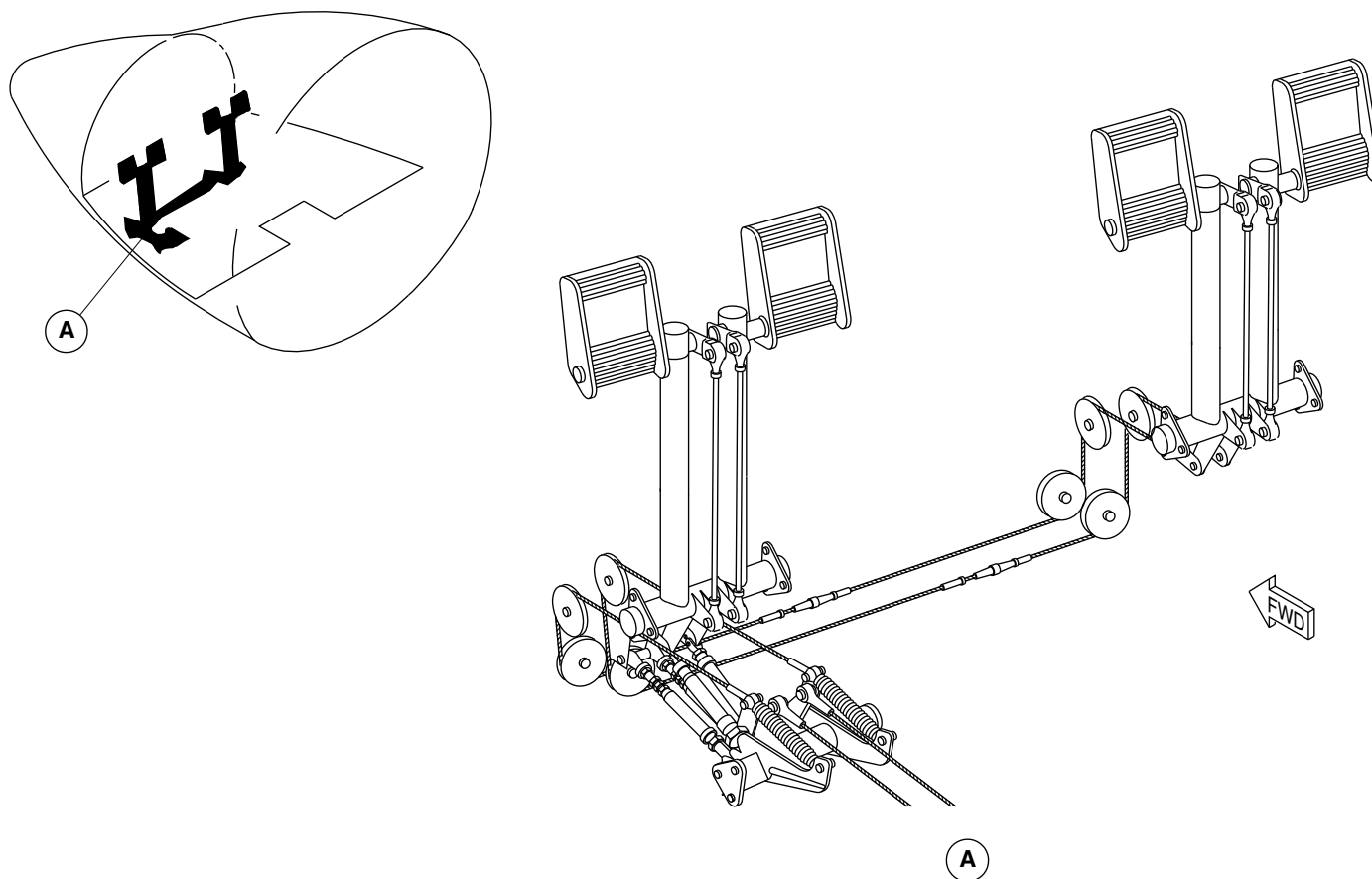
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Brake Pedals Locator
Figure 8

PSM 1-84-2A
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**ON A/C ALL

32-44-00-001

PARKING BRAKE SYSTEM

Introduction

The parking brake system is used to keep hydraulic pressure applied to the brake units on the Main Landing Gear (MLG) when the aircraft is parked. The parking brake system can also be used to stop the aircraft on the ground when the normal brake system is not available.

General Description

The parking brake system uses hydraulic pressure from the No. 2 hydraulic system. An accumulator in the brake system supplies pressure for parking brake operation when the engines are stopped. A handpump can be used to manually pressurize the accumulator when the aircraft is on the ground. Parking brake pressure is applied equally to all of the main wheel brake units.

A PARKING BRAKE caution light on the Caution and Warning panel in the flight compartment comes on when the parking brake lever is in the PARK position. A pressure indicator is installed in the flight compartment to indicate the parking brake pressure available.

The parking brake system has the components that follow:

- Parking Brake Control Lever (32-44-21)
- Parking Brake Pressure Indication (31-61-02)
- Control Cable Assembly (32-44-26)

- Parking Brake Control Valve (32-44-01)
- Handpump (32-44-06)
- Accumulator (32-44-11)
- Charge Valve and Gauge (32-44-16).

Detailed Description

[Refer to Figure 1.](#)

When the PARK/EMERG control lever is moved from the stowed position toward the PARK position, the park brake control valve is actuated by a control cable. The parking brake control valve single metering section applies progressive hydraulic pressure to the fuse/shuttle valves. The hydraulic pressure moves the shuttle valve component of the fuse/shuttle valve to close the inlet ports for the normal brake system. This lets the hydraulic pressure from the parking brake control valve go through the shuttle valves and operate the brake units.

[Refer to Figures 2 and 3.](#)

When the parking brake lever is in the PARK position, the PARKING BRAKE caution light on the Caution and Warning panel comes on. When No. 2 hydraulic pressure is not available, hydraulic accumulator pressure is shown on the PARK BRAKE pressure indicator in the flight compartment. A 28 Vdc electrical power source is needed for the PARK BRAKE pressure indicator and the pressure transmitter to operate.

[Refer to Figure 4.](#)

The parking brake is released when the release button on the parking brake lever is pushed in and the lever is moved forward to the stowed position. A spring in the parking brake control valve

PSM 1-84-2A

EFFECTIVITY:

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closes the pressure port and opens the brake lines to the return side of the hydraulic system. This releases the brakes and the PARKING BRAKE caution light goes off.

The fluid side of the hydraulic accumulator is pressurized from the No. 2 hydraulic system when the No. 2 engine is operating. After the engine has stopped, a check valve closes and hydraulic pressure is stored in the accumulator to supply six full parking brake applications. The handpump can be used to pressurize the accumulator when the No. 2 hydraulic system is not available.

Parking Brake Lever

Refer to Figure 1.

The parking brake lever is installed in the flight compartment, in the parking brake control quadrant, on left side of the center console. A cable assembly connects the parking brake lever to the parking brake control valve. A release button on the parking brake lever unlocks the lever from the PARK position.

Parking Brake Control Valve

Refer to Figures 5 and 6.

The parking brake control valve is installed in the right wing root, aft of Sta. X388.00 and is operated by the parking brake control cable. The No. 2 hydraulic system pressure is supplied to the parking brake control valve. From the control valve, pressure is metered to the fuse/shuttle valves and the brake units in proportion to the amount of parking brake lever movement. This lets the pilot apply proportional braking pressure through the park/emergency brake system when the normal braking system has failed.

When the parking brake lever is moved from the stowed to the PARK position, full hydraulic pressure is applied to the four brake units. A pressure transducer in the parking brake control valve supplies a signal to the flight compartment indication system to indicate the hydraulic pressure in the parking brake system. The parking brake control valve has a relief valve to prevent an over pressure condition when the handpump is used to pressurize the system. Two check valves are contained in the brake control valve housing. These valves are used to isolate the No. 2 hydraulic system and the handpump from the parking brake system.

Handpump

Refer to Figure 7.

A lever operated handpump is installed in the right engine nacelle. It supplies an alternate method to pressurize the accumulator to 3000 psi (20684 kPa) when the aircraft is on the ground and the engines are not operating. The handpump handle is stowed adjacent to the handpump in the right nacelle. On aircraft with Modsum 4–113723 or SB84–32–99 incorporated, the handpump handle is stowed in the fuselage nose compartment.

Charge Valve and Gauge

Refer to Figures 8 and 9.

The charge valve and gauge are installed in the right wing root and are connected to the gas pressure side of the accumulator. The gauge supplies an indication of the accumulator pressure. The charge valve is used to charge the gas side of the accumulator with nitrogen.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Accumulator

Refer to Figure 10.

The accumulator is installed in the right wing root area. It supplies hydraulic pressure for the parking brake operation when the engines are stopped, or if No. 2 hydraulic pressure fails. The accumulator is a cylindrical piston type with a gross rated volume of 50 inches³ (820 centimeters³).

The accumulator is pressurized with nitrogen on one side of the internal piston and hydraulic fluid on the other side. The gas side of the accumulator is pre-charged to 500 psi (3447 kPa). The fluid side is pressurized by the No. 2 hydraulic system or the handpump. The fluid side will hold enough pressure to supply six full parking brake applications and releases.

Training Information Points

The parking brake system uses hydraulic pressure from the No. 2 hydraulic system.

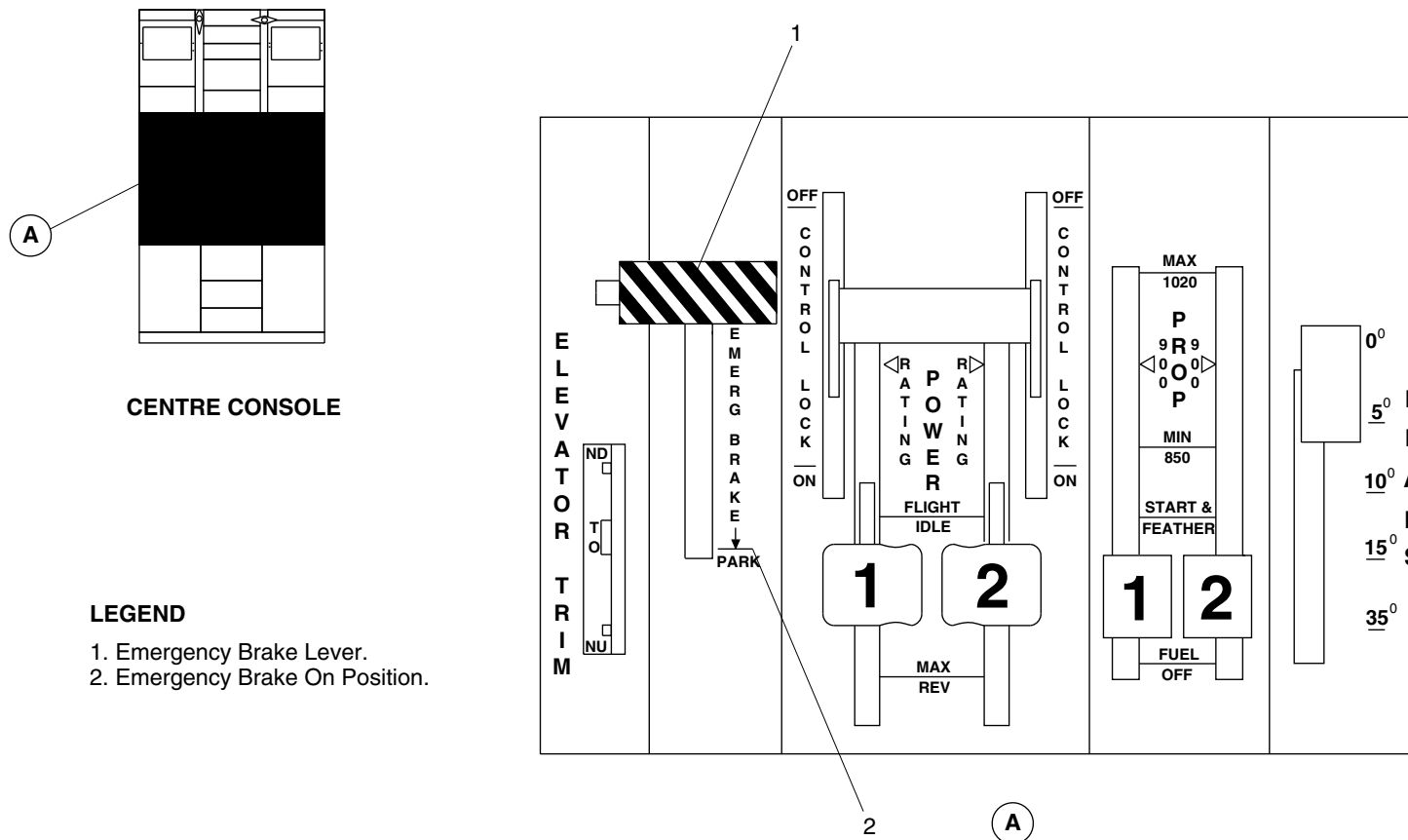
The parking brake system can be used to stop the aircraft on the ground when the normal brake system is not available.

The handpump can be used to manually pressurize the accumulator when the aircraft is on the ground and the engines are not operating.



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LEGEND

1. Emergency Brake Lever.
2. Emergency Brake On Position.

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Park Brake Control Detail
Figure 1

PSM 1-84-2A
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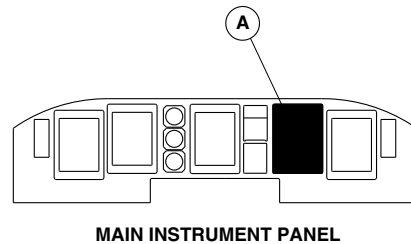
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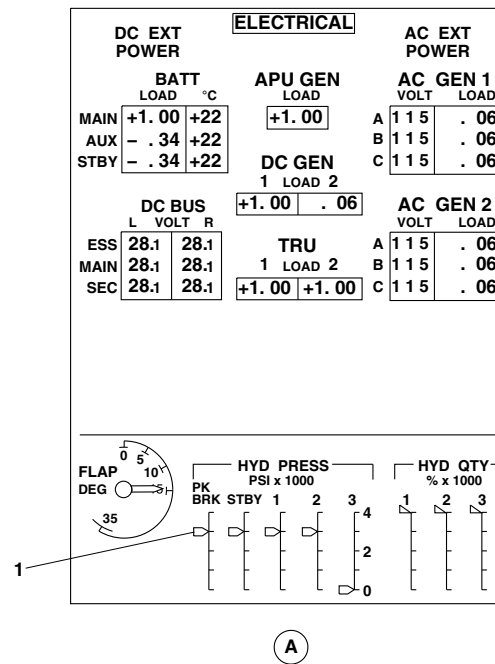
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LEGEND

1. Parking Brake Hydraulic Pressure Indicator.



Parking Brake Pressure Indication
Figure 2

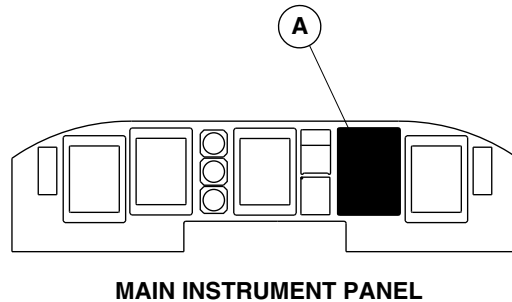
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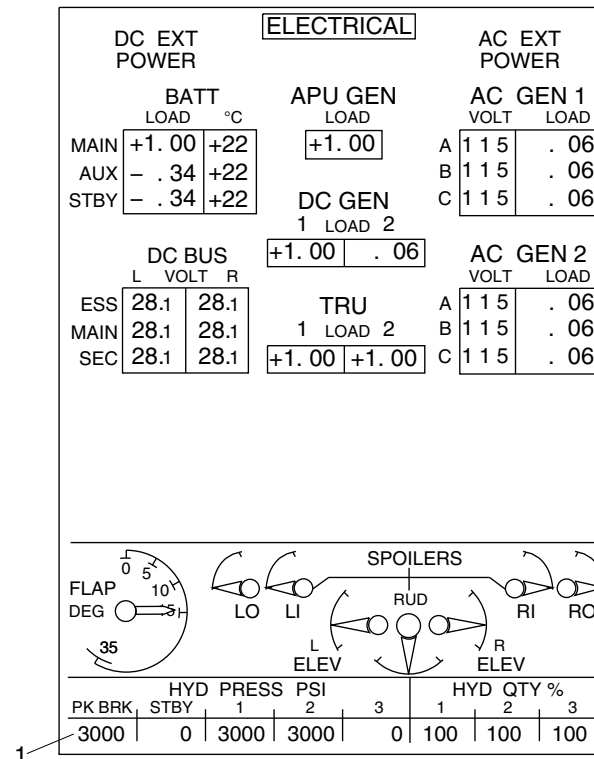


AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Parking Brake Hydraulic Pressure Indicator.



fs708a01.cgm

Parking Brake Pressure (Reversion) Indication
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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Config 001

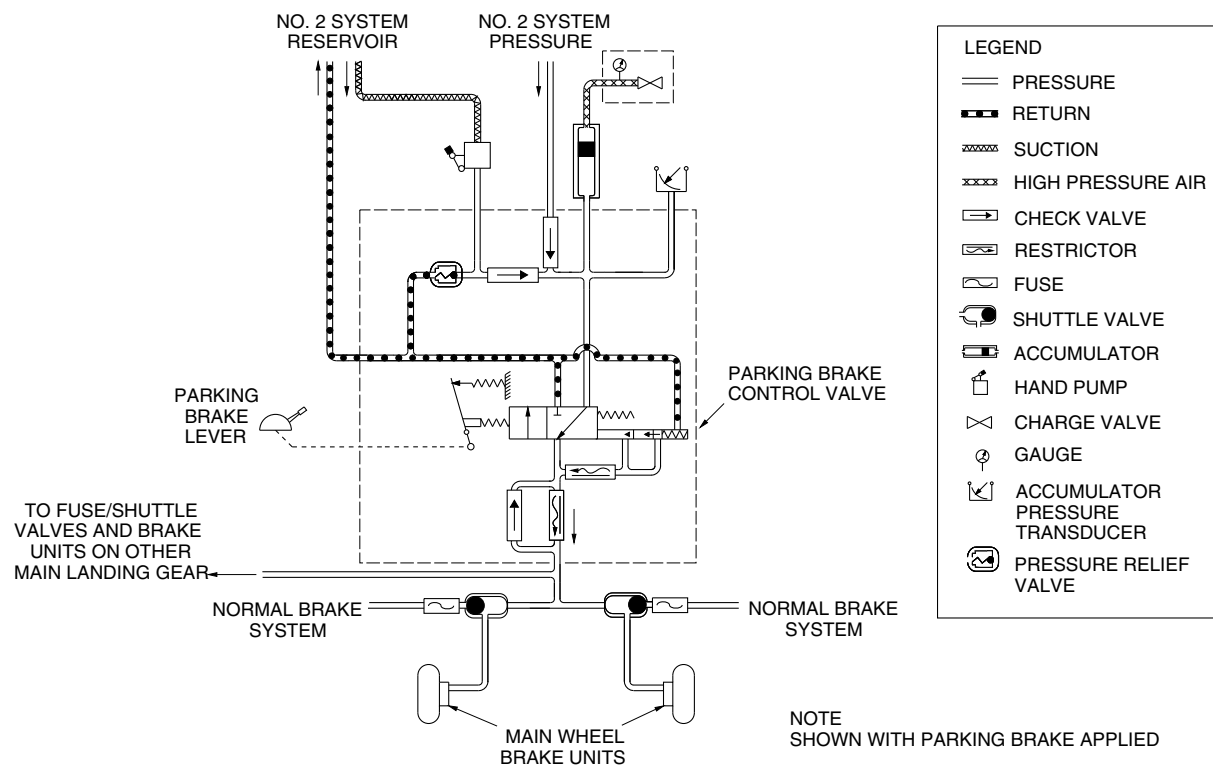
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fs665a01.cgm

Parking Brake System – Hydraulic Schematic
Figure 4

PSM 1-84-2A
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Config 001

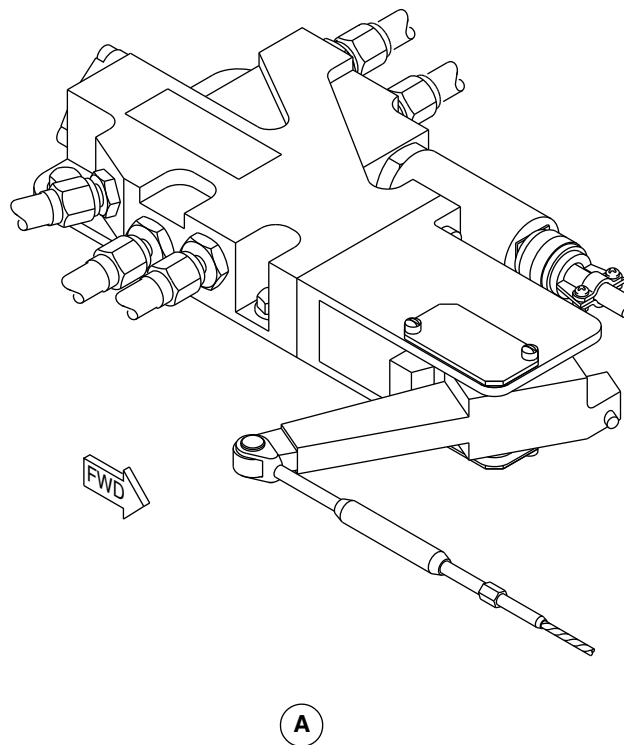
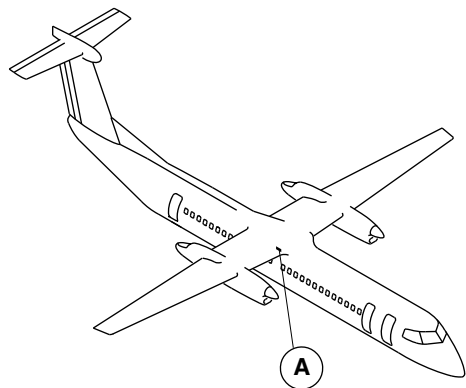
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Park Brake Control Valve Locator
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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Config 001

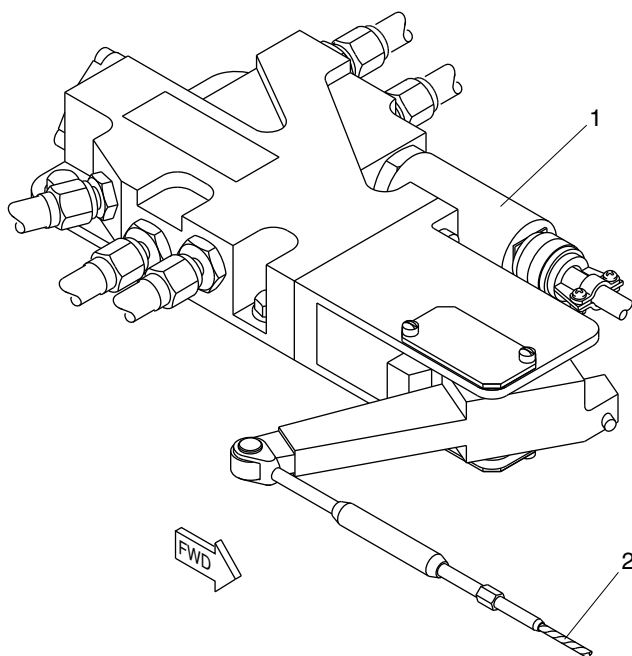
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LEGEND

- 1. Accumulator Pressure Transducer.
- 2. Parking Brake Control Cable.

fs406a01.cgm

Park Brake Control Valve Detail
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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Config 001

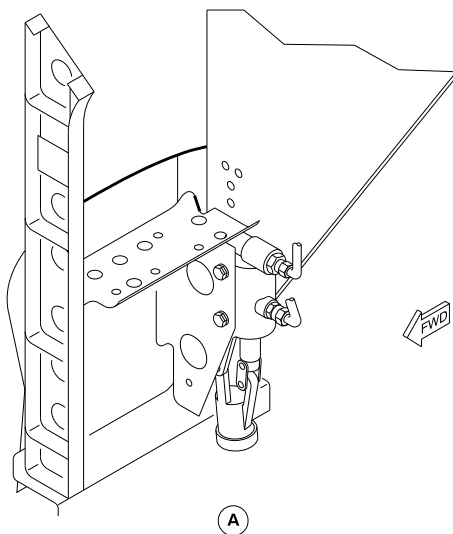
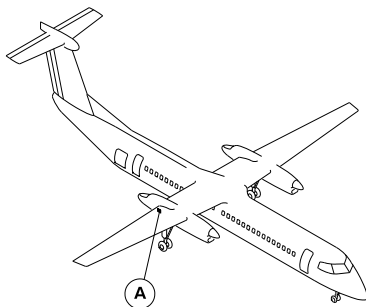
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1s153a01.dwg, 09/24/11/09

Parking Brake Handpump
Figure 7

PSM 1-84-2A
EFFECTIVITY:
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Config 001

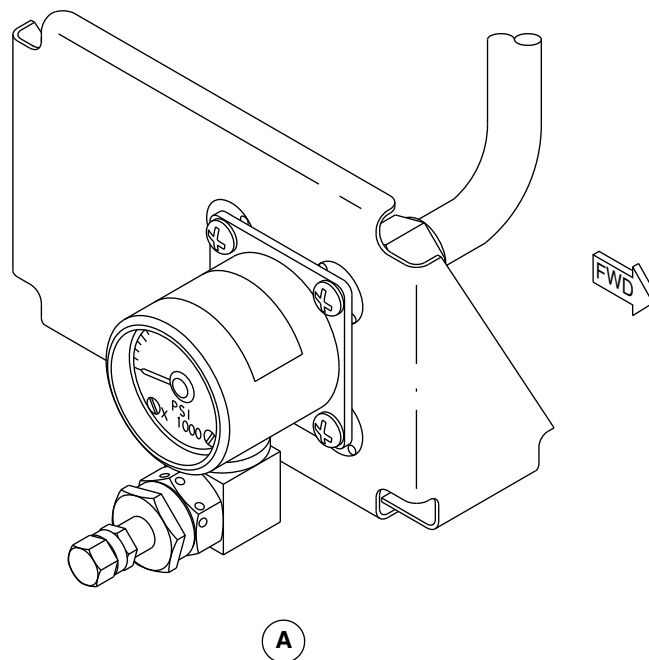
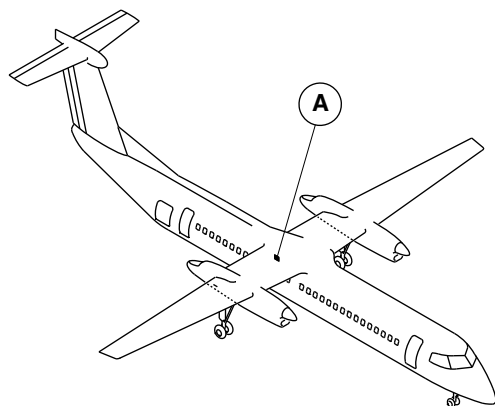
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Accumulator Pressure Gauge Locator
Figure 8

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 32-44-00
Config 001

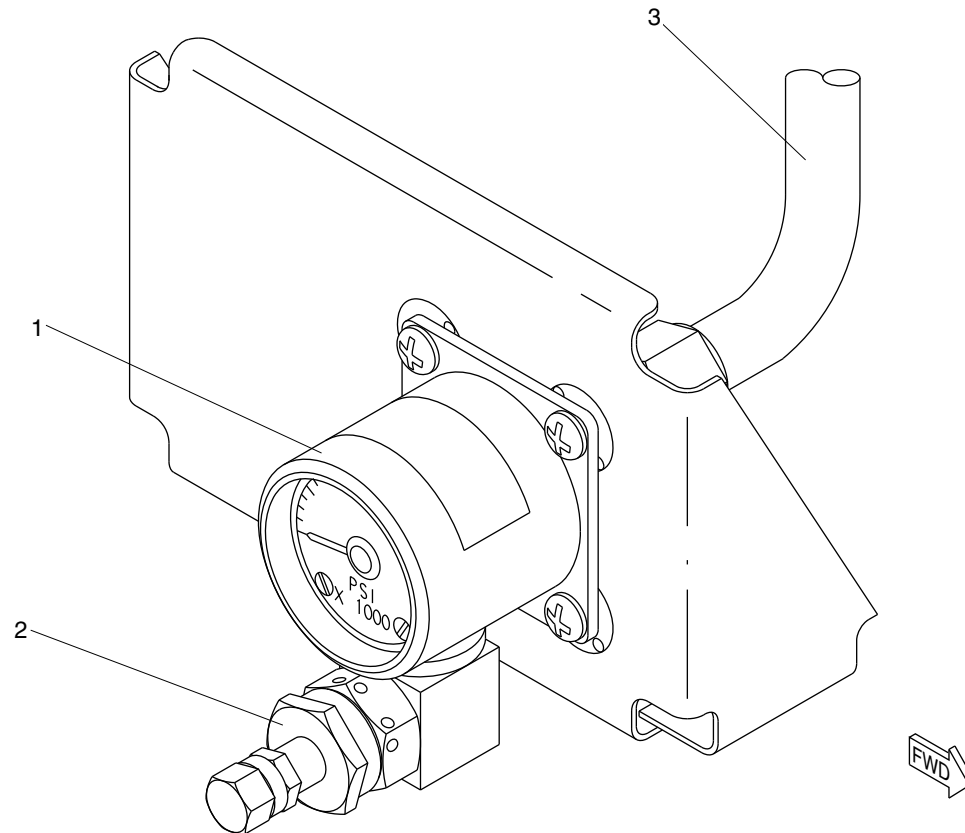
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LEGEND

- 1. Accumulator Pressure Gauge.
- 2. Charge Valve.
- 3. Accumulator Pressure Line.

fs143a01.cgm

Accumulator Pressure Gauge Detail
Figure 9

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 32-44-00
Config 001

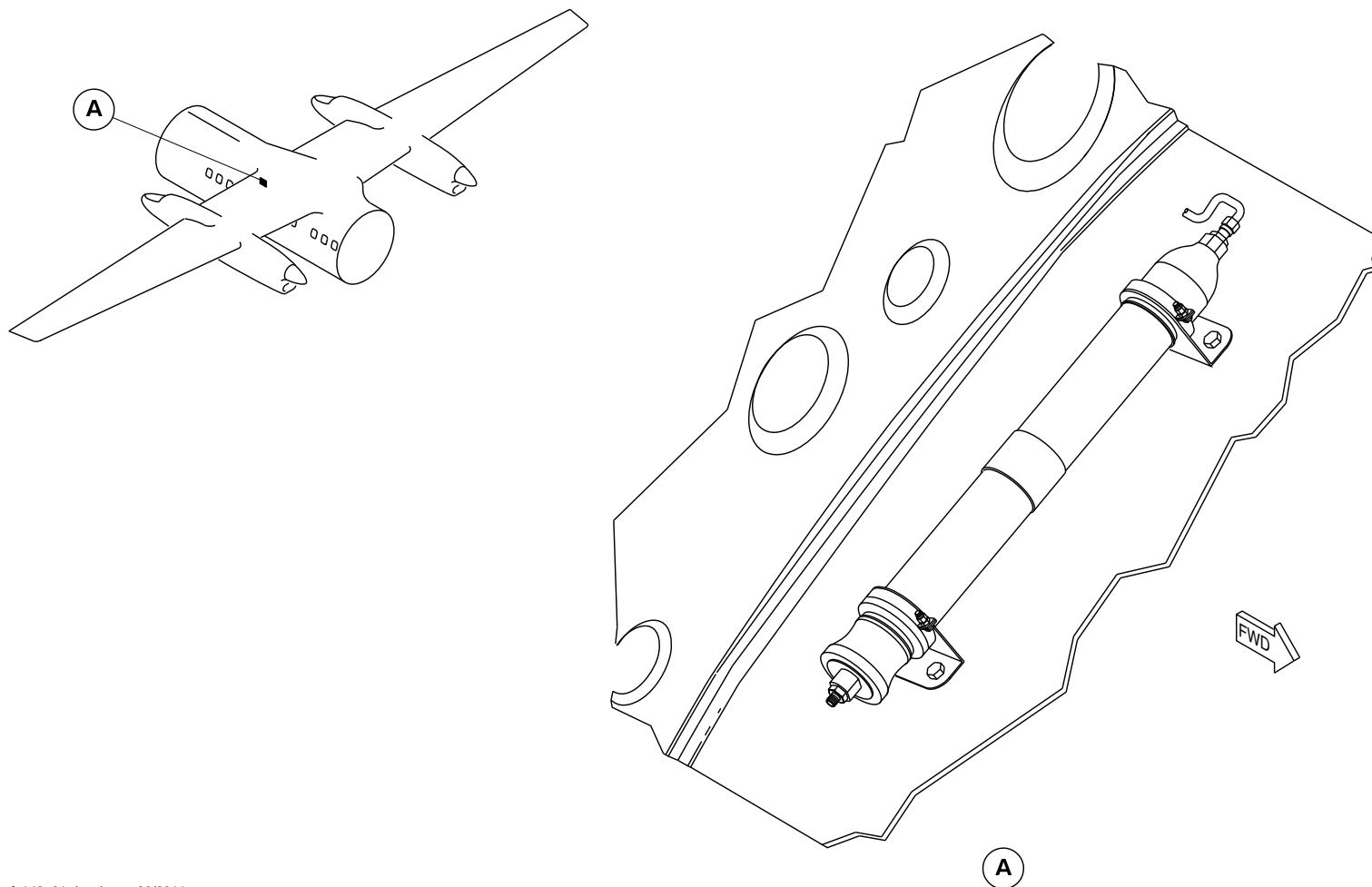
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Parking Brake Accumulator
Figure 10

PSM 1-84-2A
EFFECTIVITY:
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32-46-00-001

ANTI-SKID SYSTEM

Introduction

The anti-skid system controls hydraulic pressure to the four main wheel brake units to supply individual wheel anti-skid protection.

General Description

Refer to Figure 1.

The brake pedals control hydraulic pressure from the No. 1 hydraulic system through the brake control valve to the anti-skid control valves. The anti-skid control valves control hydraulic pressure to the fuse/shuttle valves and the brake units. When the anti-skid system senses a skid condition, it sends a signal to the applicable skid control valve. The skid control valve releases the brake pressure through the anti-skid return line. The anti-skid system supplies individual wheel anti-skid control for each main wheel. The anti-skid system operates with the normal brake system.

The anti-skid system supplies the functions that follow:

- Individual wheel anti-skid control
- Touchdown protection
- Locked wheel protection
- System monitoring.

The anti-skid system has the components that follow:

- Anti-Skid Control Unit (32-46-01)
- Wheel Speed Transducers (32-46-11)
- Skid Control Valves (32-46-06)
- Anti-Skid Switch (32-46-16).

Detailed Description

The Anti Skid Control Unit (ASCU) receives wheel speed information from the wheel speed transducers to sense wheel deceleration. When a skid is sensed, the ASCU sends a signal to the skid control valve to release the brake pressure to the non-rotating wheel. The skid control valve reacts to the signal and releases a portion of the brake pressure through the anti-skid return line. This decreases the brake pressure to the affected wheel so that it can recover from the skid. When the wheel has recovered from the skid, the ASCU removes the signal from the skid control valve. This lets the metered brake pressure from the brake control valve through to the brake unit.

Touchdown protection prevents brake pressure from being applied to the brake units before touchdown or before wheel spin-up has occurred.

The wheel control boards in the ASCU receive Weight-On-Wheel (WOW) and gear up and locked signals from the PSEU, as well as, wheel speed information from the wheel speed transducers to determine when touchdown protection is to be provided.

The ASCU supplies a full brake release signal to the skid control valve when: .

- The wheel speed is less than 30 knots (55.6 km/h)

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- The WOW signals from the PSEU indicates "AIR"
- The gear "Up and Locked" signal indicates the gear is up and locked.

When the skid control valve receives the signal from the ASCU, it will release the brake pressure through the anti-skid return line

Touchdown protection (full brake release signal) will be removed or inhibited to the skid control valve when the wheel speed exceeds 35 knots (64.86 km/h) or 5 seconds after WOW signals from the PSEU indicates "GROUND".

Locked wheel protection lets the affected wheel recover from deep skids.

The outboard wheels (1 and 4) and the inboard wheels (2 and 3) are paired for locked wheel protection. The higher of the two wheel speeds is the used as the reference velocity. The reference velocity is allowed to decelerate by 25 ft/sec² (7.62 m/sec²). A locked wheel condition exists when the wheel speed of either wheel decreases to less than 30% of the reference velocity.

When a locked wheel condition occurs, the ASCU sends a signal to the skid control valve to release brake pressure to the wheel that is locked. The skid control valve will release the brake pressure through the anti-skid return line until the wheel recovers from the deep skid. To recover from a locked wheel condition, the wheel speed must be more than 35% of the reference velocity. The ASCU will remove the full brake release signal to the skid control valve to permit wheel braking.

Anti Skid Control Unit

[Refer to Figure 2.](#)

The ASCU is installed on the right side of the aircraft, below the passenger cabin floor at the mid-wing area, on the equipment rails. The ASCU is an electronic box that has two wheel control circuit boards, one for the inboard wheels and one for the outboard wheels.

The functions of the wheel control circuit boards are for anti-skid control and Built-In-Test (BIT). Each wheel board uses a micro-controller and related circuitry to do the digital processing of the control and built-in-test software. Each wheel board also contains analog circuits for filters, wave shaping, test signal generation, valve drivers, and monitor functions.

[Refer to Figure 3.](#)

The ASCU receives 28 Vdc power from the right main DC bus through ANTISKID INBD and ANTISKID OUTBD circuit breakers, through the anti-skid switch. Both power sources are needed for the ASCU to supply full anti-skid control.

The fault data recorded by the ASCU is transmitted over the ARINC 429 data bus as a series of pages and is shown on the Audio Radio Control Display Unit (ARCDU). The number of pages transmitted depends on the number of faults. Current and past faults can show on the display unit.

[Refer to Figure 3.](#)

The ASCU sends signals to the caution and warning light and advisory system and the Central Diagnostic System (CDS) to show and record faults in the system.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The ASCU receives signals from the:

- Anti-skid switch
- Four wheel speed transducers
- Four skid control valves.

The ASCU receives signals from the Proximity Sensor Electronic Unit (PSEU) for landing protection functions.

Skid Control Valves

[Refer to Figure 4.](#)

The skid control valves are two separate valves in one common manifold. Each skid control valve contains a screen on the inlet port. The screen protects the skid control valve components from contaminants larger than 40 microns. The skid control valves receive metered pressure from the brake control valve.

The right control valve gives skid control for the inboard and outboard brake units on the right main landing gear. The left control valve gives skid control for the inboard and outboard brake units on the left main landing gear. The skid control valves are installed on the left wall of the left and right nacelle.

Wheel Speed Transducer

[Refer to Figures 5 and 6.](#)

The wheel speed transducer is an electrical generating device that supplies wheel speed information to the ASCU. A wheel speed transducer is installed in each end of the main landing gear axles.

The hubcap on the wheel engages with a clip on the wheel speed transducer. When the wheel turns, the wheel speed transducer supplies an AC signal with a frequency of 100 cycles per wheel revolution. This signal is changed to a digital signal in the ASCU for anti-skid control use.

Anti-Skid Switch

[Refer to Figure 7.](#)

The ANTI-SKID switch is located on the copilot's glareshield panel. The ANTI-SKID switch is a three position OFF/ON/TEST switch.

When the ANTI-SKID switch is set to the ON position, 28 Vdc is supplied to the inboard and outboard wheel control circuit boards in the ASCU. When set to the TEST position, 28 Vdc is supplied to the wheel control circuit boards and the system Built-In-Test function. When the ANTI-SKID switch is set to OFF, all electrical power is removed from the ASCU.

System Monitoring

The ASCU monitors the system with a built-in-test function. The built-in-test operates in two modes, continuous test mode and start up self test mode.

Continuous tests are initiated during normal operation of the anti-skid system. The tests do not cause interference with anti-skid operation. If the anti-skid system fails, the ASCU sends a signal to the Caution and Warning Panel (CAWP). Failure data is recorded in non-volatile memory in the ASCU.

The start up self test is initiated when the anti-skid switch is held to the TEST position or whenever the ASCU is energized. The start up self test is a check on the proper response of the anti-skid control



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

system functions and to determine the operational status of the ASCU, the skid control valves, and the wheel speed transducers. Normal anti-skid operation may be inhibited during the start up self test.

If a start up self test is initiated while using the anti-skid switch on the ground, the anti-skid failure lights comes on in the flight compartment for 6 seconds. They will come on for 3 seconds if the test was done while in the air. The start up self test is inhibited when the wheel speed is more than 17 knots (31.5 km/h).

If the anti-skid system passes the start up self test, normal anti-skid system operation will continue. If the anti-skid system fails the start up self test, the fail indicators on the CAWP comes on.

Fault Detection

The ASCU monitors the anti-skid system and senses operational faults. If a fault is sensed, the ASCU turns on the INBD or OUTBD ANTISKID caution lights on the CAWP. Failure data is stored in non-volatile memory in the ASCU. Past failure data is recorded and stored in non-volatile memory in the ASCU. The failure data is available to maintenance personnel through the CDS. A maximum of fifteen past faults can be recorded.

The ASCU monitors the current and voltage of the skid control valve, and checks for open and short circuits. The ASCU monitors the wheel speed transducers for open and short circuits.

Anti-Skid Fault Display on the CDS

Fault data from the ASCU is transmitted at the ARINC 429 interface as a series of pages to be shown on the display. Depending on the

number of faults, either two or three pages are transmitted. Two types of faults are included, current faults and past faults.

Current faults are always one page long. The faults are a list of names of line replaceable units shown in a fixed sequence. If a fault is corrected, it will be removed from the list when a new fault display is requested. The current faults page can show eleven names of line replaceable units.

Past faults may be one or two pages long. All past faults have a flight leg shown with the name of the line replaceable unit. Past faults are displayed in the reverse order that they occurred. The maximum number of past faults displayed is 15.

The oldest recorded faults are overwritten with new faults as they occur. For each flight leg where there was a fault recorded, one old fault will be overwritten.

The ASCU fault is always recorded in the non-volatile memory. The remaining LRU faults are recorded only when the wheel speed on both the wheels is more than 8 knots (14.82 km/h) or when the aircraft is in the air. Each faulty LRU will be recorded only once for each flight leg even if the fault has occurred multiple times. The same fault can be present in the present faults and the historical faults lists.

The flight leg zero always shows the present flight leg. The flight leg number for the stored faults (past faults) is increased by one when the sequence that follows occurs:

- Power-up or start-up test
- The air/ground switch indicates ground
- The air/ground switch indicates air
- The air/ground switch indicates ground.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Training Information Points

The anti-skid system operates with the normal brake system. It does not operate with the parking/emergency brake system.

The wheel speed transducer is an electrical generating device that supplies wheel speed information to the ASCU. A wheel speed transducer is installed in each end of the main landing gear axles.

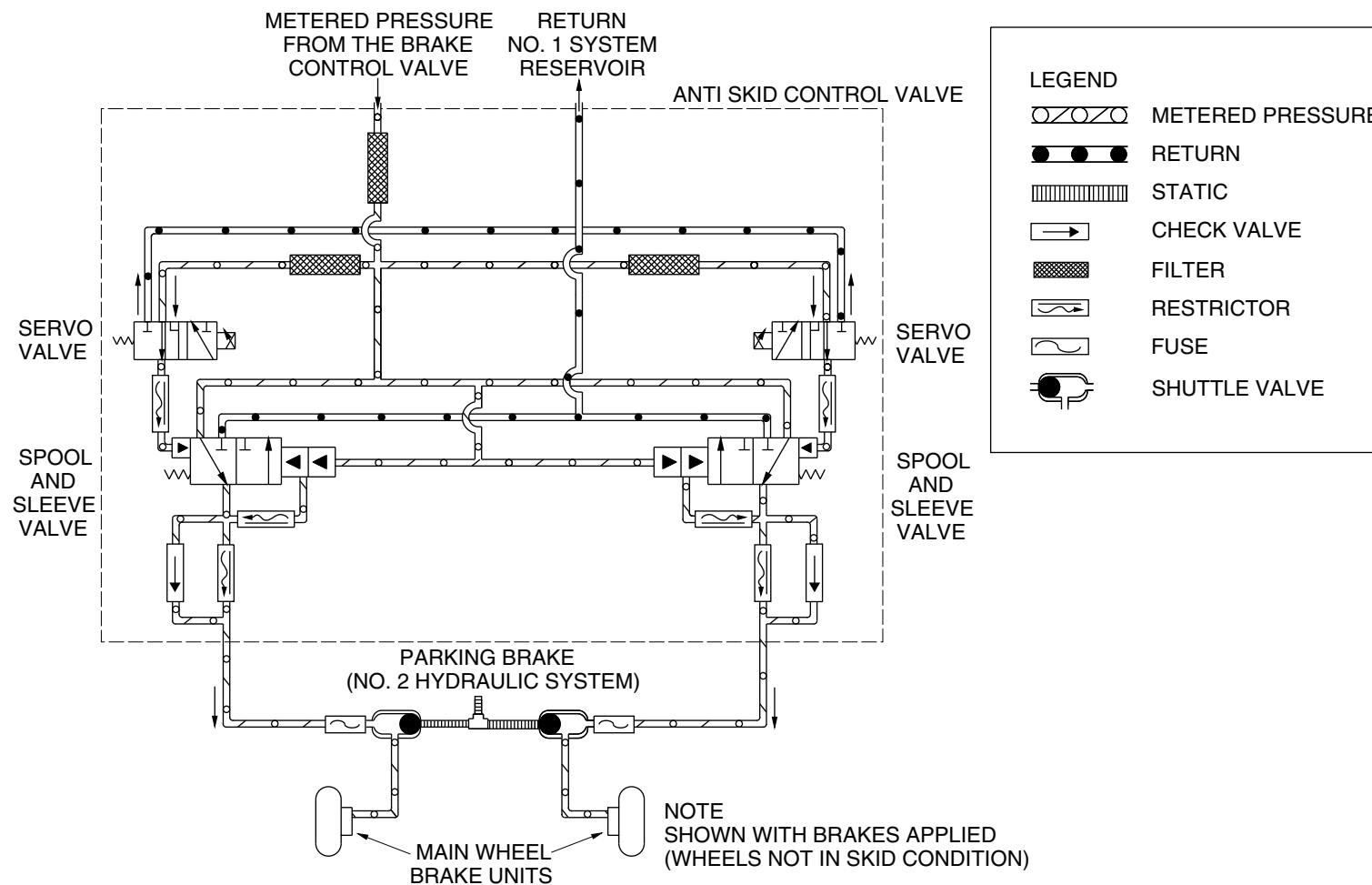
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Anti-Skid System – Hydraulic Schematic
Figure 1

PSM 1-84-2A
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Config 001

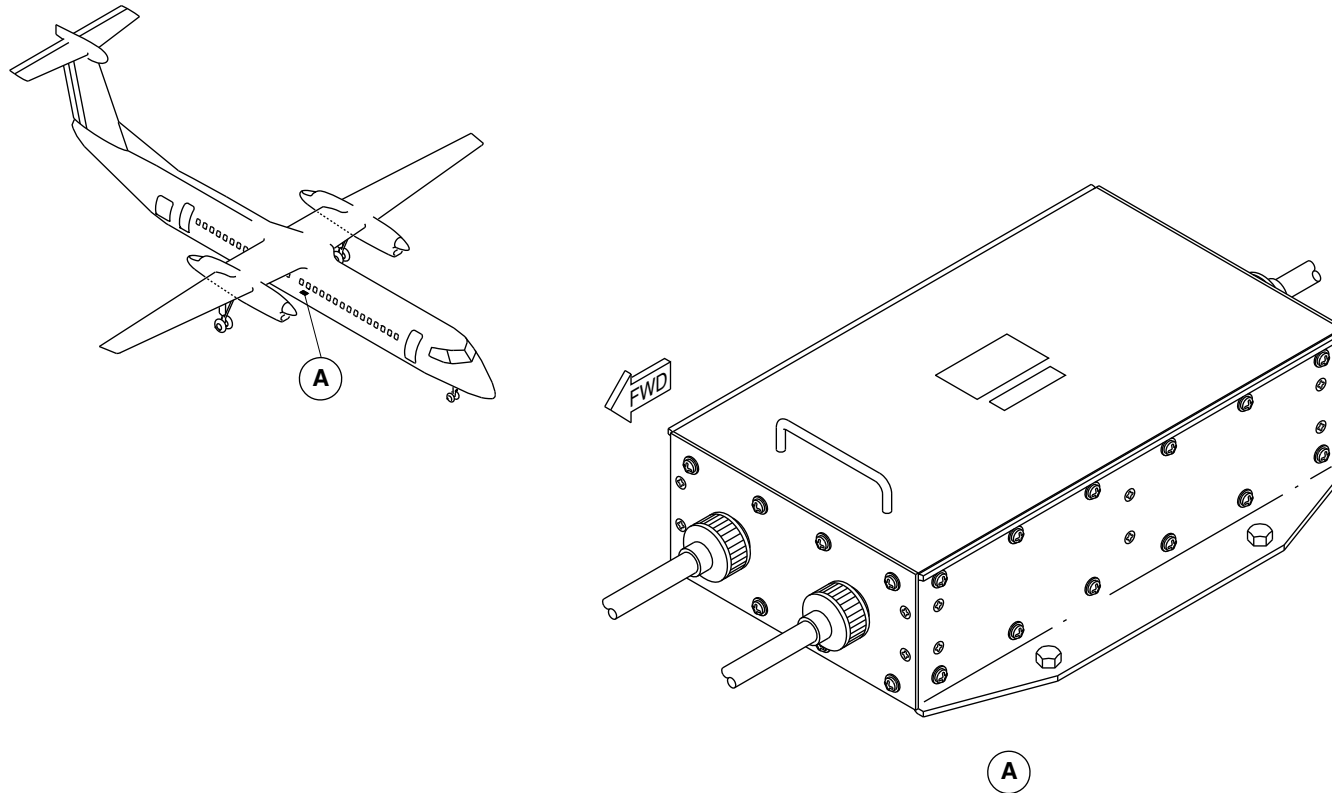
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Anti-Skid System Control Unit Locator
Figure 2

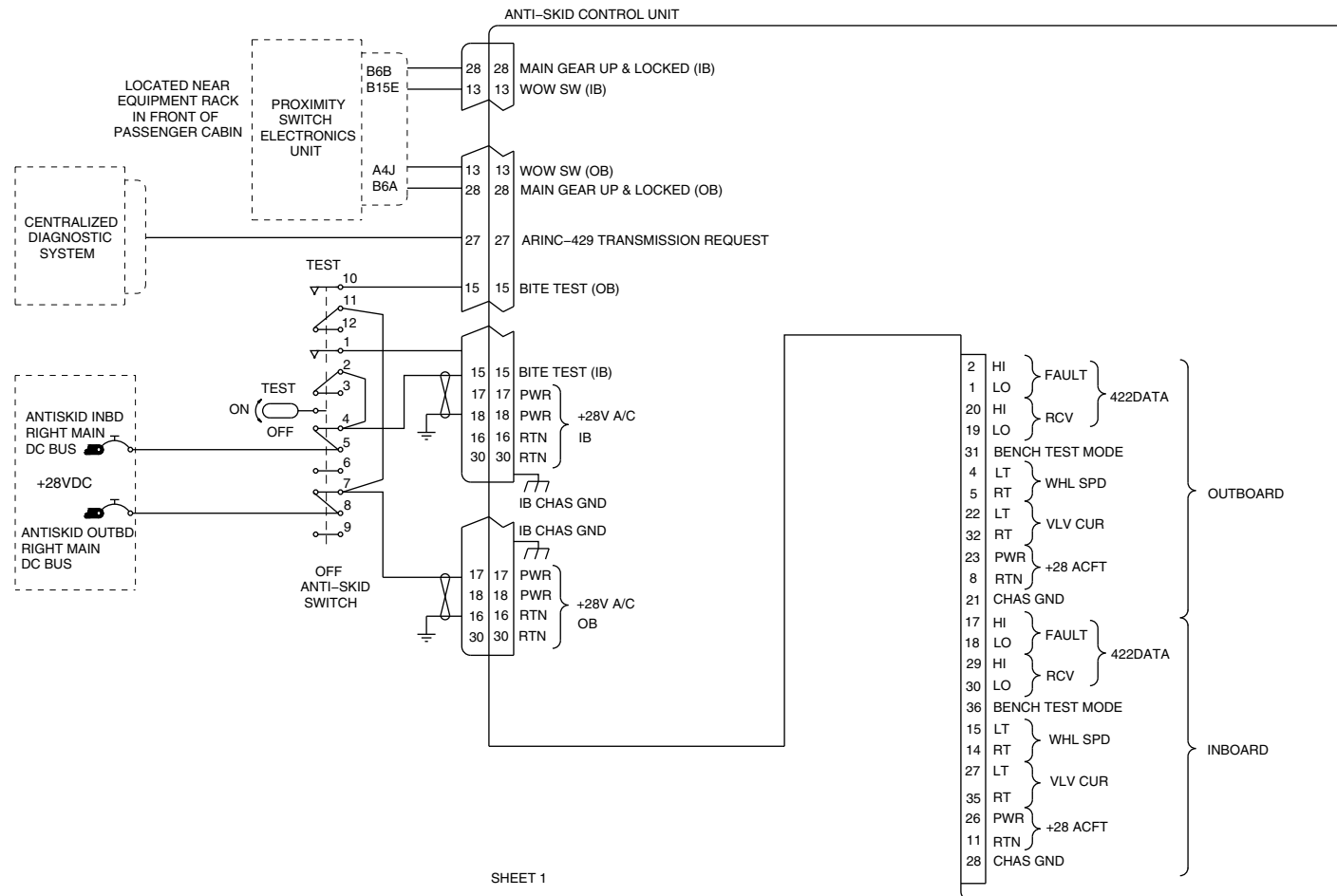
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SHEET 1

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Anti-Skid System – Electrical Schematic Page 1
Figure 3 (Sheet 1 of 2)

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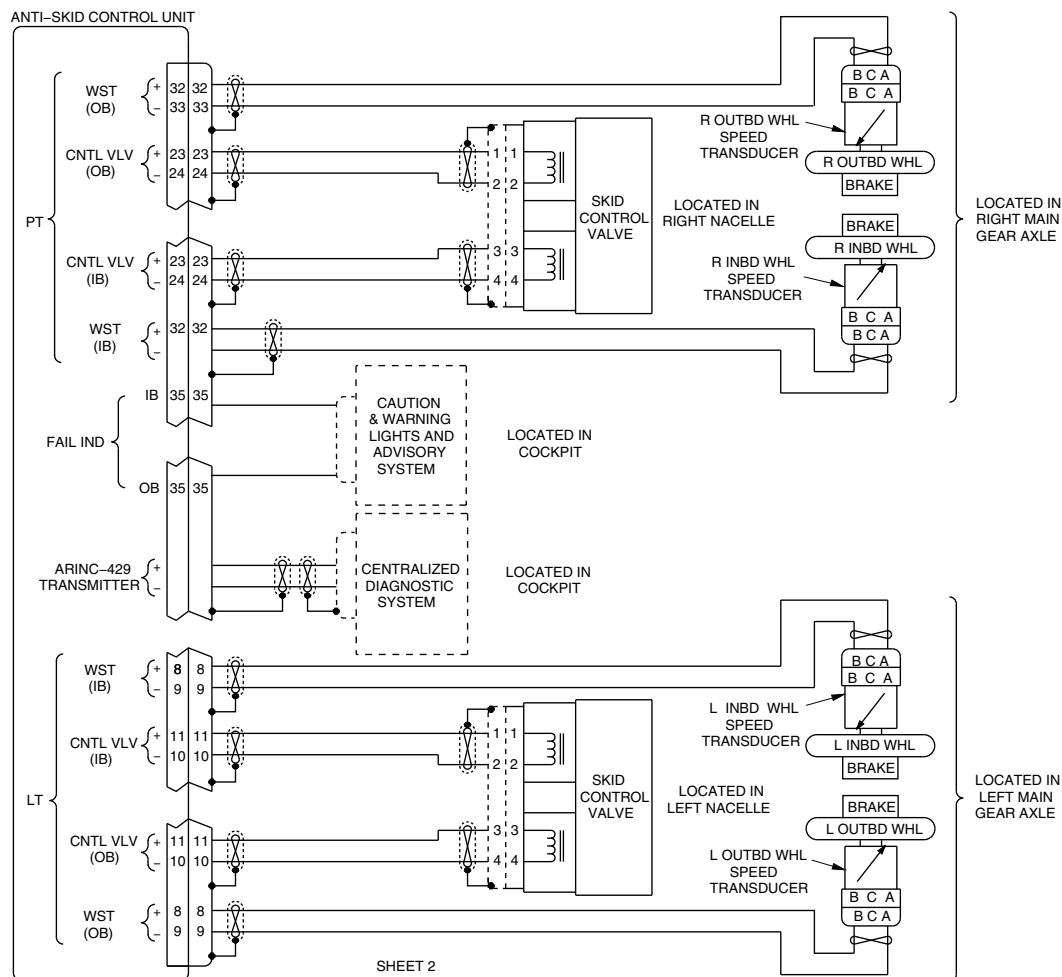
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Anti-Skid System – Electrical Schematic Page 1
Figure 3 (Sheet 2 of 2)

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EFFECTIVITY:
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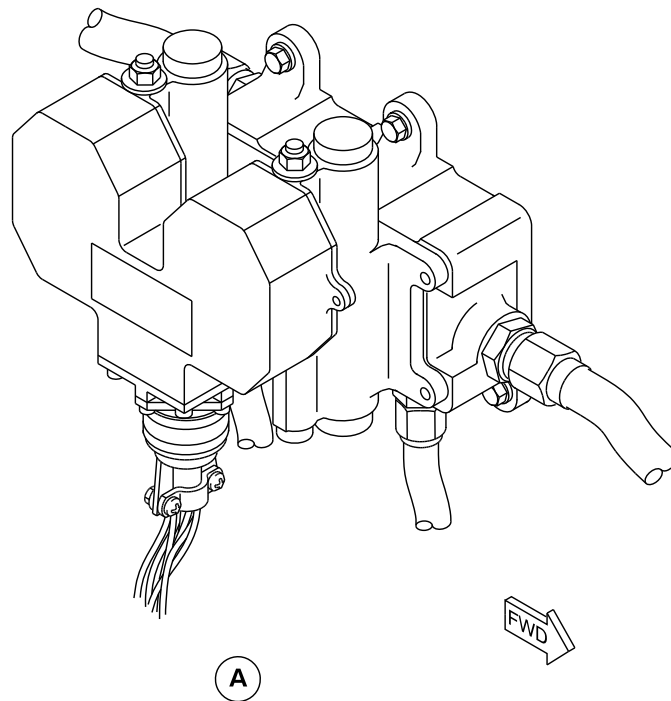
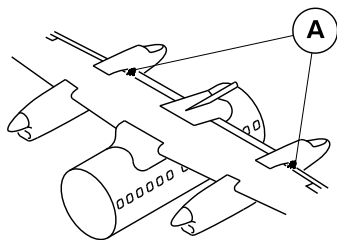
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Anti Skid Control Valve Locator
Figure 4

PSM 1-84-2A
EFFECTIVITY:
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Config 001

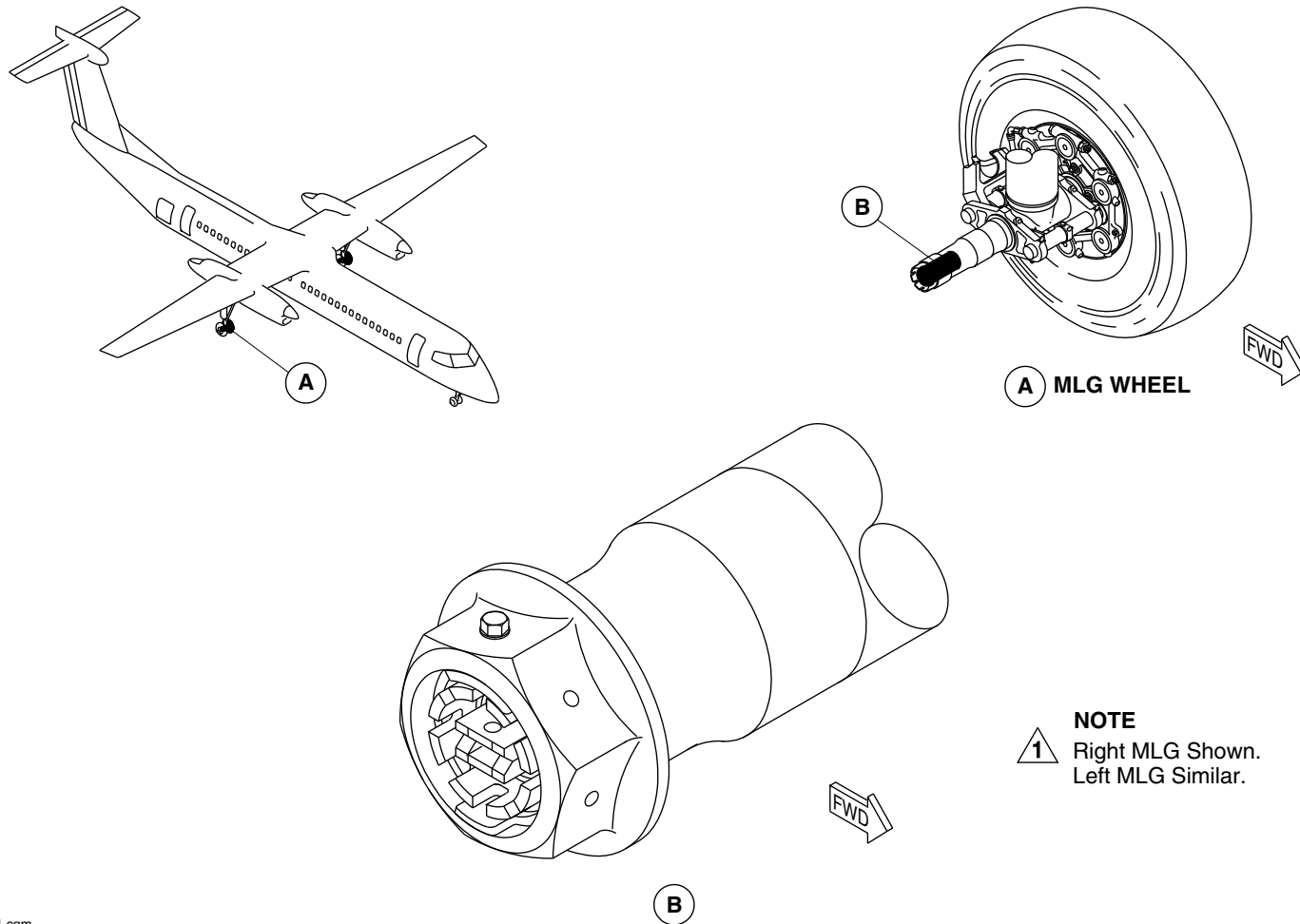
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Wheel Speed Transducer Locator
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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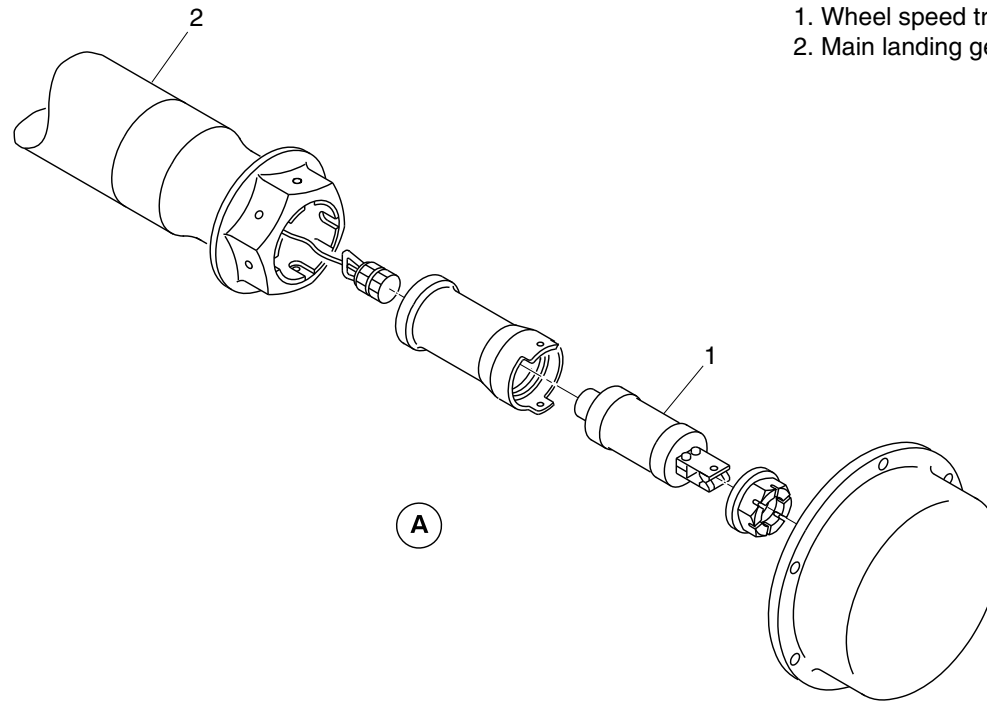
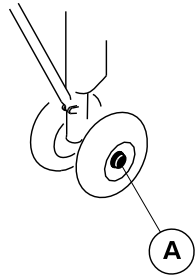
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LEGEND

- 1. Wheel speed transducer.
- 2. Main landing gear axle.

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Detail Wheel Speed Transducer
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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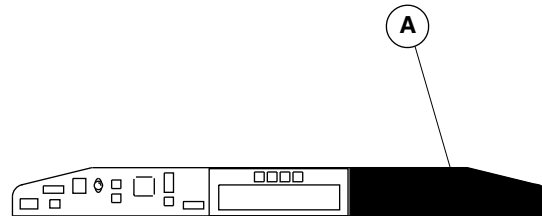
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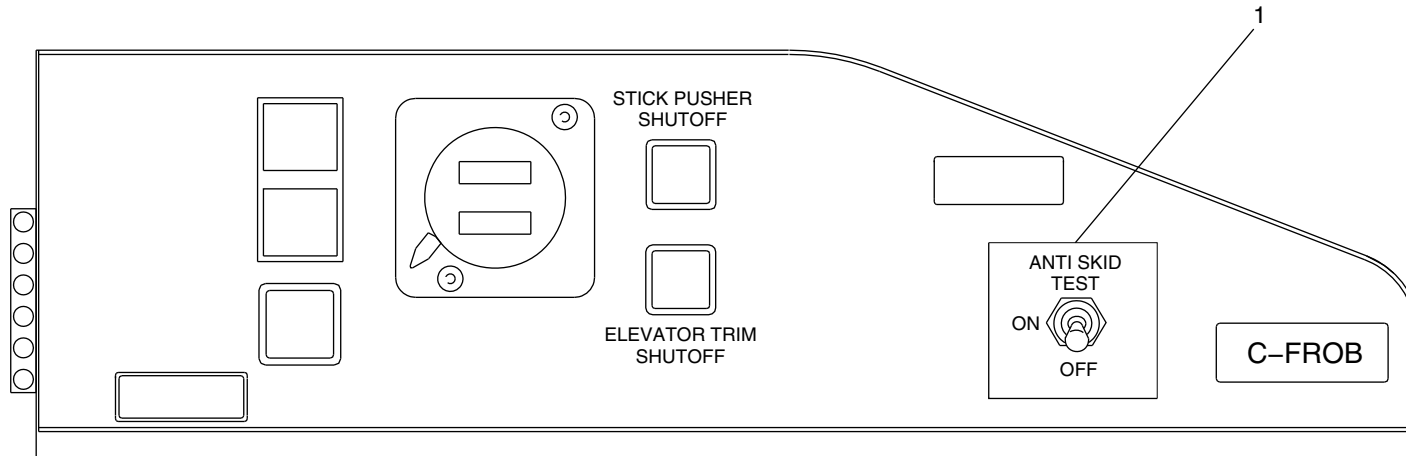
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GLARESHIELD PANEL

LEGEND

1. Antiskid Test Switch.



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Anti-Skid System, Anti Skid Switch Locator
Figure 7

PSM 1-84-2A
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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32-50-00-001

STEERING SYSTEM

Introduction

The steering system gives directional control of the aircraft while on the ground.

General Description

Refer to Figures 1 and 2.

The steering system supplies shimmy damping of the nosewheels. The STEERING switch in the flight compartment sets the steering system to the power ON mode or the power OFF mode. Hydraulic powered shimmy damping is supplied by the steering manifold when the steering system is in the power ON mode. Passive, free castor shimmy damping of the nosewheels is supplied by the nosewheel centering system when the pilot select switch is in the power OFF mode. If any component in the steering system fails, the nosewheel centering system will supply passive, free caster shimmy damping of the nosewheels. The passive, free castor mode lets the nosewheels steer $\pm 120^\circ$ to the left and to the right of the center position.

The steering system has the components that follow:

- Electronic Steering Control Unit (32-51-01)
- Hand control (32-51-06)
- Rudder pedal steering potentiometer (32-51-11)

- Nosewheel steering manifold (32-51-16)
- Nose landing gear centering mechanism (32-51-21)
- Power steering unit (32-51-26)
- Hydraulic steering motor (32-51-31)
- Steering feedback assembly (32-51-36).

Detailed Description

Refer to Figures 3, 4 and 5.

The steering system is electronically controlled by the Steering Control Unit (SCU). The SCU receives electrical power from the aircraft 28 Vdc left and right busses. The SCU does built-in-test and diagnostic functions on the components of the steering system.

The steering system is powered by the aircraft No.2 hydraulic system. Hydraulic pressure to the hydraulic motor is controlled by the steering manifold. The hydraulic motor operates a reduction gearbox which steers the nosewheels.

The SCU is an electronic box installed on the outer side of the Nose Landing Gear (NLG) wheelwell wall. Access to the SCU in the equipment bay of the forward nose fuselage is through a panel on the right side of the nose fuselage.

The SCU uses input commands from the steering tiller, the rudder pedals, and the Proximity Sensor Electronic Unit (PSEU). Input signals are received from the STEERING switch, the steering manifold, and the Rotary Variable Differential Transformers (RVDT). The SCU supplies output commands to the steering manifold and gives failure signals to the CAUTION and WARNING panel.

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The steering tiller is a self centering unit that lets the pilot steer the nosewheels for forward taxi at low speed. The steering tiller is installed in the flight compartment on the pilot's left side console. The steering tiller gives full authority steering command signals, of the nosewheel position to the SCU and can steer the nosewheels ± 70 degrees to the left and right of the center position.

Refer to Figures 6 and 7.

The rudder pedals are self centering and allow the crew to steer the nosewheels for high speed taxi, take-off, and landing purposes. The rudder pedals can steer the nosewheels to a maximum of ± 8 degrees left and right of the center position. The potentiometer for the rudder pedals is installed on the support structure of the control quadrant of the rudder system. A linkage assembly connects the potentiometer to the control quadrant of the rudder system. The potentiometer gives limited authority steering command signals of the nosewheel position to the SCU.

The steering manifold is installed on the right front side of the outer cylinder of the NLG. The steering manifold receives electrical signals from the SCU and converts the signals to supply hydraulic pressure to the hydraulic motor to steer the nosewheels. The steering manifold sends nosewheel position signals to the SCU to keep the nosewheels in the commanded steering position.

The hydraulic motor is attached to the bottom left side of the reduction gearbox on the left forward side of the outer cylinder of the NLG. The hydraulic motor receives hydraulic pressure for proportional control of the nosewheel steering from the steering manifold. The hydraulic motor uses a feedback gear to operate the reduction gearbox to steer the nosewheels.

The reduction gearbox assembly is installed on the top of the NLG outer cylinder. A feedback gear in the reduction gearbox engages

with the splines on the inner cylinder of the NLG to transmit torque to steer the nosewheels.

The RVDT assembly is installed on top of the reduction gearbox. The RVDT is connected to the reduction gearbox assembly with gears. The RVDT assembly sends signals to the SCU to give the steered position of the nosewheels.

The nosewheel centering mechanism is installed on the lower end of the NLG inner cylinder. The spring loaded centering actuator has a roller bearing that rolls on a machined race on the NLG outer cylinder. The race has a detent at the centered position to return the nosewheels to the straight ahead position. The centering mechanism centers the nosewheels when the landing gear is retracted.

Training Information Points

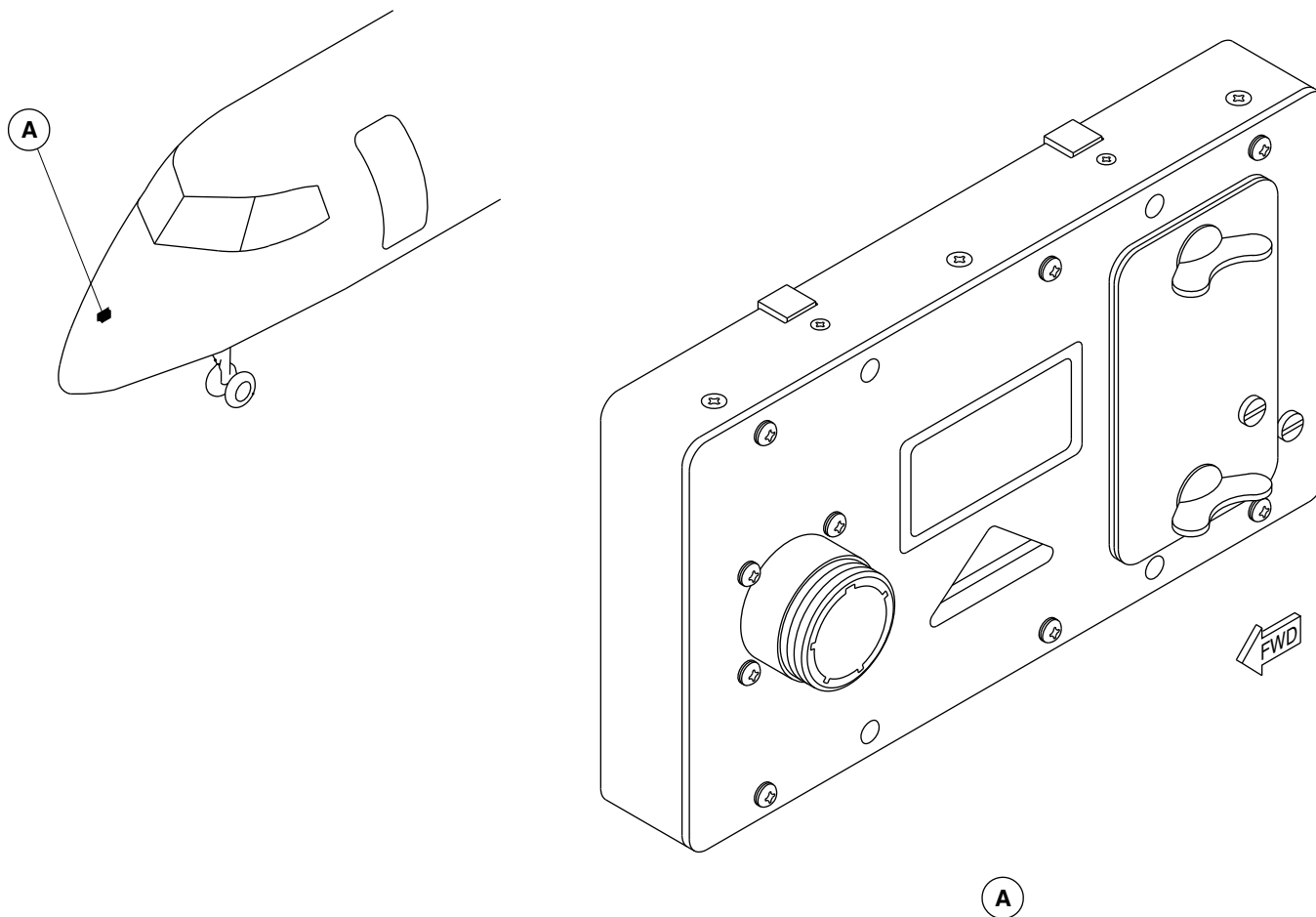
When you do maintenance on the nose landing gear make sure that the nose landing gear ground lock is engaged, the door lock pin is installed and that the hydraulic pressure is shut off. If you do not do this, you can cause injuries to persons and/or damage to the equipment.

Before you connect the tow bar to the aircraft make sure that the nosewheel steering switch is set to the off position and stays off during towing. After towing, make sure the nosewheel is centered before you disconnect the towbar. If you do not do this, you can cause injuries to persons and/or damage to the equipment.



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Steering Control Unit
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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Config 001

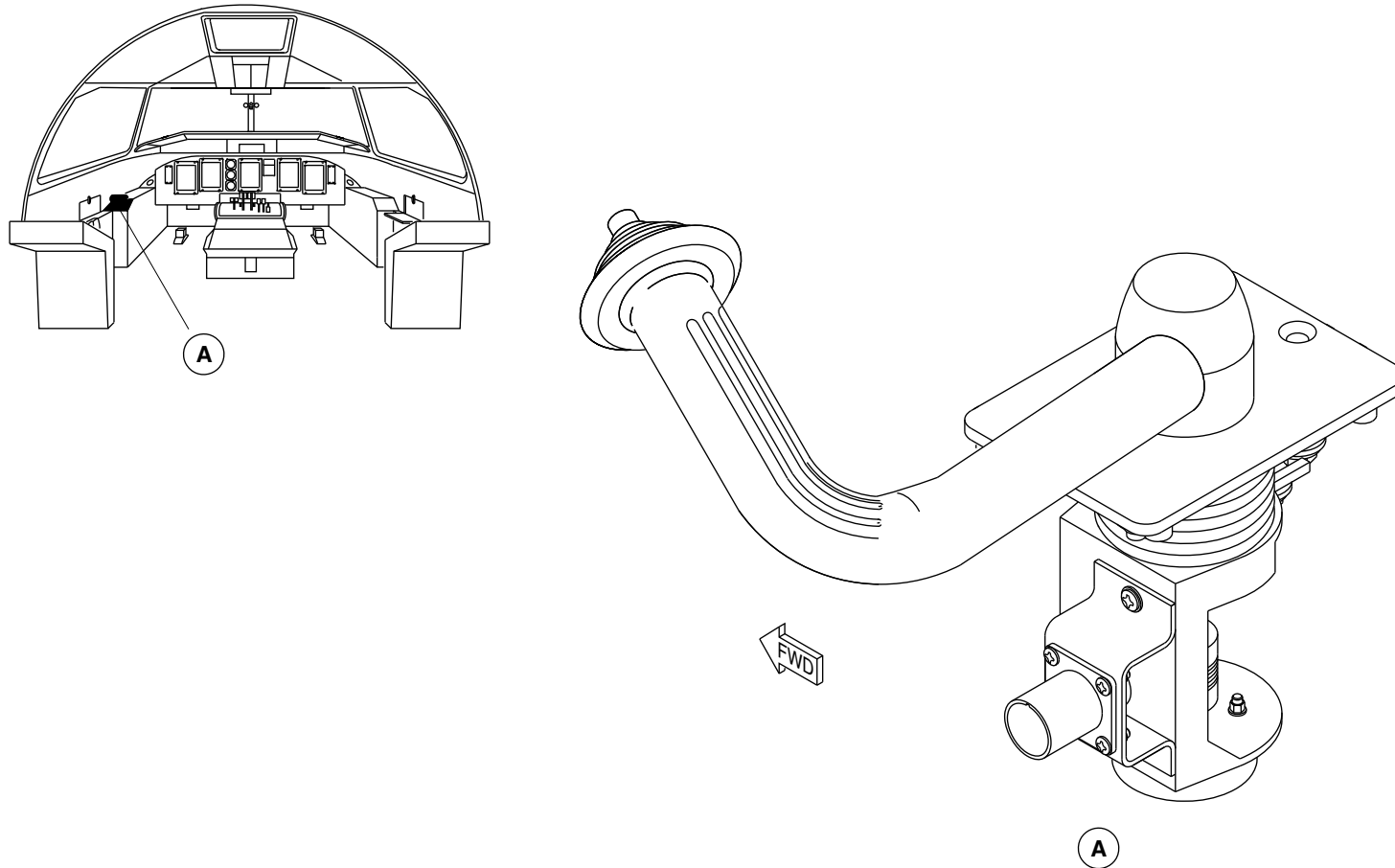
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Pilot Steering Hand Control Locator
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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Config 001

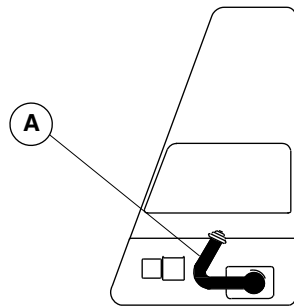
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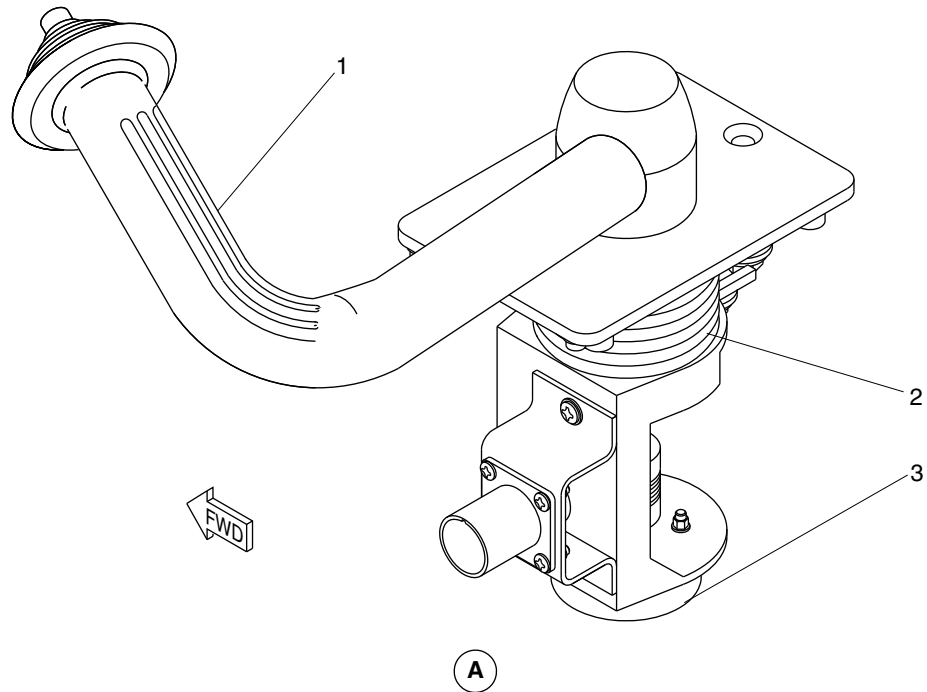
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LEFT SIDE CONSOLE

LEGEND

- 1. Pilot Hand Control.
- 2. Centering Spring.
- 3. Potentiometer.



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Pilot Steering Hand Control Detail
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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Config 001

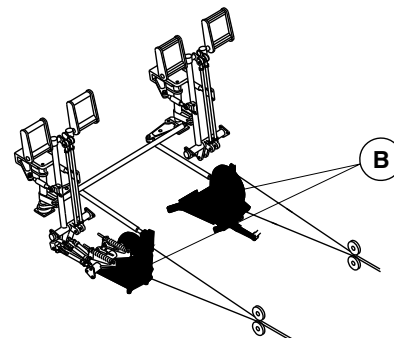
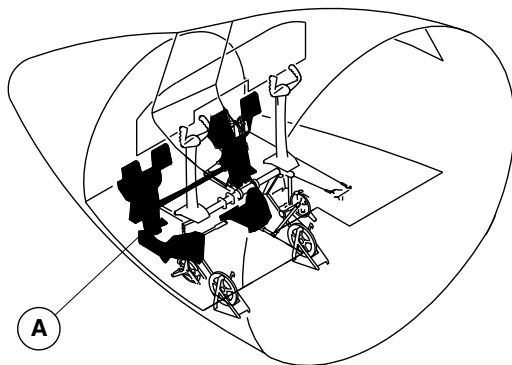
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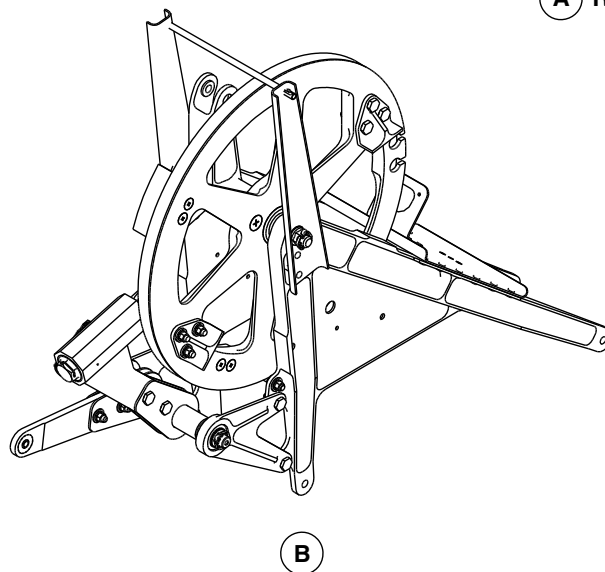


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A RUDDER FORWARD ASSEMBLY



NOTE

Right side shown.
Left side similar.

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Rudder Pedal Steering Potentiometer Locator
Figure 4

PSM 1-84-2A
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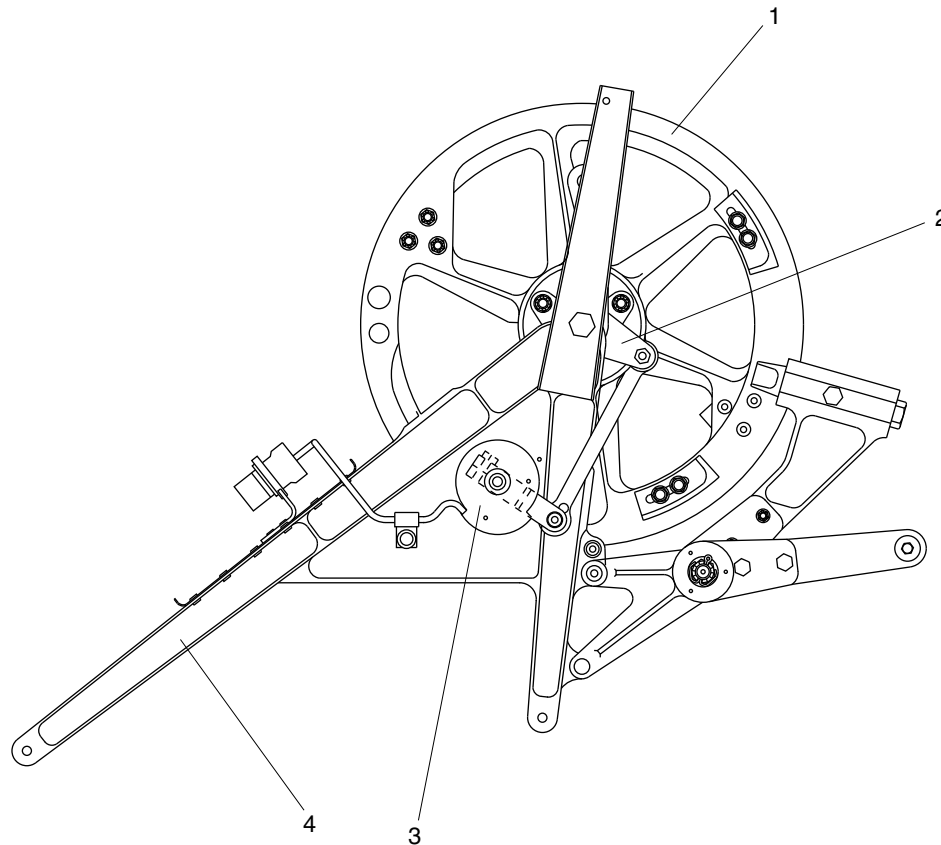
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LEGEND

- 1. Rudder Quadrant.
- 2. Linkage Assembly.
- 3. Rudder Pedal Potentiometer.
- 4. Support Structure.

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Rudder Pedal Potentiometer Detail
Figure 5

PSM 1-84-2A
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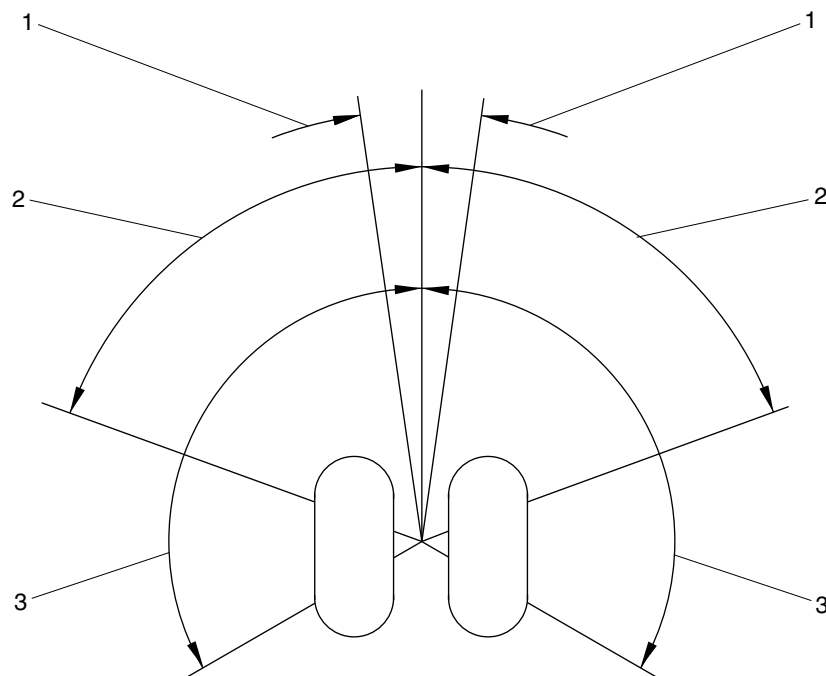
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LEGEND

1. High-Speed Taxi Mode 8 +/- 1 Degrees Left and Right from Rudder Pedals.
2. Low-Speed Taxi Mode 70 Degrees Left and Right from Steering Control Handle.
3. Unpowered (Caster) Mode 120 Degrees Left and Right.

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Nosewheel Steering Ranges
Figure 6

PSM 1-84-2A
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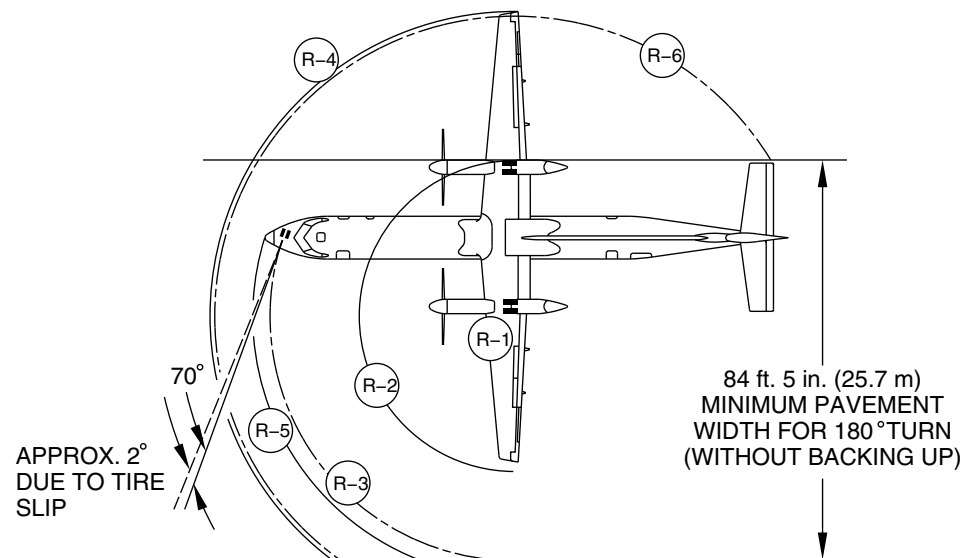
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ITEM	RADIUS
R-1 INNER GEAR	4 ft. 9 in. (1.52 m)
R-2 OUTER GEAR	33 ft. 10 in. (10.32 m)
R-3 NOSE GEAR	50 ft. 7 in. (15.41 m)
R-4 WING TIP	64 ft. 9 in. (19.74 m)
R-5 NOSE	54 ft. 10 in. (16.73 m)
R-6 ELEVATOR TIP	62 ft. 9 in. (19.13 m)

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Aircraft Turning Circles
Figure 7

PSM 1-84-2A
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NOSEWHEEL STEERING SYSTEM

Introduction

The steering system for the nosewheels gives electronically controlled and hydraulically powered steering of the Nose Landing Gear (NLG). Steering of the nosewheels is for taxi, take-off, and landing.

General Description

Refer to Figure 1.

Hydraulically powered steering of the nosewheels is available when the NLG is down and locked and the aircraft is on the ground. The STEERING switch, located in the flight compartment on the pilot's side console, must be in the STEERING position for the steering system to operate.

The Steering Control Unit (SCU) electronically controls the steering system for the nosewheels. Electrical power for the SCU is supplied through the STEERING switch controlled by the pilot. The switch receives electrical power from the 28 VDC left and right main busses. The left main bus power is through C.B. C6, right main bus power is through C.B. P5. The SCU does three types of Built-In-Test (BIT) on the components of the NWS system:

- Power-Up BIT (PBIT)
- Continuous BIT (CBIT)

- Maintenance Diagnostic BIT (MBIT).

The power steering unit has a steering manifold, a hydraulic motor, and a reduction gearbox with Rotary Variable Differential Transformers (RVDT). Hydraulic pressure to operate the steering system for the nosewheels is supplied by the No.2 hydraulic system. The steering manifold controls the hydraulic pressure to the hydraulic motor. The hydraulic motor operates a reduction gearbox which steers the nosewheels.

The nosewheel steering system includes the components that follow:

- Electronic Steering Control Unit (32–51–01)
- Pilot Hand Control (32–51–06)
- Rudder Pedal Steering Potentiometer (32–51–11)
- Nosewheel Steering Manifold (with the NWS Manifold Solenoid and NWS Pressure Switch) (32–51–16)
- NLG Centering Mechanism (32–51–21)
- Power Steering Unit (32–51–26)
- Hydraulic Steering Motor (32–51–31)
- Steering Feedback Assembly (Rotary Variable Differential Transformer) (32–51–36)
- STEERING Switch (32–51–41).

Detailed Description

Full authority steering of the nosewheels is by the pilot hand control located on the pilot's side console, in the flight compartment. The function of the pilot hand control is to steer the nosewheels for low speed forward taxiing only.

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Limited authority steering of the nosewheels is by the rudder pedals in the flight compartment. The rudder pedals control steering of the nosewheels for high speed forward taxiing, take-off, and landing.

Hydraulic powered shimmy damping of the nosewheels is supplied by the steering manifold when the STEERING switch is in the STEERING position. Passive, free castor shimmy damping of the nosewheels is supplied by the nosewheel centering system when the STEERING switch is in the OFF position. If any component in the steering system fails, the nosewheel centering system will give passive, free caster shimmy damping of the nosewheels. The passive, free castor mode lets the nosewheels move ± 120 degrees to the left and to the right of the center position.

If any component in the steering system fails, a NOSE STEERING caution light on the Caution and Warning panel in the flight compartment comes on.

Hydraulic power is removed from the NWS system when there is no Weight-On-Wheels (WOW) input from the PSEU and the NLG selector valve begins NLG retraction. The SCU delays the closing of the Solenoid Operated Valve (SOV) of the steering manifold by 5 seconds after lift-off because of a possible Weight-Off-Wheels (WOFW) false signal that can occur during the lift-off phase of flight.

NOTE

The SCU will also delay the opening of the SOV by 1 second after it gets a WOW signal from the PSEU (after touch-down) because of a possible WOW false signal that can occur during the touch-down phase of flight.

Normal Modes of NWS System Operation

Table 1 shows the different NWS modes of operation:

Table 1: NWS Modes of Operation – Table 1

PHASE	SYSTEM REQUIREMENTS	FUNCTIONS
Ground	– SCU on and no faults detected – NLG is down and locked – WOW input from the PSEU – STEERING switch in the ON position.	After a 1 second delay after touch-down, power steering is available. Directional control of the aircraft from pilot inputs (PHC or rudder pedal input).
Lift Off	– SCU on and no faults detected – NLG is down and locked – No WOW input from the PSEU – STEERING switch in the ON position.	The SCU commands a straight-ahead signal to the NLG. Pilot inputs are disregarded. After a 5 second delay, the nosewheels are centered and the SOV is de-energized.



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PHASE	SYSTEM REQUIREMENTS	FUNCTIONS
Air	– SCU on and no faults detected – NLG is not down and locked – No WOW input from the PSEU – STEERING switch in the ON position.	The SOV is de-energized. The SCU commands a straight-ahead reference signal to the NLG. Pilot inputs are disregarded.
Approach	– SCU on and no faults detected – NLG is down and locked – No WOW input from the PSEU – STEERING switch in the ON position.	The SOV is de-energized. The SCU commands a straight-ahead reference signal to the NLG. Pilot inputs are disregarded. The SCU does a validity check of the PHC and rudder pedals.

The SCU uses input commands from the PSEU, rudder pedals and the PHC. Input signals are also received from the STEERING switch, the nosewheel steering manifold and the feedback system RVDTs. The SCU gives output commands to the steering manifold and failure signals to the Caution and Warning panel.

The WOW condition signal and the landing gear down and locked condition signal are necessary for the SCU to enable the powerd steering mode of operation. The WOW condition signal will arm the STEERING switch when it is set to the STEERING position.

When the aircraft is in the air, the SCU sends a straight-ahead reference signal (zero-angle command). When the aircraft lands (and the WOW input signals and the landing gear down and locked condition signal are recieved from the PSEU), the straight-ahead reference signal is removed by the SCU. The SCU commands the SOV to energize (the SOV opens) and then starts to count to 1 second. The mode selector valve of the steering manifold is opened by the SOV. The hydraulic pressure increases to the system operating pressure in a fraction of a second and is monitored by the NWS pressure switch. At the same time, the SCU synchronizes the input commands from the PHC potentiometer and the rudder pedal potentiometer with the actual nosewheel position (by RVDT feedback). The difference between the input and feedback

Steering Control Unit

[Refer to Figure 2.](#)

The Steering Control Unit is an electronic box installed on the outer side of the Nose Landing Gear wheelwell wall. A panel on the right side of the nose fuselage gives access to the SCU in the equipment bay of the forward nose fuselage.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

commands is then gradually reduced over the 1 second period to give a smooth transition to the powered steering mode.

NOTE

This synchronization feature will not start until the steering pressure switch detects that there is sufficient hydraulic pressure available to the NWS system.

NOTE

If the SCU detects that the PHC is not between ± 8 degrees of the straight-ahead position in the moment that the system operating pressure is reached and synchronization starts, the SCU will command the SOV to de-energize. The SOV and the mode selector valve of the steering manifold will close to put the NWS system in the passive, free castor mode. At the same time, the SCU sends a signal to the Caution and Warning panel. When the signal is received, the NOSE STEERING caution light comes on.

The SCU receives steering command inputs from the potentiometers of the PHC and rudder pedals. The SCU receives signals from the RVDTs on the reduction gearbox relative to the steered position of the nosewheels. The SCU cancels the steering signal when the steered position of the nosewheels matches the position commanded by the PHC or rudder pedals. The SCU keeps the nosewheel steering angle by the hand control to between ± 70 degrees and by the rudder pedals to between ± 8 degrees. The steering system can be set to STEERING or OFF with the STEERING switch in the flight compartment.

The steering system is tested by the SCU for faults on power up. The SCU continuously monitors the system for faults while it is in operation. The SCU can record and store up to 100 failures in a non-volatile memory (NVM) for maintenance functions. Maintenance

personnel can use the push button on the SCU to review the last 18 failures. The complete failure history can be downloaded to a personal computer for analysis by maintenance personnel.

The SCU has a command path in the computer hardware that monitors input commands (from the PHC and rudder pedal potentiometers) and monitors the feedback from the RVDTs on the reduction gearbox. The logic to control the different phases of the powered steering mode are also in the command path. A separate monitor path in the computer hardware duplicates the command path. If there is a difference between the command path and the monitor path, the SCU system will prevent the operation of the NWS system and a fault will be logged in the NVM.

When a component in the steering system fails, the SCU sends a signal to the Caution and Warning panel. When the signal is received, the NOSE STEERING caution light comes on. The SCU monitors the SOV for mechanical failures and monitors the potentiometer of the PHC for proper function. The SCU monitors the Electro Hydraulic Servo Valve (EHSV) torque motor to make sure it responds correctly to input commands. The SCU monitors the Linear Variable Differential Transducer (LVDT) to make sure that it responds correctly to input commands.

A computer can be connected to terminals on the SCU for steering system rigging. The computer measures output signals from components that follow to rig the steering system:

- RVDTs
- Pilot hand control potentiometer
- Rudder potentiometer



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Nosewheel Steering Manifold

Refer to Figures 3 , 4 and 5.

The nosewheel steering manifold is a self contained assembly which is installed on the right front side of the NLG outer cylinder. The steering manifold controls hydraulic pressure to the steering hydraulic motor for powered steering of the nosewheels. Hydraulic power for the steering manifold is supplied by the No. 2 hydraulic system through the landing gear selector valve extension tubing. Hydraulic fluid is returned to the No. 2 hydraulic system through the retraction tubing of the landing gear selector valve.

Two electrical connectors supply electric power to the steering manifold. One connector supplies electrical power to the SOV, the EHSV, and the EHSV LVDT. The second connector supplies electrical power to the NWS pressure switch.

The NWS pressure switch is used to monitor hydraulic pressure status. After take-off, the pressure switch is checked by C-BIT (momentarily energizing the SOV) and if the response is no pressure (P-SW open circuit), the fault P-SW is written into non-volatile memory (NVM) of the SCU. Pressure switch P-BIT test is a test of the monitor, so simulated WOW and pressure switch signals are used to check the response of the monitor. During P-BIT the pressure switch response is not used. If the NWS pressure switch indicates that hydraulic pressure is available, the SCU will send a signal to the Caution and Warning. When the signal is received, the NOSE STEERING caution light comes on. This will be an indication that the SOV is stuck open. The pilot will then put the STEERING switch to the OFF position. If the SOV is stuck open, the NWS pressure switch gives a ground to the NOSE STEERING caution light to make it come on.

The mode selector valve stops hydraulic pressure to the EHSV when the NWS system is in the passive, free castor mode. The SOV is in the closed position until the NLG is compressed when the aircraft lands. The solenoid of the SOV is then energized by signals from the SCU to open the mode selector valve and to configure the steering system for powered steering. The EHSV gets signals from the SCU and converts them into proportional spool displacement. The spool displacement sends hydraulic pressure to the hydraulic motor to drive the reduction gearbox and to steer the nosewheels to the commanded position.

Nosewheel position signals are sent from the LVDT installed in the steering manifold, to the SCU to keep the nosewheels in the commanded steering position. The pressure switch senses any changes in hydraulic pressure that are caused by the failure of components in the steering manifold. The pressure switch sends a signal to the SCU. When the signal is received, a light on the Caution and Warning panel in the flight compartment comes on.

Reduction Gearbox

The reduction gearbox is a self contained assembly that is installed on the top of the NLG outer cylinder. The feedback gear of the reduction gearbox connects with a spline on the inner cylinder of the NLG. The feedback gear of the reduction gearbox is operated by the hydraulic motor to transmit torque to the spline on the inner cylinder of the NLG. The reduction gearbox is lubricated with a mixture of grease and oil.

Rotary Variable Differential Transformer (RVDT) Assembly

The RVDT assembly is a self contained assembly bolted to the top of the reduction gearbox. The RVDT assembly connects with the



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feedback gear of the reduction gearbox. The SCU uses signals from the RVDT assembly to give the steered position of the nosewheels. The SCU uses one RVDT to give the position of the nosewheels. The SCU uses the second RVDT to check the accuracy of the first RVDT as a fault test.

Hydraulic Motor

The hydraulic motor is installed on the bottom, left forward side of the reduction gearbox. The hydraulic motor connects with the feedback gear of the reduction gearbox. The hydraulic motor receives hydraulic pressure for proportional steering control of the nosewheel from the EHSV of the steering manifold. The hydraulic motor operates a reduction gear assembly in the reduction gearbox to steer the nosewheels to the commanded position.

Nosewheel Centering Mechanism

[Refer to Figures 6 and 7.](#)

The centering mechanism center the nosewheels when the landing gear is retracted. The nosewheel centering mechanism is installed to a collar which is attached to the lower end of the NLG inner cylinder. The spring loaded centering actuator has a roller bearing that rolls on a machined race on the NLG outer cylinder. The machined race has a detent at the centered position. The detent returns the nosewheels to the straight ahead position. The centering actuator cannot move the nosewheels to the center position. It only holds them in the center position when the steering system aligns them. Attached to the centering actuator is the NWCENT proximity sensor.

The proximity sensor sends a signal to the PSEU when the centering actuator is in the straight ahead detent of the machined race. For the

landing gear to retract, the PSEU must get an air mode signal and the NLG proximity sensor must read NEAR (NWCENT NEAR). After take-off the PSEU sends the "WOW OFF" signal to the SCU. The SCU ignores the pilot input and enters a "straight ahead mode". Then the SCU energizes the EHSV in the steering manifold and actuates the hydraulic motor to center the nose gear. When the SCU senses the nose gear is at the neutral position through two RVDTs located at the top of the steering gearbox, the hydraulic motor stops. When the centering actuator rod is fully extended in the detent, the nosewheels are in the center position. The PSEU then lets the landing-gear retraction sequence start when the landing-gear handle is moved to the UP position.

NOTE

PSEU requires the NWCENT proximity sensor to read NEAR before retraction. Any discrepancy or miss alignment of the RVDT's will make the nose wheel steering hydraulics stop before the NLG has reached center position. This will give a NWCENT UNREASONABLE FAR condition.

Pilot Hand Control

[Refer to Figures 8 and 9.](#)

The pilot hand control (PHC) is a self centering unit and is located on the left side console in the flight compartment. The PHC lets the pilot steer the nosewheels. The PHC has a potentiometer that gives input signals to the SCU in proportion to the actual PHC position. The PHC has hard stops on either side of the center position. The centering spring returns the PHC to the neutral position when the input from the pilot is removed.

The PHC gives full authority steering control of the nosewheel for low speed, forward taxiing. For reverse taxiing, the PHC is in the center



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

position. The PHC can steer the nosewheels ± 70 degrees to the left and to the right of the center position. The potentiometer of the PHC is monitored for failure on power up by the SCU PBIT function and is also monitored continuously for failure by the CBIT function.

Rudder Pedal Potentiometer

Refer to Figures 10 and 11.

The rudder pedal potentiometer is a self contained bolt-on assembly with electrical leads for remote airframe connections. The rudder pedal potentiometer is installed on the support structure of the rudder control quadrant in the equipment bay in the forward nose fuselage. The potentiometer is connected to the rudder quadrant with a linkage assembly for rotary inputs in proportion to rudder pedal movement. A panel located on the right side of the nose fuselage gives access to the potentiometer. Centering springs on the rudder pedals operate to return the pedals to the neutral position when the pilot inputs are removed.

The rudder pedals allow the crew to steer the nosewheels for high speed taxi, take-off and landing. For reverse taxiing, the rudder pedals are in the center position. The rudder pedals can steer the nosewheel ± 8 degrees to the left and to the right of the center position. The potentiometer gives input signals to the SCU. The signals are in proportion to the actual position of the rudder pedals. The potentiometer of the rudder pedal is monitored for failure on power up by the SCU PBIT function and is also monitored continuously for failure by the CBIT function.

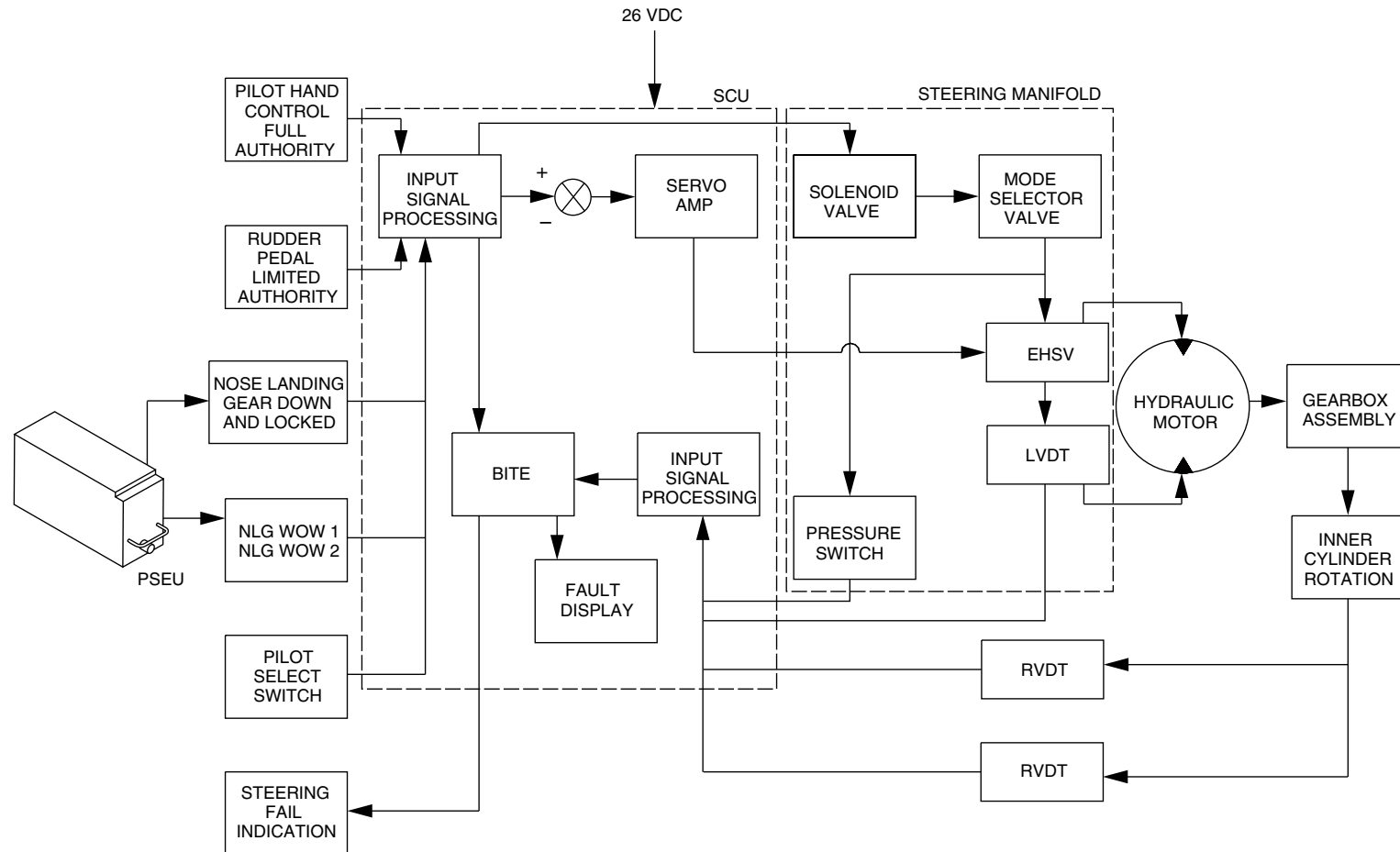
Training Information Points

When you do maintenance on the nose landing gear make sure that the nose landing gear ground lock is engaged, the door lock pin is installed and that the hydraulic pressure is shut off. If you do not do this, you can cause injuries to persons and/or damage to the equipment.

Before you connect the tow bar to the aircraft make sure that the nosewheel steering switch is set to the off position and stays off during towing. After towing, make sure the nosewheel is centered before you disconnect the towbar. If you do not do this, you can cause injuries to persons and/or damage to the equipment.



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Nosewheel Steering Block Diagram
Figure 1

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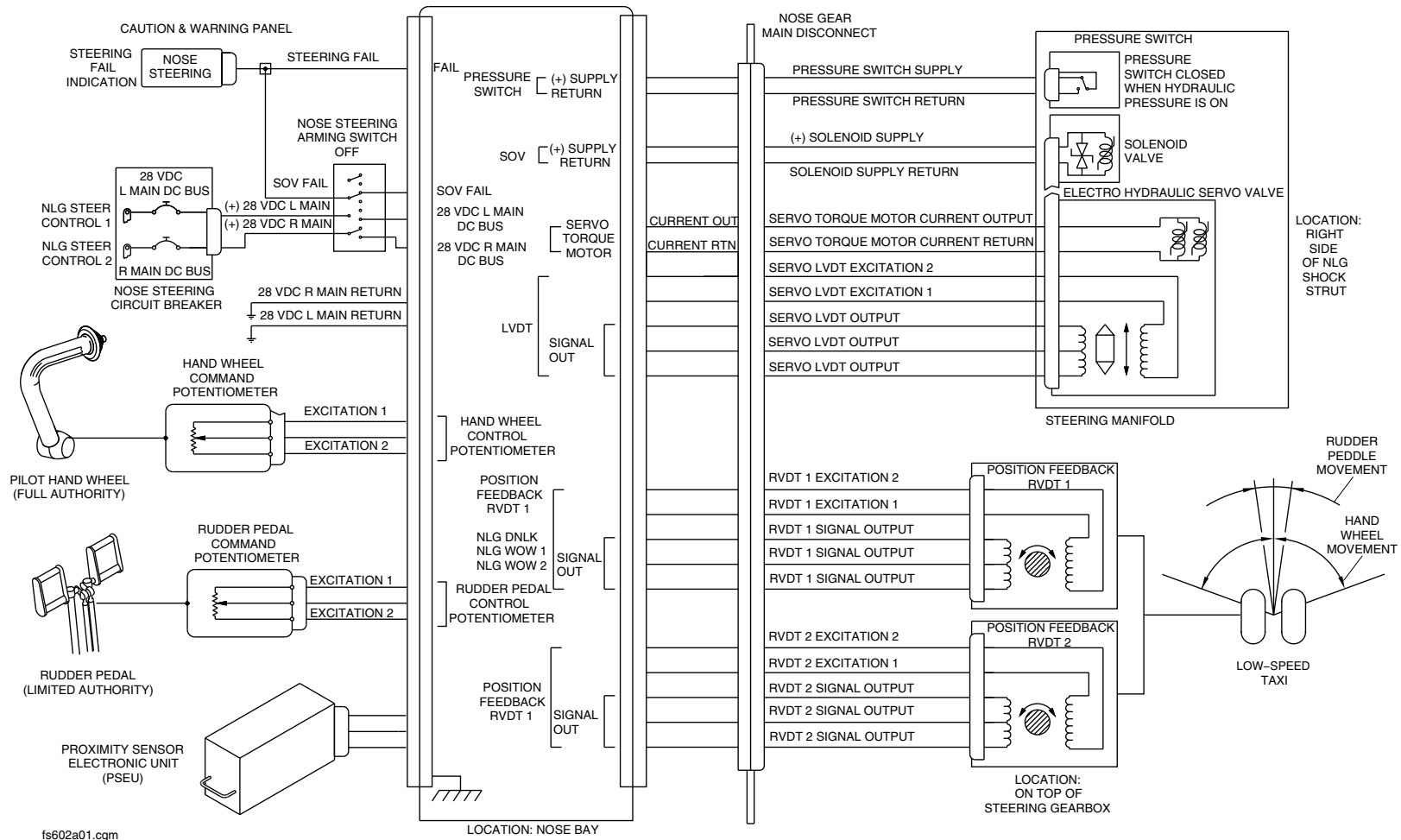
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Nosewheel Steering Electrical Synoptic
Figure 2

PSM 1-84-2A
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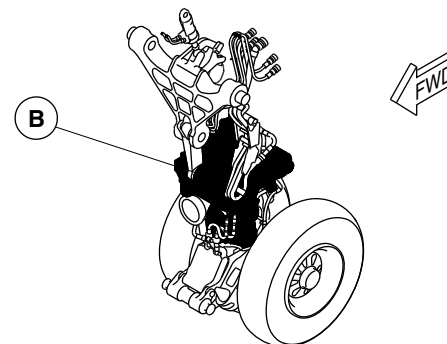
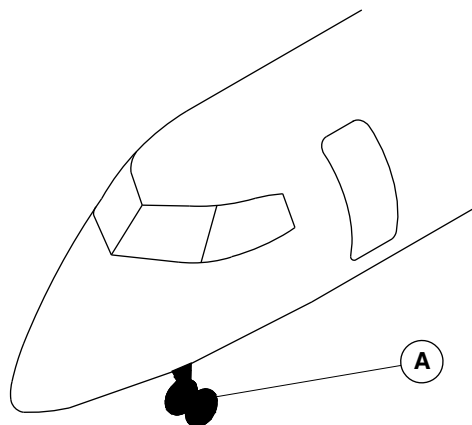
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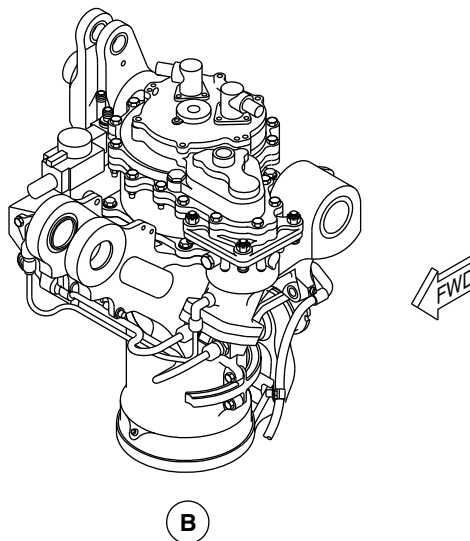


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A NOSE LANDING GEAR ASSEMBLY



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Nosewheel Steering Assembly Locator
Figure 3

PSM 1-84-2A
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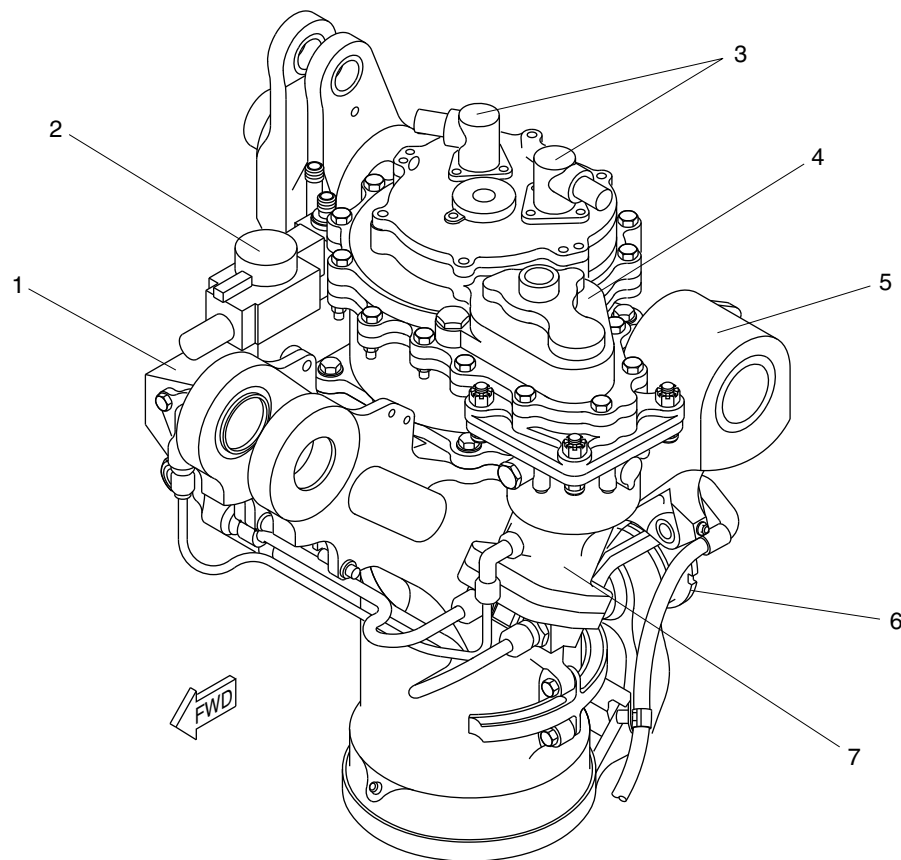
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LEGEND

- 1. Steering Manifold.
- 2. Electro Hydraulic Servo Valve.
- 3. Rotary Variable Differential Transformer.
- 4. Reduction Gearbox.
- 5. Shock Strut.
- 6. Centering Actuator.
- 7. Hydraulic Motor.

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Nosewheel Steering Assembly Detail
Figure 4

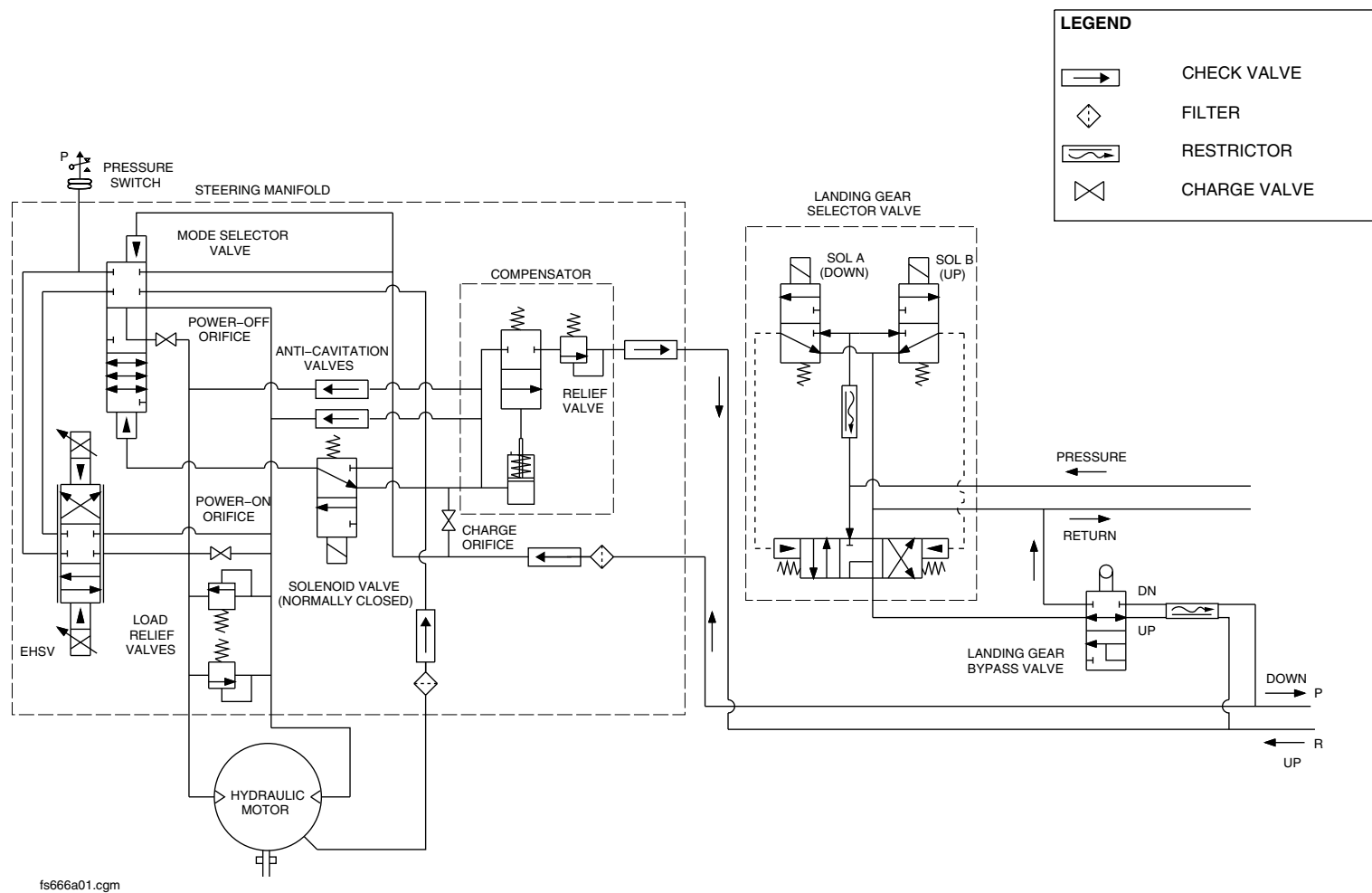
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Nosewheel Steering Hydraulic Synoptic
Figure 5

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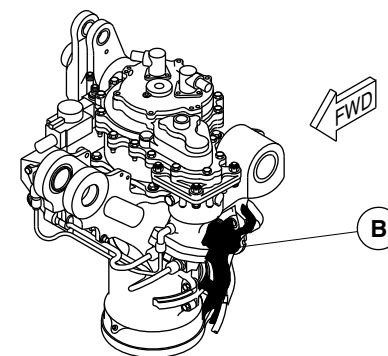
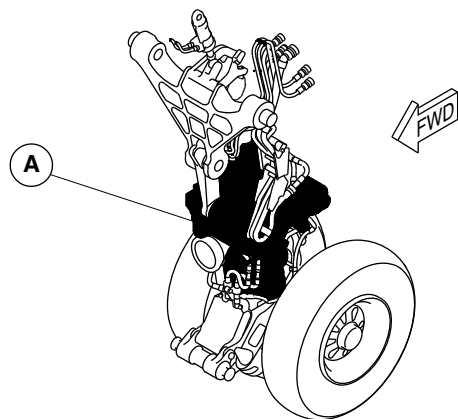
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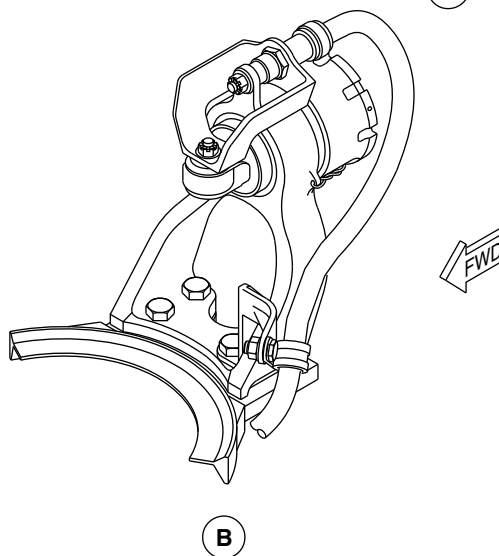


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A NOSEWHEEL STEERING ASSEMBLY



fs491a01.cgm

Nosewheel Centering Actuator Locator
Figure 6

PSM 1-84-2A
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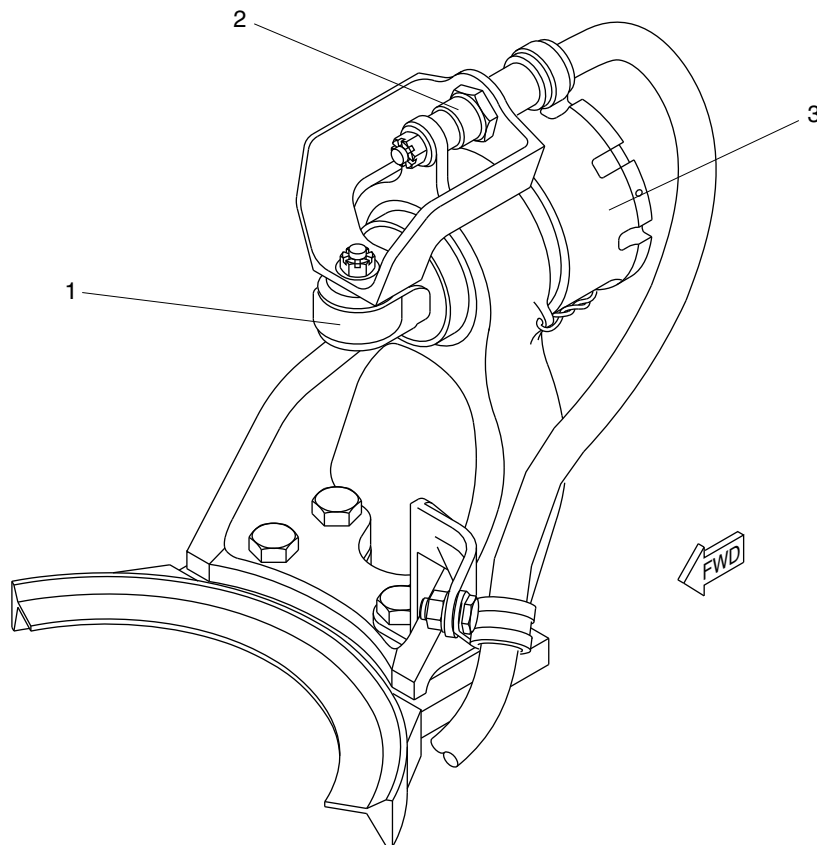
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LEGEND

- 1. Roller Bearing.
- 2. Proximity Sensor.
- 3. Centering Actuator.

fs492a01.cgm

Nosewheel Centering Actuator Detail
Figure 7

PSM 1-84-2A
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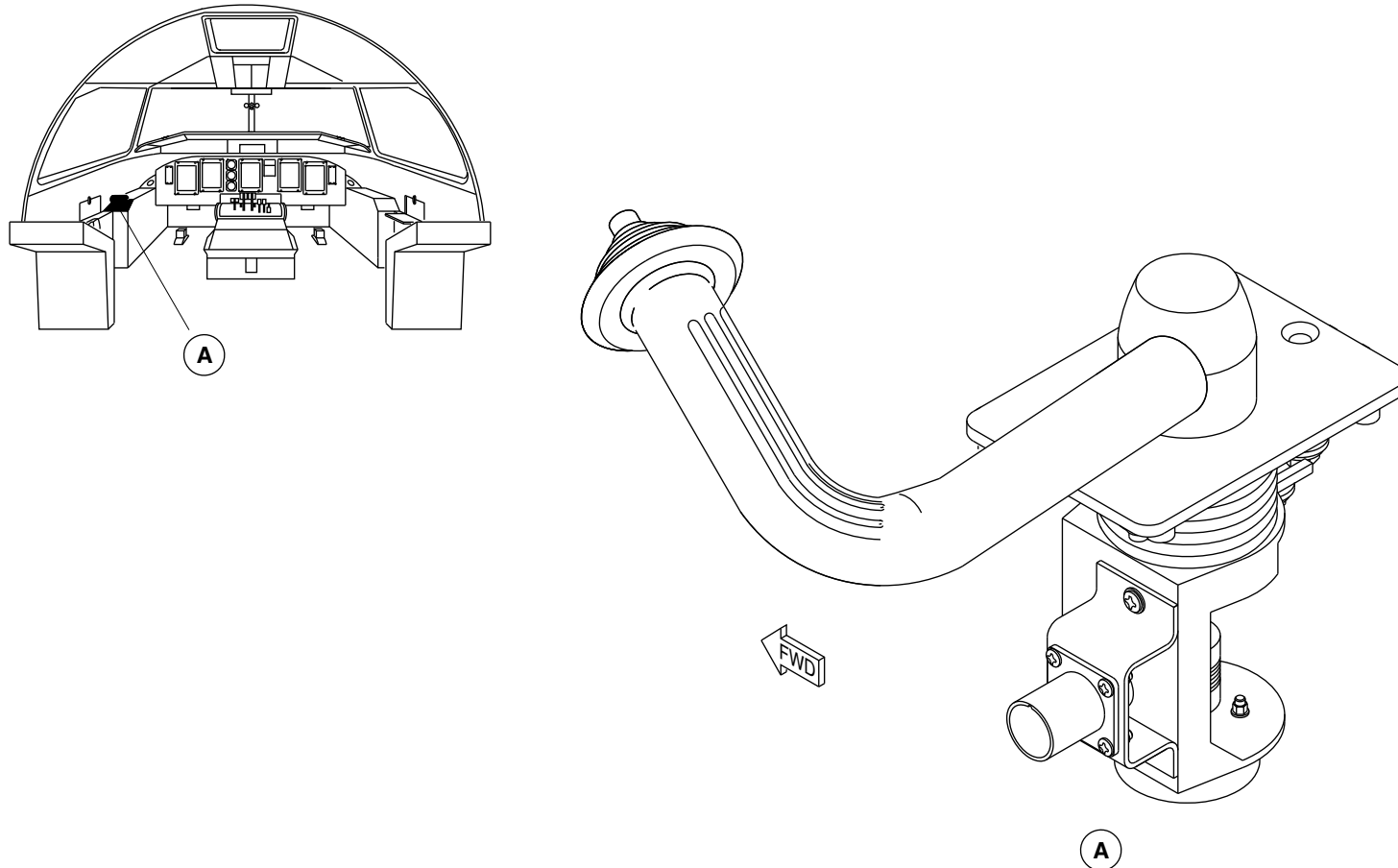
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Pilot Steering Hand Control Locator
Figure 8

PSM 1-84-2A
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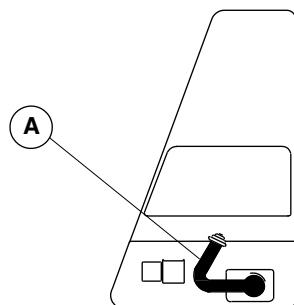
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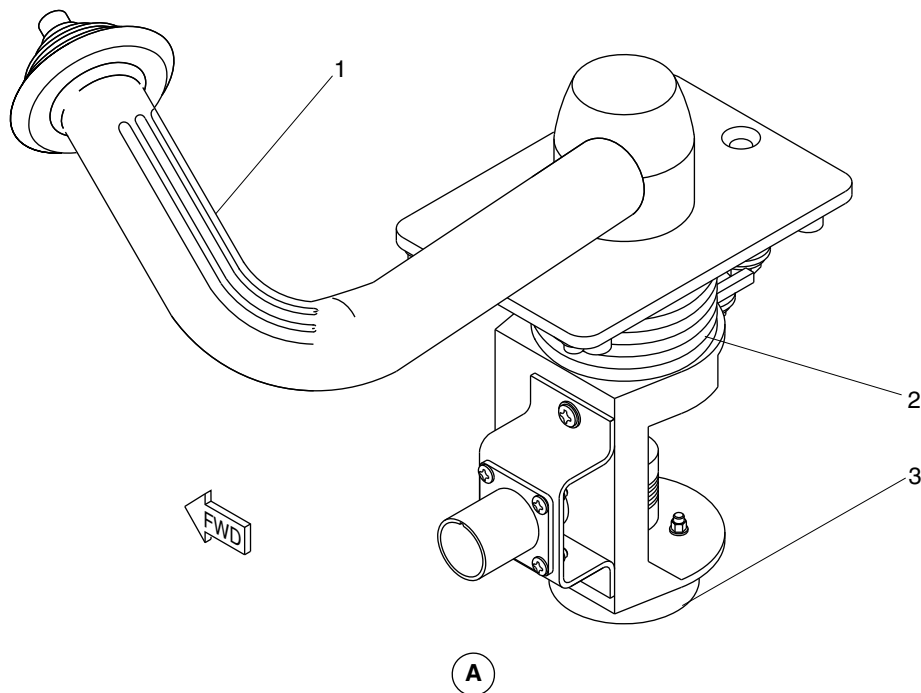
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEFT SIDE CONSOLE

LEGEND

- 1. Pilot Hand Control.
- 2. Centering Spring.
- 3. Potentiometer.



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Pilot Steering Hand Control Detail
Figure 9

PSM 1-84-2A
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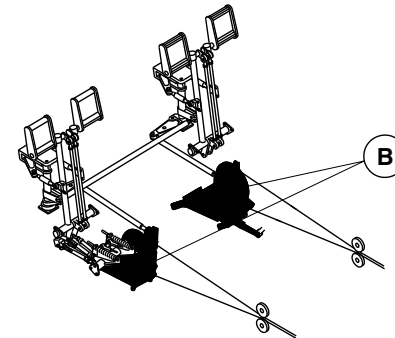
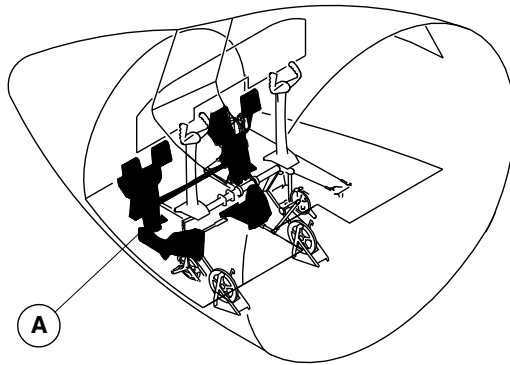
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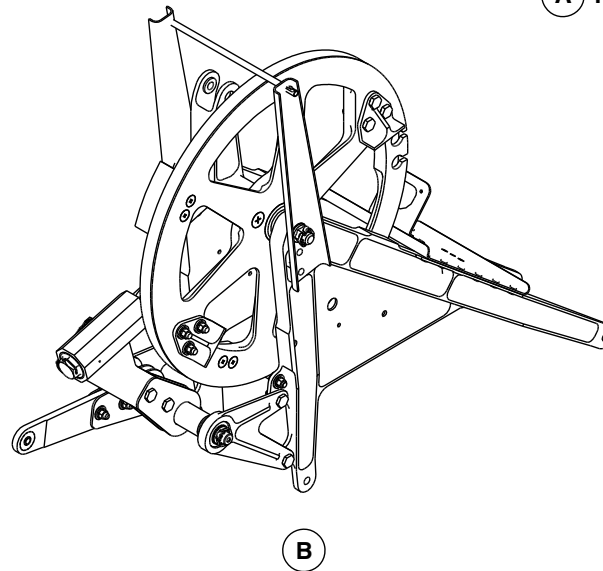


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A RUDDER FORWARD ASSEMBLY



NOTE

Right side shown.
Left side similar.

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Rudder Forward Assembly Locator
Figure 10

PSM 1-84-2A
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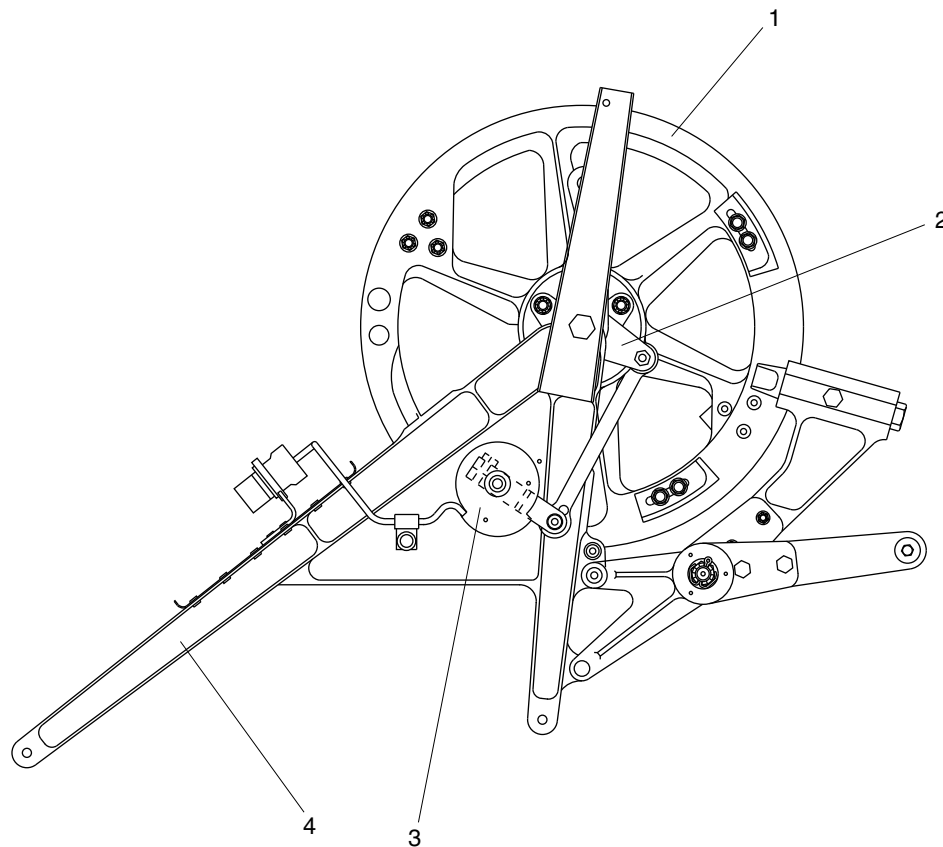
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LEGEND

- 1. Rudder Quadrant.
- 2. Linkage Assembly.
- 3. Rudder Pedal Potentiometer.
- 4. Support Structure.

fs488a01.cgm

Rudder Forward Assembly Detail
Figure 11

PSM 1-84-2A
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POSITION AND WARNING

Introduction

The landing gear position lights show the position of each landing gear and the related doors.

General Description

Refer to Figures 1 and 2.

The landing gear position lights are operated by the uplock and the downlock sensors through the Proximity Switch Electronics Unit (PSEU).

The landing gear position lights come on to show the conditions that follow:

- The landing gears are down and locked
- The landing gears are not locked up or down
- The landing gear doors stayed open after landing gear extension or retraction
- The landing gears are not in the up and locked position or the down and locked position.

An alternative indication of the landing gear extension is controlled by the PSEU (not an interdependent system) and given by the visual indicators. The visual indicators are installed in the flight compartment behind the LANDING GEAR ALTERNATE

EXTENSION door. A PARKING BRAKE caution light, on the Caution and Warning panel in the flight compartment comes on when the parking brake is moved to the PARK position.

A landing gear configuration warning is sounded in the flight compartment by the Remote Control Audio Unit (RCAU). The landing gear configuration warning will sound if all the landing gears are not in a safe condition for the landing (Refer to 32-62-00).

A HORN MUTE and TEST switch is installed on the landing gear selector panel.

Detailed Description

Refer to Figure 1.

The green LEFT, NOSE, and RIGHT lights come on as each landing gear is locked in the down position. The red LEFT, NOSE, and RIGHT lights come on as each landing gear moves from the down and locked position or the up and locked position. The amber L DOOR, N DOOR, and R DOOR lights come on as the doors open and go out after the doors have closed.

An amber light in the LANDING GEAR selector lever comes on when the landing gear is in transition from either the up and locked or down and locked position. This amber light and the three red lights in the landing gear selector panel are on at the same time.

Landing Gear Position Indication

The landing position lights are installed on the LANDING GEAR selector panel in the flight compartment, on the right side of the engine instrument panel. An amber light is supplied for each set of doors. The lights are identified as L DOOR, N DOOR, and R DOOR.

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A green light and a red light are supplied for each landing gear. The lights are identified as LEFT, NOSE, and RIGHT.

The green LEFT, NOSE, and RIGHT lights come on as each landing gear is locked in the down position. The red LEFT, NOSE, and RIGHT lights come on as each landing gear moves from the down and locked position or the up and locked position. The amber L DOOR, N DOOR, and R DOOR lights come on for as long as the doors stay open and go out after the doors have closed.

The DIM/BRT switch is used to adjust the light intensity of the landing gear selector panel lights. The TEST CAUT/ADVSY switch is used to test the LANDING GEAR selector panel lights.

An amber light in the landing gear selector lever comes on when the landing gear is in transition from either the up and locked or down and locked position. This amber light and the three red lights in the landing gear selector panel are on at the same time.

Sensors and Targets

Refer to Figure 4.

The proximity sensors are stable conductors. The inductance of a sensor changes when the ferromagnetic stainless steel target comes near. The targets move in relation to the sensors when the landing gears move during extension and retraction. When the related part of the landing gear is in the closed or locked position, the target is "near" the sensor.

The target increases the inductance value of the sensor above the PSEU sensor circuit switch point. Each sensor is connected to a sensor circuit in the PSEU. The sensor circuit is switched on when the inductance of the sensor increases to the switch point. This occurs when the target is brought within approximately 0.050 inch

(1.27 millimeter) from the sensor face. This is a target "near" condition. The reverse occurs when the target is taken away, a target "far" condition. The PSEU logic circuits operate from the "near" or "far" switching signals.

Alternate Indication System

Refer to Figure 3.

Verification of the alternate indication of the landing gear down and locked condition is as follows:

- The LANDING GEAR ALTERNATE EXTENSION door in the flight compartment floor is opened
- The landing gear downlock verification switch is set
- The three green landing gear downlock verification lights are monitored. If the landing gear is down and locked, all three lights will be on.

Training Information Points

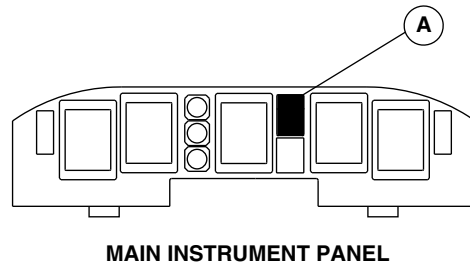
Remove all electrical power from the aircraft before you do maintenance. Put warning placards at the external power receptacle and in the flight compartment. This is necessary to prevent electrical shock to persons and/or damage to the equipment.

Obey all the electrostatic-discharge safety precautions when you do maintenance on or near electrostatic-sensitive discharge devices. If you do not do this, you can cause damage to equipment.



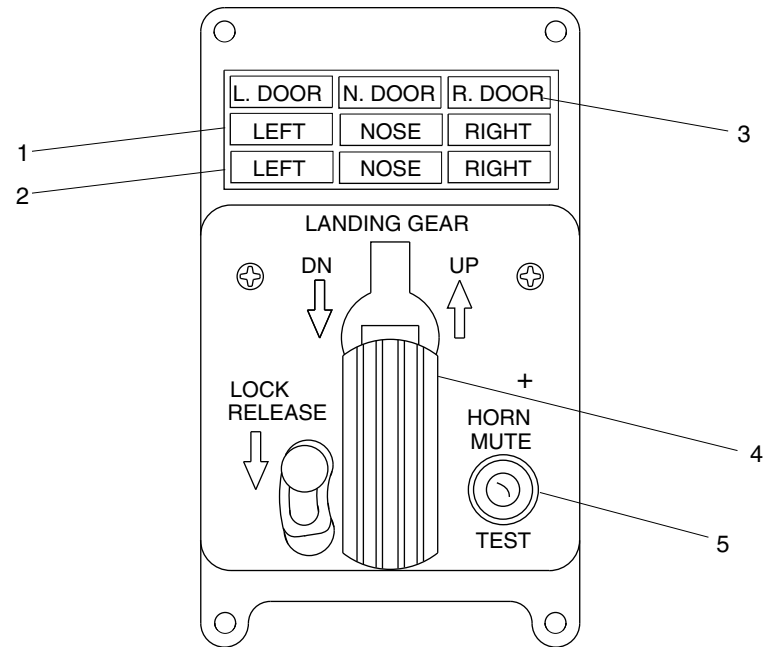
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LEGEND

1. "Green" Gear Downlocked.
2. "Red" Gear Not Uplocked Or Downlocked.
3. "Amber" Door Open.
4. Handle "Amber" (Concurrent W/"Red" Unsafe Lights) Gear In-Transit.
5. Warning Horn Mute/Test Switch.



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Landing Gear Selector Panel
Figure 1

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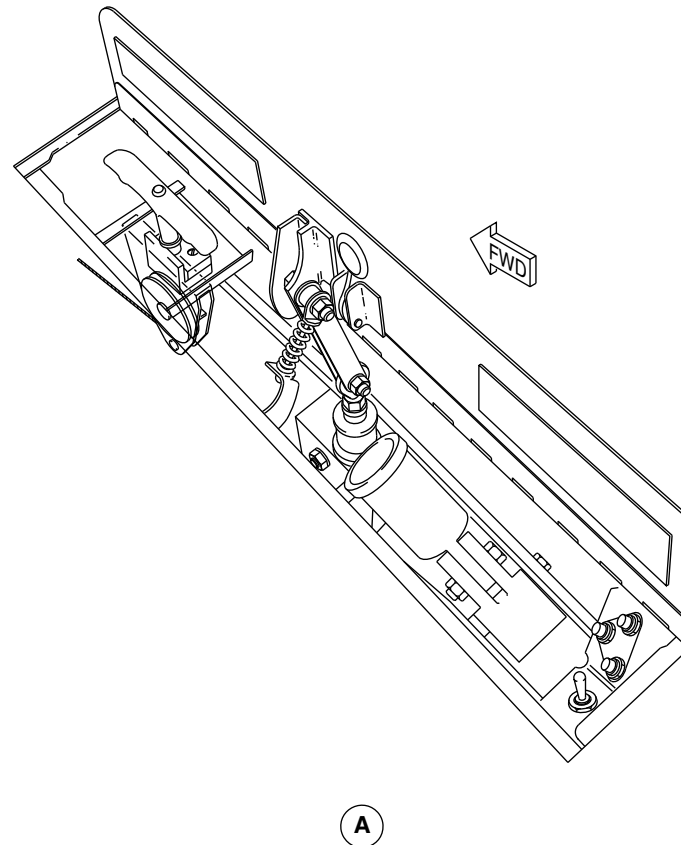
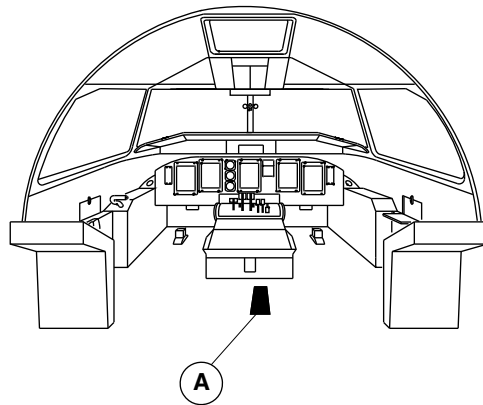
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Position and Warning, Landing Gear Alternate Extension Door Locator
Figure 2

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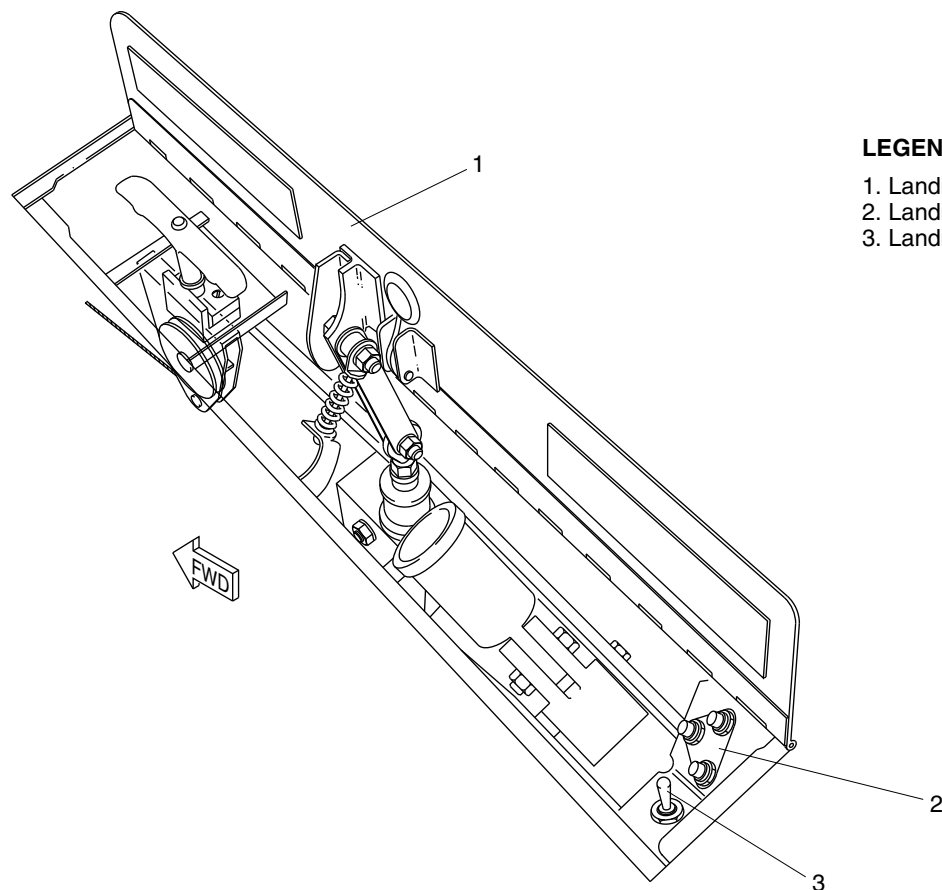
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LEGEND

- 1. Landing Gear Alternate Extension Door.
- 2. Landing Gear Downlock Verification Lights.
- 3. Landing Gear Downlock Verification Switch.

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Landing Gear Alternate Extension Door Detail
Figure 3

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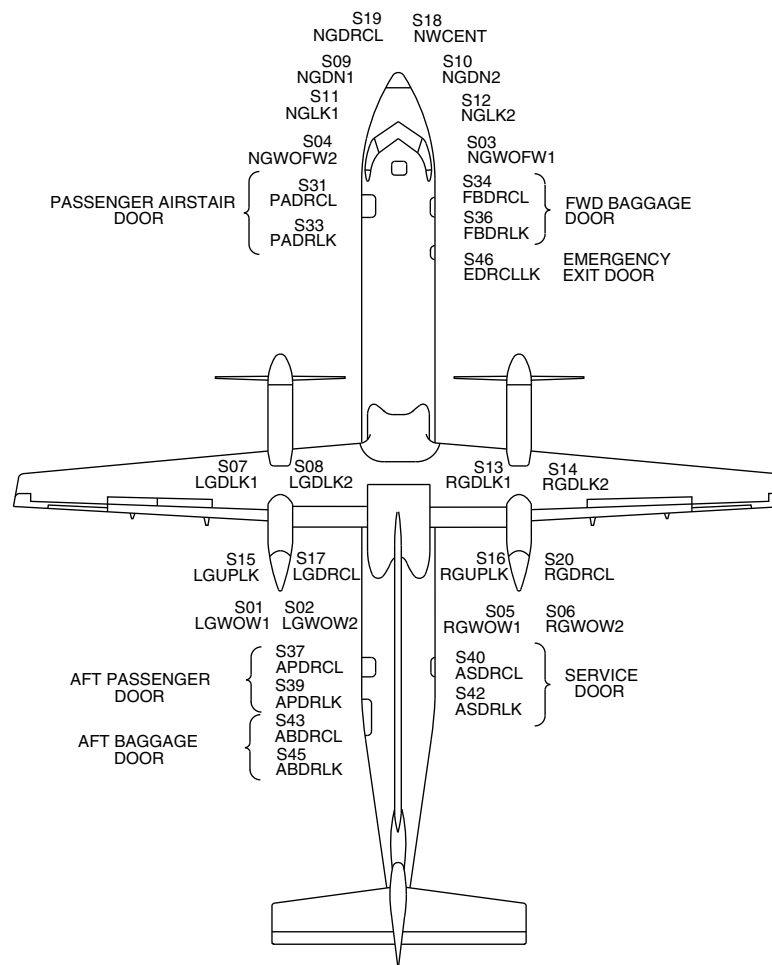
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Proximity Switch Locations
Figure 4

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PROXIMITY SENSOR SYSTEM

Introduction

The Proximity Sensor System (PSS) is a part of the landing gear system. The system gives:

- Landing Gear Control (LGC)
- Landing Gear status Indication (LGI)
- Alternate Downlock Indication (ADI)
- Weight–On–Wheels indication (WOW1 and WOW2)
- Fuselage Door Status Indication (FDSI) to the flight compartment and other aircraft systems.

General Description

The Proximity Sensor System (PSS) has a Proximity Sensor Electronics Unit (PSEU) (Refer to SDS32–61–01) and proximity sensors.

[Refer to Figure 1.](#)

The PSS uses 31 proximity sensors and 5 discrete inputs to give the position of the landing gear, the gear doors and the fuselage doors. The outputs of the sensors are processed by the PSEU. The PSEU gives 104 identified discrete outputs to control the landing gear. They also show the position of the landing gear components, aircraft on ground, gear doors and fuselage doors.

A logic equation controls each discrete output that fully explains its operation. The logic equation uses sensor and discrete inputs to control the outputs. There are 51 logic equations used by the PSEU to control the discrete outputs (Refer to SDS32–61–01).

Detailed Description

[Refer to Figure 2.](#)

The PSS is a 28 V DC energized, microprocessor based proximity system. All power inputs and outputs are sent through the Proximity Switch Electronics Unit (PSEU). The system has signal conditioning, proximity sensors, discrete monitoring and control signals, interface wiring and a Built–In–Test Equipment (BITE) keypad and display.

The four discrete inputs transmitted to the PSS are as follows:

- Landing gear extension
- Landing gear retraction
- ALTERNATE EXTENSION INHIBIT switch, set to INHIBIT
- Landing–gear lamp test set.

System Functions

Landing Gear Control (LGC)

LGC Overview

The Landing Gear Control Function supplies discrete outputs to control the movement of the landing gear (extend or retract) and landing gear doors (open or close). Twenty proximity sensors and three discrete inputs are used by the PSEU to give seven discrete

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outputs that supply the landing gear control function. The three discrete inputs are:

- Extend
- Retract
- Landing Gear Inhibit.

The outputs of the PSEU control the aircraft functions that follow:

- Landing Gear Doors Closure
- Extend/Retract Landing Gear

LGC Functions

Landing Gear Doors Closure

On aircraft without SB 84–32–66 and ModSum 4–126420 and ModSum 4–110293 incorporated: The PSEU gives three output signals which independently control the closure of each landing gear door. There is one output for each of the landing gear doors. The specified Door Sequence Valve for the main and nose gear controls the closure of the landing gear doors. The gear doors are closed when the landing gear is down and locked, and when the gear is up and locked. The gear doors are only open when the gear is in transit.

On aircraft with SB 84–32–66 or ModSum 4–126420 or ModSum 4–110293 incorporated: The PSEU gives three output signals which independently control the closure of each landing gear door. There is one output for each of the landing gear doors. The specified Door Sequence Valve for the main and nose gear controls the closure of the landing gear doors. The gear doors will start to close 10 seconds after the landing gear is down and locked, and the transition of the doors to fully closed condition is in 1 second or less after the 10 seconds delay. When the landing gear is up and locked, the gear

doors close immediately without delay. The gear doors are only open when the gear is in transit.

For the Nose Gear Door Sequence Valve output, the primary and alternate Nose Gear Down and Locked proximity sensors (S09 to 12) are used to close the landing Gear doors. When the gear is down and locked (as shown by the proximity sensors) and the Landing Gear Handle is in the “Extend” position, the output will initiate the door closure. When the Landing Gear Handle is in the “Retract” position, and the sensors show that the nose gear is down and not locked, the output will initiate the door closure. If the Landing Gear Inhibit signal is activated, the PSEU does not control the door sequence valve.

To close their related gear doors the right and left main gears use the proximity sensors that follow:

- Right Gear Downlock #1 (S13)
- Right Gear Downlock #2 (S14)
- Right Gear Uplock (S16)
- Left Gear Downlock #1 (S07)
- Left Gear Downlock #2 (S08)
- Left Gear Uplock (S15).

The gear door solenoid valve energizes and the doors close when you get the conditions that follow:

- When the Landing Gear Handle is in the “Extend” position
- The proximity sensors show that the main gear is down and locked.

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The gear sequence valve will close the gear door when you get the conditions that follow:

- When the Landing Gear Handle is in the "Retract" position
- The uplock proximity sensors show that the gear is up and locked.

Should the Landing Gear Inhibit signal come on, the door valves will not be controlled by the PSEU

Landing Gear Extend

The PSEU has two discrete inputs (the Extend Command and the Landing Gear Inhibit) to generate the Extend Selector Valve output. When the Landing Gear Handle is in the Extend position and the LGI switch is not on, an output signal will extend the landing gears.

Should the Inhibit signal become active, the Extend Selector Valve output will deactivate.

An additional design feature is that the Extend discrete output goes back through a switch in the gear handle. Thus, only when the handle is in the Extend position does the power go to the extend valve. This safety feature makes sure that the handle and the PSEU have the same conditions before the gear can move.

Landing Gear Retract

To supply the Retract Selector Valve output, the PSEU uses the input from many proximity sensors. The sensors used and the areas that they come from are as follows:

- Weight on Wheels (WOW1 and 2); sensors S01 to S06
- Landing Gear Indication
- Landing Gear Doors closed; sensors S17, S19 and S20

- Nose Gear Locked; sensor S11
- Main Gear Uplocked; sensors S15 and S16
- Alternate Downlock (Nose Gear Locked); sensor S12
- Landing gear Control (Nose Centered); sensor S18.

Proximity sensor S18 (NWCENT) monitors if the nosewheel is in its center position. You can find sensor S18 on the nosewheel shock strut. This sensor prevents gear retraction if the nosewheel is not in its center position. The Retract Selector Valve output also uses two discrete input signals; Retract Command and Landing Gear Inhibit. The WOW signals prevent the retraction of the landing gear while on the ground. The Retract Selector Valve discrete output is activated only when you get all of the conditions that follow:

- The Landing Gear Control Handle is in the Retract position
- The WOW sensors report that the aircraft is in the air
- The nosewheel is in the center position
- The Landing Gear Control is not inhibited
- A landing gear door is open or a landing gear component is not in the uplock condition.
- The retract output is deactivated 3 seconds after the last gear door is closed.

The same design property for the Extend discrete output is also used for the Retract output. The PSEU Retract output goes back through the gear handle switch. Thus, only when the handle is in the Retract position does the power go to the retract valve. This safety property makes sure that the handle and the PSEU have the same conditions before the gear can move.



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Landing Gear Indication (LGI)

LGI Overview

The Landing Gear Indication function supplies discrete outputs to the Landing Gear Control and Indication Panel (LGCIP). This gives the primary indication to the crew of the position of the landing gear and gear doors. The LGI function monitors the conditions that follow:

- The Right, Left and Nose Gear Door Position (open or closed)
- The Right, Left and Nose Gear Safe (down and locked)
- The Right, Left and Nose Gear Unsafe Position (gear in transit).

The Landing Gear Indication function uses 13 proximity sensors to monitor the position of the landing gear and the landing gear doors. These positions are the gear down and locked, the gear up and locked and the gear doors closed.

The outputs of the sensors are processed by the PSEU. The PSEU uses four discrete inputs for this function. One discrete input is from the Advisory Control Unit (lamps test) and two are from the Landing Gear handle (Retract and Extend). Landing Gear Inhibit is the fourth discrete input used by the PSEU for this function. The PSEU supplies sixteen discrete outputs to the components that follow:

- The Input Output Modules (IOM1 and 2) in the Integrated Flight Cabinets (IFC 1 and 2) (Flight Data Recorder)
- The Landing Gear Control and Indication Panel (LGCIP)
- The Caution and Warning Panel (CAWP)

- The Antiskid/Brake Control Units (ACUs) (Inboard and Outboard).

LGI Functions

Left, Right and Nose Landing Gear Door Position Monitoring

One proximity sensor for each set of gear doors monitors the left, right and nose landing gear doors status (open or closed). The right MLG door position is monitored by sensor S20 (RGDRCL). The left MLG door position is monitored by sensor S17 (LGDRCL). The nose gear door position is monitored by sensor S19 (NGDRCL). When the proximity sensor senses a target FAR condition with a landing gear door, an amber indicator light comes on, on the LGCIP. This shows that the door is open.

You can find the sensors for the left and right MLG doors on the MLG door mechanisms. You will find the sensor for the NLG doors on the NLG door mechanism.

Left and Right Landing Gear Position Monitoring

The left and right main landing gears are monitored for two conditions. These conditions are landing gear safe (down and locked) and landing gear unsafe (the gear position does not agree with the handle position). The LGCIP gives an indication for gear safe and gear unsafe conditions.

One (primary) sensor monitors the left and right landing gear to make sure of the landing gear's downlock status. Whenever a downlock proximity sensor senses a target NEAR condition, a green "safe" light comes on for that landing gear. These lights come on only when the landing gear is down and locked. You can find the left (sensor S07, LGDLK1) and right (sensor S13, RGDLK1) sensors, for the landing gear down lock status, on the stabilizer brace lock link assembly.



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These proximity sensors also make sure that the landing gear's position agrees with the position of the landing gear handle. For example, if the gear handle is in the Extend position, the red "unsafe" light comes on if the specified gear is not down and locked. Also, if the gear handle is in the retract position, the red light comes on if the specified gear is not up and locked.

One (primary) sensor monitors the left and right landing gear to make sure of the landing gear's uplock status. You can find the left (sensor S15, LGUPLK1) and right (sensor S16, RGPUPLK1) sensors, for the landing gear uplock status, on the uplock assembly. There is one sensor for each set of gear. During the normal extension sequence, if one of the three gear "safe" lights are not on, the Alternate Downlock Indication is consulted. Or if one of the three gear "unsafe" lights come on, the Alternate Downlock Indication is consulted.

Nose Gear Monitoring

Two proximity sensors monitor the NLG position status. One sensor senses if the landing gear is down, and the other one senses if the landing gear is locked. You can lock the nose gear in the extended or the retracted position. If the sensors show that the nose gear is in the down and locked position, then the green "safe" light, on the LGCIP, will come on. This will occur regardless of the position of the Landing Gear Control Handle.

However, if the handle is in the Extend position and the nose gear is not down and locked, the red "unsafe" light on the LGCIP will come on. If the Landing Gear Handle is in the Retract position and the sensors show that the gear is not up and locked, the red "unsafe" light will come on. You can find the nose gear down sensor (S09) on the nose gear wheel compartment airframe bracket. You can find the nose gear locked sensor (S11) on the nose gear drag strut assembly.

Alternate Downlock Indication (ADI)

ADI Overview

The Alternate Downlock Indication function is not an independent system. The alternate downlock indication function is supplied from the PSEU, providing another visual indication of the downlock status of the landing gears. The crew consults the ADI if the primary Landing Gear Indication function shows "unsafe" gear. Since the ADI is not an isolated function, it uses the same sensors and mostly the same PSC cards within the PSEU to process inputs and supply outputs.

The Alternate Downlock Indication function monitors the conditions that follow:

- Right gear down and locked
- Left gear down and locked
- Nose gear down and locked.

The left gear and the right gear each have two sensors that provide signals, reporting the landing gear is down and locked. The nose gear has four proximity sensors. Two sensors provide signals for the down position and two provide signals reporting the NLG is locked. The sensors used by the ADI function are the same sensors used in the primary LGI function.

The PSEU uses eight proximity sensors and one discrete input from the Advisory Control Unit to supply outputs for the aircraft systems that follow:

- Alternate Downlock Lamp Panel
- Nosewheel Steering System
- Taxi Light Contactor

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- Integrated Flight Cabinet 1 (IFC1), Input Output Modules 1 and 2 (IOMs 1 and 2)

There is one discrete input from the Advisory Control Unit for the ADI function. This discrete input is a lamp test. If a crew member pushes the Lamp Test 2 button on the ACU, all three lights on the Alternate Downlock Lamp Panel come on. This procedure is only a test for the lamp operation.

ADI Functions

Left and Right Landing Gear Down and Locked

There is one proximity sensor on the left MLG and one on the right MLG to make sure that the landing gear is down and locked. You can find the sensor for the right landing gear (S12, RGDLC2) on the right stabilizer brace lock link assembly. You can find the sensor for the left landing gear (S08, LGDLK2) on the left stabilizer brace lock link assembly.

When the sensor shows that the gear is down and locked, the PSEU supplies a discrete output to the ADI panel. This output turns on a green light for each main gear.

Nose Gear Down and Locked

To make sure that the nose gear is down and locked the PSS uses two proximity sensors. You can find the sensor S10 (NGDN2) on the nose gear compartment airframe bracket. This sensor makes sure that the landing gear is down. The other sensor makes sure that the landing gear is locked. You can find this sensor (S12, NGLK2) on the nose gear drag strut assembly.

When the nose gear is down and locked, the PSEU supplies a discrete output which turns on a green light on the ADI Panel for the nose gear.

Weight-on-Wheels (WOW)

WOW Function Overview

The Weight-on-Wheels (WOW) function supplies many discrete outputs to other aircraft systems. WOW discrete signals are active if the aircraft is on the ground. Weight-off-Wheels (WOFW) discrete signals are active if the aircraft is in the air. The WOW function uses six proximity sensors and the PSEU to tell you if the aircraft is on or off the ground.

Two not connected channels, Weight-on-Wheels 1 (WOW1) and Weight-on-Wheels 2 (WOW2), supply the weight-on-wheels discrete outputs. WOW1 and WOW2 have three proximity sensors at each channel. These sensors sense the compression of the landing-gear shock struts to find if the aircraft is in the air (Air Mode) or on the ground (Ground Mode). WOW1 and WOW2 are also used to supply a redundant check of the aircraft status if a proximity sensor fails.

The PSEU puts together the six proximity sensor outputs of WOW1 and WOW2 to give many WOW discrete outputs to other aircraft systems. Each WOW channel also supplies an output that is unique to that channel. The conditions monitored by this function are:

- Left Gear Weight-on-Wheels
- Nose Gear Weight-off-Wheels
- Right Gear Weight-on-Wheels



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WOW Functions

Left Gear Weight-on-Wheels

Two proximity sensors monitor the Left Gear Weight-on-Wheels status. You can find the WOW1 sensor (S01, LGWOW1) and the WOW2 sensor (S02, LGWOW2) side by side on the left gear shock strut. These sensors monitor the compression of the shock strut to find if there is weight on the left landing gear wheels. The PSEU uses output from these sensors to give many signals.

Nose Gear Weight-off-Wheels

Two proximity sensors monitor the Nose Gear Weight-off-Wheels status. You can find the WOW1 sensor (S03, NGWOFW1) and the WOW2 sensor (S04, NGWOFW2) side by side on the nose-gear shock strut. These sensors monitor the compression of the shock strut to find if there is weight off the nose landing-gear wheels. The PSEU uses output from these sensors to give many signals.

Right Gear Weight-on-Wheels

Two proximity sensors monitor the Right Gear Weight-on-Wheels status. You can find the WOW1 sensor (S05, RGWOW1) and the WOW2 sensor (S026, RGWOW2) side by side on the right gear shock strut. These sensors monitor the compression of the shock strut to find if there is weight on the right landing gear wheels. The PSEU uses output from these sensors to give many signals.

Weight-on-Wheels Signals

The PSEU supplies five types of signals that are used in WOW1 and in the WOW2 channels. Many aircraft systems use the five signals that follow as inputs:

- Weight-on-Wheels “Ground Mode” (WOW). See logic equation 36 (Refer to SDS32-61-01).
- Weight-on-Wheels “Main Gear” (MGWOW). See logic equation 37 (Refer to SDS32-61-01).
- Weight-off-Wheels “Main Gear” (MGWOFW). See logic equation 38 (Refer to SDS32-61-01).
- Weight-on-Wheels “Nose Gear” (NGWOW). See logic equation 39 (Refer to SDS32-61-01).
- Weight-off-Wheels “Nose Gear” (NGWOFW). See logic equation 40 (Refer to SDS32-61-01).

WOW Signal

The PSEU uses the WOW signal to make sure that there is weight on all of the landing gear wheels.

The PSEU uses all six WOW proximity sensor outputs. The WOW outputs are active in the conditions that follow:

- If all six WOW sensors operate correctly, then five of the six sensors are necessary to show that the aircraft is on the ground.
- If one of the sensors becomes defective, then four of the remaining five sensors are necessary to show that there is weight on the wheels.
- If two WOW sensors from the same WOW system (WOW1 or WOW2) are defective, then three of the remaining four sensors will show that the aircraft is on the ground.



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- If one full WOW system is defective (three WOW sensors from the same WOW system), then two of the three sensors from the other system are necessary to show that there is weight on the wheels.
- If one or more WOW sensors are defective on the two WOW systems (WOW1 and WOW2), the WOW outputs will default to the inactive, fail-safe state (air mode).

MGWOW Signal

The MGWOW signal uses four proximity sensor inputs. Two inputs are from the left gear and two are from the right gear. They show if the main gear is on the ground (WOW). The MGWOW outputs are active in the conditions that follow:

- If all four of the main gear WOW sensors operate correctly, then three of the four sensors are necessary to show that the aircraft is on the ground.
- If one of the main gear WOW sensors becomes defective, then two of the remaining three sensors have to show that there is weight on the wheels.
- If one full WOW system is defective (two WOW sensors from the same WOW system), then one of the two sensors from the other system is necessary to show that there is weight on the wheels.
- If one or more main gear WOW sensors are defective on the two WOW systems (WOW1 and WOW2), the main gear WOW outputs will default to the inactive, fail-safe state (air mode).

MGWOFW Signal

The MGWOFW outputs use four WOW proximity sensor inputs. Two inputs are from the left gear and two are from the right gear. They show if the main gear is off the ground (WOFW). The MGWOW outputs are active in the conditions that follow:

- If all four of the main gear WOW sensors operate correctly, then three of the four sensors are necessary to show that the aircraft is in the air.
- If one of the main gear WOW sensors becomes defective, then two of the remaining three sensors have to show that there is no weight on the wheels.
- If one full WOW system is defective (two WOW sensors from the same WOW system), then one of the two sensors from the other system is necessary to show that the aircraft is in the air.
- If one or more main gear WOW sensors are defective on the two WOW systems (WOW1 and WOW2), the main gear WOW outputs will default to the inactive, fail-safe state (air mode).

NGWOW Signal

The NGWOW outputs use the two nose gear WOFW proximity sensors to show if the nose gear is on the ground (WOW). The NGWOW outputs are active in the conditions that follow:

- If the two nose gear WOFW sensors operate correctly, then the two sensors are necessary to show that the aircraft is on the ground.
- If one of the nose gear WOFW sensors becomes defective, then the other sensor has to show that there is weight on the wheels of the nose gear.



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- If the two nose gear WOFW sensors are defective, the NGWOW outputs will default to the inactive, fail-safe state (air mode).

NGWOFW Signal

The NGWOFW outputs use input from the two nose gear WOFW proximity sensors to show if the nose gear is in the air (WOFW). The NGWOFW outputs are active in the conditions that follow:

- If the two nose gear WOFW sensors operate correctly, then the two sensors are necessary to show that the aircraft is on the ground.
- If one of the nose gear WOFW sensors becomes defective, then the other sensor has to show that there is no weight on the wheels of the landing gear.
- If the two nose gear WOFW sensors are defective, the NGWOW outputs will default to the inactive, fail-safe state (air mode).

WOW (1 and 2) Operational

There are two discrete outputs supplied by the PSEU that are special to WOW1 and WOW2 systems. These signals are Weight-on-Wheels 1 Operational (DOT13E) and Weight-on-Wheels 2 Operational (DOT13A), respectively. These two signals are supplied in the same way (Refer to SDS32-61-01 for logic equations 41 and 42).

The CAWP use these two discrete outputs to show that the WOW function operates correctly. If one of the two signals becomes deactivated, on the CAWP, the WT ON WHEELS caution light comes

on. For the caution light to come on, the WOW Operational discrete output will deactivate in the conditions that follow:

- There is a disagreement between two of the WOW sensors (in the same WOW system) that the aircraft is either on the ground or in the air.
- A minimum of one proximity sensor becomes defective in each of the WOW1 and WOW2 systems at the same time.

Fuselage Doors Status Indications (FDSI)

FDSI Overview

The Fuselage Doors Status Indication function gives a visual indication in the flight compartment when a fuselage door “open” or “unlocked” condition occurs. The doors that are monitored by this function are as follows:

- Passenger Airstair Door
- Aft Passenger Door
- Forward Baggage Door
- Aft Baggage Door
- Aft Service Door
- Emergency Exit Door

Eleven proximity sensors monitor the door status. The PSEU monitors the outputs of the sensors. PSEU Door Status outputs are sent to the Cabin Pressure Control System (CPCS) and the Integrated Flight Cabinet 1 and 2 (IFC 1 and 2). The IFC sends these signals to the Electronic Flight Information System (EFIS) and the Caution and Warning Panel (CAWP).



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FDSI Monitoring Functions

Passenger Airstair Door Status Monitoring

Two proximity sensors monitor the passenger airstair door. You can find one sensor on the top aft side of the door. You will find the other sensor installed adjacent to the door seal pressurizing valve.

Sensor S31 (PADRCL) makes sure that the door is closed. Sensor S33 (PADRLK) makes sure that the door is locked when the mechanism is fully over center and in safety. When either sensor detects a “sensor far” condition, the FUSELAGE DOORS warning light, on the CAWP, comes on. You can make sure that the airstair door is closed and locked with the position of the internal handle.

Aft Passenger Door Status Monitoring

Two proximity sensors monitor the aft passenger door. You can find one sensor (S37) at the aft upper roller guide on the fuselage frame. You will find the other sensor (S39) installed at the door handle.

Sensor S37 (APDRCL) makes sure that the door is closed. Sensor S39 (APDRLK) makes sure that the door is locked. When either sensor detects a “sensor far” condition, the FUSELAGE DOORS warning light, on the CAWP, comes on. You can make sure that the aft passenger door is closed and locked when the position of the external handle is flush with the door skin.

Forward Baggage Door Status Monitoring

Two proximity sensors monitor the forward baggage door. You can find one sensor (S34) at the aft upper roller guide on the fuselage frame. You will find the other sensor (S36) installed in the door handle mechanism.

Sensor S34 (FBDRCL) makes sure that the door is closed. Sensor S36 (FBDRLK) makes sure that the door is locked. When either sensor detects a “sensor far” condition, the FUSELAGE DOORS warning light, on the CAWP, comes on. You can make sure that the forward baggage door is closed and locked when the position of the external handle is flush with the door skin.

Aft Baggage Door Status Monitoring

Two proximity sensors monitor the aft baggage door. You can find one sensor (S43) on the lower fuselage frame. You will find the other sensor (S45) installed at the door handle. Sensor S43 (ABDRCL) makes sure that the door is closed. Sensor S45 (ABDRLK) makes sure that the door is locked. When either sensor detects a “sensor far” condition, the FUSELAGE DOORS warning light, on the CAWP, comes on.

You can make sure that the aft baggage door is closed and locked when the door seal inflates. The door must be closed and locked for the door seal to pressurize.

Aft Service Door Status Monitoring

Two proximity sensors monitor the aft service door. You can find one sensor (S40) at the upper fixed roller position. You will find the other sensor (S42) installed in the door handle mechanism.

Sensor S40 (ASDRCL) makes sure that the door is closed. Sensor S42 (ASDRLK) makes sure that the door is locked. When either sensor detects a “sensor far” condition, the FUSELAGE DOORS warning light, on the CAWP, comes on. You can make sure that the aft passenger door is closed and locked when the position of the external handle is flush with the door skin.

Emergency Exit Door Status Monitoring

PSM 1-84-2A

EFFECTIVITY:

See first effectivity on page 2 of 32-61-00

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One proximity sensor monitors the emergency exit door. You can find the sensor (S46, EDRCLK) on the upper part of the door frame. When a sensor detects a “sensor far” condition, the FUSELAGE DOORS warning light, on the CAWP, comes on. You can make sure that the aft passenger door is closed and locked when the position of the outer door skin is flush with the aircraft.

FDSI Outputs

Doors Status Monitoring Function Input to the Pressurization System

The PSEU uses all fuselage door proximity sensors to supply a discrete output to the Cabin Pressure Controller (CPC). This signal (DOT13B) is “All fuselage doors closed and locked”. This signal tells the CPC to inhibit the automatic pressurization modes of the aircraft until all doors are closed and locked. When the proximity sensors show that all of the fuselage doors are closed and locked, the PSEU will change the output to let the automatic pressurization modes operate (Refer to SDS32–61–01 for logic equation 49). There is a 2 second persistence time on this output such that nuisance faults are minimized when the aircraft moves down the runway. This means that the output will stay active for 2 seconds after a door is opened or unlocked.

Doors Status Monitoring Function Input to the Caution and Warning Panel

The PSEU uses all fuselage door proximity sensors to supply a discrete output (all fuselage doors closed and locked) to the Caution and Warning Panel (CAWP). The FUSELAGE DOORS warning light, on the CAWP, comes on if one of the aircraft doors is not closed and locked. The Master Warning indicator also comes on.

The PSEU sends one discrete output (DOT10B) to the CAWP for all doors closed and locked (Refer to SDS32–61–01 for logic equation 49). There is a 2 second persistence time on this output such that nuisance faults are minimized when the aircraft moves down the runway. This means that the output will stay active for 2 seconds after a door is opened or unlocked.

Doors Status Monitoring Function Input to the Integrated Flight Cabinet

Each fuselage door uses its specified sensors to supply a discrete output to make sure that the door is closed and locked. These outputs go through the Integrated Flight Cabinet (IFC) to show on the Engine and System Integrated Display (ESID). The Fuselage Doors page of the ESID highlights the open or unlocked doors in amber. This lets the crew see which door is open or unlocked when the CAWP warning indicator comes on. There is a 2 second persistence time on this output such that nuisance faults are minimized when the aircraft moves down the runway. This means that the output will stay active for 2 seconds after a door is opened or unlocked. The output signals and their logic equation numbers are as follows:

NOTE

Refer to SDS32–61–01 for the logic equations in this table.

FUSELAGE DOOR	DISCRETE OUTPUT	LOGIC EQUATION
Aft Baggage Door	DOT11B	46
Aft Passenger Door	DOT12D	44
Passenger Airstair Door	DOT11A	43



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FUSELAGE DOOR	DISCRETE OUTPUT	LOGIC EQUATION
Forward Baggage Door	DOT12B	45
Emergency Exit Door	DOT12A	48
Aft Service Door	DOT12E	47

FDSI Annunciation

After a 2 second delay, you will get the indications that follow when a fuselage door is not closed and locked:

- The FUSELAGE DOORS warning light on the CAWP and the “Master Warning” light on the glareshield flash on
- The Fuselage Doors page on the ESID shows the affected door in “amber”.

The FUSELAGE DOORS warning light, on the CAWP, flashes on if one of the aircraft doors is not closed and locked. The Master Warning indicator also flashes. When the pilot pushes the Master Warning indicator to acknowledge the warning, the Master Warning indicator goes off. The FUSELAGE DOORS warning light stays on but does not flash. If the open or unlocked condition continues, the warning light will stay on. If a subsequent door open or unlock condition occurs, then the FUSELAGE DOORS warning light and the Master Warning indicator will flash after a 2 second delay.

System Power Input

The PSS is powered by 28 V DC. All power inputs and outputs are through the PSEU. The PSEU is powered from the right and the left main 28 V DC busses, and with the right and the left essential 28 V DC busses.

Each Proximity Sensing Card (PSC) in the PSEU has two power inputs. If one PSC input fails, power continues to be available from the second power supply. This will make sure that there is no interruption in the PSS operation. The table that follows shows the relation between the voltage input buses and the PSEU PSC circuit cards.

Table 1: Electrical Power Supply to PSEU Cards

28 V DC INPUT BUS (CB)	PSEU BOARD POWER	J1 PIN NO.
RIGHT MAIN (N6)	PSC D, PSC E	B4D, B4E
RIGHT ESSENTIAL (M6)	PSC D, PSC E, BITE	B4A, B4B
LEFT MAIN #1 (D6)	PSC B, PSC C	A7K
LEFT MAIN #2 (E6)	PSC A, BITE	A3F, A3G, A3H, A3J, A3K
LEFT ESSENTIAL #1 (G6)	PSC B, PSC C	A7F
LEFT ESSENTIAL #2 (F6)	PSC A	A5F, A5G, A5H, A5J, A5K



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The logic circuits of the PSEU use the "near" and "far" condition of the proximity sensor circuits to operate the applicable landing gear and door indications.

Refer to Figure 3.

The PSEU has four Proximity Signal Conditioning (PSC) circuit cards. A fifth card and a BITE circuit card can be installed to connect hardware and filtering components. Each PSC circuit card can do a Built-In-Test (BIT) function. Each circuit card can send a maximum of thirty discrete outputs, receive six discrete inputs and receive inputs from a maximum of eight proximity sensors.

Each PSC has two channels. Each channel has four proximity sensors, three discrete inputs, twelve discrete outputs, and two 28 V DC power buses. Each output has a hardware controlled, fail safe condition for computer failure or power loss. If one power input fails, electrical power continues to be available from a second power supply to make sure there is no interruption in the PSS operation.

Data shows on the front panel or remote display unit through an external, memory mapped RS232 Interface. Channel interrogation discretely are output through a discrete output control register. The outputs are requests for the proximity data from an applicable signal conditioner channel.

Built-In-Test Equipment

The Proximity Sensor System PSEU contains Built in Test (BIT) and BITE functions. These functions let the PSEU do the maintenance and functional testing of the full operation of the PSS.

The BIT function is contained on each PSEU PSC circuit card. It senses faults on that card or on that card's outputs. Included are sensor faults (short or open), discrete output faults (short, open,

failed) and any circuit faults on that board. Each PSC card then communicates this BIT data to the BITE card where it is kept. BIT is done continuously by the PSC cards.

BITE is contained on an isolated circuit card in the PSEU. This separation between the BITE and the remaining PSCs makes sure that there will not be a loss of the proximity sensor or discrete functions. A loss of functionality can be due to a BITE function failure.

The BITE supplies an individual signal for each of the PSCs that is a request for BIT data. An active signal on this line requests the Status and Current Fault data from the corresponding PSC. This data is kept in a non-volatile memory location by the BITE until the data can be saved. If there is a power interruption, the data will stay in the non-volatile memory. On investigation by maintenance personnel, the Status and Current Fault data can be used to make sure of the fault location.

The non-volatile memory of the PSEU records all failure and fault data which occurs in the Retraction Control System (RCS). The non-volatile memory can keep the fault history for thirty one flight legs. The "Fault History" sub-menu display will display the current flight leg "#0". The PSEU will display a complete list of reported faults (20) in the "Fault History" menu for each flight leg. The non-volatile memory contains the data that follows:

- An entry which contains alphanumeric messages for each failure and fault that occurred in the RCS
- An entry which shows that the non-volatile memory content was shown to the operator
- An entry for each Line Replaceable Unit (LRU) that is replaced and which has an LRU replacement code.



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Maintenance personnel enter this data and it is recorded in chronological sequence in the non-volatile memory. The non-volatile memory satisfies the system requirements for system diagnostics and maintenance interface.

The BITE also monitors six power inputs to the PSEU. Two discrete output signals, PSEU Operational (DOT2BT) and PSS Maintenance Check Required (DOT1BT), are supplied by the BITE board. This controls the condition of two indicators, PSEU OPER and PSS MAINT. You can find these indicators on the Maintenance Panel above the wardrobe. The PSEU Operational signal keeps the PSEU OPER lamp on when there is a minimum of one active power bus to all of the PSEU components. Also, the lamp will stay on if there are no current PSEU faults recorded. When the PSS MAINT Check Required lamp comes on (the PSEU Operational lamp goes off) it shows that there is a fault that has been recorded.

[Refer to Figures 4 and 5.](#)

You can use the test panel on the front face of the PSEU to get BITE data. You can select from six push buttons on the BITE keypad. Four more commands are reserved for ground support equipment inputs which can only be given by the BITE keypad. You can find the BITE INSTRUCTIONS label below the test panel which gives the procedures on how to operate the test panel. A list of codes that the operator uses in the test procedure is also given.

[Refer to Figure 6.](#)

The ON/OFF button turns the BITE display ON or OFF. When you change from OFF to ON, the BITE display always starts at a PRESENT FAULTS? prompt. If BITE is in the DETAILED STATUS ENABLE mode, you can push the ON/OFF button to end that mode.

You will push the menu button to return to the most recent, higher level prompt. The UP button gives you access to the previous fault or status display when you get a list of displays. The DOWN button sequences to the subsequent fault or status display on the list. When you push the YES button, the response is affirmative. When you push the NO button, the response is negative.

The Ground Support Equipment (GSE) uses the ERASE REQUEST to erase the recorded contents of the non-volatile memory. The HISTORY DUMP command causes a dump of all recorded fault history and rigging history data. The DETAILED STATUS ENABLE command starts the transmission of all PSC channel data through the BITE GSE connector. The BITE display is turned OFF (or stays OFF if not turned ON), and the BITE begins a real time transmission of all the PSC channel data. The DETAILED STATUS DISABLE command completes the transmission of the real time PSC channel data.

The PSEU uses the proximity sensors to receive input of the "near" or "far" position of the targets. The position of the targets of the different sensors give the landing-gear position data and the WOW data to the PSEU.

The PSEU uses the inputs from the components that follow:

- The landing gear
- The forward and aft passenger entrance doors
- The forward and aft baggage compartment doors
- The emergency exit door
- The service door.



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PSEU Fault Detection

The fault detection system finds the circuit logic of thirty one proximity sensor inputs and two discrete inputs from the LANDING GEAR selector lever. The circuit logic is based on the current PSEU mode of operation. This function senses other types of proximity sensor fault conditions, which possibly are not a proximity sensing fault.

The PSEU software uses the proximity sensor inputs from all the PSC channels to find out which PSEU mode of operation to use. If one or more PSC channel faults do not give valid information to the BITE, or is in interrogation mode, the circuit logic test will not be done. If this occurs, the PSEU will use the PSEU modes of operation that follow:

Table 2: PSEU Operating Mode Determination

PSEU OPERATING MODE	CONDITION
Weight-on-wheels	At least four of any of the six conditions that follow: LGWOW1 = True LGWOW2 = True RGWOW1 = True RGWOW2 = True NGWOFW1 = False NGWOFW2 = False
Gear down and locked	One of the two conditions that follow: # 1: Any one down and locked indication from all three landing gears: (LGDLK1+LGDLK2) (RGDLK1+RGDLK2) (NGDLK1*NGDLK1+LGDK2*LGDK2) #2: All sensors from any two gears indicate down and locked.



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PSEU OPERATING MODE	CONDITION
Gear in transit	At least two of any of the four conditions that follow: LGUPLK = False RGUPLK = False NGDLK1 = False NGDLK2 = False and at least four of any of the six conditions that follow: LGDLK1 = False LGDLK2 = False RGDLK1 = False RGDLK2 = False NGDN1 = False NGDN2 = False
Gear up and locked	No other PSEU mode of operation found

The software shows each circuit logic fault for each defined system component which fails to agree with the circuit logic condition for 30 ± 5.0 seconds in the PSEU operating mode. Circuit logic faults are not reported on the components that are defective. The PSEU can record fault data for a maximum of thirty one flight legs. Data older than that is overwritten.

The PSEU monitors the sensors and the six discrete inputs to find the status of the power input to the PSEU. An input power bus is defective if it stays inactive for 90 ± 10 seconds. The software prevents fault messages which are connected to communication failures to PSC cards that are not energized.

[Refer to Figure 7.](#)

For faults kept in the PSEU memory, a PSS MAINT light comes on, on the maintenance panel above the wardrobe. This shows that

there are faults recorded by the PSEU. The PSEU OPER light shows power is received by all internal PSEU components. There are no present internal PSEU faults when the PSEU OPER light is on.

Proximity Sensors and Targets

[Refer to Figure 8.](#)

There are thirty one proximity sensors installed in the aircraft. Each main landing gear has six proximity sensors and senses the conditions that follow:

- Two for WOW
- One for the uplock
- Two for the downlock
- One for the wheel bay doors closed.

The nose landing gear has eight proximity sensors and senses the conditions that follow:

- Two for WOW
- Two for the uplock
- Two for the downlock
- One for the wheel bay doors closed
- One for the nosewheel centered.

There are eleven more proximity sensors that monitor the closed and locked status of the passenger entrance door, the baggage compartment doors, the emergency exit doors, and the service door. The PSEU collects the output from the proximity sensors to do retraction control management functions.



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Twenty sensors, S01 through S20, and their targets are installed on the landing gear. Sensors S03, S04, S09 through S12, S18, and S19 are installed on the nose landing gear. Sensors S01, S02, S07, S08, S15, and S17 are installed on the left main landing gear. Sensors S05, S06, S13, S14, S16, and S20 are installed on the right main landing gear.

Eleven sensors and their targets are installed on the passenger, service, baggage, and emergency exit doors. S31 and S33 are installed on the forward passenger door. S34 and S36 are installed on the forward baggage door.

Sensors S37 and S39 are installed on the aft passenger door. Sensors S43 and S45 are installed on the aft baggage door. Sensors S40 and S42 are installed on the service door. Sensor S46 is installed on the emergency exit door.

Refer to Figure 9.

The proximity sensors are sealed, variable reluctance, devices. When a target moves into a position near the sensor face, the inductance of the sensor increases to a value where the PSEU finds a "near" target condition. When the target moves away from the sensor, the inductance of the sensor decreases to a value where the PSEU finds a "far" target condition. The PSEU uses the "near" and "far" sensor condition to logically find the condition of, and give, the discrete outputs from the PSEU.

The proximity sensors can become unserviceable in one of three modes:

- An open circuit with no output
- A short circuit output
- An output of serviceable range "far" because of drift.

You must replace a proximity sensor when it is unserviceable. You cannot repair the sensors. If a proximity sensor becomes unserviceable the PSEU will do one of the two:

- Default that sensor to its fail safe mode (air mode)
- Ignore that sensor and use the remaining non-faulted sensors to do the function when redundant proximity sensor inputs are available.

General System Troubleshooting

General system troubleshooting starts with indication faults that are found by the flight crew. The procedure to find the primary cause starts with the flight crew report of the problem. It then goes to the interrogation of the PSEU BITE panel to find the correct cause of the fault. When you use this panel and the data it contains, it will identify the defective component in the most satisfactory manner.

NOTE

You do all troubleshooting with the aircraft on the ground, the gear extended and the weight-on-wheels.

Indication Faults

The PSEU supplies a number of discrete outputs for the flight compartment indication. The outputs that follow are included:

- Landing Gear Control and Indication Panel (LGCIP)
- Alternate Downlock Indication (ADI)
- WOW Fail Caution and Warning Panel Indication
- LDG INOP Caution and Warning Panel Indication
- Fuselage Doors Caution and Warning Panel Indication



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Gear Control and Sequencing Faults

The PSEU supplies gear control and the sequence for correct operation of the landing gear from the inputs of the landing gear handle. This includes the aft gear doors. A brief description of the usual extension and retraction sequences are as follows:

Extend Sequence

- Handle is moved to the Extend (Down) position
- PSEU senses the handle's Extend position
- PSEU senses that the Landing Gear Inhibit Switch is not active
- PSEU turns on the Extend output which moves the Extend Selector Valve
- Hydraulic pressure opens the main and nose gear doors based on the selector valve position
- Gear extends to the downlock position
- PSEU finds that the downlock sensors are target "near"
- PSEU then activates the output to the Main and Nose Gear Door Sequence Valve
- Gear doors then close.

Retract Sequence

- Handle is moved to the Retract (Up) position
- PSEU senses the handle's Retract position
- PSEU senses that the Landing Gear Inhibit Switch is not active

- PSEU senses that the aircraft is in Air Mode (not weight-on-wheels)
- PSEU senses that the nose gear is in its center position (NWCENT sensor target is near)
- PSEU turns on the Retract output which energizes the Retract Selector Valve
- Hydraulic pressure opens the main and nose gear doors based on the selector valve position
- Gear retracts to the uplock position
- PSEU finds that the uplock sensors are target "near"
- PSEU then deactivates the output to the Main and Nose Gear Door Sequence Valve
- Gear doors then close.
- PSEU deactivates the Retract output 3 seconds after it finds the main and nose gear door sensors are target "near"

Basic troubleshooting procedures make sure that the PSEU gets the correct inputs and that it supplies the correct outputs.

Main Gear Door Cycling problems are caused by one of the two:

- Incorrect rigging of the gear door sensors. You must adjust the doors before you adjust the sensors.
- Incorrect rigging of the uplock sensors.

A typical scenario for this problem is as follows:

- The gear is retracted after take-off
- The aft gear doors close correctly



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- The aft gear door sensor is just barely target “near “
- Aft gear door sensor goes target “far” due to:
 - Hydraulic pressure turned off and the door relaxes
 - Air pressure on the door causes the door to move slightly
- PSEU turns on the Retract selector valve to the supply hydraulic pressure to the system
- Aft gear doors are forced closed (doors not open at this time)
- Hydraulic pressure hits the uplock assembly
- Uplock sensor goes target “far”
- Hydraulic pressure cycles the doors

NOTE

A “far” condition of the left or right sensor can cause those doors to cycle, the opposite doors to cycle or the two sets of doors to cycle.

Proximity Sensor Faults

Most of the proximity sensor faults are because of the aircraft wiring or in the rigging of the sensor. Before you replace a sensor make sure that you examine the wiring and the rigging.

The three basic fault messages for a proximity sensor fault are as follows:

- Proximity Sensor “Short”

This message shows that there is a short circuit in the wiring from the PSEU to the sensor or in the sensor itself. This short circuit can be a short of one of the two conductors to ground, or a short between the two conductors. Since the proximity sensor is a potted assembly, it is very unusual for a short to be in the sensor.

- Proximity Sensor “Open”

This message shows that there is an open circuit in the wiring from the PSEU to the sensor or in the sensor itself. Since the proximity sensor is a potted assembly, it is very unusual for an open to be in the sensor.



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– Proximity Sensor Unreasonable

The PSEU finds four aircraft operating modes based on all the proximity sensor inputs. The four modes are as follows:

- Gear Up and Locked – The aircraft is in the air with the gear in the uplocks (normal cruise)
- Gear Down and Locked – The aircraft is in the air with the gear down locked (approach, take-off)
- Gear In Transit – The aircraft is in the air and the gear is moving between uplock and downlock
- Weight-on-Wheels – The aircraft is on the ground with the gear compressed

If one proximity sensor does not match the aircraft mode, it is flagged as an “Unreasonableness” fault. For example, if the aircraft is in normal cruise and an uplock sensor becomes target “far”, a fault is logged. This is because a target far is not a “reasonable” condition for this sensor. Another example is when the aircraft is on the ground and one WOW sensor show Air Mode, this is Unreasonable.

NOTE

An Unreasonable proximity sensor is not a short or an open sensor. The sensor operates correctly, but is in the incorrect condition. Incorrect sensor rigging is the most probable cause for this type of fault.

BITE Message Explanation

The PSEU has a front panel BITE display and a keypad that interfaces to the BITE circuit card. The BITE circuit card gathers

status from each PSC card and keeps the faults in NVRAM. The PSEU functions are not disrupted when you turn on and use the BITE panel. You can use it when the aircraft is in flight or while the gear cycles when the aircraft is on jacks.

BITE User Interface

The BITE User Interface is responsible to show the fault, rigging and status data. It uses the keypad to scroll through the different menus. The BITE display shows the top level menus that follow:

- Present Faults
- Fault History
- Rigging Help
- Rigging History
- System Status
- Fault History Erase

When you turn the BITE display on, the “Present Faults” menu prompt is displayed. With each keypad selection of Yes, No, Up, Down, ON/OFF or Menu is interpreted. The next menu is then chosen based on the menu flow shown in the Top Level Menu Flow table that follows:



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Table 3: TOP LEVEL MENU FLOW

CURRENT MENU	NEXT MENU FOLLOWING BUTTON PUSH			
	YES	NO	UP	DOWN
PRESENT FAULTS?	Enter Present Faults	FAULT HISTORY?	ERASE FAULT HISTORY?	FAULT HISTORY?
FAULT HISTORY?	Enter Fault History	RIGGING HELP?	PRESENT FAULTS?	RIGGING HELP?
RIGGING HELP?	Enter Rigging Help	RIGGING HISTORY?	FAULT HISTORY?	RIGGING HISTORY?
RIGGING HISTORY?	Enter Rigging History	SYSTEM STATUS?	RIGGING HELP?	SYSTEM STATUS?
SYSTEM STATUS?	Enter System Status	ERASE FAULT HISTORY?	RIGGING HISTORY?	ERASE FAULT HISTORY?
ERASE FAULT HISTORY?	Begin Erasure of Fault History	PRESENT FAULTS?	SYSTEM STATUS?	PRESENT FAULTS?

If a keypad entry is not a valid response to the display prompt, the message “Button Inactive” is shown for a minimum of one second. After the “Button Inactive” message, the display prompt goes back to the message that was first on the display. The “Menu” keypad button returns you to the previous top level menu display. The “ON/OFF”

switch turns the BITE Display on and off. If there is no Valid PSC Data, all data from the specified channel is shown as dashes (– –).

If the display has been on for 15 ± 1 minutes, without a keypad entry, the BITE display will turn off.

Present Faults Menu Item

The “Present Faults” sub-menu shows faults which are currently present on the aircraft. These are hard faults that the PSEU sees at that time. The BITE system will show the full list of reported faults [Refer to Figure 10](#).

NOTE

There is no flight leg or indication of an intermittent fault shown for “Present Faults”.

If a shown fault clears and there is no keypad button activity, the display will be updated with the message “Fault No Longer Present” within 10 seconds. If “Fault No Longer Present” shows and the fault returns while there is no keypad button activity, the fault message will reappear on the display.

Fault History Menu Item

The “Fault History” sub-menu shows faults that occurred during previous flight legs and since the last Fault History Erase. The data that follows is also included:

- The number of the flight leg in which the fault occurred (leg #0 is the current flight leg....up to #31)
- An indication to show if the fault is intermittent.

The software can show the full list of reported faults [Refer to Figure 10](#). You will see only Flight Legs which contain faults.



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Rigging Help Menu Item

The “Rigging Help” sub-menu gives assistance during the installation and maintenance of the proximity sensors. Once you select a sensor, the first display line indicates “*****_XXXXXXXXXXXX”. The “*” characters show the abbreviation for that proximity sensor. The “X” characters show one of the messages that follow:

- OK – Valid Target Near condition
- TARGET_FAR – Valid Target Far condition
- TOO_CLOSE – Target Near condition with sensor/target gap too small
- TOO_FAR – Target Near condition with sensor/target gap too large

The second display line gives an indication of the relative position of the proximity sensor to the target.

Rigging History Menu Item

The “Rigging History” sub-menu gives a summary of all the logged instances of pending failure conditions for each proximity sensor. These failure conditions show that one sensor installation will change. This could be because of wear or because of damage. For example, the installation has a Valid Near condition, but the target is too near to the sensor’s face. A Rigging History fault is recorded to tell the mechanic to examine the installation of that sensor.

Only sensors which have one or more instances of a Rigging Fault kept in NVRAM are shown in the “Rigging History” list.

The first line for each proximity sensor with a Rigging Fault shows “*****_###”. The “*” characters show the abbreviation for that proximity sensor. The “###” characters show the number of Flight Legs during which this Rigging Fault was recorded.

The second line of the display shows the type of Rigging Fault identified with one of the messages that follow:

- NEAR_GAP_TOO_SMALL_ – Target Near condition with sensor/target gap too small
- NEAR_GAP_TOO_LARGE_ – Target Near condition with sensor/target gap too large
- FAR_GAP_TOO_SMALL_ – Target Far condition with sensor/target gap too small

System Status Menu Item

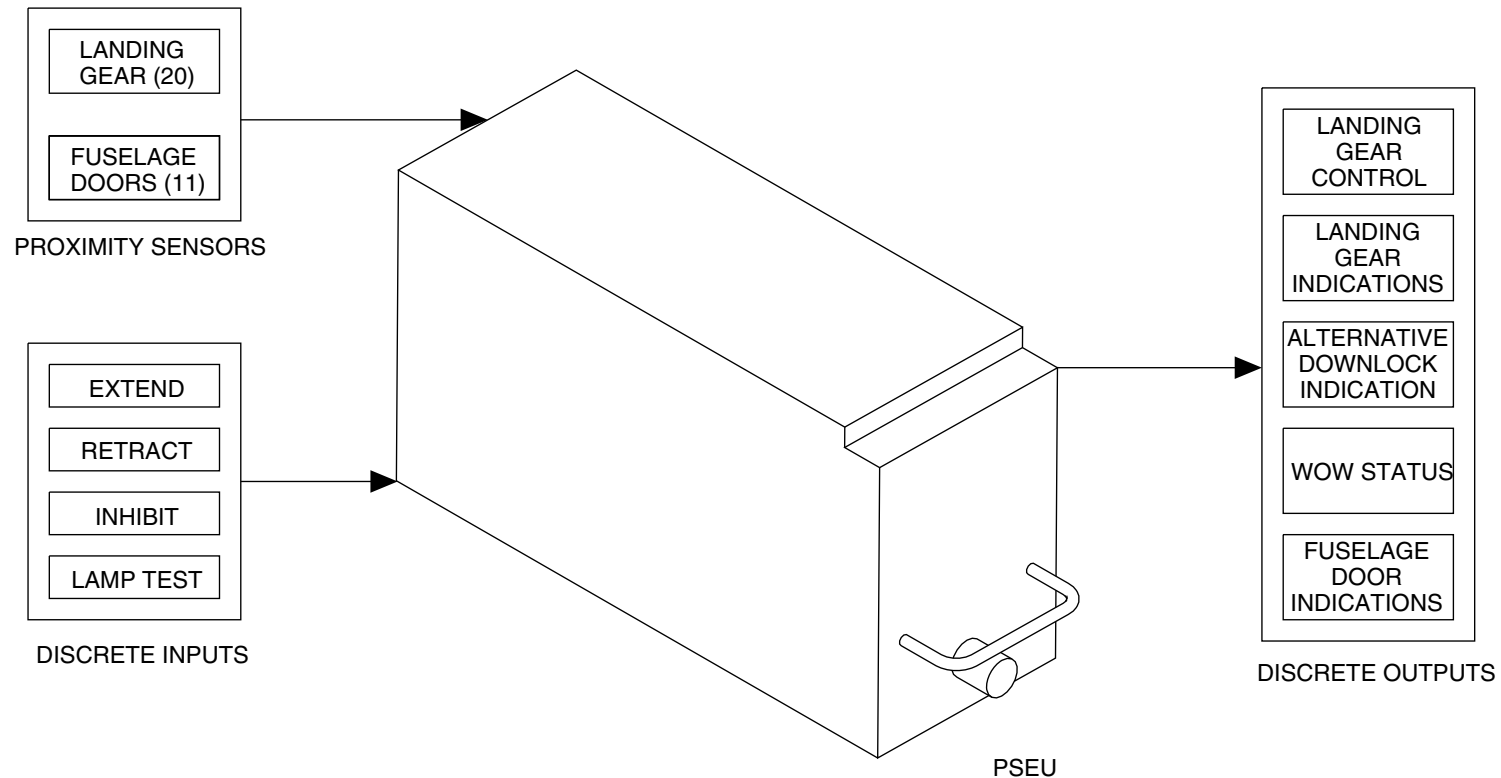
The System Status menu and sub-menus give the full status of all discrete inputs, discrete outputs and proximity sensors for each PSC card and for the BITE card. You can use this for detailed aircraft level troubleshooting. [Refer to Figure 11.](#) for a list of the System Status BITE messages.

Fault History Erase Menu Item

You use the Fault History Erase feature to erase the fault history from NVRAM. Before you erase the faults, it is necessary to confirm the erase function. With the confirmation prompt on the display, and you push the YES button on the keypad, you erase the NVRAM. When the NVRAM is clear of faults, the display then shows the ERASE COMPLETE prompt.



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Proximity Sensor System Inputs and Outputs
Figure 1

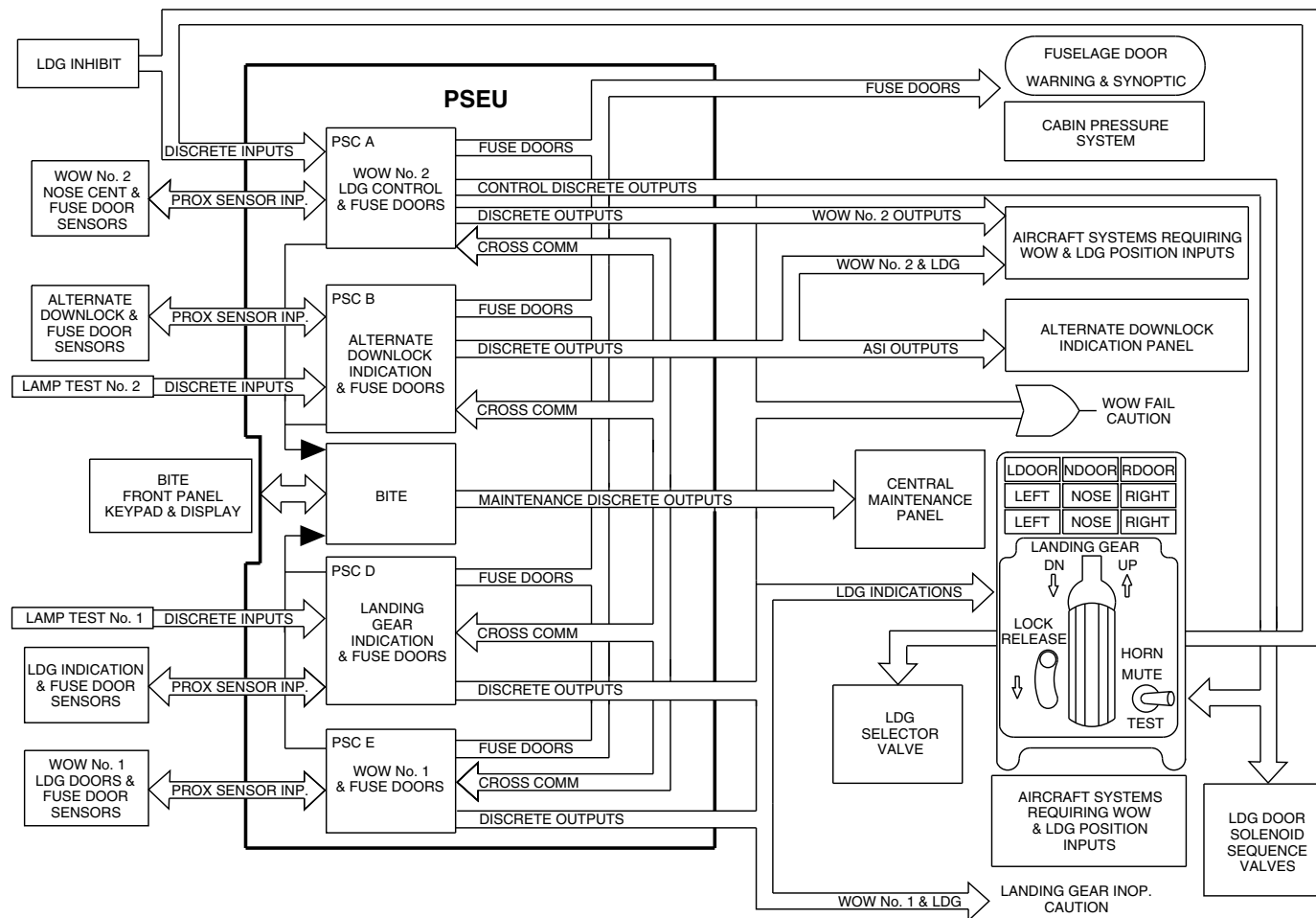
PSM 1-84-2A
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PSEU Inputs and Outputs Block Diagram
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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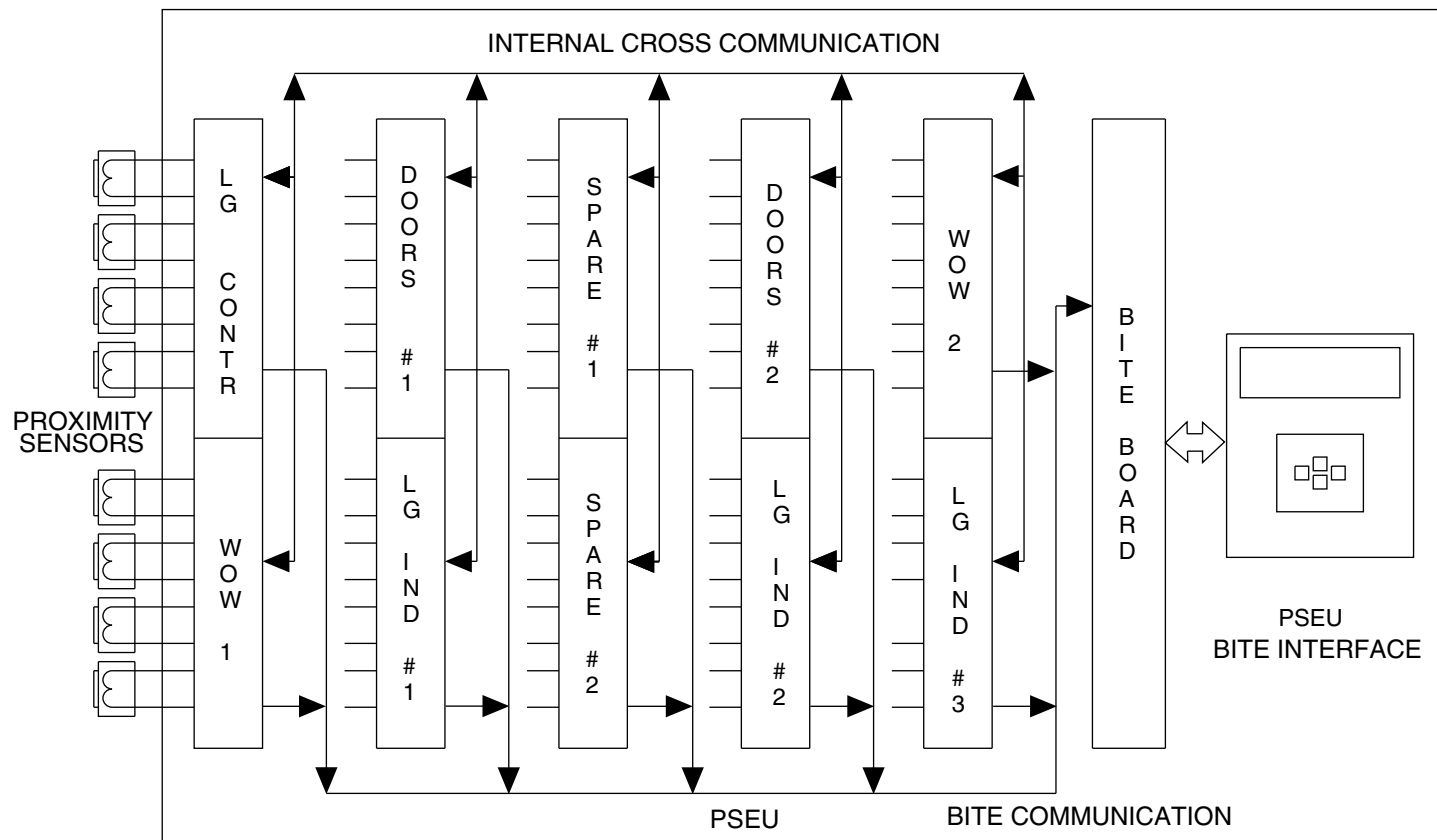
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PSEU Cards
Figure 3

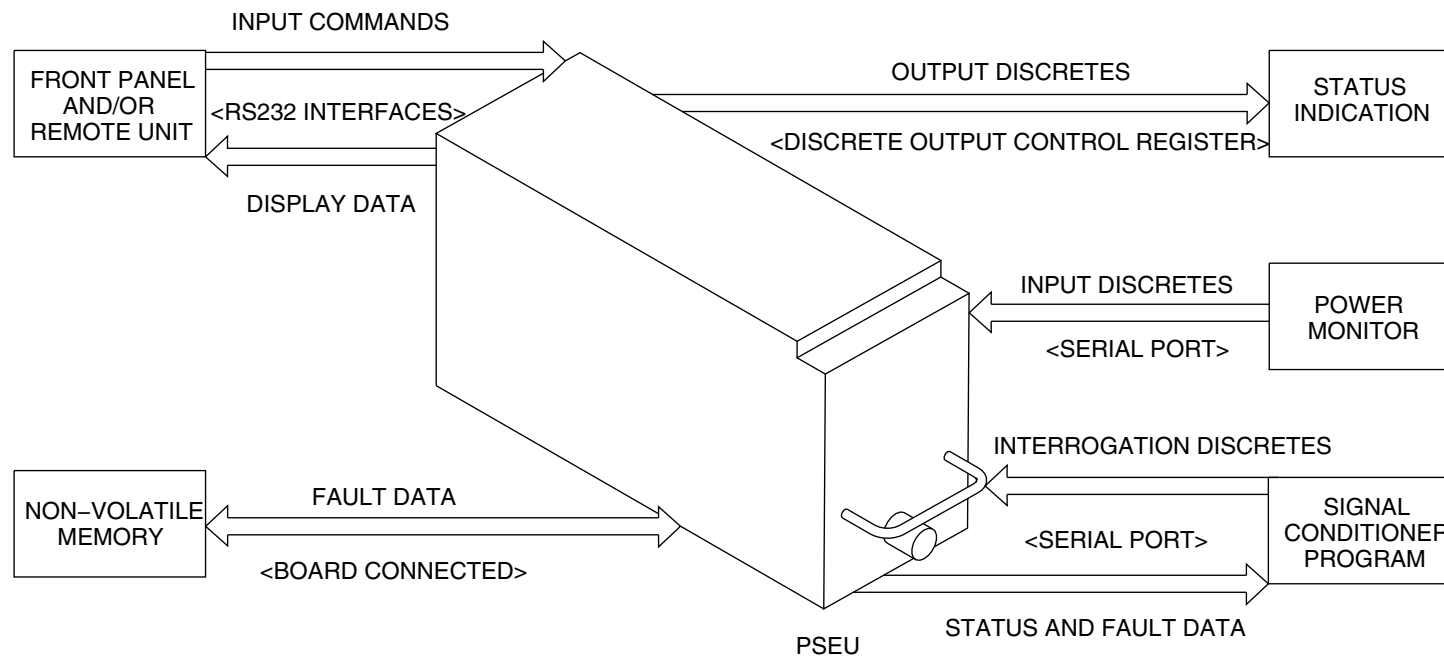
PSM 1-84-2A
EFFECTIVITY:
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BITE Interfaces, Inputs and Outputs
Figure 4

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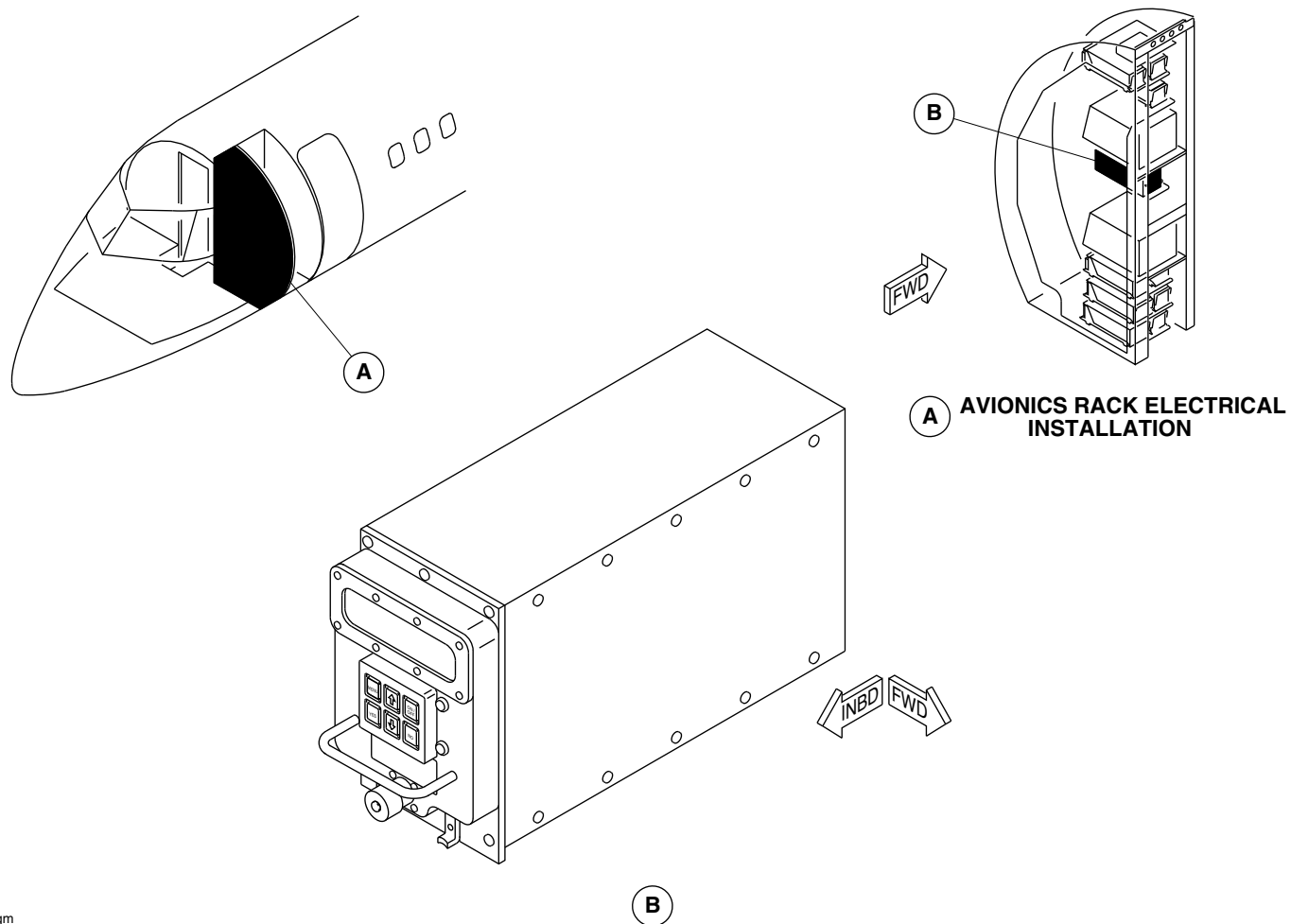
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PSEU Locator
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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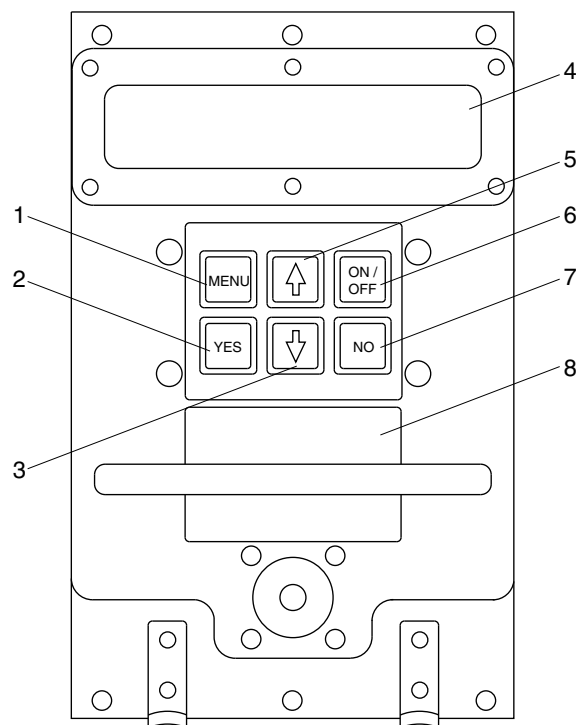
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LEGEND

- 1. Menu Button.
- 2. Yes Button.
- 3. Down Button.
- 4. LCD.
- 5. Up Button.
- 6. On/Off Button.
- 7. No Button.
- 8. Information Placard.

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PSEU Selector Buttons
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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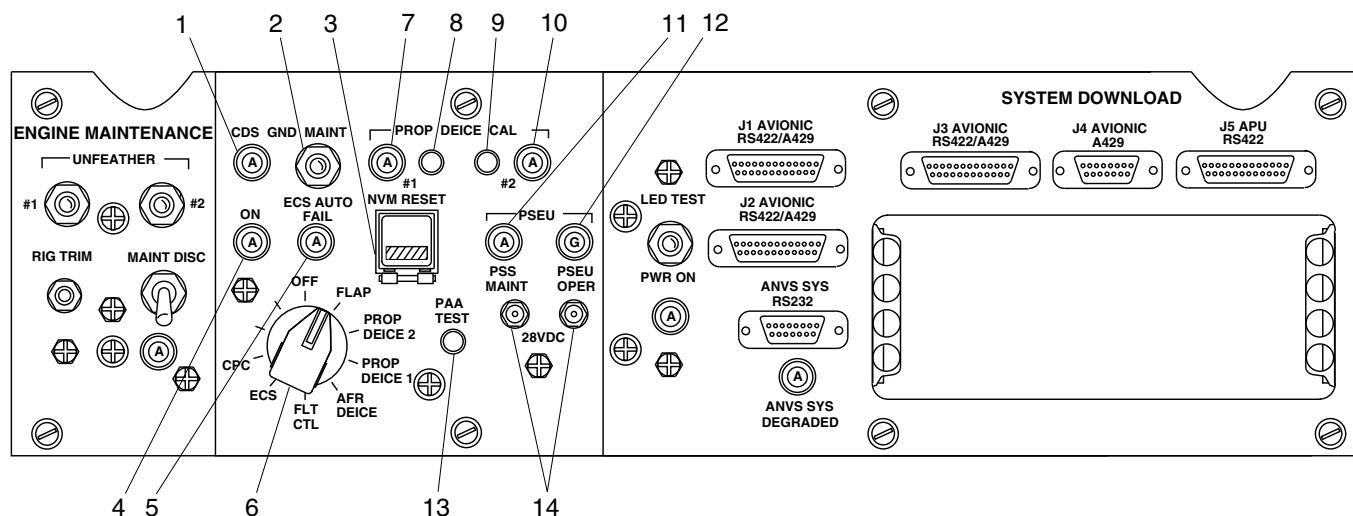


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LEGEND

- | | |
|-------------------------------------|-----------------------------------|
| 1. CDS GND MAINT light. | 8. #1 PROP DEICE CAL push-button. |
| 2. CDS GND MAINT toggle switch. | 9. #2 PROP DEICE CAL push-button. |
| 3. NVM RESET (Guarded) push-button. | 10. #2 PROP DEICE CAL light. |
| 4. System selector light. | 11. PSS MAINT light. |
| 5. ECS AUTO FAIL light. | 12. PSEU OPER light. |
| 6. System selector rotary switch. | 13. PAA TEST push-button. |
| 7. #1 PROP DEICE CAL light. | 14. 28 Vdc test jacks. |



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Maintenance Panel – Systems Maintenance Panel
Figure 7

PSM 1-84-2A
EFFECTIVITY:
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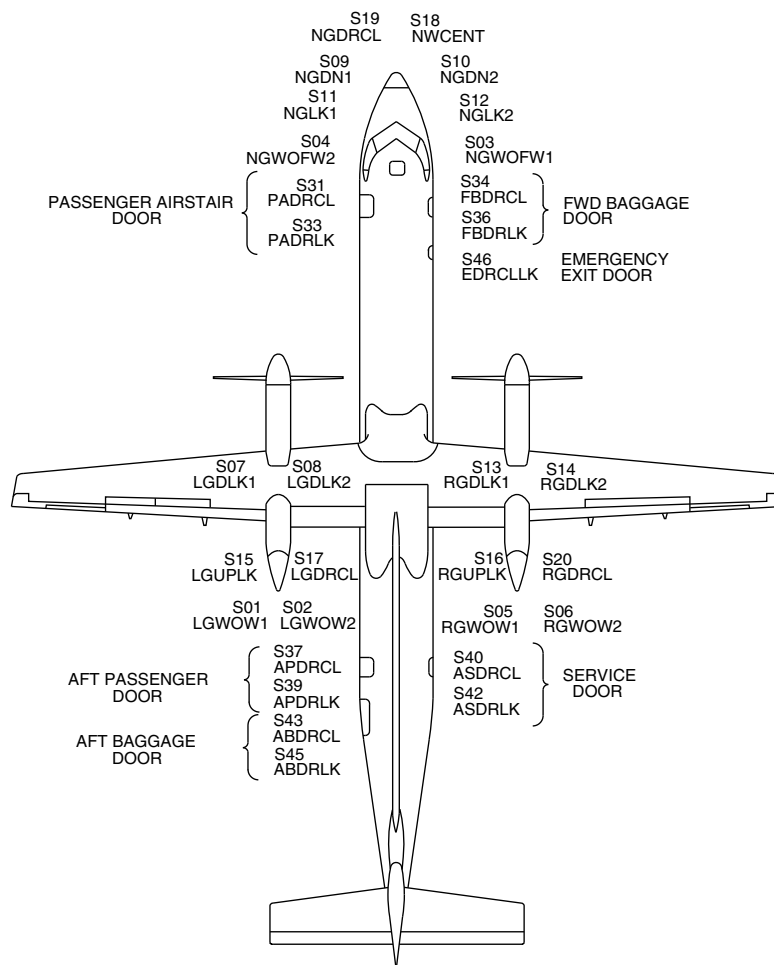
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Proximity Sensors Locations
Figure 8

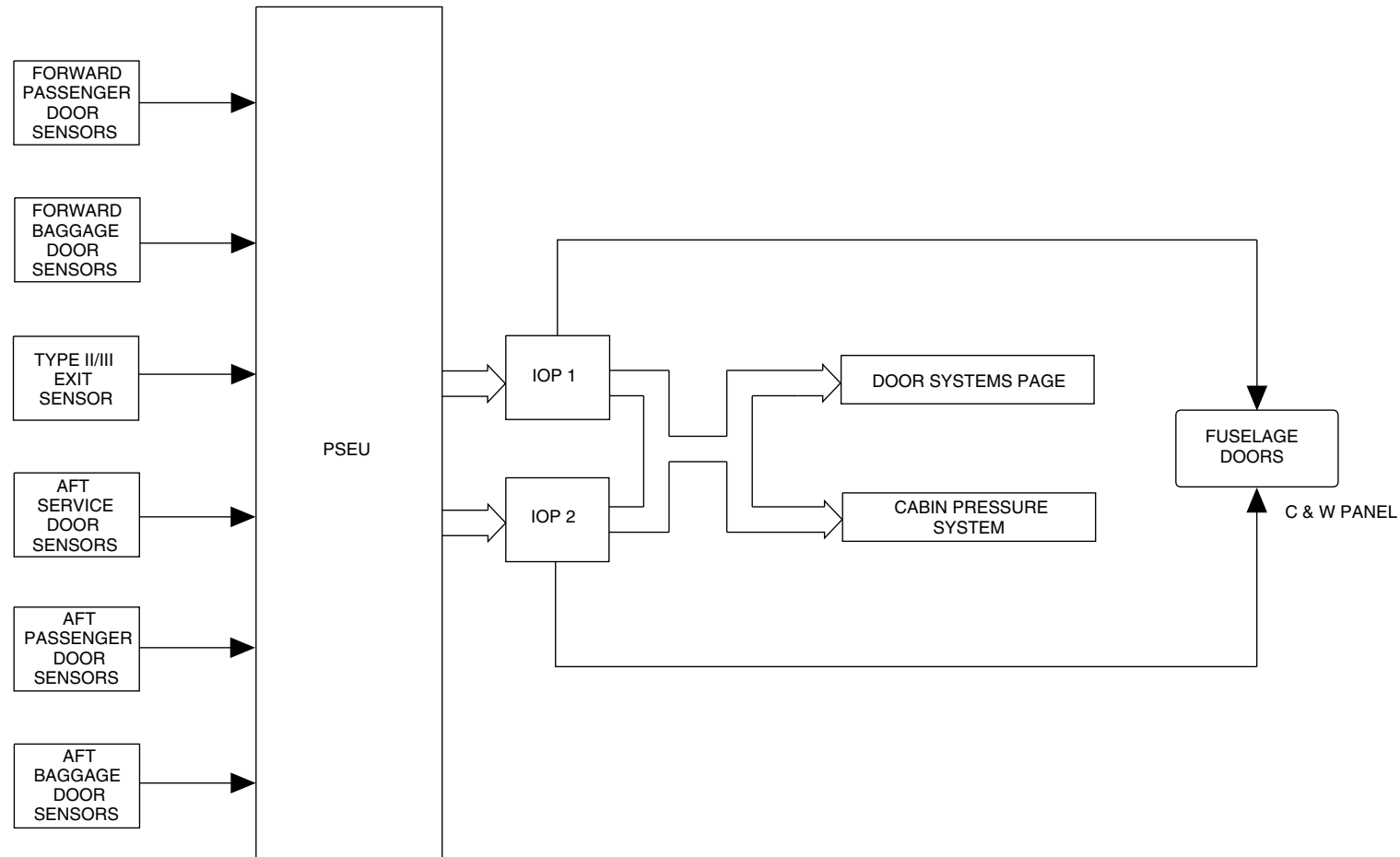
PSM 1-84-2A
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Door Sensor Inputs to PSEU
Figure 9



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Fault Definition	Top Line Display	Bottom Line Display
Present Faults or Fault History Faults		
1. BITE Non-volatile Memory Fault	FAULT_MEMORY_FAIL	_____ – ## ²
2. PSEU Power Input – Left Main 1	_LEFT_MAIN_1_POWER_ ¹	FAILED_____ – ## ²
– Left Main 2	_LEFT_MAIN_2_POWER_ ¹	FAILED_____ – ## ²
– Right Main	RIGHT_MAIN_POWER_ ¹	FAILED_____ – ## ²
–Left Essential 1	_LEFT_ESS_1_POWER_ ¹	FAILED_____ – ## ²
–Left Essential 2	_LEFT_ESS_2_POWER_ ¹	FAILED_____ – ## ²
–Right Essential	RIGHT_ESS_POWER_ ¹	FAILED_____ – ## ²
3. PSC Channel Fault	PSEU_CHAN_z_FAIL_ ^{1 4}	(One of the following messages:)
–PSC-BITE Communications Fault		COMMUNICATION_FTL – ## ²
–PSC Self-Determined Failure		CHANNEL_INOP_____ – ## ²
–Signal Conditioner Failure		SIG_COND_0n_FAIL_ – ## ^{2 6}
–Discrete Output Fault		DOTnnz_FAIL_____ – ## ^{2 4 5}
4. Proximity Sensor Open Fault	*****_FAULT_____ ^{1 3}	PROX_SENS_OPEN_____ – ## ²
5. Proximity Sensor Short Fault	*****_FAULT_____ ^{1 3}	PROX_SENS_SHORT_____ – ## ²
6. Proximity Sensor Unreasonable Fault	*****_FAULT_____ ^{1 3}	UNREASONABLE_aaaa – ## ^{2 7}
7. External Output Discrete Fault	DOTnnz_____ ^{1 4}	(One of the following messages:)
–Open Fault		LOAD_FAULT_____ – ## ²
–Short or Overcurrent Fault		OVERCURRENT_FAULT – ## ²
8. External Input Discrete Fault	DINnnz_____ ^{1 4}	FAILED_____ – ## ²
Rigging History Reported Faults		
9. Rigging Faults	*****_RIGGING ____&& ^{3 8}	(One of the following messages:)
–Target Near, Gap Too Small		NEAR_GAP_TOO_SMALL_
–Target Near, Gap Too Large		NEAR_GAP_TOO_LARGE_
–Target Far, Gap Too Small		FAR_GAP_TOO_SMALL_

Notes: Underscores represents blank characters.

1 "I" displayed if fault is intermittent, blank otherwise. This indication is displayed for Fault History faults only

2 "##" represents Flight Leg 00 – 31 (decimal). This indication is displayed for Fault History faults only.

3 "*****" represents the Proximity Sensor abbreviation as specified in Paragraph 3.1. of the PSS Interface Control Document.

4 "z" represents the PSC Channel identifier "A" – "E" (alpha).

5 "nn" represents the Discrete Output signal number "1" – "30" (decimal).

6 "n" represents the Signal Conditioner number "1" – "8" (decimal).

7 "aaaa" represents NEAR OR FAR.

8 "&&" represents the number of Flight Legs during which this rigging fault was logged.

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BITE Reported Faults
Figure 10

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Status Definition	Top Line Display	Bottom Line Display
1. BITE Software Version	BITE_SOFTWARE_____	VERSION_REV_____hh ²
2. PSEU Operating Mode	PSEU_OPERATING_MODE_	(One of the following:) WEIGHT_ON_WHEELS____ GEAR_DOWN_AND-LOCKED GEAR_IN_TRANSIT GEAR_UP_AND_LOCKED
3. PSEU Power Input – Left Main 1	_LEFT_MAIN_1_POWER_	ACTIVE or FAILED
–Left Main 2	_LEFT_MAIN_2_POWER_	ACTIVE or FAILED
–Right Main	RIGHT_MAIN_POWER_	ACTIVE or FAILED
–Left Essential 1	_LEFT_ESS_1_POWER_	ACTIVE or FAILED
–Left Essential 2	_LEFT_ESS_2_POWER_	ACTIVE or FAILED
–Right Essential	RIGHT_ESS_POWER_	ACTIVE or FAILED
4. End of BITE Status Display	END_OF_____	BITE_STATUS
5. PSC Software Version	PSC-CHAN_z_SOFTWARE_1	VERSION_REV_____hh ²
6. PSC Data Version	PSC_CHAN_z_DATA_____1	VERSION_REV_____hh ²
7. PSC ID Discretes	PSC_CHAN_Z_ID_____	DISCRETES_____hh ²
8. Proximity Sensor Status & Inductance	*****_TARGET_aaaa_3 4	L_=_###.###_mH_____5
9. Proximity Sensor Status & Resistance	*****_TARGET_aaaa_3 4	R_=_###.###_ohms_____6
10. Discrete Input Status	DINnnz_____1 7	ACTIVE or INACTIVE
11. Discrete Output Status	DOTnnz_____1 7	ACTIVE or INACTIVE
12. End of PSC Status Display	END_OF_____	PSC_z_STATUS_____1

Notes: Underscores represents blank characters,

1 "z" represents the PSC Channel identifier "A" – "E" (alpha).

2 "hh" represents a number from "00" – "FF" (hexadecimal).

3 "*****" represents the Proximity Sensor abbreviation as specified in Paragraph 3.1. of the PSS Interface Control Document.

4 "aaaa" represents NEAR OR FAR_.

5 "###.###" represents the sensor inductance in millihenries.

6 "###.###" represents the sensor resistance in ohms.

7 "nn" represents the signal number "1" – "30" (decimal).

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System Status BITE Messages
Figure 11

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PROXIMITY SENSOR ELECTRONIC UNIT

Introduction

The primary functions of the Proximity Sensor Electronic Unit (PSEU) are as follows:

- Monitor and operate the indicators of the landing gear extension and retraction system
- Supply sequence signals to operate the landing-gear door sequence valves
- Monitor the uplock, downlock, and WOW data to other systems
- Give the landing gear position and WOW data to other systems
- Give an indication when the passenger and/or baggage doors are not closed correctly.

The PSEU uses 31 variable reluctance proximity sensors to get the input about the position of the landing-gear components, gear doors and fuselage doors.

Description

Refer to SDS32-61-00 for a description of the PSEU.

Operation

The logic circuits in the PSEU control the operation of the PSEU system. Equations that show the logic of the PSEU logic circuits give the relation of the input for the different operations. Each logic equation has a number. The logic equation numbers are used on the PSEU system schematic to identify the conditions that give an output at that specified location on the schematic.

Table 1: Proximity Sensor Identification

SENSOR NO.	FUNCTION	FUNCTION ABBREV.	FUNCTIONS USED IN (SEE NOTE 1)	NORMAL SENSOR TRUE POSITION
S01	LEFT GEAR Weight on Wheels #1	LGWOW1	WOW1, WOW2, LGC	NEAR
S02	LEFT GEAR Weight on Wheels #2	LGWOW2	WOW1, WOW2, LGC	NEAR
S03	NOSE GEAR Weight off Wheels #1	NGWOFW1	WOW1, WOW2, LGC	FAR
S04	NOSE GEAR Weight off Wheels #2	NGWOFW2	WOW1, WOW2, LGC	FAR
S05	RIGHT GEAR Weight on Wheels #1	RGWOW1	WOW1, WOW2, LGC	NEAR



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SENSOR NO.	FUNCTION	FUNCTION ABBREV.	FUNCTIONS USED IN (SEE NOTE 1)	NORMAL SENSOR TRUE POSITION
S06	RIGHT GEAR Weight on Wheels #2	RGWOW2	WOW1, WOW2, LGC	NEAR
S07	LEFT GEAR DOWNLOCK#1	LGDLK1	LGC, LGI, ADI	NEAR
S08	LEFT GEAR DOWNLOCK#2	LGDLK2	LGC, LGI, ADI	NEAR
S09	NOSE GEAR DOWN#1	NGDN1	LGC, LGI, ADI	NEAR
S10	NOSE GEAR DOWN#2	NGDN2	LGC, LGI, ADI	NEAR
S11	NOSE GEAR LOCK#1	NGLK1	LGC, LGI, ADI	NEAR
S12	NOSE GEAR LOCK#2	NGLK2	LGC, LGI, ADI	NEAR
S13	RIGHT GEAR DOWNLOCK#1	RGDLK1	LGC, LGI, ADI	NEAR
S14	RIGHT GEAR DOWNLOCK#2	RGDLK2	LGC, LGI, ADI	NEAR
S15	LEFT GEAR UPLOCK	LGUPLK	LGC, LGI	NEAR
S16	RIGHT GEAR UPLOCK	RGUPLK	LGC, LGI	NEAR

SENSOR NO.	FUNCTION	FUNCTION ABBREV.	FUNCTIONS USED IN (SEE NOTE 1)	NORMAL SENSOR TRUE POSITION
S17	LEFT GEAR DOORS CLOSED	LGDRCL	LGC, LGI	NEAR
S18	NOSE WHEEL CENTERED	NWCENT	LGC	NEAR
S19	NOSE GEAR DOORS CLOSED	NGDRCL	LGC, LGI	NEAR
S20	RIGHT GEAR DOORS CLOSED	RGDRCL	LGC, LGI	NEAR
S31	PASSENGER AIRSTAIR DOOR CLOSED	PADRCL	FDSI	NEAR
S33	PASSENGER AIRSTAIR DOOR LOCKED	PADRLK	FDSI	NEAR
S34	FORWARD BAGGAGE DOOR CLOSED	FBDRCL	FDSI	NEAR
S36	FORWARD BAGGAGE DOOR LOCKED	FBDRLK	FDSI	NEAR
S37	AFT PASSENGER DOOR CLOSED	APDRCL	FDSI	NEAR
S39	AFT PASSENGER DOOR LOCKED	APDRLK	FDSI	NEAR

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SENSOR NO.	FUNCTION	FUNCTION ABBREV.	FUNCTIONS USED IN (SEE NOTE 1)	NORMAL SENSOR TRUE POSITION
S40	AFT SERVICE DOOR CLOSED	ASDRCL	FDSI	NEAR
S42	AFT SERVICE DOOR LOCKED	ASDRLK	FDSI	NEAR
S43	AFT BAGGAGE DOOR CLOSED	ABDRCL	FDSI	NEAR
S45	AFT BAGGAGE DOOR LOCKED	ABDRLK	FDSI	NEAR
S46	EMERG. EXIT CLOSED/LOCKED	EDRCLLK	FDSI	NEAR

Note 1:

WOW	Weight on Wheels
LGC	Landing Gear Control
LGI	Landing Gear Indication
ADI	Alternate Downlock Indication
FDSI	Fuselage Door Status Indication

Logic Equations

The Proximity Sensor Electronics Unit (PSEU) does aircraft functions with inputs from the aircraft. Logic equations use these inputs to supply discrete outputs. The PSEU receives the inputs that follow:

- Discrete Inputs
- Proximity Sensor Inputs.

The discrete inputs include the landing gear handle (extend and retract), the landing gear inhibit, and the lamp test inputs.

The proximity sensor inputs include the proximity sensor “near” and “far” conditions determined by each Proximity Signal Conditioner (PSC) circuit card (Refer to [Refer to Figure 1.](#)).

The PSEU uses an ARINC 600 connector (J1). this connector is a 400 pin connector that is divided into three separate bays. Each bay has a letter given to it by ARINC 600 (A, B or C). The keying for PSEU P/N 30145–0101 and 30145–0202 is pattern No. 1. The keying for PSEU P/N 30145–0301, 30145–0401 and 30145–0501 is pattern No. 22. The keying for PSEU P/N 30145–0601 and 30145–0602 is pattern No. 32.

NOTE

You do not have to measure the output directly at the output pins. You can use the BITE panel System Status menu to show the status of each discrete output.

Logic Equation Notes

All equation terms shown use their operational mnemonic (LGDLK1). These names show the status reported by the component when it is in the NEAR (for sensors) or the ACTIVE (for inputs) condition. For example, when the Left Gear Downlock #1 sensor is NEAR, the term



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LGDLK1=TRUE. If the sensor shows FAR, then LGDLK1=FALSE. Similarly, if the Extend Command discrete input to the PSEU is ACTIVE, then the term EXTCMD=TRUE.

Terms in these equations with the suffix _F (LGDLK1_F) show the faulted condition of their component. For example, if the Left Gear Downlock #1 sensor shows a fault condition (Open or Shorted), the term LGDLK1_F=TRUE. If that sensor is not faulted, the term LGDLK1_F=FALSE.

Terms in these equations with the suffix _P (FBDRCL_P) show the previous status of that component. Before you determine the current status of a component in the proximity system, the previously given status of each component is stored. This stored, or previous, status can then be used in these equations to sense event transitions. Terms with no suffix show the present status of the component. The status of each component in the system is calculated nominally every 125 msec.

In the equations that follow, the symbol (·) shows the function AND. The symbol (+) shows the function OR.

Equation numbers 6, 7, 8 and 26 thru 31 are ignored. These equations are only used for Flight Test Instrumentation (FTI) and not for normal system operation.

Equation No. 1 EXTEND SELECTOR VALVE (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

EXTCMD · NOT LDGINH

Equation Explanation: The Extend Selector Valve discrete output is activated anytime the landing-gear handle is in the EXTEND position (as reported by the EXTCMD discrete input). This will occur unless

the Landing-Gear Control is inhibited as reported by an active LDGINH discrete input.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT01A	A1F	LGSV

Result: When you get the conditions of the equation, the PSEU (at card PSC A) gives an output of 28 V DC.

Equation No. 2 RETRACT SELECTOR VALVE (On aircraft without SB84-32-143 and SB84-32-149 and MODSUM 4-124247 and MODSUM 4-126663 incorporated and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501)

RETCMD · NOT WOW · (NOT DOORS + NOT NGLK1 + NOT NGLK2 + NOT LGUPLK + NOT RGUPLK) · NOT LDGINH · NWCENT

Equation Explanation: The Retract Selector Valve discrete output is activated only when all of the conditions that follow are true:

- The landing-gear handle is in the RETRACT position (as reported by the RETCMD discrete input)
- The WOW sensors show that the aircraft is in the air
- Landing-Gear Control is not inhibited
- The nosewheel is centered
- A landing-gear door is open or a landing-gear component is not in the uplock condition.



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DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT02A	A1G	LGSV

Result: When you get the conditions of the equation, the PSEU (at card PSC A) gives an output of 28 V DC.

Equation No. 2 RETRACT SELECTOR VALVE (On aircraft with MODSUM 4-124247 or 4-126663 or SB84-32-143 or SB84-32-149 incorporated and on aircraft installed with PSEU P/N 30145-0601 or 30145-0602)

$\{RETCMD \cdot MGWOFW \cdot NGWOFW \cdot (NOT\ DOORS + NOT\ NGLK1 + NOT\ NGLK2 + NOT\ LGUPLK + NOT\ RGUPLK) \cdot NOT\ LDGINH \cdot NWCENT\} \cdot NOT\ \{[LGDRCL \cdot NOT\ LGUPLK \cdot (NOT\ LGDNLK1 + NOT\ LGDNLK2)] + [RGDRCL \cdot NOT\ RGUPLK \cdot (NOT\ RGDNLK1 + NOT\ RGDNLK2)] + [NGDRCL \cdot (NOT\ NGLK1 + NOT\ NGLK2) \cdot (NOT\ NGDN1 + NOT\ NGDN2)]\}$

Equation Explanation: The Retract Selector Valve discrete output is activated only when all of the conditions that follow are true:

- The landing–gear handle is in the RETRACT position (as reported by the RETCMD discrete input)
- Inhibits gear retract until the gear are off the ground
- Landing–Gear Control is not inhibited
- The nosewheel is centered
- A landing–gear door is open or a landing–gear component is not in the uplock condition.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT02A	A1G	LGSV

Result: When you get the conditions of the equation, the PSEU (at card PSC A) gives an output of 28 V DC.

Equation No. 3 NG DOOR SEQUENCE VALVE (On aircraft without SB 84-32-66 and MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0501 or 30145-0601 or 30145-0602)

$[(RETCMD \cdot (NOT\ NGLK1 + NGDN1) \cdot (NOT\ NGLK2 + NGDN2)) + (EXTCMD \cdot NGLK1 \cdot NGDN1 \cdot NGLK2 \cdot NGDN2)] \cdot NOT\ LDGINH$

Equation Explanation: The Nose Gear Door Sequence Valve controls the closure of the Nose Gear Door depending on the state of the Extend and Retract Selector Valves. If the Extend Selector Valve is active, then an active NG Door Sequence Valve output will close the Nose Gear Door. If the Retract Selector Valve is active, then an inactive NG Door Sequence Valve output will close the Nose Gear Door. Thus, the Nose Gear Door Sequence Valve discrete output is as follows:

- Activated when the landing–gear handle is in the EXTEND position and the Nose Gear is in the Down and Locked position, and
- Deactivated when the landing–gear handle is in the RETRACT position and the Nose Gear is in the Up and Locked position
- Deactivated when the Landing Gear Control is inhibited by an active LDGINH discrete input.



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DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT04A	A1J	LGDSV

Result: When you get the conditions of the equation, the PSEU (at card PSC A) gives an output of 28 V DC to the nose landing-gear door sequence valve.

Equation No. 3 NG DOOR SEQUENCE VALVE (On aircraft with SB 84-32-66 or MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0401)

$$[(RETCMD \cdot (NOT\ NGLK1 + NGDN1) \cdot (NOT\ NGLK2 + NGDN2)) + (EXTCMD \cdot NGLK1 \cdot NGDN1 \cdot NGLK2 \cdot NGDN2)] \cdot NOT\ LDGINH$$

Equation Explanation: The Nose Gear Door Sequence Valve controls the closure of the Nose Gear Door depending on the state of the Extend and Retract Selector Valves. If the Extend Selector Valve is active, then an active NG Door Sequence Valve output will close the Nose Gear Door. If the Retract Selector Valve is active, then an inactive NG Door Sequence Valve output will close the Nose Gear Door. Thus, the Nose Gear Door Sequence Valve discrete output is as follows:

- Activated when the landing-gear handle is in the EXTEND position and 10 seconds after the Nose Gear is in the Down and Locked position, and
- Deactivated when the landing-gear handle is in the RETRACT position and the Nose Gear is in the Up and Locked position
- Deactivated when the Landing Gear Control is inhibited by an active LDGINH discrete input.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT04A	A1J	LGDSV

Result: When you get the conditions of the equation, the PSEU (at card PSC A) gives an output of 28 V DC to the nose landing-gear door sequence valve.

Equation No. 4 RG DOOR SEQUENCE VALVE (On aircraft without SB 84-32-66 and MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0501 or 30145-0601 or 30145-0602)

$$((RETCMD \cdot NOT\ RGUPLK) + (EXTCMD \cdot RGDLK1 \cdot RGDLK2)) \cdot NOT\ LDGINH$$

Equation Explanation: The Right Gear Door Sequence Valve controls the closure of the Right Gear Door depending on the state of the Extend and Retract Selector Valves. If the Extend Selector Valve is active, then an active RG Door Sequence Valve output will close the Right Gear Door. If the Retract Selector Valve is active, then an inactive RG Door Sequence Valve output will close the Right Gear Door. Thus, the Right Gear Door Sequence Valve discrete output is:

- Activated when the landing-gear handle is in the EXTEND position and the Right Gear is in the Down and Locked position, and
- Deactivated when the landing-gear handle is in the RETRACT position and the Right Gear is in the Up and Locked position
- Deactivated when the Landing Gear Control is inhibited by an active LDGINH discrete input.



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DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT05A	A1K	LGDSV

Result: When you get the conditions of the equation, the PSEU (at card PSC A) gives an output of 28 V DC to the right landing-gear door sequence valve.

Equation No. 4 RG DOOR SEQUENCE VALVE (On aircraft with SB 84-32-66 or MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0401)

$((\text{RETCMD} \cdot \text{NOT RGUPLK}) + (\text{EXTCMD} \cdot \text{RGDLK1} \cdot \text{RGDLK2})) \cdot \text{NOT LDGINH}$

Equation Explanation: The Right Gear Door Sequence Valve controls the closure of the Right Gear Door depending on the state of the Extend and Retract Selector Valves. If the Extend Selector Valve is active, then an active RG Door Sequence Valve output will close the Right Gear Door. If the Retract Selector Valve is active, then an inactive RG Door Sequence Valve output will close the Right Gear Door. Thus, the Right Gear Door Sequence Valve discrete output is:

- Activated when the landing-gear handle is in the EXTEND position and 10 seconds after the Right Gear is in the Down and Locked position, and
- Deactivated when the landing-gear handle is in the RETRACT position and the Right Gear is in the Up and Locked position
- Deactivated when the Landing Gear Control is inhibited by an active LDGINH discrete input.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT05A	A1K	LGDSV

Result: When you get the conditions of the equation, the PSEU (at card PSC A) gives an output of 28 V DC to the right landing-gear door sequence valve.

Equation No. 5 LG DOOR SEQUENCE VALVE (On aircraft without SB 84-32-66 and MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0501 or 30145-0601 or 30145-0602)

$((\text{RETCMD} \cdot \text{NOT LGUPLK}) + (\text{EXTCMD} \cdot \text{LGDLK1} \cdot \text{LGDLK2})) \cdot \text{NOT LDGINH}$

Equation Explanation: The Left Gear Door Sequence Valve controls the closure of the Left Gear Door depending on the state of the Extend and Retract Selector Valves. If the Extend Selector Valve is active, then an active LG Door Sequence Valve output will close the Left Gear Door. If the Retract Selector Valve is active, then an inactive LG Door Sequence Valve output will close the Left Gear Door. Thus, the Left Gear Door Sequence Valve discrete output is:

- Activated when the landing-gear handle is in the EXTEND position and the Left Gear is in the Down and Locked position, and
- Deactivated when the landing-gear handle is in the RETRACT position and the Left Gear is in the Up and Locked position
- Deactivated when the Landing Gear Control is inhibited by an active LDGINH discrete input.



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DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT03A	A1H	LGDSV

Result: When you get the conditions of the equation, the PSEU (at card PSC A) gives an output of 28 V DC to the left landing–gear door sequence valve.

Equation No. 5 LG DOOR SEQUENCE VALVE (On aircraft with SB 84–32–66 or MODSUM 4–126420 incorporated and on aircraft installed with PSEU P/N 30145–0401)

$((\text{RETCMD} \cdot \text{NOT LGUPLK}) + (\text{EXTCMD} \cdot \text{LGDLK1} \cdot \text{LGDLK2})) \cdot \text{NOT LDGINH}$

Equation Explanation: The Left Gear Door Sequence Valve controls the closure of the Left Gear Door depending on the state of the Extend and Retract Selector Valves. If the Extend Selector Valve is active, then an active LG Door Sequence Valve output will close the Left Gear Door. If the Retract Selector Valve is active, then an inactive LG Door Sequence Valve output will close the Left Gear Door. Thus, the Left Gear Door Sequence Valve discrete output is:

- Activated when the landing–gear handle is in the EXTEND position and 10 seconds after the Left Gear is in the Down and Locked position, and
- Deactivated when the landing–gear handle is in the RETRACT position and the Left Gear is in the Up and Locked position
- Deactivated when the Landing Gear Control is inhibited by an active LDGINH discrete input.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT03A	A1H	LGDSV

Result: When you get the conditions of the equation, the PSEU (at card PSC A) gives an output of 28 V DC to the left landing–gear door sequence valve.

Equation No. 9 NG SAFE INDICATION (On aircraft without SB 84–32–66 and MODSUM 4–126420 incorporated and on aircraft installed with PSEU P/N 30145–0201 or 30145–0301 or 30145–0501)

$(\text{NGLK1} \cdot \text{NGDN1}) + \text{LMPTST1}$

Equation Explanation: The Nose Gear Safe Indication discrete output is activated when the Landing–Gear Indication Function senses that the Nose Gear is Down and Locked. Also when the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT07D	B3D	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing–gear handle.



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Equation No. 9 NG SAFE INDICATION (On aircraft with SB 84-32-66 or MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0401)

$\{EXTCMD \cdot [(NGDN1 \cdot NOT\ NGDN1_F) + (NGDN2 \cdot NOT\ NGDN2_F)] \cdot [(NGLK1 \cdot NOT\ NGLK1_F) + (NGLK2 \cdot NOT\ NGLK2_F)]\} + LMPTST1$

Equation Explanation: The Nose Gear Safe Indication discrete output to the Landing Gear Control and Indication Panel (LGCIP) is activated when the EXTEND is selected and one or both unfaulted nose gear down sensors detect DOWN and one or both unfaulted lock sensors detect LOCKED and one or both not faulted. Also when the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT07D	B3D	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 9 NG SAFE INDICATION (On aircraft with SB84-32-143 or SB84-32-149 or MODSUM 4-124247 or MODSUM 4-126663 incorporated and on aircraft installed with PSEU P/N 30145-0601 or 30145-0602)

$\{EXTCMD \cdot [(NGDN1 + NGDN1_F) \cdot (NGDN2 + NGDN2_F) \cdot NGLK1 + NGLK1_F) \cdot (NGLK2 + NGLK2_F)] \cdot [(NGDN1 \cdot NGLK1) + (NGDN2 \cdot NGLK2) + (NGDN1 \cdot NGLK2) + (NGDN2 \cdot NGLK1)]\} + LMPTST1$

Equation Explanation: The Nose Gear Safe Indication discrete output to the Landing Gear Control and Indication Panel (LGCIP) is activated when the EXTEND is selected and one or both unfaulted nose gear down sensors detect DOWN and one or both unfaulted lock sensors detect LOCKED and one or both not faulted. Also when the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT07D	B3D	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 10 RG SAFE INDICATION (On aircraft without SB 84-32-66 and MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-501)

RGDLK1 + LMPTST1

Equation Explanation: The Right Gear Safe Indication discrete output is activated when the Landing-Gear Indication Function senses that the Right Gear is Down and Locked. Also when the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT08D	B3C	LGCIP

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Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 10 RG SAFE INDICATION (On aircraft with SB 84-32-66 or SB84-32-143 or SB84-32-149 or MODSUM 4-124247 or MODSUM 4-126663 or MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0401 or 30145-0601 or 30145-0602)

$$\{EXTCMD \cdot [(RGDLK1 + RGDLK1_F) \cdot (RGDLK2 + RGDLK2_F) \cdot (NOT RGDLK1_F + NOT RGDLK2_F)]\} + LMPTST1$$

Equation Explanation: The Right Gear Safe Indication discrete output to LGCIP is activated when the EXTEND has been selected and either both unfaulted right main gear Downlock sensors are detected down, or either of the right main gear Downlock sensors is detected down and is not faulted while the other sensor reports faulted. The two sensors cannot be faulted. It can also be activated when the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT08D	B3C	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 11 LG SAFE INDICATION (On aircraft without SB 84-32-66 and MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0501)

LGDLK1 + LMPTST1

Equation Explanation: The Left Gear Safe Indication discrete output is activated when the Landing-Gear Indication Function senses that the Left Gear is Down and Locked. Also when the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT06D	B3E	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 11 LG SAFE INDICATION (On aircraft with SB 84-32-66 or SB84-32-143 or SB84-32-149 or MODSUM 4-124247 or MODSUM 4-126663 or MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0401 or 30145-0601 or 30145-0602)

$$\{EXTCMD \cdot [(LGDLK1 + LGDLK1_F) \cdot (LGDLK2 + LGDLK2_F) \cdot (NOT LGDLK1_F + NOT LGDLK2_F)]\} + LMPTST1$$

Equation Explanation: The Left Gear Safe Indication discrete output to LGCIP is activated when the EXTEND has been selected and either both unfaulted left main gear Downlock sensors are detected down, or either of the left main gear Downlock sensors is detected down and is not faulted while the other sensor reports faulted. The

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two sensors cannot be faulted. It can also be activated when the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT06D	B3E	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 12 NG UNSAFE INDICATION (On aircraft without SB 84-32-66 and MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0501)

$(RETCMD \cdot (NOT\ NGLK1 + NGDN1)) + (EXTCMD \cdot (NOT\ NGLK1 + NOT\ NGDN1)) + LMPTST1$

Equation Explanation: The Nose Gear Unsafe Indication discrete output is active when the Nose Gear status is different from the position set by the landing-gear handle. Thus, the output activates as follows:

- When UP is set but the Nose Gear does not show Up and Locked
- When DOWN is set but the Nose Gear does not show Down and Locked
- When the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT10D	B3A	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 12 NG UNSAFE INDICATION (On aircraft with SB 84-32-66 or MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0401)

$\{RETCMD \cdot (((NOT\ NGLK1 \cdot NOT\ NGLK1_F) + (NOT\ NGLK2 \cdot NOT\ NGLK2_F)) + ((NGDN1 \cdot NOT\ NGDN1_F) + (NGDN2 \cdot NOT\ NGDN2_F))) + \{EXTCMD \cdot NOT\ [(NGDN1 \cdot NOT\ NGDN1_F) + (NGDN2 \cdot NOT\ NGDN2_F)] \cdot [(NGLK1 \cdot NOT\ NGLK1_F) + (NGLK2 \cdot NOT\ NGLK2_F)] + [(NGDN1_F \cdot NGDN2_F) + (NGLK1_F \cdot NGLK2_F)] + LMPTST1$

Equation Explanation: The Nose Gear Unsafe Indication discrete output is active when the Nose Gear status is different from the position set by the landing-gear handle. Thus, the output activates as follows:

- When RETRACT is selected but the Nose Gear is not reporting Locked by one or both unfaulted lock sensors or one or both unfaulted down sensors reporting down
- When EXTEND is selected but the Nose Gear is not reporting Down by one or both of unfaulted down sensors and Nose Gear reporting Locked by one or both of the unfaulted lock sensors

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- When both down sensors or lock sensors are faulted
- When the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT10D	B3A	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 12 NG UNSAFE INDICATION (On aircraft with SB84-32-143 or SB84-32-149 or MODSUM 4-124247 or 4-126663 incorporated and on aircraft installed with PSEU P/N 30145-0601 or 30145-0602)

$$\{RETCMD \cdot [(NOT\ NGLK1 \cdot (NOT\ NGLK1_F)) + (NOT\ NGLK2 \cdot NOT\ NGLK2_F) + (NGDN1 \cdot NOT\ NGDN1_F) + (NGDN2 \cdot NOT\ NGDN2_F)]\} + \{EXTCMD \cdot NOT\ [(NGDN1 + NGDN1_F) \cdot (NGDN2 + NGDN2_F) \cdot (NGLK1 + NGLK1_F) \cdot (NGLK2 + NGLK2_F)] \cdot [(NGDN1 \cdot NGLK1) + (NGDN2 \cdot NGLK2) + (NGDN1 \cdot NGLK2) + (NGDN2 \cdot NGLK1)]\} + [(NGDN1_F \cdot NGDN2_F) + (NGLK1_F \cdot NGLK2_F)] + LMPTST1$$

Equation Explanation: The Nose Gear Unsafe Indication discrete output is active when the Nose Gear status is different from the position set by the landing-gear handle. Thus, the output activates as follows:

- When RETRACT is selected but the Nose Gear is not reporting Locked by one or both unfaulted lock sensors or one or both unfaulted down sensors reporting down

- When EXTEND is selected but the Nose Gear is not reporting Down by all the Nose Gear Down, the Nose Gear Lock inputs are Near or faulted and any one or both of Nose Gear Down/Lock sensors are Near.
- When both down sensors or lock sensors are faulted
- When the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT10D	B3A	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 13 RG UNSAFE INDICATION (On aircraft without SB 84-32-66 and MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0501)

$$(RETCMD \cdot (NOT\ RGUPLK + RGD LK1)) + (EXTCMD \cdot (RGUPLK + NOT\ RGD LK1)) + LMPTST1$$

Equation Explanation: The Right Gear Unsafe Indication discrete output is active when the Right Gear status is different from the position set by the landing-gear handle. Thus, the output activates as follows:

- When RETRACT is set but the Right Gear does not show Up and Locked
- When EXTEND is set but the Right Gear does not show Down and Locked

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- When the Lamp Test 1 discrete input is active.

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DOT11D	B4C	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 13 RG UNSAFE INDICATION (On aircraft with SB84-32-66 or MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0401)

$[RETCMD \cdot NOT \text{RGUPLK}] + \{EXTCMD \cdot NOT [(RGDLK1 \cdot (RGDLK2 + NOT \text{RGUPLK})) + (RGDLK2 \cdot (RGDLK1 + NOT \text{RGUPLK}))]\} + [RGDLK1_F \cdot RGDLK2_F] + LMPTST1$

Equation Explanation: The Right Gear Unsafe Indication discrete output is active when the Right Gear status is different from the position set by the landing-gear handle. Thus, the output activates as follows:

- When RETRACT is selected but the Right Gear does not report UP by the uplock sensor
- When EXTEND is selected but the Right Gear does not report DOWNLOCK by one downlock sensor or when both downlock sensors are faulted
- When the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT11D	B4C	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 13 RG UNSAFE INDICATION (On aircraft with SB84-32-143 or SB84-32-149 or MODSUM 4-124247 or 4-126663 incorporated and on aircraft installed with PSEU P/N 30145-0601 or 30145-0602)

$[RETCMD \cdot NOT \text{RGUPLK}] + \{EXTCMD \cdot NOT [(RGDLK1 + RGDLK1_F) \cdot (RGDLK2 + RGDLK2_F) \cdot (NOT \text{RGDLK1_F} + NOT \text{RGDLK2_F})]\} + (RGDLK1_F \cdot RGDLK2_F) + LMPTST1$

Equation Explanation: The Right Gear Unsafe Indication discrete output is active when the Right Gear status is different from the position set by the landing-gear handle. Thus, the output activates as follows:

- When RETRACT is selected but the Right Gear does not report UP by the uplock sensor
- When EXTEND is selected but the Right Gear does not report DOWNLOCK by one downlock sensor and either the remaining downlock sensor or the uplock sensor reporting NOT UP or when both downlock sensors are faulted
- When the Lamp Test 1 discrete input is active.

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DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT11D	B4C	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 14 LG UNSAFE INDICATION (On aircraft without SB84-32-66 and MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0501)

$(RETCMD \cdot (NOT LGUPLK + LGDLK1)) + (EXTCMD \cdot (LGUPLK + NOT LGDLK1)) + LMPTST1$

Equation Explanation: The Left Gear Unsafe Indication discrete output is active when the Left Gear status is different from the position set by the landing-gear handle. Thus, the output activates as follows:

- When RETRACT is set but the Left Gear does not show Up and Locked
- When EXTEND is set but the Left Gear does not show Down and Locked
- When the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT09D	B3B	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 14 LG UNSAFE INDICATION (On aircraft with SB84-32-66 or MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0401)

$[RETCMD \cdot NOT LGUPLK] + \{EXTCMD \cdot NOT [(LGDLK1 \cdot (LGDLK2 + NOT LGUPLK)) + (LGDLK2 \cdot (LGDLK1 + NOT LGUPLK))]\} + [LGDLK1_F \cdot LGDLK2_F] + LMPTST1$

Equation Explanation: The Left Gear Unsafe Indication discrete output is active when the Left Gear status is different from the position set by the landing-gear handle. Thus, the output activates as follows:

- When RETRACT is selected but the Left Gear does not report UP by the uplock sensor
- When EXTEND is selected but the Left Gear does not report DOWNLOCK by one downlock sensor and either the remaining downlock sensor or the uplock sensor reporting NOT UP or when both downlock sensors are faulted
- When the Lamp Test 1 discrete input is active.



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DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT09D	B3B	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 14 LG UNSAFE INDICATION (On aircraft with SB84-32-143 or SB84-32-149 or MODSUM 4-124247 or 4-126663 incorporated and on aircraft installed with PSEU P/N 30145-0601 or 30145-0602)

$$[RETCMD \cdot NOT \text{LGUPLK}] + \{EXTCMD \cdot NOT [(LGDLK1 + LGDLK1_F) \cdot (LGDLK2 + LGUPLK2_F) \cdot (NOT \text{LGDLK1_F} + NOT \text{LGDLK2_F})]\} + LGDLK1_F \cdot LGDLK2_F] + LMPTST1$$

Equation Explanation: The Left Gear Unsafe Indication discrete output is active when the Left Gear status is different from the position set by the landing-gear handle. Thus, the output activates as follows:

- When RETRACT is selected but the Left Gear does not report UP by the uplock sensor
- When EXTEND is selected but the Left Gear does not report DOWNLOCK by one downlock sensor or when both downlock sensors are faulted
- When the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT09D	B3B	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 15 NG DOORS NOT CLOSED (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

NOT NGDRCL + LMPTST1

Equation Explanation: The Nose Gear Doors Not Closed discrete output activates when the Nose Gear Doors are not closed or when the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT10E	B11D	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC E) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 16 RG DOORS NOT CLOSED (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

NOT RGDRCCL + LMPTST1



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Equation Explanation: The Right Gear Doors Not Closed discrete output activates when the Right Gear Doors are not closed or when the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT11E	B12E	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC E) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 17 LG DOORS NOT CLOSED (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

NOT LGDRCL + LMPTST1

Equation Explanation: The Left Gear Doors Not Closed discrete output activates when the Left Gear Doors are not closed or when the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT09E	B11E	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC E) sends a ground signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 18 GEAR UNSAFE (On aircraft without SB 84-32-66 and MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0501)

$$[RETCMD \cdot (NOT\ RGUPLK + NOT\ LGUPLK + NOT\ NGLK1 + NGDN1 + RGD LK1 + LGDLK1)] + [EXTCMD \cdot (NOT\ RGD LK1 + NOT\ LGDLK1 + NOT\ NGLK1 + NOT\ NGDN1 + RGUPLK + LGUPLK)] + LMPTST1$$

Equation Explanation: The Gear Unsafe discrete output activates when a landing-gear position disagrees with the landing-gear handle position. Thus, it activates when RETRACT is set but not all the gear show Up and Locked, and when EXTEND is set but not all the gear show Down and Locked. It also activates when the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT05D	B2A	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a 28 V DC signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 18 GEAR UNSAFE (On aircraft with SB 84-32-66 or MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0401)

$$\{RETCMD \cdot [(NOT\ RGUPLK) + (NOT\ LGUPLK) + (NOT\ NGLK1 \cdot NOT\ NGLK2)]\} + \{EXTCMD \cdot NOT\ [((NGDN1 \cdot NOT\ NGDN1_F) + (NGDN2 \cdot NOT\ NGDN2_F)) \cdot ((NGLK1 \cdot NOT\ NGLK1_F) + (NGLK2 \cdot NOT\ NGLK2_F)) + NOT\ (((RGDLK1 \cdot (RGDLK2 + NOT\ RGUPLK)) +$$



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$$(RGDLK2 \cdot (RGDLK1 + NOT\ RGUPLK))) + (((LGDLK1 \cdot (LGDLK2 + NOT\ LGUPLK)) + (LGDLK2 \cdot (LGDLK1 + NOT\ LGUPLK)))) + [(NGDN1_F \cdot NGDN2_F) + (NGLK1_F \cdot NGLK2_F) + (RGDLK1_F \cdot RGDLK2_F) + (LGDLK1_F \cdot LGDLK2_F)] + LMPTST1$$

Equation Explanation: The Gear Unsafe discrete output activates when a landing-gear position disagrees with the landing-gear handle position. Thus, it activates as follows:

- When RETRACT is set but not all the gears show Up and Locked by their sensors (both lock sensors for nose gear or an uplock sensor for right or left main gears)
- When EXTEND is set but not all the gear show Down and Locked by the unfaulted sensors
- When down, lock or downlock sensors are simultaneously faulted
- It also activates when the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT05D	B2A	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a 28 V DC signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 18 LG In Transit (On aircraft with SB84-32-143 or SB84-32-149 or MODSUM 4-124247 or MODSUM 4-126663 incorporated and on aircraft installed with PSEU P/N 30145-0601 or 30145-0602)

$$[(NOT\ (NGDN1 \cdot NGLK1) + (NGDN2 \cdot NGLK2)) \cdot (NGDN1 + NOT\ NGLK1) + (NGDN2 + NOT\ NGLK2)] + [(NOT\ (LGDNLK1 + LGDNLK2) \cdot (NOT\ LGDNLK1_F + NOT\ LGDNLK2_F)) \cdot [(RGDNLK1 + RGDNLK2) \cdot (NOT\ RGDNLK1_F + NOT\ RGDNLK2_F)] \cdot (NOT\ LGUPLK + NOT\ RGUPLK)] + LMPTST1$$

Equation Explanation: The LG In Transit activates when the gear downlock and gear uplock is not satisfied or the Lamp Test 1 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT05D	B2A	LGCIP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a 28 V DC signal to the Advisory Control Unit. The output signal then goes to the Landing Gear Control and Indication Panel above the landing-gear handle.

Equation No. 19 ALL LANDING-GEAR DOWNLOCK (On aircraft without SB84-32-66 and MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0501)

$$NGDN1 \cdot NGLK1 \cdot LGDLK1 \cdot RGDLK1$$

Equation Explanation: The All Landing-Gear Downlock discrete output activates only when primary downlock sensors for all three gears show the Down and Locked condition.



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DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT16D	B6D	IOM2

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to IOM2 (Refer to WM31-41-00).

Equation No. 19 ALL LANDING-GEAR DOWNLOCK (On aircraft with SB84-32-66 or SB84-32-143 or SB84-32-149 or MODSUM 4-126420 or MODSUM 4-124247 or MODSUM 4-126663 incorporated and on aircraft installed with PSEU P/N 30145-0401 or 30145-0601 or 30145-0602)

$$[(\text{NGDN1} + \text{NGDN2}) \cdot (\text{NOT NGDN1_F} + \text{NOT NGDN2_F})] \cdot [(\text{NGLK1} + \text{NGLK2}) \cdot (\text{NOT NGLK1_F} + \text{NOT NGLK2_F})] \cdot (\text{RGDLK1} + \text{RGDLK2}) \cdot (\text{NOT RGDLK1_F} + \text{NOT RGDLK2_F}) \cdot (\text{LGDLK1} + \text{LGDLK2}) \cdot (\text{NOT LGDLK1_F} + \text{NOT LGDLK2_F})$$

Equation Explanation: The All Landing-Gear Downlock discrete output activates only when all three gears show the Down and Locked condition as reported by the down, lock and downlock sensors measurements as follows:

- Nose Gear – One or both unfaulted down sensors report DOWN and one or both unfaulted lock sensors reports DOWN
- Right Gear – One or both downlock sensors report DOWNLOCK and minimum one sensor is not faulted
- Left Gear – One or both downlock sensors report DOWNLOCK and minimum one sensor is not faulted.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT16D	B6D	IOM2

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to IOM2 (Refer to WM31-41-00).

Equation No. 20 NOSE GEAR DOWNLOCK (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602 incorporated)

$$(\text{NGDN1} \cdot \text{NGLK1}) + (\text{NGDN2} \cdot \text{NDLK2})$$

Equation Explanation: The Nose Gear Downlock discrete output activates when the Primary Indication or the Alternate Downlock Indication System shows the Nose Gear is Down and Locked.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT08B	A11F	SCU
DOT09B	A11G	Contacto
DOT14B	C2F	IFC1, IFC2

Result: When you get the conditions of the equation, the PSEU (at card PSC B) sends a ground signal to the Steering Control Unit (Refer to WM32-51-00). From the connector J1 pin A11G (at card PSC B) a ground signal is sent to the taxi light contactor (Refer to WM33-41-00). A ground signal is also sent to IOM1 and IOM2



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(Refer to , WM31-41-00) from the connector J1 pin C2F (at card PSC B). IOM1 and IOM2 are found in their related IFCs.

Equation No. 21 NOSE GEAR NOT UPLOCK (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602 incorporated)

$(\text{NGDN1} + \text{NOT NGLK1}) + (\text{NGDN2} + \text{NOT NGLK2})$

Equation Explanation: The Nose Gear Not Uplock discrete output activates when the Primary Indication or the Alternate Downlock Indication System shows the Nose Gear is not up and Locked.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT14D	B5B	IFC1, IFC2

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to IOM1 and IOM2 (Refer to WM31-41-00). IOM1 and IOM2 are found in their related IFCs.

Equation No. 22 MAIN GEAR DOWNLOCK (On aircraft without SB84-32-66 and MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0501)

$(\text{LGDLK1} \cdot \text{RGDLK1}) + (\text{LGDLK2} \cdot \text{RGDLK2})$

Equation Explanation: The Main Gear Downlock discrete output activates when the Primary Indication or the Alternate Downlock Indication System shows the Main gear is Down and Locked.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT18B	C2K	IFC1, IFC2, TCAS

Result: When you get the conditions of the equation, the PSEU (at card PSC B) sends a ground signal to IOM1 and IOM2 (Refer to WM31-41-00). IOM1 and IOM2 are found in their related IFCs. A ground signal is also sent from pin C2K to the TCAS (Refer to , WM34-43-00).

Equation No. 22 MAIN GEAR DOWNLOCK (On aircraft with SB 84-32-66 or SB84-32-143 or SB84-32-149 or MODSUM 4-126420 or MODSUM 4-124247 or MODSUM 4-126663 incorporated and on aircraft installed with PSEU P/N 30145-0401 or 30145-0601 or 30145-0602)

$[(\text{LGDNLK1} + \text{LGDNLK2}) \cdot (\text{NOT LGDNLK1_F} + \text{NOT LGDNLK2_F})] \cdot [(\text{RGDNLK1} + \text{RGDNLK2}) \cdot (\text{NOT RGDNLK1_F} + \text{NOT RGDNLK2_F})]$

Equation Explanation: The Main Gear Downlock discrete output activates when the Primary System Downlock sensor or the Alternate System Downlock sensor shows the main gear is Down and Locked for each main gear, and minimum one downlock sensor of each Main Gear is not faulted.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT18B	C2K	IFC1, IFC2
DOT19B	C3F	TCAS

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Result: When you get the conditions of the equation, the PSEU (at card PSC B) sends a ground signal to IOM1 and IOM2 (Refer to WM31-41-00). IOM1 and IOM2 are found in their related IFCs. A ground signal is also sent from pin C3F to the TCAS (Refer to , WM34-43-00).

Equation No. 23 MAIN GEAR UPLOCK (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

LGUPLK · RGUPLK

Equation Explanation: The Main Gear Uplock discrete output activates when the Main Gear is Up and Locked.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT18D	B6B	ACU
DOT19D	B6A	ACU

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to the ACU (Refer to WM32-46-00).

Equation No. 24 MAIN GEAR NOT UPLOCK (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

NOT LGUPLK + NOT RGUPLK

Equation Explanation: The Main Gear Not Uplock discrete output activates when the Main Gear is not Up and Locked.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT17D	B6C	IFC1, IFC2

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to IOM1 and IOM2 (Refer to WM31-41-00). IOM1 and IOM2 are found in their related IFCs.

Equation No. 25 LANDING GEAR CONTROL OPERATIONAL (On aircraft without SB84-32-66 and MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0501)

$$[NOT\ DOT01A_F \cdot NOT\ DOT02A_F \cdot NOT\ DOT03A_F \cdot NOT\ DOT04A_F \cdot NOT\ DOT05A_F] \cdot [NOT\ LDGINH \cdot ((EXTCMD \cdot NOT\ RETCMD) + (NOT\ EXTCMD \cdot RETCMD))] \cdot [EXTCMD \cdot (NOT\ LGDLK1_F + LGDLK2) \cdot (NOT\ LGDLK2_F + LGDLK1) \cdot (NOT\ RGD LK1_F + RGD LK2) \cdot (NOT\ RGD LK2_F + RGD LK1) \cdot (NOT\ NGLK1_F + NGLK2) \cdot (NOT\ NGLK2_F + NGLK1) \cdot (NOT\ NGDN1_F + NGDN2) \cdot (NOT\ NGDN2_F + NGDN1)] + RETCMD]$$

Equation Explanation: The Landing Gear Control Operational output deactivates if one or more of the inputs or conditions that follow are true:

- Extend Selector Valve Discrete Output Fault
- Retract Selector Valve Discrete Output Fault
- NG Door Sequence Valve Discrete Output Fault
- RG Door Sequence Valve Discrete Output Fault
- LG Door Sequence Valve Discrete Output Fault
- LDGINH discrete input is TRUE

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- The EXTCMD and RETCMD discrete inputs are not in complimentary states
- In the presence of a downlock sensor fault, the opposite functional sensor does not show NEAR when the EXTCMD is TRUE. For example, if the Primary Left Gear Downlock sensor is open or EXTEND has been selected, this output will deactivate (and thus light the LDG INOP light) if the Alternate Left Gear Downlock sensor does not show NEAR.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT13D	B5C	CAWP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) removes a ground signal from pin B5C through the Relay Junction Box No. 2. It then goes to the Caution and Warning Panel to light the LDG GEAR INOP caution light.

Equation No. 25 LANDING GEAR CONTROL OPERATIONAL (On aircraft with SB 84-32-66 or MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0401)

$$[NOT\ DOT01A_F \cdot NOT\ DOT02A_F \cdot NOT\ DOT03A_F \cdot NOT\ DOT04A_F \cdot NOT\ DOT05A_F] \cdot [NOT\ LDGINH \cdot ((EXTCMD \cdot NOT\ RETCMD) + (NOT\ EXTCMD \cdot RETCMD))] \cdot [[EXTCMD \cdot (NOT\ LGDLK1_F + LGDLK2) \cdot (NOT\ LGDLK2_F + LGDLK1) \cdot (NOT\ RGD LK1_F + RGD LK2) \cdot (NOT\ RGD LK2_F + RGD LK1) \cdot (NOT\ NGLK1_F + NGLK2) \cdot (NOT\ NGLK2_F + NGLK1) \cdot NOT\ (NGLK1_F \cdot NGLK2_F) \cdot (NOT\ NGDN1_F + NGDN2) \cdot (NOT\ NGDN2_F + NGDN1) \cdot NOT\ (NGDN1_F \cdot NGDN2_F)] + RETCMD] \cdot (NOT\ RGUPLK_F \cdot NOT\ LGUPLK_F \cdot NOT\ RGDRCL_F \cdot NOT$$

$$LGDRCL_F \cdot NOT\ NGDRCL_F) \cdot (NOT\ NWCENT_F + NOT\ DOT08E)$$

Equation Explanation: The Landing Gear Control Operational output deactivates if one or more of the inputs or conditions that follow are true:

- Extend Selector Valve Discrete Output Fault
- Retract Selector Valve Discrete Output Fault
- NG Door Sequence Valve Discrete Output Fault
- RG Door Sequence Valve Discrete Output Fault
- LG Door Sequence Valve Discrete Output Fault
- LDGINH discrete input is TRUE
- The EXTCMD and RETCMD discrete inputs are not in complimentary states or in the presence of a downlock sensor fault or Nose Gear down or lock sensor fault, the opposite unfaulted functional sensor does not show NEAR when the EXTCMD is TRUE. For example, if the Primary Left Gear Downlock sensor is open and EXTEND has been selected, this output will deactivate (and thus light the LDG INOP light) if the Alternate Left Gear Downlock sensor does not show Downlock and not faulted
- When any Uplock or gear door sensor is faulted
- When fault is detected in the nose wheel centering sensor and the steering system is operational

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DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT13D	B5C	CAWP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) removes a ground signal from pin B5C through the Relay Junction Box No. 2. It then goes to the Caution and Warning Panel to light the LDG GEAR INOP caution light.

Equation No. 25 LANDING GEAR CONTROL OPERATIONAL (On aircraft with SB84-32-143 or SB84-32-149 or MODSUM 4-124247 or MODSUM 4-126663 incorporated and on aircraft installed with PSEU P/N 30145-0601 or 30145-0602)

$$[NOT DOT01A_F \cdot NOT DOT02A_F \cdot NOT DOT03A_F \cdot NOT DOT04A_F \cdot NOT DOT05A_F] \cdot [NOT LDGINH \cdot ((EXTCMD \cdot NOT RETCMD) + (NOT EXTCMD \cdot RETCMD))] \cdot [EXTCMD \cdot (NOT LGDLK1_F + LGDLK2) \cdot (NOT LGDLK2_F + LGDLK1) \cdot (NOT RGD LK1_F + RGD LK2) \cdot (NOT RGD LK2_F + RGD LK1) \cdot (NOT NGLK1_F + NGLK2) \cdot (NOT NGLK2_F + NGLK1) \cdot NOT (NGLK1_F \cdot NGLK2_F) \cdot (NOT NGDN1_F + NGDN2) \cdot (NOT NGDN2_F + NGDN1) \cdot NOT (NGDN1_F \cdot NGDN2_F)] + RETCMD] \cdot (NOT RGUPLK_F \cdot NOT LGUPLK_F \cdot NOT RGDRCL_F \cdot NOT LGDRCL_F \cdot NOT NGDRCL_F) \cdot (NOT NWCENT_F + NOT DOT08E) \cdot NOT (LGUPLK \cdot (((RGWOW1 + RGWOW1_F) \cdot (RGWOW2 + RGWOW2_F) \cdot (LGWOW1 + LGWOW1_F)] + [(RGWOW1 + RGWOW1_F) \cdot (RGWOW2 + RGWOW2_F) \cdot (LGWOW2 + LGWOW2_F)] + [(RGWOW1 + RGWOW1_F) \cdot (LGWOW1 + LGWOW1_F) \cdot (LGWOW2 + LGWOW2_F)] + [(RGWOW2 + RGWOW2_F) \cdot (LGWOW1 + LGWOW1_F) \cdot (LGWOW2 + LGWOW2_F)]) \cdot [(NOT RGWOW1_F \cdot NOT LGWOW1_F) + (NOT RGWOW2_F \cdot NOT LGWOW2_F)]) +$$

$$RGUPLK \cdot (((RGWOW1 + RGWOW1_F) \cdot (RGWOW2 + RGWOW2_F) \cdot (LGWOW1 + LGWOW1_F)] + [(RGWOW1 + RGWOW1_F) \cdot (RGWOW2 + RGWOW2_F) \cdot (LGWOW2 + LGWOW2_F)] + [(RGWOW1 + RGWOW1_F) \cdot (LGWOW1 + LGWOW1_F) \cdot (LGWOW2 + LGWOW2_F)] + [(RGWOW2 + LGWOW2_F) \cdot (RGWOW2 + LGWOW2_F) \cdot (LGWOW1 + LGWOW1_F) \cdot (LGWOW2 + LGWOW2_F)]) \cdot [(NOT RGWOW1_F \cdot NOT LGWOW1_F) + (NOT RGWOW2_F \cdot NOT LGWOW2_F)]) \cdot NOT (RETCMD \cdot \{(LGDRCL \cdot NOT LGUPLK \cdot (NOT LGDLK1 + NOT LGDLK2)) + (RGDRCL \cdot NOT RGUPLK \cdot (NOT RGD LK1 + NOT RGD LK2)) + (NGDRCL \cdot (NOT NGLK1 + NOT NGLK2) \cdot (NOT NGDN1 + NOT NGDN2))\})$$

Equation Explanation: The Landing Gear Control Operational output deactivates if one or more of the inputs or conditions that follow are true:

- Extend Selector Valve Discrete Output Fault
- Retract Selector Valve Discrete Output Fault
- NG Door Sequence Valve Discrete Output Fault
- RG Door Sequence Valve Discrete Output Fault
- LG Door Sequence Valve Discrete Output Fault
- LDGINH discrete input is TRUE
- The EXTCMD and RETCMD discrete inputs are not in complimentary states or in the presence of a downlock sensor fault or Nose Gear down or lock sensor fault, the opposite unfaulted functional sensor does not show NEAR when the EXTCMD is TRUE. For example, if the Primary Left Gear Downlock sensor is open and EXTEND has been selected, this output will deactivate (and thus light the LDG INOP light) if the Alternate Left Gear Downlock sensor does not show Downlock and not faulted

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- When any Uplock or gear door sensor is faulted
- When fault is detected in the nose wheel centering sensor and the steering system is operational
- When one or two MLG uplocks are engaged on ground
- When one of the LG door closed before the Uplock is detected.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT13D	B5C	CAWP

Result: When you get the conditions of the equation, the PSEU (at card PSC D) removes a ground signal from pin B5C through the Relay Junction Box No. 2. It then goes to the Caution and Warning Panel to light the LDG GEAR INOP caution light.

Equation No. 32 NG SAFE (ADI) (On aircraft without SB84-32-66 and MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0501 or 30145-0601 or 30145-0602)

$(NGLK2 \cdot NGDN2) + LMPTST2$

Equation Explanation: The Nose Gear Safe (Alternate Downlock Indication) discrete output is activated when the ADI Function finds that the Nose Gear is Down and Locked, or when the Lamp Test 2 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT02B	C1G	ADIP

Result: When you get the conditions of the equation, the PSEU (at card PSC B) gives an output of 28 V DC to the NG SAFE light below the flight compartment floor.

Equation No. 32 NG SAFE (ADI) (On aircraft with SB 84-32-66 or MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0401)

$[(NGDN1 \cdot NOT\ NGDN1_F) + (NGDN2 \cdot NOT\ NGDN2_F)] \cdot [(NGLK1 \cdot NOT\ NGLK1_F) + (NGLK2 \cdot NOT\ NGLK2_F)] + LMPTST2$

Equation Explanation: The Nose Gear Safe (Alternate Downlock Indication) discrete output is activated with the conditions that follow:

- When EXTEND is selected and when one or both of the Nose Gear down sensors report Down and these sensors are not faulted and also when one or both of the Nose Gear lock sensors report Locked and these sensors are not faulted
- When the Lamp Test 2 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT02B	C1G	ADIP

Result: When you get the conditions of the equation, the PSEU (at card PSC B) gives an output of 28 V DC to the NG SAFE light below the flight compartment floor.



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Equation No. 33 RG SAFE (ADI) (On aircraft without SB 84-32-66 and MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0501 or 30145-0601 or 30145-0602)

RGDLK2 + LMPTST2

Equation Explanation: The Right Gear Safe (Alternate Downlock Indication) discrete output is activated when the ADI Function senses that the Right Gear is Down and Locked, or when the Lamp Test 2 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT03B	C1H	ADIP

Result: When you get the conditions of the equation, the PSEU (at card PSC B) gives an output of 28 V DC to the RG SAFE light below the flight compartment floor.

Equation No. 33 RG SAFE (ADI) (On aircraft with SB84-32-66 or MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0401)

$$[RGDLK1 \cdot (RGDLK2 + NOT\ RGUPLK)] + [RGDLK2 \cdot (RGDLK1 + NOT\ RGUPLK)] + LMPTST2$$

Equation Explanation: The Right Gear Safe (Alternate Downlock Indication) discrete output is activated with the conditions that follow:

- When both of the right Main Gear downlock sensors report Down

- When minimum one of the downlock sensor reports Down and the right Main Gear Uplock does not report target NEAR (for reasonableness)
- When the Lamp Test 2 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT03B	C1H	ADIP

Result: When you get the conditions of the equation, the PSEU (at card PSC B) gives an output of 28 V DC to the RG SAFE light below the flight compartment floor.

Equation No. 34 LG SAFE (ADI) (On aircraft without SB84-32-66 and MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0501 or 30145-0601 or 30145-0602)

LGDLK2 + LMPTST2

Equation Explanation: The Left Gear Safe (Alternate Downlock Indication) discrete output is activated when the ADI Function senses that the Left Gear is Down and Locked, or when the Lamp Test 2 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT01B	C1F	ADIP

Result: When you get the conditions of the equation, the PSEU (at card PSC B) gives an output of 28 V DC to the LG SAFE light below the flight compartment floor.



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Equation No. 34 LG SAFE (ADI) (On aircraft with SB84-32-66 or MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0401)

$$[LGDLK1 \cdot (LGDLK2 + NOT LGUPLK)] + [LGDLK2 \cdot (LGDLK1 + NOT LGUPLK)] + LMPTST2$$

Equation Explanation: The Left Gear Safe (Alternate Downlock Indication) discrete output is activated with the conditions that follow:

- When both of the left Main Gear downlock sensors report Down
- When minimum one of the downlock sensor reports Down and the left Main Gear Uplock does not report target NEAR (for reasonableness)
- When the Lamp Test 2 discrete input is active.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT01B	C1F	ADIP

Result: When you get the conditions of the equation, the PSEU (at card PSC B) gives an output of 28 V DC to the LG SAFE light below the flight compartment floor.

Equation No. 35 ALL LANDING GEAR DOWNLOCK (ADI) (On aircraft without SB84-32-66 and MODSUM 4-126420 incorporated and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0501)

$$NGDN2 \cdot NGLK2 \cdot LGDLK2 \cdot RGD LK2$$

Equation Explanation: The All Landing Gear Downlock (Alternate Downlock Indication) discrete output activates only when ADI

downlock sensors for all three gear show the Down and Locked condition.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT16B	C2H	IOM1

Result: When you get the conditions of the equation, the PSEU (at card PSC B) sends a ground signal to IOM1 (Refer to WM31-41-00).

Equation No. 35 ALL LANDING GEAR DOWNLOCK (ADI) (On aircraft with SB84-32-66 or SB84-32-143 or SB84-32-149 or MODSUM 4-126420 or MODSUM 4-124247 or MODSUM 4-126663 incorporated and on aircraft installed with PSEU P/N 30145-0401 or 30145-0601 or 30145-0602)

$$[(NGDN1 + NGDN2) \cdot (NOT NGDN1_F + NOT NGDN2_F)] \cdot [(NGLK1 + NGLK2) \cdot (NOT NGLK1_F + NOT NGLK2_F)] \cdot [(RGDLK1 + RGDLK2) \cdot (NOT RGDLK1_F + NOT RGDLK2_F)] \cdot [(LGDLK1 + LGDLK2) \cdot (NOT LGDLK1_F + NOT LGDLK2_F)]$$

Equation Explanation: The All Landing Gear Downlock (Alternate Downlock Indication) discrete output activates only when all three gears are down and locked as determined by the down, lock and downlock sensor conditions that follow:

- Nose Gear – One or both unfaulted down sensors report Down and one or both unfaulted lock sensors report Down
- Right Gear – One or both downlock sensors report Downlock and minimum one of the sensors is not faulted
- Left Gear – One or both downlock sensors report Downlock and minimum one of the sensors is not faulted.

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DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT16B	C2H	IOM1

Result: When you get the conditions of the equation, the PSEU (at card PSC B) sends a ground signal to IOM1 (Refer to WM31-41-00).

Equation No. 36 WEIGHT-ON-WHEELS (WOW) GROUND MODE
(On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

$$\begin{aligned}
 &[(RGWOW1 + RGWOW1_F) \cdot (RGWOW2 + RGWOW2_F) \cdot \\
 &(LGWOW1 + LGWOW1_F) \cdot (LGWOW2 + LGWOW2_F) \cdot (NOT \\
 &NGWOFW1 + NGWOFW1_F)] + [(RGWOW1 + RGWOW1_F) \cdot \\
 &(RGWOW2 + RGWOW2_F) \cdot (LGWOW1 + LGWOW1_F) \cdot \\
 &(LGWOW2 + LGWOW2_F) \cdot (NOT NGWOFW2 + NGWOFW2_F)] + \\
 &[(RGWOW1 + RGWOW1_F) \cdot (RGWOW2 + RGWOW2_F) \cdot \\
 &(LGWOW1 + LGWOW1_F) \cdot (NOT NGWOFW1 + NGWOFW1_F) \cdot \\
 &(NOT NGWOFW2 + NGWOFW2_F)] + [(RGWOW1 + RGWOW1_F) \cdot \\
 &(RGWOW2 + RGWOW2_F) \cdot (LGWOW2 + LGWOW2_F) \cdot (NOT \\
 &NGWOFW1 + NGWOFW1_F) \cdot (NOT NGWOFW2 + NGWOFW2_F)] \\
 &+ [(RGWOW1 + RGWOW1_F) \cdot (LGWOW1 + LGWOW1_F) \cdot \\
 &(LGWOW2 + LGWOW2_F) \cdot (NOT NGWOFW1 + NGWOFW1_F) \cdot \\
 &(NOT NGWOFW2 + NGWOFW2_F)] + [(RGWOW2 + RGWOW2_F) \cdot \\
 &(LGWOW1 + LGWOW1_F) \cdot (LGWOW2 + LGWOW2_F) \cdot (NOT \\
 &NGWOFW1 + NGWOFW1_F) \cdot (NOT NGWOFW2 + \\
 &NGWOFW2_F)] \cdot [(NOT RGWOW1_F \cdot NOT LGWOW1_F \cdot NOT \\
 &NGWOFW1_F) + (NOT RGWOW2_F \cdot NOT LGWOW2_F \cdot NOT \\
 &NGWOFW2_F)]
 \end{aligned}$$

Equation Explanation: The WOW equation outputs are active in the conditions that follow:

- If all six WOW sensors operate correctly, five of those six sensors show that the aircraft is on the ground
- If one WOW sensor is faulted, four of the remaining five sensors show that the aircraft is on the ground
- If two WOW sensors from the same WOW System (WOW1 or WOW2) are faulted, three of the remaining four sensors show that the aircraft is on the ground
- If all three WOW sensors from the same WOW System are faulted (or one full WOW System is inoperative), two of the remaining three sensors on the non-failed WOW System show that the aircraft is on the ground

If one or more sensors are faulted on the two WOW Systems (WOW1 and WOW2) of the same gear, the WOW output will default to the inactive (fail-safe) state.

Table 2: Pre Modsum 4-126276

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT20A	A6K	CPC
DOT21A	A7G	Throttle (MTOP)
DOT23A	A8F	Avionics Cooling Fans
DOT25A	A8H	AHRS1, ADU1

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DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT04B	C1J	n/a
DOT05B	C1K	n/a
DOT17B	C2J	Hydraulic System
DOT23B	C3K	No Smoking sign
DOT24B	C4F	ARCDU
DOT25B	C4G	APU
DOT26B	C4H	RCAU
DOT27B	C4J	TCAS, WXR
DOT04D	B2B	n/a
DOT23D	B7B	ECU
DOT24D	B7A	ARCDU2
DOT25D	B8E	APU
DOT27D	B8C	APU Fire Protection Panel
DOT30D	B9D	EPCU

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT04E	B1B	n/a
DOT20E	C2C	CVR
DOT21E	C2B	TMU
DOT25E	C3B	AHRS2, ADU2
DOT30E	C4A	FCU

Results:

- When you get the conditions of the equation, the PSEU (at card PSC A) sends a ground signal to the CPC (Refer to WM21–31–00).
- The connector J1 pin A7G (at card PSC A) sends a ground signal to the throttle (Refer to WM21–31–00).
- The connector J1 pin A8F (at card PSC A) sends a ground signal to the avionics cooling fans (Refer to WM21–25–00).
- The connector J1 pin A8H (at card PSC A) sends a ground signal to the AHRS1 (Refer to WM34–21–00).
- At the connector J1 pins C1J and C1K (at card PSC B) are tagged and stowed.
- The connector J1 pin C2J (at card PSC B) sends a ground signal to the main hydraulic system No. 3 (Refer to WM29–12–00–1).



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- From the connector J1 pin C3K (at card PSC B) a ground signal is sent to the No Smoking signs at the PSUs (Refer to WM33–24–00).
- At the connector J1 pin C4F (at card PSC B) a ground signal is sent to the ARCDU1 (Refer to WM23–81–00).
- At the connector J1 pin C4G (at card PSC B) a ground signal is sent to the APU Relay Panel (Refer to WM26–23–00).
- At the connector J1 pin C4H (at card PSC B) a ground signal is sent to the RCAU (Refer to WM23–51–00).
- A ground signal is sent from the connector J1 pin C4J (at card PSC B) to the TCAS and the Weather Radar Systems.
- The wire from connector J1 pin B2B (at card PSC D) is not used; it is tagged and stowed.
- At the connector J1 pin B7B (at card PSC E), a ground signal is sent to the Temperature Control and Indication System (ECU) (Refer to WM21–61–00).
- At the connector J1 pin B7A (at card PSC A) a ground signal is sent to the ARCDU2 (Refer to WM23–81–00).
- A ground signal is sent from the connector J1 pin B8E (at card PSC D) to the APU.
- A ground signal is also sent from the connector J1 pin B8C (at card PSC D) to the APU Fire Protection System.
- At the connector J1 pin B9D (at card PSC E) a ground signal is sent to the EPCU (Refer to WM24–31–00).

- When you get the conditions of the equation, the PSEU gives an output of 28 V DC. This output is from the connector J1 pin B1B to the Auxiliary WOW Relay 1.
- At the connector J1 pin C2C (at card PSC E) a ground signal is sent to the CVR (Refer to WM23–71–00).
- At the connector J1 pin C2B (at card PSC E) a ground signal is sent to the TMU (Refer to WM30–61–00).
- At the connector J1 pin C3B (at card PSC E) a ground signal is sent to the AHRS2 (Refer to WM34–21–00).
- At the connector J1 pin C4A (at card PSC E) a ground signal is sent to the FCU (Refer to WM27–51–00).

Table 3: Post Modsums 4–126276 and 4–126420

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT20A	A6K	CPC
DOT21A	A7G	Throttle (MTOP)
DOT23A	A8F	Avionics Cooling Fans
DOT25A	A8H	AHRS1, ADU1
DOT05B	C1K	n/a
DOT17B	C2J	Hydraulic System
DOT23B	C3K	No Smoking sign

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DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT24B	C4F	ARCDU
DOT23D	B7B	ECU
DOT24D	B7A	ARCDU2
DOT27D	B8C	APU Fire Protection Panel
DOT04E	B1B	n/a
DOT20E	C2C	CVR
DOT21E	C2B	TMU
DOT25E	C3B	AHRS2, ADU2
DOT30E	C4A	FCU

Results:

- When you get the conditions of the equation, the PSEU (at card PSC A) sends a ground signal to the CPC (Refer to WM21–31–00).
- The connector J1 pin A7G (at card PSC A) sends a ground signal to the throttle (Refer to WM21–31–00).
- The connector J1 pin A8F (at card PSC A) sends a ground signal to the avionics cooling fans (Refer to WM21–25–00).

- The connector J1 pin A8H (at card PSC A) sends a ground signal to the AHRS1 (Refer to WM34–21–00).
- At the connector J1 pins C1K (at card PSC B) are tagged and stowed.
- The connector J1 pin C2J (at card PSC B) sends a ground signal to the main hydraulic system No. 3 (Refer to WM29–12–00–1).
- From the connector J1 pin C3K (at card PSC B) a ground signal is sent to the No Smoking signs at the PSUs (Refer to WM33–24–00).
- At the connector J1 pin C4F (at card PSC B) a ground signal is sent to the ARCDU1 (Refer to WM23–81–00).
- At the connector J1 pin B7B (at card PSC D), a ground signal is sent to the Temperature Control and Indication System (ECU) (Refer to WM21–61–00).
- At the connector J1 pin B7A (at card PSC D) a ground signal is sent to the ARCDU2 (Refer to WM23–81–00).
- A ground signal is also sent from the connector J1 pin B8C (at card PSC D) to the APU Fire Protection System.
- When you get the conditions of the equation, the PSEU gives an output of 28 V DC. This output is from the connector J1 pin B1B to the Auxiliary WOW Relay 1.
- At the connector J1 pin C2C (at card PSC E) a ground signal is sent to the CVR (Refer to WM23–71–00).
- At the connector J1 pin C2B (at card PSC E) a ground signal is sent to the TMU (Refer to WM30–61–00).



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- At the connector J1 pin C3B (at card PSC E) a ground signal is sent to the AHRS2 (Refer to WM34–21–00).
- At the connector J1 pin C4A (at card PSC E) a ground signal is sent to the FCU (Refer to WM27–51–00).

Equation No. 37 WEIGHT-ON-WHEELS (MGWOW) MAIN GEAR
(On aircraft installed with PSEU P/N 30145–0201 or 30145–0301 or 30145–0401 or 30145–0501 or 30145–0601 or 30145–0602)

$$((RGWOW1 + RGWOW1_F) \cdot (RGWOW2 + RGWOW2_F) \cdot (LGWOW1 + LGWOW1_F)) + ((RGWOW1 + RGWOW1_F) \cdot (RGWOW2 + RGWOW2_F) \cdot (LGWOW2 + LGWOW2_F)) + ((RGWOW1 + RGWOW1_F) \cdot (LGWOW1 + LGWOW1_F) \cdot (LGWOW2 + LGWOW2_F)) + ((RGWOW2 + RGWOW2_F) \cdot (LGWOW1 + LGWOW1_F) \cdot (LGWOW2 + LGWOW2_F)) \cdot ((NOT RGWOW1_F \cdot NOT LGWOW1_F) + (NOT RGWOW2_F \cdot NOT LGWOW2_F))$$

Equation Explanation: The MGWOW equation outputs are active in the conditions that follow:

- If all four Main Gear WOW sensors operate correctly, three of those four sensors show that the aircraft is on the ground
- If one Main Gear WOW sensor is faulted, two of the remaining three sensors show that the aircraft is on the ground
- If the two Main Gear WOW sensors from the same WOW System are faulted (or one full WOW System is inoperative), one of the remaining two sensors on the non-faulted WOW System show that the aircraft is on the ground

If one or more Main Gear WOW sensors are faulted on the two WOW Systems (WOW1 and WOW2) of the same gear, the MGWOW output will default to the inactive (fail-safe) state.

Table 4: Pre Modsum 4–126276

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT15A	A4K	SCV
DOT16A	A6F	SCV
DOT17A	A6G	FCEU
DOT18A	A6H	PEC
DOT19A	A6J	PEC
DOT22A	A7J	MFD1, ED
DOT20B	C3G	IOM1, IOM2
DOT21B	C3H	ANVS
DOT22B	C3J	Co-Pilot Clock
DOT22D	B7C	Pilot Clock
DOT26D	B8D	IFC1, IFC2
DOT15E	C1D	SCV

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DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT16E	C1C	SCV
DOT17E	C1B	FCEU
DOT18E	C1A	PEC
DOT19E	C2D	PEC
DOT22E	C2A	PFD1/2, MFD2

Result:

- When you get the conditions of the equation, the PSEU (at card PSC A) gives an output of 28 V DC to the SCV (Refer to WM27–62–00).
- At the connector J1 pin A6F (at card PSC A) a ground signal is sent to the SCV (Refer to WM27–62–00).
- At the connector J1 pin A6G (at card PSC A), a ground signal is sent to the FCEU (Refer to WM27–62–00).
- The connector J1 pins A6H (card PSC B) and 6J (card PSC A) send a ground signal to the PEC (Refer to WM61–20–00).
- At the connector J1 pin A7J (at card PSC A), a ground signal is sent to the MFD1 and the ED (Refer to WM31–61–00).
- The connector J1 pin C3G is a Spare.

- At the connector J1 pin C3H (at card PSC B), a ground signal is sent to the ANVS (Refer to WM23–35–00).
- At the connector J1 pin C3J (at card PSC B), a ground signal is sent to the Co–Pilot Clock (Refer to WM31–21–00).
- At the connector J1 pin B7C (at card PSC D), a ground signal is sent to the Pilot Clock (Refer to WM31–21–00).
- When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal from to IOM1 and IOM2 (Refer to WM31–41–00). IOM1 and IOM2 are found in their related IFCs.
- At the connector J1 pins C1D and 1C (at card PSC E) a ground signal is sent to the SCV (Refer to WM27–62–00).
- At the connector J1 pin C1B (at card PSC E), a ground signal is sent to the FCEU (Refer to WM27–62–00).
- The connector J1 pins C1A (card PSC A) and 2D (card PSC E) send a ground signal to the PEC (Refer to WM61–20–00).
- At the connector J1 pin C2A (at card PSC E), a ground signal is sent to the PFD1/2 and the MFD2 (Refer to WM31–61–00).

Table 5: Post Modsum 4–126276, Pre Modsum 4–126420 and Pre



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DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT15A	A4K	SCV
DOT16A	A6F	SCV
DOT17A	A6G	FCEU
DOT18A	A6H	PEC
DOT19A	A6J	PEC
DOT22A	A7J	MFD1, ED
DOT04B	C1J	n/a
DOT20B	C3G	IOM1, IOM2
DOT21B	C3H	ANVS
DOT22B	C3J	Co–Pilot Clock
DOT25B	C4G	APU
DOT26B	C4H	RCAU
DOT27B	C4J	TCAS, WXR
DOT22D	B7C	Pilot Clock

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT25D	B8E	APU
DOT26D	B8D	IFC1, IFC2
DOT30D	B9D	EPCU
DOT15E	C1D	SCV
DOT16E	C1C	SCV
DOT17E	C1B	FCEU
DOT18E	C1A	PEC
DOT19E	C2D	PEC
DOT22E	C2A	PFD1/2, MFD2
DOT26E	C3A	FGCOM1, IOP1, FGMON1, SPM1
DOT27E	C4D	FGCOM2, IOP2, FGMON2, SPM2

Results:

- When you get the conditions of the equation, the PSEU (at card PSC A) gives an output of 28 V DC to the SCV (Refer to WM27–62–00).
- At the connector J1 pin A6F (at card PSC A) a ground signal is sent to the SCV (Refer to WM27–62–00).

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- At the connector J1 pin A6G (at card PSC A), a ground signal is sent to the FCEU (Refer to WM27–62–00).
- The connector J1 pins A6H (card PSC B) and 6J (card PSC A) send a ground signal to the PEC (Refer to WM61–20–00).
- At the connector J1 pin A7J (at card PSC A), a ground signal is sent to the MFD1 and the ED (Refer to WM31–61–00).
- At the connector J1 pins C1J (at card PSC B) are tagged and stowed.
- The connector J1 pin C3G is a Spare.
- At the connector J1 pin C3H (at card PSC B), a ground signal is sent to the ANVS (Refer to WM23–35–00).
- At the connector J1 pin C3J (at card PSC B), a ground signal is sent to the Co–Pilot Clock (Refer to WM31–21–00).
- At the connector J1 pin C4G (at card PSC B) a ground signal is sent to the APU Relay Panel (Refer to WM26–23–00).
- At the connector J1 pin C4H (at card PSC B) a ground signal is sent to the RCAU (Refer to WM23–51–00).
- A ground signal is sent from the connector J1 pin C4J (at card PSC B) to the TCAS and the Weather Radar Systems.
- At the connector J1 pin B7C (at card PSC D), a ground signal is sent to the Pilot Clock (Refer to WM31–21–00).
- A ground signal is sent from the connector J1 pin B8E (at card PSC D) to the APU.
- When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal from to IOM1 and IOM2 (Refer to WM31–41–00). IOM1 and IOM2 are found in their related IFCs.
- At the connector J1 pin B9D (at card PSC E) a ground signal is sent to the EPCU (Refer to WM24–31–00).
- At the connector J1 pins C1D and 1C (at card PSC E) a ground signal is sent to the SCV (Refer to WM27–62–00).
- At the connector J1 pin C1B (at card PSC E), a ground signal is sent to the FCEU (Refer to WM27–62–00).
- The connector J1 pins C1A (card PSC A) and 2D (card PSC E) send a ground signal to the PEC (Refer to WM61–20–00).
- At the connector J1 pin C2A (at card PSC E), a ground signal is sent to the PFD1/2 and the MFD2 (Refer to WM31–61–00).
- At the connector J1 pin C3A (at card PSC E), a ground signal is sent to the FGCOM1, IOP1, FGMON1 and the SPM1.
- At the connector J1 pin C4D (at card PSC E), a ground signal is sent to the FGCOM2, IOP2, FGMON2 and the SPM2.

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Table 6: Post Modsum 4–126420 and Post Service Bulletin 84–32–66

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT15A	A4K	SCV
DOT16A	A6F	SCV
DOT17A	A6G	FCEU
DOT18A	A6H	PEC, Rad Alt #2
DOT19A	A6J	PEC, Rad Alt #2
DOT22A	A7J	MFD1, ED
DOT26A	A8J	FGCOM1, IOP1, FGMON1, SPM1
DOT27A	A8K	IOP2, FGMON2, SPM2
DOT04B	C1J	n/a
DOT20B	C3G	IOM1, IOM2
DOT21B	C3H	ANVS
DOT22B	C3J	Co–Pilot Clock
DOT25B	C4G	APU

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT26B	C4H	RCAU
DOT27B	C4J	TCAS, WXR
DOT22D	B7C	Pilot Clock
DOT25D	B8E	APU
DOT26D	B8D	IFC1, IFC2
DOT30D	B9D	EPCU
DOT15E	C1D	SCV
DOT16E	C1C	SCV
DOT17E	C1B	FCEU
DOT18E	C1A	PEC, Rad Alt #1
DOT19E	C2D	PEC, Rad Alt #1
DOT22E	C2A	PFD1/2, MFD2
DOT26E	C3A	FGCOM1, IOP1, FGMON1, SPM1
DOT27E	C4D	FGCOM2, IOP2, FGMON2, SPM2

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Results:

- When you get the conditions of the equation, the PSEU (at card PSC A) gives an output of 28 V DC to the SCV (Refer to WM27–62–00).
- At the connector J1 pin A6F (at card PSC A) a ground signal is sent to the SCV (Refer to WM27–62–00).
- At the connector J1 pin A6G (at card PSC A), a ground signal is sent to the FCEU (Refer to WM27–62–00).
- The connector J1 pins A6H (card PSC B) and 6J (card PSC A) send a ground signal to the PEC and the Radio Altimeter System (Refer to WM61–20–00).
- At the connector J1 pin A7J (at card PSC A), a ground signal is sent to the MFD1 and the ED (Refer to WM31–61–00).
- At the connector J1 pin A8J (at card PSC A), a ground signal is sent to the FGCOM1, IOP1, FGMON1 and the SPM1.
- At the connector J1 pin A8K (at card PSC A), a ground signal is sent to the IOP2, FGMON2 and the SPM2.
- At the connector J1 pins C1J (at card PSC B) are tagged and stowed.
- The connector J1 pin C3G is a Spare.
- At the connector J1 pin C3H (at card PSC B), a ground signal is sent to the ANVS (Refer to WM23–35–00).
- At the connector J1 pin C3J (at card PSC B), a ground signal is sent to the Co–Pilot Clock (Refer to WM31–21–00).
- At the connector J1 pin C4G (at card PSC B) a ground signal is sent to the APU Relay Panel (Refer to WM26–23–00).
- At the connector J1 pin C4H (at card PSC B) a ground signal is sent to the RCAU (Refer to WM23–51–00).
- A ground signal is sent from the connector J1 pin C4J (at card PSC B) to the TCAS and the Weather Radar Systems.
- At the connector J1 pin B7C (at card PSC D), a ground signal is sent to the Pilot Clock (Refer to WM31–21–00).
- A ground signal is sent from the connector J1 pin B8E (at card PSC D) to the APU.
- When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal from to IOM1 and IOM2 (Refer to WM31–41–00). IOM1 and IOM2 are found in their related IFCs.
- At the connector J1 pin B9D (at card PSC E) a ground signal is sent to the EPCU (Refer to WM24–31–00).
- At the connector J1 pins C1D and 1C (at card PSC E) a ground signal is sent to the SCV (Refer to WM27–62–00).
- At the connector J1 pin C1B (at card PSC E), a ground signal is sent to the FCEU (Refer to WM27–62–00).
- The connector J1 pins C1A (card PSC A) and 2D (card PSC E) send a ground signal to the PEC and the Radio Altimeter System (Refer to WM61–20–00).
- At the connector J1 pin C2A (at card PSC E), a ground signal is sent to the PFD1/2 and the MFD2 (Refer to WM31–61–00).

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- At the connector J1 pin C3A (at card PSC E), a ground signal is sent to the FGCOM1, IOP1, FGMON1 and the SPM1.
- At the connector J1 pin C4D (at card PSC E), a ground signal is sent to the FGCOM2, IOP2, FGMON2 and the SPM2.

Equation No. 38 WEIGHT-OFF-WHEELS (MGWOFW) MAIN GEAR (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

$$[(\text{NOT RGWOW1} + \text{RGWOW1_F}) \cdot (\text{NOT RGWOW2} + \text{RGWOW2_F}) \cdot (\text{NOT LGWOW1} + \text{LGWOW1_F})] + [(\text{NOT RGWOW1} + \text{RGWOW1_F}) \cdot (\text{NOT RGWOW2} + \text{RGWOW2_F}) \cdot (\text{NOT LGWOW2} + \text{LGWOW2_F})] + [(\text{NOT RGWOW1} + \text{RGWOW1_F}) \cdot (\text{NOT LGWOW1} + \text{LGWOW1_F}) \cdot (\text{NOT LGWOW2} + \text{LGWOW2_F})] + [(\text{NOT RGWOW2} + \text{RGWOW2_F}) \cdot (\text{NOT LGWOW1} + \text{LGWOW1_F}) \cdot (\text{NOT LGWOW2} + \text{LGWOW2_F})]$$

Equation Explanation: The MGWOW equation outputs are active in the conditions that follow:

- If all four Main Gear WOW sensors operate correctly, three of those four sensors show that the aircraft is in the air
- If one Main Gear WOW sensor is faulted, two of the remaining three sensors show that the aircraft is in the air

- If the two Main Gear WOW sensors from the same WOW System are faulted (or one full WOW System is inoperative), one of the remaining two sensors on the non-failed WOW System show that the aircraft is in the air

If one or more Main Gear WOW sensors are faulted on the two WOW Systems (WOW1 and WOW2) of the same gear, the MGWOFW output will default to the inactive (fail-safe) state.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT14A	A4J	ACU
DOT14E	B15E	ACU

Result: When you get the conditions of the equation, the PSEU (at card PSC A) sends a ground signal to the ACU. A ground signal is also sent from the connector J1 pin B15E (at card PSC E) to the ACU (Refer to WM32-46-00).

Equation No. 39 WEIGHT-ON-WHEELS (NGWOW) NOSE GEAR (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

$$(\text{NOT NGWOFW1} + \text{NGWOFW1_F}) \cdot (\text{NOT NGWOFW2} + \text{NGWOFW2_F}) \cdot (\text{NOT NGWOFW1_F} + \text{NOT NGWOFW2_F})$$

Equation Explanation: The NGWOW equation outputs are active in the conditions that follow:

- If the two Nose Gear WOFW sensors operate correctly, the two sensors show that the aircraft is on the ground
- If one Nose Gear WOFW sensor is faulted, the other sensor shows that the aircraft is on the ground

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If the two Nose Gear WOFW sensors are faulted, the NGWOW output will default to the inactive (fail-safe) state.

Table 7: Pre Modsum 4-126276

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT08A	A2H	SCU
DOT24A	A8G	PFD1/2, MFD2
DOT26A	A8J	FGCOM1, IOP1, FGMON1, SPM1
DOT27A	A8K	IOP2, FGMON2, SPM2
DOT08E	B10D	SCU
DOT24E	C3C	MFD1, ED
DOT26E	C3A	FGCOM1, IOP1, FGMON1, SPM1
DOT27E	C4D	FGCOM2, IOP2, FGMON2, SPM2

Result:

- When you get the conditions of the equation, the PSEU (at card PSC A) sends a ground signal to the SCU (Refer to WM32-51-00).

- At the connector J1 pin A8G (at card PSC A), a ground signal is sent to the PFD1/2 and the MFD2 (Refer to WM31-61-00).
- At the connector J1 pin A8J (at card PSC A), a ground signal is sent to the FGCOM1, IOP1, FGMON1 and the SPM1.
- At the connector J1 pin A8K (at card PSC A), a ground signal is sent to the IOP2, FGMON2 and the SPM2.
- At the connector J1 pin B10D (at card PSC E) a ground signal from the connector A pin 2H to the SCU (Refer to WM32-51-00).
- At the connector J1 pin C3C (at card PSC E) a ground signal is sent to the MFD1 and ED (Refer to WM77-21-00).
- At the connector J1 pin C3A (at card PSC E), a ground signal is sent to the FGCOM1, IOP1, FGMON1 and the SPM1.
- At the connector J1 pin C4D (at card PSC E), a ground signal is sent to the FGCOM2, IOP2, FGMON2 and the SPM2.

Table 8: Post Modsum 4-126276, Pre Modsum 4-126420 and Pre Service Bulletin 84-32-66

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT08A	A2H	SCU
DOT24A	A8G	PFD1/2, MFD2

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DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT26A	A8J	FGCOM1, IOP1, FGMON1, SPM1
DOT27A	A8K	IOP2, FGMON2, SPM2
DOT04D	B2B	n/a
DOT08E	B10D	SCU
DOT24E	C3C	MFD1, ED

Results:

- When you get the conditions of the equation, the PSEU (at card PSC A) sends a ground signal to the SCU (Refer to WM32–51–00).
- At the connector J1 pin A8G (at card PSC A), a ground signal is sent to the PFD1/2 and the MFD2 (Refer to WM31–61–00).
- At the connector J1 pin A8J (at card PSC A), a ground signal is sent to the FGCOM1, IOP1, FGMON1 and the SPM1.
- At the connector J1 pin A8K (at card PSC A), a ground signal is sent to the IOP2, FGMON2 and the SPM2.
- The wire from connector J1 pin B2B (at card PSC D) is not used; it is tagged and stowed.

- At the connector J1 pin B10D (at card PSC E) a ground signal from the connector A pin 2H to the SCU (Refer to WM32–51–00).
- At the connector J1 pin C3C (at card PSC E) a ground signal is sent to the MFD1 and ED (Refer to WM77–21–00).

Table 9: Post Modsum 4–126420 and Post Service Bulletin 84–32–66

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT08A	A2H	SCU
DOT24A	A8G	PFD1/2, MFD2, IFC1/2
DOT04D	B2B	n/a
DOT08E	B10D	SCU
DOT24E	C3C	MFD1, ED, IFC1/2

Results:

- When you get the conditions of the equation, the PSEU (at card PSC A) sends a ground signal to the SCU (Refer to WM32–51–00).
- At the connector J1 pin A8G (at card PSC A), a ground signal is sent to the PFD1/2, MFD2 and the IFC1/2 (Refer to WM31–61–00).
- The wire from connector J1 pin B2B (at card PSC D) is not used; it is tagged and stowed.



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- At the connector J1 pin B10D (at card PSC E) a ground signal from the connector A pin 2H to the SCU (Refer to WM32–51–00).
- At the connector J1 pin C3C (at card PSC E) a ground signal is sent to the MFD1, ED and the IFC1/2 (Refer to WM77–21–00).

Equation No. 40 WEIGHT–OFF–WHEELS (NGWOFW) NOSE GEAR (On aircraft installed with PSEU P/N 30145–0201 or 30145–0301 or 30145–0401 or 30145–0501 or 30145–0601 or 30145–0602)

$(NGWOFW1 + NGWOFW1_F) \cdot (NGWOFW2 + NGWOFW2_F) \cdot (NOT\ NGWOFW1_F + NOT\ NGWOFW2_F)$

Equation Explanation: The NGWOFW equation outputs are active in the conditions that follow:

- If the two Nose Gear WOFW sensors operate correctly, the two sensors show that the aircraft is in the air
- If one Nose Gear WOFW sensor is faulted, the other sensor shows that the aircraft is in the air

If the two Nose Gear WOFW sensors are faulted, the NGWOFW output will default to the inactive (fail–safe) state.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT06A	A2F	IOM1
DOT06E	B9E	IOM2

Result: When you get the conditions of the equation, the PSEU (at card PSC A) sends a ground signal to the IOM1 (Refer to WM31–41–00). A ground signal is also sent from the connector J1 pin B9E (at card PSC E) to the IOM2 (Refer to , WM31–41–00).

Equation No. 41 WEIGHT–ON–WHEELS 1 OPERATIONAL (On aircraft installed with PSEU P/N 30145–0201 or 30145–0301 or 30145–0401 or 30145–0501 or 30145–0601 or 30145–0602)

$(([RGWOW1 + RGWOW1_F] \cdot [LGWOW1 + LGWOW1_F] \cdot (NOT\ NGWOFW1 + NGWOFW1_F)) + [(NOT\ RGWOW1 + RGWOW1_F) \cdot (NOT\ LGWOW1 + LGWOW1_F) \cdot (NGWOFW1 + NGWOFW1_F)]) \cdot (NOT\ RGWOW1_F \cdot NOT\ LGWOW1_F \cdot NOT\ NGWOFW1_F) + (NOT\ RGWOW2_F \cdot NOT\ LGWOW2_F \cdot NOT\ NGWOFW2_F)$

Equation Explanation: The Weight–On–Wheels 1 Operational output discrete will deactivate in the conditions that follow:

- Disagreement of aircraft mode (Air versus Ground) of any one non–faulted WOW1 sensor with the other non–faulted WOW1 sensors, or
- A minimum of one Proximity Sensor fault in each WOW system (WOW1 and WOW2) at the same time

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT13E	B14E	CAWP

Result: When you get the conditions of the equation, the PSEU (at card PSC E) removes a ground signal to the CAWP to light the WT ON WHEELS caution light.



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Equation No. 42 WEIGHT-ON-WHEELS 2 OPERATIONAL (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

$$(((RGWOW2 + RGWOW2_F) \cdot (LGWOW2 + LGWOW2_F) \cdot (NOT\ NGWOFW2 + NGWOFW2_F)) + [(NOT\ RGWOW2 + RGWOW2_F) \cdot (NOT\ LGWOW2 + LGWOW2_F) \cdot (NGWOFW2 + NGWOFW2_F)]) \cdot (NOT\ RGWOW1_F \cdot NOT\ LGWOW1_F \cdot NOT\ NGWOFW1_F) + (NOT\ RGWOW2_F \cdot NOT\ LGWOW2_F \cdot NOT\ NGWOFW2_F)$$

Equation Explanation: The Weight-On-Wheels 2 Operational output discrete will deactivate in the conditions that follow:

- Disagreement of aircraft mode (Air versus Ground) of any one non-faulted WOW2 sensor with the other non-faulted WOW2 sensors, or
- A minimum of one Proximity Sensor fault in each WOW system (WOW1 and WOW2) at the same time

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT13A	A4H	CAWP

Result: When you get the conditions of the equation, the PSEU (at card PSC A) removes a ground signal to the CAWP to light the WT ON WHEELS caution light.

Equation No. 43 PASSENGER AIRSTAIR DOOR (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

PADRCL · PADRLK

Equation Explanation: The Passenger Airstair Door discrete output activates when the Passenger Airstair Door is Closed and Locked.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT11A	A4F	IFC1, IFC2

Result: When you get the conditions of the equation, the PSEU (at card PSC A) sends a ground signal to IOM1 and IOM2 (Refer to WM31-41-00). IOM1 and IOM2 are found in their related IFCs.

Equation No. 44 AFT PASSENGER DOOR (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

APDRCL · APDRLK

Equation Explanation: The Aft Passenger Door discrete output activates whenever the Aft Passenger Door is Closed and Locked.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT12D	B5D	IFC1, IFC2

Result: When you get the conditions of the equation, the PSEU (at card PSC D) sends a ground signal to IOM1 and IOM2 (Refer to WM31-41-00). IOM1 and IOM2 are found in their related IFCs.

Equation No. 45 FORWARD BAGGAGE DOOR (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

FBDRCL · FBDRLK



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Equation Explanation: The Forward Baggage Door discrete output activates when the Forward Baggage Door is Closed and Locked.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT12B	A14F	IFC1, IFC2

Result: When you get the conditions of the equation, the PSEU (at card PSC B) sends a ground signal to IOM1 and IOM2 (Refer to WM31-41-00). IOM1 and IOM2 are found in their related IFCs.

Equation No. 46 AFT BAGGAGE DOOR (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

ABDRCL · ABDRLK

Equation Explanation: The Aft Baggage Door discrete output activates when the Aft Baggage Door is Closed and Locked.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT11B	A13F	IFC1, IFC2

Result: When you get the conditions of the equation, the PSEU (at card PSC B) sends a ground signal to IOM1 and IOM2 (Refer to WM31-41-00). IOM1 and IOM2 are found in their related IFCs.

Equation No. 47 AFT SERVICE DOOR (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

ASDRCL · ASDRLK

Equation Explanation: The Aft Service Door discrete output activates when the Aft Service Door is Closed and Locked.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT12E	B13E	IFC1, IFC2

Result: When you get the conditions of the equation, the PSEU (at card PSC E) sends a ground signal to IOM1 and IOM2 (Refer to WM31-41-00). IOM1 and IOM2 are found in their related IFCs.

Equation No. 48 EMERGENCY EXIT DOOR (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

EDRCLLK

Equation Explanation: The Emergency Exit Door discrete output activates when the Emergency Exit Door is Closed and Locked.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT12A	A4G	IFC1, IFC2

Result: When you get the conditions of the equation, the PSEU (at card PSC A) sends a ground signal to IOM1 and IOM2 (Refer to WM31-41-00). IOM1 and IOM2 are found in their related IFCs.



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Equation No. 49: ALL FUSELAGE DOORS CLOSED AND LOCKED (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

$[PADRCL \cdot PADRLK \cdot APDRCL \cdot APDRLK \cdot FBDRCL \cdot FBDRLK \cdot$
 $ABDRCL \cdot ABDRLK \cdot ASDRCL \cdot ASDRLK \cdot EDRCLLK] +$
 $[PADRCL_P \cdot NOT\ PADRCL] + [PADRLK_P \cdot NOT\ PADRLK] +$
 $[APDRCL_P \cdot NOT\ APDRCL] + [APDRLK_P \cdot NOT\ APDRLK] +$
 $[FBDRCL_P \cdot NOT\ FBDRCL] + [FBDRLK_P \cdot NOT\ FBDRLK] +$
 $[ABDRCL_P \cdot NOT\ ABDRCL] + [ABDRLK_P \cdot NOT\ ABDRLK] +$
 $[ASDRCL_P \cdot NOT\ ASDRCL] + [ASDRLK_P \cdot NOT\ ASDRLK] +$
 $[EDRCLLK_P \cdot NOT\ EDRCLLK]$

Equation Explanation: The All Fuselage Doors Closed and Locked discrete output activates when all six fuselage doors show the Closed and Locked condition, or when a fuselage door proximity sensor transitions from NEAR to FAR.

NOTE

When this output activates due to a sensor change from NEAR to FAR, the output will only be active for one cycle (125 msec, nominally). After the change, the output will return to the inactive condition.

NOTE

A fuselage door sensor that transitions from FAR to NEAR will not affect the output unless it is the last sensor to change. This results in all fuselage door sensors in the NEAR condition.

Table 10: On aircraft without SB 84-32-66 and Modsum 4-126420 and Modsum 4-110293 incorporated

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT10B	A12F	CAWP
DOT13B	A15F	CPC

Result: When you get the conditions of the equation, the PSEU (at card PSC B) removes a ground signal to the CAWP to light the FUSELAGE DOORS warning light. At the connector J1 pin A15F (at card PSC B) the PSEU sends a ground signal to the CPC.

Table 11: On aircraft with SB 84-32-66 and Modsum 4-126420 and Modsum 4-110293 incorporated

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT10B	A12F	CAWP

Result: When you get the conditions of the equation, the PSEU (at card PSC B) removes a ground signal to the CAWP to light the FUSELAGE DOORS warning light.

Equation No. 50: PSS MAINTENANCE CHECK REQUIRED (DOT1BT) (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

AnyFaultHistoryFault



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Equation Explanation: This BITE-generated discrete output is active when there is a Fault recorded in Fault History. Rigging History messages are not Fault History messages and will therefore not activate this discrete output.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT1BT	B9B	CMP

Result: When you get the conditions of the equation, the PSEU (at the BITE card) sends a ground signal to the CMP (Refer to WM31-34-00).

Equation No. 51: PSEU OPERATIONAL (DOT2BT) (On aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

$((\text{NOT LeftEss1_F} + \text{NOT LeftMain1_F}) \cdot (\text{NOT LeftEss2_F} + \text{NOT LeftMain2_F}) \cdot (\text{NOT RightEss_F} + \text{NOT RightMain_F})) \cdot (\text{NOT NVM_F} \cdot \text{NOT PSCA_F} \cdot \text{NOT PSCB_F} \cdot \text{NOT PSCD_F} \cdot \text{NOT PSCE_F})$

Equation Explanation: This BITE-generated discrete output is active when there is a minimum of one active power bus to all internal PSEU components, and there are no current PSEU faults recorded in Present Faults.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT2BT	B9A	CMP

Result: When you get the conditions of the equation, the PSEU (at the BITE card) sends a ground signal to the CMP (Refer to WM31-34-00).

Equation No. 52: FUSELAGE DOORS CABIN PRESSURE ENABLE (DOT13B) (On aircraft with SB84-32-66 or MODSUM 4-126420 and on aircraft installed with PSEU P/N 30145-0201 or 30145-0301 or 30145-0401 or 30145-0501 or 30145-0601 or 30145-0602)

$[(\text{PADRCL} \cdot \text{PADRLK}) \cdot (\text{ABDRCL} \cdot \text{ABDRLK}) \cdot \text{EDRCLLK}] + [\text{PADRCL_P} \cdot \text{NOT PADRCL}] + [\text{PADRLK_P} \cdot \text{NOT PADRLK}] + [\text{ABDRCL_P} \cdot \text{NOT ABDRCL}] + [\text{ABDRLK_P} \cdot \text{NOT ABDRLK}] + [\text{EDRCLLK_P} \cdot \text{NOT EDRCLLK}]$

Equation Explanation: The Fuselage Doors Cabin Pressure Enable discrete output activates when all unvented fuselage doors show the Closed and Locked condition. The equation allows for temporary NEAR to FAR door sensor transitions when the doors have been closed or locked previously.

DISCRETE OUTPUT	PSEU CONNECTOR PIN NUMBER	INTERFACING LRU
DOT13B	A15F	CPC

Result: When you get the conditions of the equation, at the connector J1 pin A15F (at card PSC B) the PSEU sends a ground signal to the CPC.



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PSEU Logic Conditions vs. Sensor Target Position

FUNCTION	TARGET POSITION	CONDITION
_DRCL (Door Closed)	Near	Door Closed
	Far	Door NOT Closed
_DRLK (Door Locked)	Near	Door Locked
	Far	Door NOT Locked
EDRCLK (Emerg. Door Closed and Locked)	Near	Door Closed and Locked
	Far	Door NOT Closed and Locked
_GDLK# (Gear Downlock)	Near	Gear Down and Locked
	Far	Gear NOT Down and Locked
_GUPLK (Gear Uplock)	Near	Gear Up and Locked
	Far	Gear NOT Up and Locked
NGDN (Nose Gear Down)	Near	Nose Gear Down
	Far	Nose Gear NOT Down

FUNCTION	TARGET POSITION	CONDITION
NGLK (Nose Gear Locked)	Near	Nose Gear Locked
	Far	Nose Gear NOT Locked
_WOW# (Weight-on-Wheels)	Near	Weight is ON Wheels
	Far	Weight is OFF Wheels
NGWOFW# (Weight-off-Wheels)	Near	Weight is OFF Wheels
	Far	Weight is ON Wheels
NWCENT (Nose Wheel Centered)	Near	Wheel Centered
	Far	Wheel NOT Centered

NOTE

An underscore (_) where present, is replaced by an alpha character showing the gear or door location. A number symbol (#) where present, is replaced by a number.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LRU Abbreviations

MNEMONIC	NAME
ACU	Antiskid/Brake Control Unit
ADIP	Alternate Downlock Indication Panel
ADU	Air Data Unit
AHRS	Attitude Heading Reference System
ANVS	Active Noise and Vibration Suppression
APU	Auxiliary Power Unit
ARCDU	Audio Radio Control Display Unit
ATC	Air Traffic Control
CAWP	Caution & Warning Panel
CMP	Central Maintenance Panel
CPC	Cabin Pressure Controller
CVR	Cockpit Voice Recorder
ECS	Environmental Control System
ECU	Environmental Control Unit

MNEMONIC	NAME
ED	Engine Display
EIS	Electronic instrument System
EPCU	Electronic Power Control Unit
FCEU	Flight Control Electronic Unit
FCU	Flap Control Unit
FDR	Flight Data Recorder
FGCMD	Flight Guidance Command
FGMON	Flight Guidance Monitor
IFC	Integrated Flight Cabinet
IOM	Input Output Module
IOP	Input Output Processor
IRPS	Ice / Rain Protection System
LGCIP	Landing Gear Control & Indication Panel
LGDSV	Landing Gear Door Sequence Valve
LGSV	Landing Gear Selector Valve

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MNEMONIC	NAME
MFD	Multi Function Display
NWS	Nose Wheel Steering
PEC	Propeller Electronic Controller
PFD	Primary Flight Display
RCAU	Remote Control Audio Unit
SCP	Signal Conditioning Program
SCU	Steering Control Unit
SCV	Spoiler Control Valve
SPM	Stall Protection Module
TCAS	Traffic Collision Avoidance System
TOC	Take-Off Configuration
TMU	Timer Monitor Unit
WXR	Weather Radar



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

PSC A – Powered by Left Main 2 and Left ESS

Resource Allocation (1)	Function (2)	J1 Pin # (3)	Type (4)	Name (5)	LRU(s) (6)	Logic Diagram (7)	Time Delay (8)	Load (max) (9)	Load Test (10)
DOT01A	LDG CNTL	A1F	1	LDG Selector Valve A EXT	LGSV	1		0.75	1
DOT02A	LDG CNTL	A1G	1	LDG Selector Valve B RET	LGSV	2	3 sec	0.75	1
DOT03A	LDG CNTL	A1H	1	LG Door Sequence Valve	LGDSV	5		1.5	2
DOT04A	LDG CNTL	A1J	1	NG Door Sequence Valve	LGDSV	3		1.5	2
DOT05A	LDG CNTL	A1K	1	RG Door Sequence Valve	LGDSV	4		1.5	2
DOT06A	WOW2	A2F	2B	TOC, IFC1, NGWOFW	IOMI	40		3mA	1
DOT07A	LDG CNTL	A2G	2B	Flight Data Recorder, EXTCMD	IFC1, IFC2	6		3mA	1
DOT08A	WOW2	A2H	2B	Nosewheel Steering, NGWOW2	SCU	39		5mA	1
DOT09A	LDG CNTL	A2J	2A	No Smoking Sign, EXTCMD		6		50mA	1
DOT10A	LDG CNTL	A2K	2B	Landing Gear Inhibit (FTI)	n/a	8		10mA	1
DOT11A	FUSE DRS	A4F	2B	Pass Airstair Door CL/LK	IFC1, IFC2	43	2 sec	3mA	1
DOT12A	FUSE DRS	A4G	2B	Emg Exit Door CL/LK	IFC1, IFC2	48	2 sec	3mA	1
DOT13A	WOW2	A4H	2B	WOW2 Operational	CAWP	42	20 sec	4mA	1
DOT14A	WOW2	A4J	2B	Antiskid Outboard, MGWOW2	ACU	38		3mA	1
DOT15A	WOW2	A4K	2A	Flt Splrs, outbd valve 2 MGWOW	SCV	37		1.2A	1
DOT16A	WOW2	A6F	2A	Flt Splrs, inbd valve 2, MGWOW	SCV	37		1.2A	1
DOT17A	WOW2	A6G	2B	Flt Cntrl Elec Unit, MGWOW2	FCEU	37		10mA	1
DOT18A	WOW2	A6H	2B	Prop Elec Cntrl Left, MGWOW2	PEC	37		10mA	1
DOT19A	WOW2	A6J	2B	Prop Elec Cntrl Right, MGWOW2	PEC	37		10mA	1
DOT20A	WOW2	A6K	2B	Cabin Pres Cntrl, WOW	CPC	36		5mA	1
DOT21A	WOW2	A7G	2B	Cabin Pres Cntrl Throttle, WOW	Throttle	36		5mA	1
DOT22A	WOW2	A7J	2B	Elec Instr Sys, MGWOW2	MFD1, ED	37		3mA	1
DOT23A	WOW2	A8F	2B	Cooling Fan, WOW	tbd	36		100mA	1
DOT24A	WOW2	A8G	2B	Elec Instr Sys, NGWOW2	PFD1&2, MFD2	39		3mA	1
DOT25A	WOW2	A8H	2B	Att Hd Ref Sys/ADU 1, WOW	AHRS1, ADU1	36		3mA	1
DOT26A	WOW2	A8J	2B	POST IFC1, MGWOW2	FGCOM1, IOP1 FGMON1, SPM1	37		3mA	1
DOT27A	WOW2	A8K	2B	POST IFC2, MGWOW2	IOP2, FGMON2, SPM2	37		3mA	1
DOT28A	LDG CNTL	A9F	2B	LDG Lever Down (FTI)	n/a	6		10mA	1
DOT29A	LDG CNTL	A9G	2B	LDG Lever Up (FTI)	n/a	7		10mA	1
DOT30A	Spare	A9H	2C	n/a	n/a	n/a	n/a	n/a	n/a

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PSC Cards Discrete Outputs (Typical)
Figure 1 (Sheet 1 of 5)

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PSC B – Powered by Left Main 1 and Left ESS 1

Resource Allocation (1)	Function (2)	J1 Pin # (3)	Type (4)	Name (5)	LRU(s) (6)	Logic Diagram (7)	Time Delay (8)	Load (max) (9)	Load Test (10)
DOT01B	ALT DLK	C1F	1	LG Safe Indication # 2	ADIP	34		70mA	1
DOT02B	ALT DLK	C1G	1	NG Safe Indication # 2	ADIP	32		70mA	1
DOT03B	ALT DLK	C1H	1	RG Safe Indication # 2	ADIP	33		70mA	1
DOT04B	WOW2	C1J	1	Aux Relay # 2, MGWOW	n/a	37		70mA	1
DOT05B	WOW2	C1K	1	Aux Relay # 4, WOW	n/a	36		70mA	1
DOT06B	Spare	A10F	2B	n/a	n/a	n/a	n/a	n/a	n/a
DOT07B	Spare	A10G	2B	n/a	n/a	n/a	n/a	n/a	n/a
DOT08B	ALT DLK	A11F	2B	Nose Wheel Steering, NGDLK	SCU	20		5mA	1
DOT09B	ALT DLK	A11G	2A	Taxi Light Contactor, NGDLK	Contactor	20		350mA	1
DOT10B	FUSE DRS	A12F	2B	All Doors CL/LK	CAWP	49	2 sec	4mA	1
DOT11B	FUSE DRS	A13F	2B	Aft Bagg Door CL/LK	IFC1, IFC2	46	2 sec	3mA	1
DOT12B	FUSE DRS	A14F	2B	Fwd Bagg Door CL/LK	IFC1, IFC2	45	2 sec	3mA	1
DOT13B	FUSE DRS	A15F	2B	Cabin Pres Cntrl, All Doors CL/LK	CPC	49		5mA	1
DOT14B	ALT DLK	C2F	2B	Flight Data Recorder, NGDLK	IFC1, IFC2	20		3mA	1
DOT15B	Spare	C2G	2B	n/a	n/a	n/a	n/a	n/a	n/a
DOT16B	ALT DLK	C2H	2B	LDG Horn, All Gear DNLK	IOM1	35		3mA	1
DOT17B	WOW2	C2J	2A	Hydraulic System, WOW		36		70mA	1
DOT18B	ALT DLK	C2K	2B	Flight Data Recorder, MGDLC	IFC1, IFC2	22		3mA	1
DOT19B	ALT DLK	C3F	2B	TCAS, MGDLC (no connection)	TCAS	22		n/a	n/a
DOT20B	WOW2	C3G	2B	Intergrated Flight Cabinet, MGWOW	IOM1, IOM2	37		3mA	1
DOT21B	WOW2	C3H	2B	Active Noise and Vibration Suppression, ANVS WOW	ANVS	37		2mA	1
DOT22B	WOW2	C3J	2B	Co-Pilot Clock, MGWOW		37		3mA	1
DOT23B	WOW2	C3K	2A	No Smoking Sign, WOW		36		50mA	1
DOT24B	WOW2	C4F	2B	Audio Radio Cntrl Display Unit 1, WOW	ARCDU	36		3mA	1
DOT25B	WOW2	C4G	2B	Aux Power Unit, MGWOW	APU	37		10mA	1
DOT26B	WOW2	C4H	2B	Remote Cntrl Audio Unit, MGWOW	RCAU	37		3mA	1
DOT27B	WOW2	C4J	2B	Traffic Coll Avoid Sys, MGWOW	TSAC, WXR	37		3mA	1
DOT28B	Spare	C4K	2B	n/a	n/a	n/a	n/a	n/a	n/a
DOT29B	Spare	C5F	2B	n/a	n/a	n/a	n/a	n/a	n/a
DOT30B	Spare	C5G	2C	n/a	n/a	n/a	n/a	n/a	n/a

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PSC Cards Discrete Outputs (Typical)
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

PSC D – Powered by Right Main and Right ESS

Resource Allocation (1)	Function (2)	J1 Pin # (3)	Type (4)	Name (5)	LRU(s) (6)	Logic Diagram (7)	Time Delay (8)	Load (max) (9)	Load Test (10)
DOT01D	Spare	B2E	1	n/a	n/a	n/a	n/a	n/a	n/a
DOT02D	Spare	B2D	1	n/a	n/a	n/a	n/a	n/a	n/a
DOT03D	Spare	B2C	1	n/a	n/a	n/a	n/a	n/a	n/a
DOT04D	WOW1	B2B	1	Aux Relay #3, NGWOW	n/a	39		70mA	1
DOT05D	LDG IND	B2A	1	LDG Unsafe Indication	LGCIP	18		100mA	1
DOT06D	LDG IND	B3E	2A	LG Safe Indication #1	LGCIP	11		100mA	1
DOT07D	LDG IND	B3D	2A	NG Safe Indication #1	LGCIP	9		100mA	1
DOT08D	LDG IND	B3C	2A	RG Safe Indication #1	LGCIP	10		100mA	1
DOT09D	LDG IND	B3B	2A	LG Not Safe Indication	LGCIP	14		100mA	1
DOT10D	LDG IND	B3A	2A	NG Not Safe Indication	LGCIP	12		100mA	1
DOT11D	LDG IND	B4C	2A	RG Not Safe Indication	LGCIP	13		100mA	1
DOT12D	FUSE DRS	B5D	2B	Aft Pass Door CL/LK	IFC1, IFC2	44	2 sec	3mA	1
DOT13D	LDG IND	B5C	2A	LDG Control Operational	CAWP	25	20 sec	4mA	1
DOT14D	LDG IND	B5B	2B	Flight Data Recorder, NOT NGUPLK	IFC1, IFC2	21		3mA	1
DOT15D	LDG IND	B6E	2B	Right Gear Uplock (FTI)	n/a	27		10mA	1
DOT16D	LDG IND	B6D	2B	LDG Horn, All Gear Downlock	IOM2	19		3mA	1
DOT17D	LDG IND	B6C	2B	Flight Data Rec, NOT MGUPLK	IFC1, IFC2	24		2mA	1
DOT18D	LDG IND	B6B	2B	Antiskid Inboard, MGUPLK	ACU	23		3mA	1
DOT19D	LDG IND	B6A	2B	Antiskid Outboard, MGUPLK	ACU	23		3mA	1
DOT20D	Spare	B7E	2B	n/a	n/a	n/a	n/a	n/a	n/a
DOT21D	Spare	B7D	2B	n/a	n/a	n/a	n/a	n/a	n/a
DOT22D	WOW1	B7C	2B	Pilot Clock, MGWOW		37		3mA	1
DOT23D	WOW1	B7B	2B	Air Conditioner Bleed, WOW	ECU	36		20mA	1
DOT24D	WOW1	B7A	2B	Aud Rad Cntrl Dply Unit 2, WOW	ARCDU2	36		3mA	1
DOT25D	WOW1	B8E	2B	Aux Power Unit, MGWOW	APU	37		10mA	1
DOT26D	WOW1	B8D	2B	Flight Data Recorder, MGWOW	IFC1, IFC2	37		3mA	1
DOT27D	WOW1	B8C	2B	APU Fire Protect, WOW		36		10mA	1
DOT28D	LDG IND	B8B	2B	Left Gear Uplock (FTI)	n/a	28		10mA	1
DOT29D	LDG IND	B8A	2B	Nose Gear Uplock (FTI)	n/a	26		10mA	1
DOT30D	WOW1	B9D	2C	Elect. Pwr. Cntrl. Unit, MGWOW	EPCU	37	n/a	10mA	1

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PSC Cards Discrete Outputs (Typical)
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PSC E – Powered by Right Main and Right ESS

Resource Allocation (1)	Function (2)	J1 Pin # (3)	Type (4)	Name (5)	LRU(s) (6)	Logic Diagram (7)	Time Delay (8)	Load (max) (9)	Load Test (10)
DOT01E	Spare	B1E	1	n/a	n/a	n/a	n/a	n/a	n/a
DOT02E	Spare	B1D	1	n/a	n/a	n/a	n/a	n/a	n/a
DOT03E	Spare	B1C	1	n/a	n/a	n/a	n/a	n/a	n/a
DOT04E	WOW1	B1B	1	Aux Relay #1, WOW	n/a	39		70mA	1
DOT05E	Spare	B1A	1	n/a	n/a	n/a	n/a	n/a	n/a
DOT06E	WOW1	B9E	2B	TOC, IFC2, NGWOFW	IOM2	40		3mA	1
DOT07E	LDG IND	B10E	2B	RG Door Closed (FTI)	n/a	30		10mA	1
DOT08E	WOW1	B10D	2B	Nosewheel Steering, NGWOW1	SCU	39		5mA	1
DOT09E	LDG IND	B11E	2A	LG Doors Not Closed	LGCIP	17		100mA	1
DOT10E	LDG IND	B11D	2A	NG Doors Not Closed	LGCIP	15		100mA	1
DOT11E	LDG IND	B12E	2A	RG Doors Not Closed	LGCIP	16		100mA	1
DOT12E	FUSE DRS	B13E	2B	Aft Service Door CL/LK	IFC1, IFC2	47	2 sec	3mA	1
DOT13E	WOW1	B14E	2B	WOW1 Operational	CAWP	41	20 sec	4mA	1
DOT14E	WOW1	B15E	2B	Antiskid, Inboard, MGWOFW1	ACU	38		3mA	1
DOT15E	WOW1	C1D	2A	Flight Spoilers Outboard Valve 1, MGWOW	SCV	37		1.2A	1
DOT16E	WOW1	C1C	2A	Flight Spoilers Inboard Valve 1, MGWOW	SCV	37		1.2A	1
DOT17E	WOW1	C1B	2B	Flight Cntrl Elect Unit, MGWOW1	FCEU	37		10mA	1
DOT18E	WOW1	C1A	2B	Prop Elect Cntrl, Left, MGWOW1	PEC	37		10mA	1
DOT19E	WOW1	C2D	2B	Prop Elect Cntrl, Right, MGWOW1	PEC	37		10mA	1
DOT20E	WOW1	C2C	2A	Cockpit Voice Recorder, WOW	CVR	36		65mA	1
DOT21E	WOW1	C2B	2B	Ice/Rain Protection, WOW	TMU	36		6mA	1
DOT22E	WOW1	C2A	2B	Elect Instrument Sys, MGWOW1	PFD1 & PFD2, MFD2	37		3mA	1
DOT23E	WOW1	C3D	2B	Spare	n/a	n/a	n/a	n/a	n/a
DOT24E	WOW1	C3C	2B	Elect Instrument Sys, NGWOW1	MFD1, ED	39		3mA	1
DOT25E	WOW1	C3B	2B	Att Head Ref Sys/Air Data Unit 2, WOW	AHRS2, ADU2	36		10mA	1
DOT26E	WOW1	C3A	2B	POST, IFC1, MGWOW1	FGCOM1, IOP1, FGMON1, SPM1	37		3mA	1
DOT27E	WOW1	C4D	2B	POST, IFC2, MGWOW1	FGCOM2, IOP2, FGMON2, SPM2	37		3mA	1
DOT28E	LDG IND	C4C	2B	LG Door Closed (FTI)	n/a	31		10mA	1
DOT29E	LDG IND	C4B	2B	NG Door Closed (FTI)	n/a	29		10mA	1
DOT30E	WOW1	C4A	2C	Flap Control Unit, WOW	FCU	36		100mA	1

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PSC Cards Discrete Outputs (Typical)
Figure 1 (Sheet 4 of 5)

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BITE PSEU Outputs

Resource Allocation (1)	Function (2)	J1 Pin # (3)	Type (4)	Name (5)	LRU(s) (6)	Logic Diagram (7)	Time Delay (8)	Load (max) (9)	Load Test (10)
DOT1BT	n/a	B9B	2A	PSS Maintenance Check Required	CMP	50		70mA	n/a
DOT2BT	n/a	B9A	2A	PSEU Operational	CMP	51		70mA	n/a

NOTES:

This data is applicable to the 30145–0301 PSEU.

- (1): Resource Allocation lists the abbreviation used for each of the 30 discrete outputs from each PSC channel. "DOT" refers to "Discrete Output" and the ending letter refers to the PSC where the output is located.
- (2): Function is one of the six functional blocks from which the output is generated.
- (3): J1 Pin # is the ARINC 600 I/O connector pin number.
- (4): Type 1 is a 28V/Open output; Type 2A is a Gnd/Open high current output; Type 2B is a Gnd/Open low current output; Type 2C is a 28V/Open low current output.
- (5): Name is the signal name. Where "(FTI)" is indicated, the signal was designated for Flight Test Instrumentation Only and is not used.
- (6): LRU is the interfacing LRU. In some cases, more than one LRU receives the output.
- (7): Logic Diagrams are shown in Appendix A.
Time delay, from input change, is 375mSec maximum unless otherwise noted. Specific times listed apply to transition from active to inactive states with a tolerance of ± 375 mSec.
- (8): Load is maximum current drawn by each LRU load (source or sink).
- (9): Load Test 1: overcurrent faults detected.
- (10): Load Test 2: load faults and overcurrent faults detected.

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PSC Cards Discrete Outputs (Typical)
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LANDING GEAR CONFIGURATION WARNING SYSTEM

Introduction

The landing gear configuration warning system supplies an aural warning to the flight crew if a landing is attempted with the aircraft not in the correct landing configuration (landing gears not down and locked).

General Description

Refer to Figures 1 and 2.

The landing gear warning system logic is divided into three sets of conditions. Each condition depends on different flight phases and related parameters. In event of the aural warning the system logic provides sufficient time for the landing gears to extend down and locked or otherwise, the pilot can make a go-around. The system logic also prevents false warnings. A HORN MUTE and TEST switch is also installed on the landing gear selector panel.

Detailed Description

Refer to Figures 2 and 3.

The continuous 800 Hz landing gear configuration warning tone is generated by two Warning Tone Generator (WTG). The WTG's are the part of the Flight Data Processing System (FDPS) (Refer to Chapter 31-40-00). The WTG's are physically housed in the Input

Output Processor (IOP) 1 and IOP 2 located in the two Integrated Flight Cabinets (IFC) 1 and IFC 2. The ARINC 429 digital data from different systems is directly input to IOP's. The analog and discrete data is input to the IOP's through the Input Output Module (IOM) 1 and IOM 2.

The IOP 1 and IOP 2 operate the WTG in a master slave configuration with IOP 1 as the master. The IOP 2 takes the command in case of a disagreement between generated warning tones. The WTG provides tone signals to the Remote Control Audio Unit (RCAU) through the IOM 1. The IOM 1 supplies the tone selection and modulation based on signals from the WTG. The IOM 2 monitors the output of the IOM 1.

The landing gear configuration warning is prioritized by the WTG with respect to other aural warnings. The RCAU amplifies the signal and sends the tone to the flight compartment speakers and to the pilot's and co-pilot's headsets.

The continuous 800 Hz landing gear configuration warning sounds in the flight deck, if the conditions that follow are sensed:

- If the HORN toggle switch, on the landing gear panel, is set to the TEST position
- OR
- If all the landing gears are not down and locked.
- AND

Condition 1 (Flaps and Engine torque):

- The flap lever is set to more than 8.5 degree
- AND

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- ENG1 torque OR ENG2 torque is less than 50%
AND
- PLA 1 AND PLA 2 are not in the rating power detent.
In this condition warning can not be muted until the conditions are removed. This logic prevents false warning. As during a take-off or a go-around the PLA's are set to the rating power detent which disables the warning.

OR

Condition 2 (Airspeed, Radio Altitude and PLA setting):

- Indicated airspeed less than 156 Knots (288 km/h), AND
RAD ALT 1 Invalid
OR
- Indicated airspeed less than 156 Knots (288 km/h), AND
RAD ALT 1 Valid, AND Above Ground Level (AGL) altitude
is less than 1053 feet
AND
- PLA1 AND PLA2 are less than FLIGHT IDLE plus 12
degrees.
In this condition warning can not be muted until the conditions are removed. This logic prevents false warning. As during a take-off or a go-around the PLA's are set to the rating power detent (above FLIGHT IDLE plus 12 degrees) which disables the warning.

OR

Condition 3 (Airspeed during single engine operation and PLA setting):

- Indicated airspeed less than 156 Knots (288 km/h), AND
RAD ALT 1 Invalid
OR
- Indicated airspeed less than 156 Knots (288 km/h), AND
RAD ALT 1 Valid, AND Above Ground Level (AGL) altitude
is less than 1053 feet
AND
- PLA1 OR PLA2 is less than FLIGHT IDLE plus 12 degrees
AND
- HORN MUTE switch is not latched to MUTE
AND
- PLA 1 AND PLA 2 are less than the rating power detent.
In this condition warning can be muted in special simulated conditions like pilot training exercise. This logic prevents false warning. As during a take-off or a go-around the PLA's are set to the rating power detent, which disables the warning.

If the landing gear configuration warning is muted, the mute will reset in the conditions that follow:

- The down lock indication indicates all gear down and
locked
OR



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- The alternate down lock indication indicates all gear down and locked
OR
- PLA 1 AND PLA 2 are greater than OR less than FLIGHT IDLE plus 12 degrees
OR
- All the landing gears are not down and locked
AND
The Flap lever is set to more than 8.5 degree
AND
ENG1 torque OR ENG2 torque is less than 50%
AND
PLA 1 AND PLA 2 are not in the rating power detent.

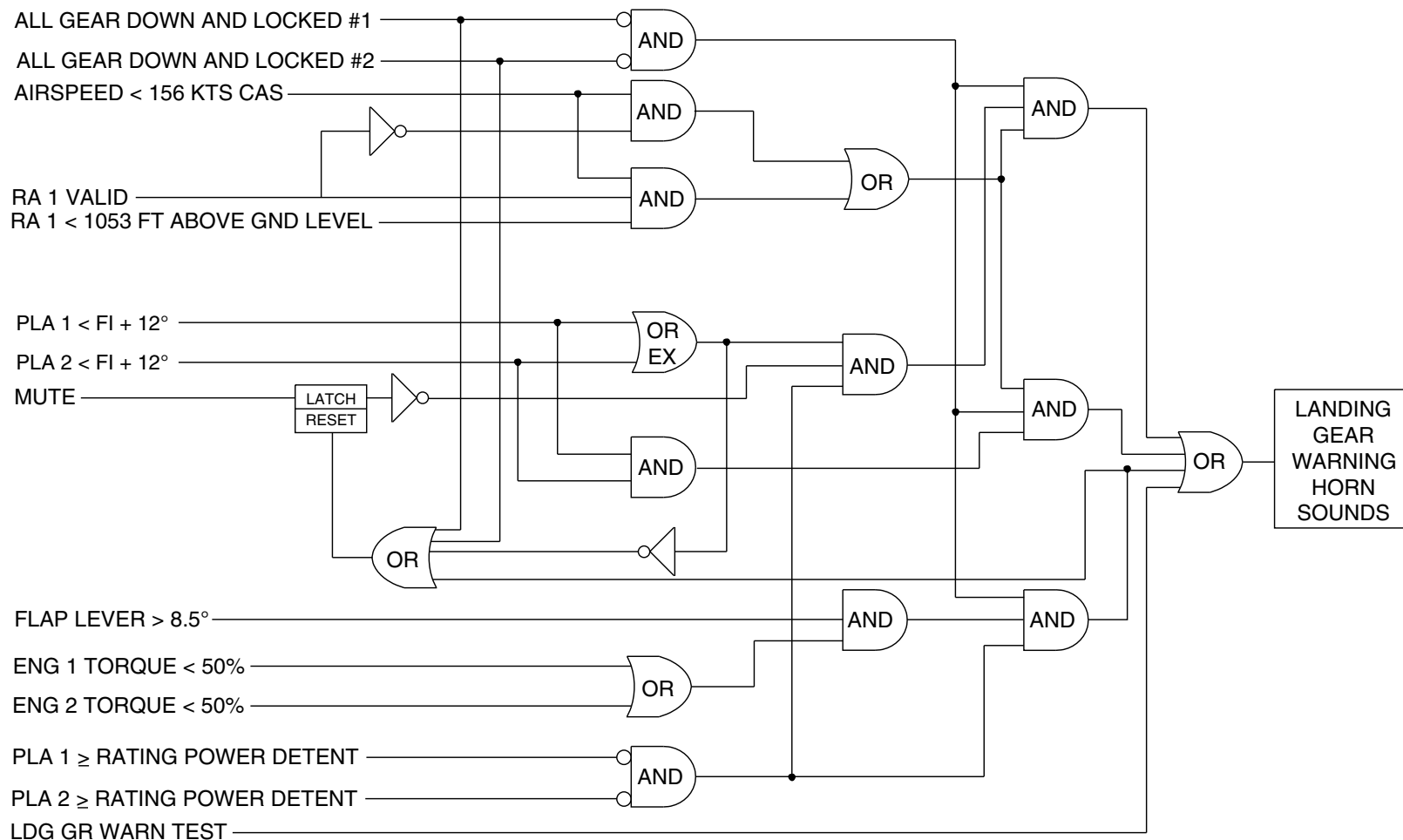
The landing gear configuration warning has interfaces with other aircraft systems as follows:

Signal	Source
GEARS DOWN AND LOCKED 1	PSEU
GEARS DOWN AND LOCKED 2	PSEU
IAS < 156 ±5 Knots	ADU1, ADU2
AG L < 1053 feet	RA1

PLA 1 < FLIGHT IDLE + 12 degree	FADEC
PLA 2 < FLIGHT IDLE + 12 degree	FADEC
HORN MUTE	LANDING GEAR CONTROL PANEL
FLAP LEVER > 8.5°	FCU
ENG 1 TORQUE < 50%	FADEC
ENG 2 TORQUE < 50%	FADEC
PLA1 ≥ RATING POWER DETENT	FADEC
PLA2 ≥ RATING POWER DETENT	FADEC
LDG WARN TEST	LANDING GEAR CONTROL PANEL



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Landing Gear Warning System Logic
Figure 1

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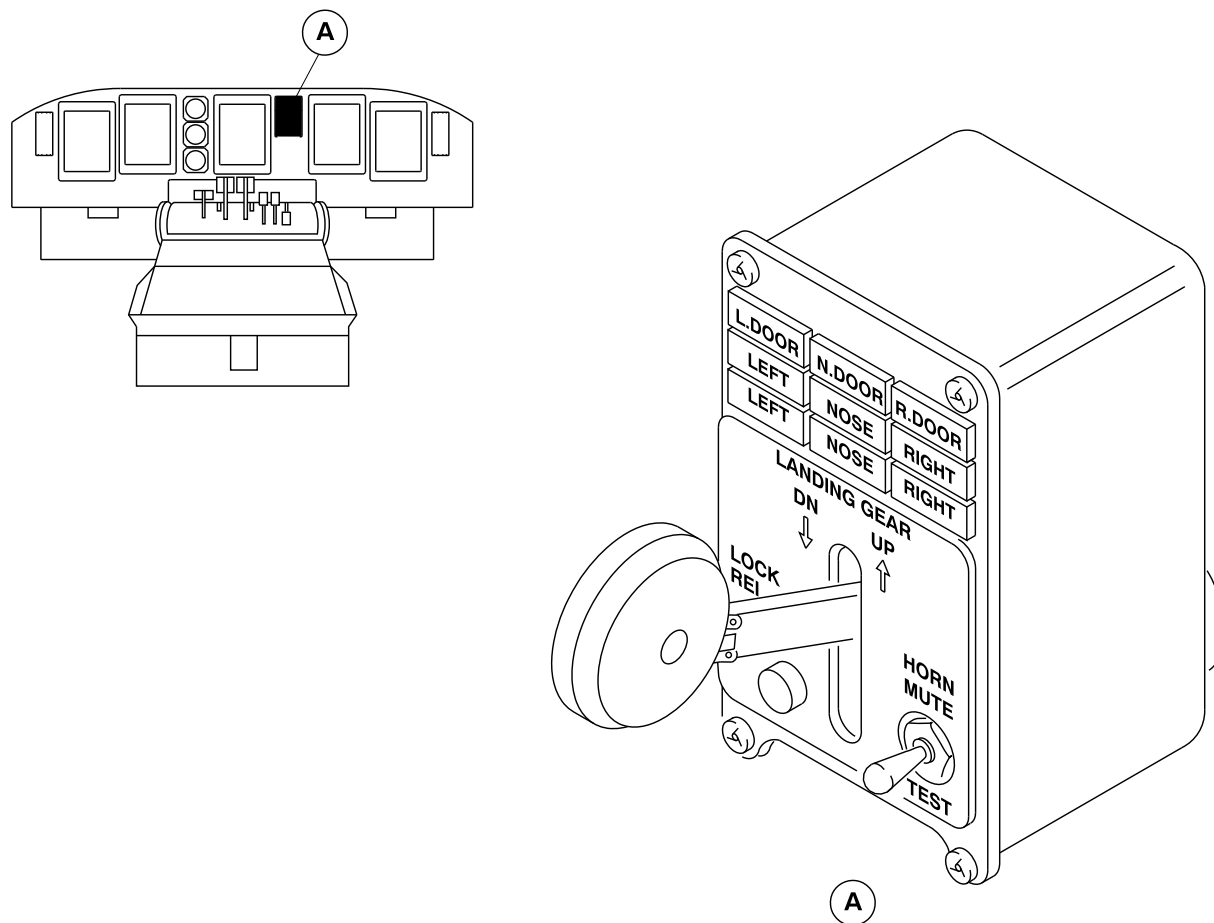
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Landing Gear Control Panel
Figure 2

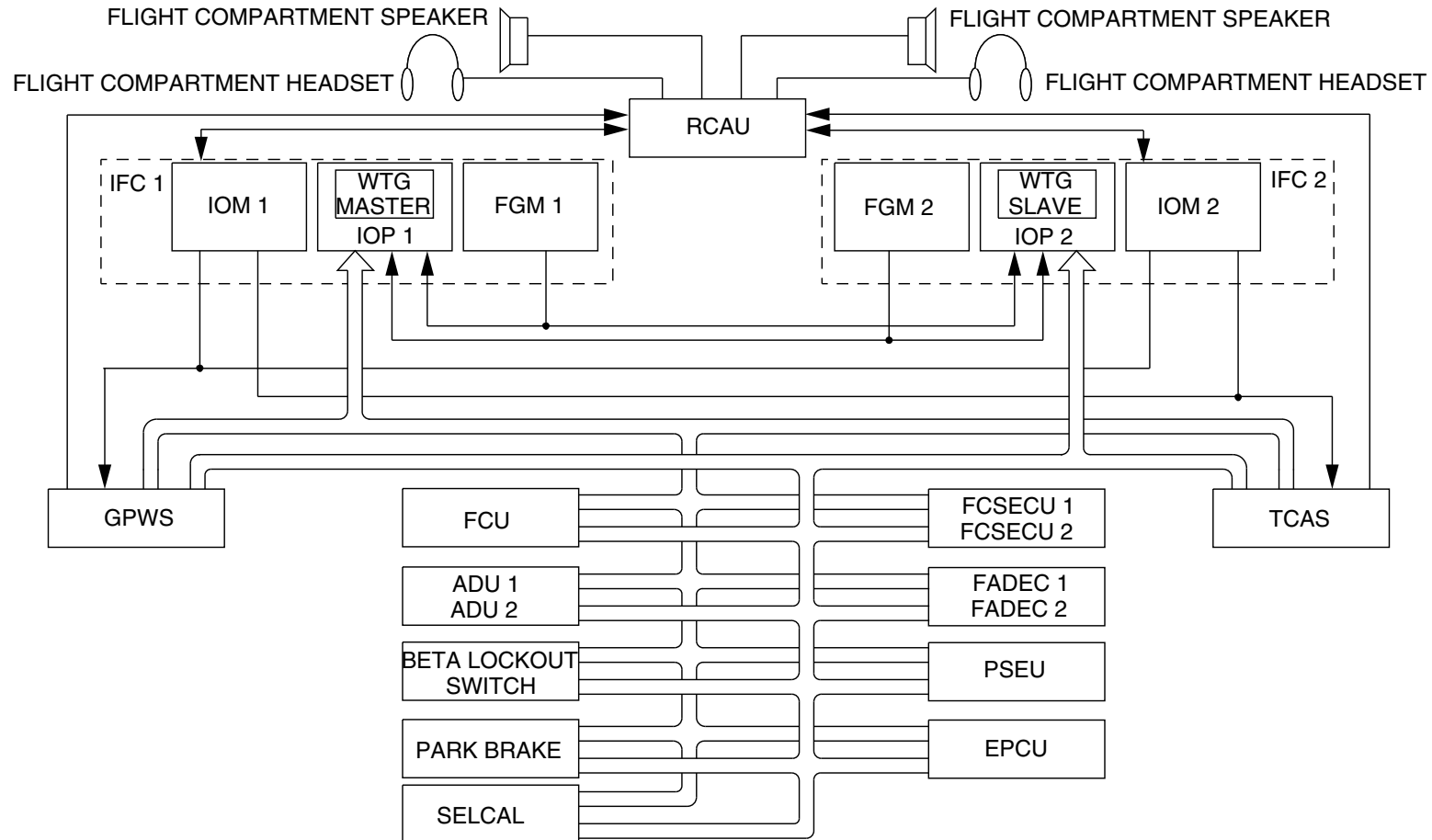
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Landing Gear Warning System
Figure 3

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